Analysis II

Lecture Transcript & Notes

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CHAPTER 1

Monday 16. February 2026

Course Overview, $\mathbb{R}^n$ as Euclidean Space, and Metric Spaces

Course Context & Overview

This is the opening lecture of **Analysis II: mehrere Variablen** (Spring 2026) at ETH Zürich, taught by Prof. Joaquim Serra. The course extends single-variable calculus (**Analysis I**) to multivariable functions, Euclidean spaces $\mathbb{R}^n$, differential topology, partial derivatives, surface integration, and vector calculus leading up to the Gauss–Green–Stokes Divergence Theorem.

Key Lecture Highlights:

- **1D FTC vs. Multivariable Calculus:** Framing the 3D Divergence Theorem as the natural higher-dimensional counterpart to the 1D Fundamental Theorem of Calculus.
- **Course Architecture & Learning Goals:** The five main pillars of Analysis II, pedagogical philosophy on proofs, multi-tiered proof mastery, Lean, and AI tools.
- **$\mathbb{R}^n$ as Euclidean Space:** Vector structure, Euclidean norm, scalar product, Euclidean distance, and formal proofs of the Triangle Inequality and Cauchy–Schwarz Inequality.
- **Metric Spaces:** Abstract metric space definition (X, d), three metric axioms, and canonical examples ($\mathbb{R}^n$, Manhattan metric, S^2, $C([a,b])$ with supremum and L^2 metrics).

Course Administration and Logistics

Mündliches Protokoll [00:00:00 - 00:02:30]

Good morning. Good morning. Thank you. Please, please, please.

Good morning and welcome to Analysis 2 (i.e., Analysis II: mehrere Variablen). I have here a projection with my name, Joaquim Serra, the name of the coordinator, Enric Florit-Simón. Now, this is my introduction. And then you know the times of the lectures, obviously, because you are here.

And I wanted to share, before starting with the mathematical content, some practical information that is very important because we are a large group and the communication and everything needs to be done in a more or less structured way. Otherwise — otherwise we collapse.

So, as a general rule, everything that applied to Analysis 1 applies to Analysis 2, unless otherwise stated. We will try to follow the same rules, and then you know what to expect. In particular, you will have this webpage on the ETH portal that — I can see that from up there is a bit difficult to see, right? — but I will try to make it bigger, yeah.

Course Portal Displayed

The lecturer projects the course website for **Analysis II: mehrere Variablen Spring 2026** on the screen, detailing lecture times (Mondays 08:15–10:00 in ETA F 5, Wednesdays 08:15–10:00 in HG F 7, Thursdays 16:15–18:00 in ETA F 5), exercise classes, exam information, and the bonus program.

Mündliches Protokoll [00:02:30 - 00:05:07]

So, you know that you need to register for the exercise classes. For the communication, please try to use whenever possible the forum. We will look at the forum and reply, and your peers can also reply to questions, because then we can use the collective knowledge, right?

There will be recordings of the lecture, but not until later on, okay? Because it is important for the lecture and for you, but mainly — I mean, for you it is your interest if you want to come or not —, but for the lecture it is very good that you come to the lecture, you ask questions, you stop me when I'm going too fast, these kinds of things. You know, it's very important. So, it is where you meet with your peers and you can communicate, ask questions to your peers. It's better for the learning.

We will tell you more about the exam later on, right? Today I will tell you a bit more what is important to be successful in the course, what to expect about this course, and the details and the rules of the exam will come later on — you don't need to know now. The bonus — there is a 0.25 bonus — will be in questions that you will answer in some exercise classes. We will tell you in advance which ones, and essentially you need to be in the exercise class and answer it. And do not worry, because the rules will be thought out — in any case it's just 0.25 points —, but the rules will be thought out so that if you miss 1 day or casuistics like this, it's not that you get 0 points, right?

Lecture notes: there will be lecture notes similar to those in Analysis 1, and we will put them on this page, hopefully by the end of each week, because I update them after the lecture, mainly depending on your questions, depending on your feedback.

And that's it! I think that the rules are mainly those, and the organization. Is there some question about organization? So, essentially, as I said, if you have doubts, it is like in Analysis 1, unless I say something that is surprising to you, as in Analysis 1. Is this okay? Yeah.

Okay, now I will switch — I have some other things I want to project later, but now, because we cannot record at the same time blackboard and projection, I will switch one moment to the blackboard.

Analysis I vs. Analysis II: High-Level Overview

Mündliches Protokoll [00:05:07 - 00:07:22]

Okay, now, just to start, we will give a very quick — so quick that it's almost a cartoon — but a very quick overview of what Analysis 1 and Analysis 2 are, right? It's very, very quick.

For me, it's very difficult to write big. So, if you don't see something, I will try to say out loud everything I write.

Analysis 1 is calculus of 1 variable (i.e., single-variable calculus). And then, if we got to choose a representative theorem of Analysis 1 that represents the course a bit, what would you choose?

Can you tell me at least 2 answers? So, 2 people can tell me: what is one representative theorem of Analysis 1? Yeah?

Student Answer

The Fundamental Theorem of Calculus.

Mündliches Protokoll [00:07:22 - 00:08:35]

Yeah, I agree! He said the Fundamental Theorem of Calculus. Makes sense, right? Is there some other guess? Yeah, there?

Student Answer

The Intermediate Value Theorem.

Mündliches Protokoll [00:08:35 - 00:10:32]

The Intermediate Value Theorem, okay. I think we need to use this (i.e., catch-box microphone) for the recording; otherwise, I will repeat. Later on, when I get better with throwing the catch-box, we will use that.

So, we said 2 theorems that are indeed important. Perhaps the most representative — I would agree, and I saw many people saying yes — is the Fundamental Theorem of Calculus (FTC), right? And it is this:

$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

Is it correct? Yeah, looks like it, doesn't it?

Now, to unfold, to write, and to understand every detail about this equation over here, you needed to introduce a lot of things. You needed to define what is that: derivative, right? What is this: the Riemann sum, integral. And even — and you spent a lot of time, right? — what is that: real number. It was not easy. Yeah, is there some mistake?

Student Spotting Equation Typo

Shouldn't it be $f(b) - f(a)$ on the right side? You wrote $f(a) - f(b)$.

Mündliches Protokoll [00:10:32 - 00:11:45]

Good! Well spotted! *[corrects the board from $f(a) - f(b)$ to $f(b) - f(a)$]*

$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

Now? Good.

Okay, so this is a fairly compact theorem, yet the ingredients you had to learn to unfold all of this and to understand the meaning of every symbol here were a bit of work.

Now, what will Analysis 2 look like?

Fundamental Theorem of Calculus (Analysis I Baseline)

In single-variable calculus (**Analysis I**), the central, unifying result connecting differentiation and integration is the Fundamental Theorem of Calculus.

Satz 1.1 (*Fundamental Theorem of Calculus (1D)*): Let $f\colon [a,b] \to \mathbb{R}$ be continuously differentiable on $[a,b]$. Then:

$$\int_a^b f'(x)\,dx = f(b) - f(a). \tag{1.1}$$

Unfolding the 1D Fundamental Theorem: Fully understanding Eq. (1.1) requires three foundational pillars established in Analysis I:

(1) **The Real Continuum ($\mathbb{R}$):** Construction of real numbers via Dedekind cuts or Cauchy sequences to guarantee completeness and the existence of limits.

(2) **Differential Calculus ($f'(x)$):** Definition of local rates of change via difference quotients and limits:

$$f'(x) := \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}.$$

(3) **Integral Calculus ($\int_a^b \ldots dx$):** Construction of Riemann sums, partitions, upper/lower Darboux sums, and taking limits as partition mesh sizes vanish.

Multivariable Calculus: The Generalized Divergence Theorem

Mündliches Protokoll [00:11:45 - 00:13:30]

This will be calculus in more than 1 variable — at least 2 variables, because otherwise it's already covered by Analysis 1.

And a representative theorem that would be the equivalent, the counterpart of this one, is the Gauss–Stokes–Divergence–Green Theorem. It has many names, and it's always the same theorem. So, it's the Gauss / Divergence / Stokes / Green Theorem.

Now I will write the theorem, and you will not understand exactly what each thing means. I will tell you intuitively, and then the goal of the course will be that by the end you will be able to understand in full precision everything that I wrote here, okay?

So, the theorem goes as follows.

Mündliches Protokoll [00:13:30 - 00:16:15]

So, we have the space — let me say I will put it in the space $\mathbb{R}^3$, three coordinates of space, right? This thing: x, y, z, or x_1, x_2, x_3, right?

And then in this space we are given a domain, we are given some surface that encloses some interior. So, this is the interior of the surface, and the exterior. And this is a surface S.

What is a surface? It could be a sphere, it could be a sort of the skin of a potato — it's a surface, a two-dimensional object. The interior of the surface — so the filling of the potato, the white part — is Ω; I call it Ω.

And then the equivalent of this theorem in this setup is this: given a vector field $f = (f_1, f_2, f_3)$ — these are three functions of space, right? At each point of space you have three numbers — this is a vector field. And how do we draw vector fields? We draw it like this, right? Like a field of some cereal, but at every point you have an arrow instead of a plant.

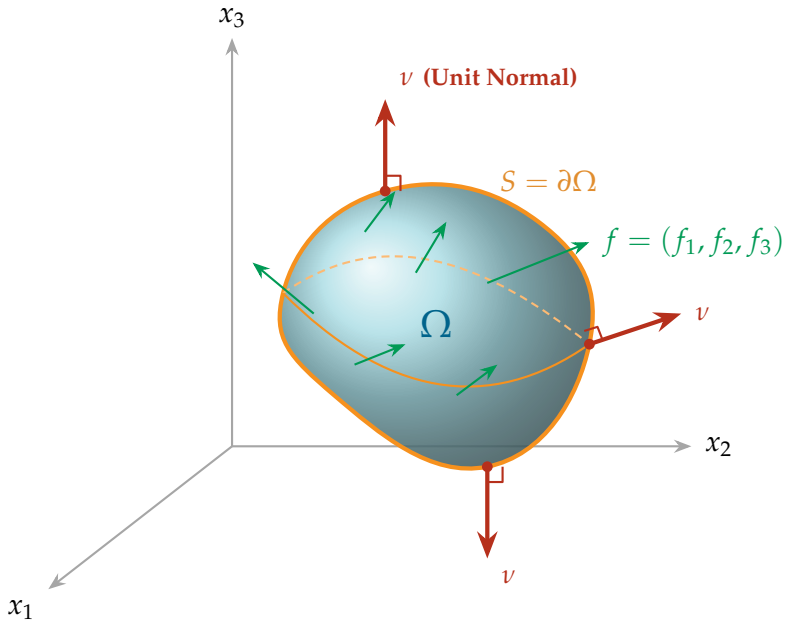
Okay, so this is the vector field f. And the theorem says that the integral in Ω of the derivative of the field, the i-th component with respect to the x_i variable, summed over i (from $i = 1$ to 3), is equal to the flux of the field across the boundary, across S:

$$\int_\Omega \sum_{i=1}^{3} \frac{\partial f_i}{\partial x_i}\, dx = \int_S f \cdot \nu \, dS.$$

Satz 1.2 (*Gauss–Green–Stokes Divergence Theorem in* $\mathbb{R}^3$): Let $\Omega \subset \mathbb{R}^3$ be a bounded domain enclosed by a smooth surface $S = \partial\Omega$, and let $\nu \colon S \to \mathbb{R}^3$ denote the outward-pointing unit normal vector field on S. For any C^1-vector field $f = (f_1, f_2, f_3) \colon \overline{\Omega} \to \mathbb{R}^3$, the total divergence over the volume equals the net flux across the boundary surface:

$$\int_\Omega \sum_{i=1}^{3} \frac{\partial f_i}{\partial x_i}\, dx = \int_S f \cdot \nu \, dS, \tag{1.2}$$

where $\sum_{i=1}^{3} \frac{\partial f_i}{\partial x_i} = \operatorname{div}(f) = \nabla \cdot f$ represents the volume divergence rate, and dS denotes the two-dimensional surface area element.



Structural Correspondence: 1D FTC vs. 3D Divergence Theorem: Notice the exact structural parallel between Theorem 1.1 and Theorem 1.2:

- **Left-Hand Side (Interior Integration of Derivatives):** In 1D, we integrate the derivative $f'(x)$ over the 1D interval domain (a, b). In 3D, we integrate the sum of partial derivatives $\sum_{i=1}^{3} \frac{\partial f_i}{\partial x_i}$ (the divergence) over the 3D volume domain Ω.
- **Right-Hand Side (Boundary Evaluation):** In 1D, the boundary of (a, b) consists of two discrete points $\{a, b\}$, yielding the boundary evaluation $f(b) - f(a)$. In 3D, the boundary of the volume Ω is the 2D surface $S = \partial\Omega$, yielding the boundary flux integral $\int_S f \cdot \nu \, dS$.

Mündliches Protokoll [00:16:15 - 00:18:20]

This is the same theorem you got here in 1 dimension, in several variables. Now the small problem is that maybe some of you in physics or in previous courses have seen some of these notations — probably not, I don't expect you to —, and we will need to introduce each of these notations: these partial derivatives, what is a domain, what is a surface? What is this surface mathematically? How do you model a surface? How do you compute an integral over a surface?

You know how to compute an integral over the real line, right? Over a segment. But over a surface? This will need some definitions and some work. And these objects that are the partial derivatives.

So, here in 1 equation you have a bit of the goal and the syllabus of the course. How will we proceed?

Structure and Syllabus of Analysis II

Mündliches Protokoll [00:18:20 - 00:21:00]

First, we will start looking at the space. In Analysis 1, you did everything in $\mathbb{R}$. Now we will need to introduce $\mathbb{R}^3$, but we are mathematicians, so we will... well, some of us, or some of you are physicists or proto-physicists, who would like to become physicists... so we will use $\mathbb{R}^n$, because we don't want to spend time proving the same theorem in the plane $\mathbb{R}^2$, then in $\mathbb{R}^3$ in space, and then when you go to do general relativity you need $\mathbb{R}^4$, right? We do everything in $\mathbb{R}^n$, where n is the dimension.

So, what is $\mathbb{R}^n$? It is the set of n-tuples of real numbers. This is a definition, okay?

$$\mathbb{R}^n := \{x = (x_1, x_2, \dots, x_n) \mid x_i \in \mathbb{R}\}.$$

Now, this is a point, and these x_i are real numbers. This you know probably because you are doing or have done linear algebra, right? So, this should be quite familiar, but still it's good that we recall.

Mündliches Protokoll [00:21:00 - 00:23:23]

Now, the first part of the course we will be studying $\mathbb{R}^n$ and the properties of $\mathbb{R}^n$. A bit as in $\mathbb{R}$ you studied the axiom of the supremum, right? Or the fact that you can take limits of Cauchy sequences, these kinds of facts, right? The same we will need to study in $\mathbb{R}^n$.

Then there will be some new things, or things that you could also study in $\mathbb{R}$ but become very trivial and now will be much more interesting: the topology of $\mathbb{R}^n$.

So, first: **1.** $\mathbb{R}^n$ as Euclidean space. **2.** Topology.

Then **3.** This object over here: partial derivatives. When you have several variables, you can differentiate a quantity with respect to each of them, right? Differentiation of functions of several variables.

Then **4.** Integration in $\mathbb{R}^n$ here, in a domain of $\mathbb{R}^n$, and on surfaces. But because we want to learn also how to integrate on surfaces, we will do a bit of an introduction to differential geometry: introduction to surfaces and submanifolds. Now, I know it's a lot of words, but we will introduce them one by one.

A surface is a usual 2-dimensional object, right? Typically smooth. When it is a 3-dimensional object in $\mathbb{R}^4$, for instance, we call this a submanifold. You can also call it a surface, but the word submanifold is more precise. Or a 2-dimensional object in a 5-dimensional space — any combination of dimensions.

And then when you know this, you will need to know how to integrate: **5.** Integration over $\mathbb{R}^n$ and surfaces.

And hopefully, then you will know all of these elements. This element — this is the normal vector; I can tell you now what this is: this is just the vector, the unit vector, the vector of length 1 that is normal to the surface, right? If you have a surface, this ν is just the normal to the surface. So, this is easy, but still we will define this in detail.

And after we know all of this, we will be able to prove that equation over here, and work with it, and see why it is useful for many things, okay? So, this is a rough plan of the course.

Syllabus and Topic Architecture for Analysis II

The course **Analysis II: mehrere Variablen** is organized around five core mathematical modules:

1. **$\mathbb{R}^n$ as Euclidean Space:** Formalization of n-dimensional real vector spaces equipped with inner products, Euclidean norms, and metric properties:
$$\mathbb{R}^n := \{x = (x_1, x_2, \dots, x_n) \mid x_i \in \mathbb{R} \text{ for all } i \in \{1, \dots, n\}\}. \tag{1.3}$$
2. **Topology of $\mathbb{R}^n$:** Open and closed sets, interior/boundary/closure, sequences, completeness, compactness (Heine–Borel Theorem), and connectedness.
3. **Differential Calculus in Several Variables (Partial Derivatives):** Directional derivatives, partial derivatives $\frac{\partial f}{\partial x_i}$, total Jacobians, chain rule, implicit function theorem, and constrained optimization (Lagrange multipliers).
4. **Geometry of Surfaces and Submanifolds:** Parametric surfaces, k-dimensional submanifolds embedded in $\mathbb{R}^n$, tangent spaces, and unit normal fields ν.
5. **Multivariable Integration and Vector Calculus:** Riemann integration over n-dimensional domains, change of variables (Jacobian determinants), line/surface integrals, differential forms, and the integral theorems of Green, Gauss, and Stokes.

The Role and Philosophy of Mathematical Proofs

Mündliches Protokoll [00:23:23 - 00:26:22]

Now, I wanted to do a very small survey about your attitude towards proofs, okay? This is important to mention.

I will put two columns. The left column is: I find proofs — so, by proofs I mean rigorous, long, very careful mathematical proofs with all details, okay? This is what I mean by proofs.

Here I will put: I find proofs essential, illuminating, useful to clarify, the best part of any course.

And on this column over here I will put: I find the proof unnecessary, confusing, mysterious, no clear rules of the game.

Now, okay, this is extreme — I'm making this on purpose, this is extreme, right? This is extreme. I know that most of you would be maybe in the middle. But I would like to know which of you lean more towards this opinion, or more towards this opinion, okay?

So, let's start with this one. Who is here? Who loves proofs? Okay.

And here? Okay, it's interesting.

So, it's roughly... there are a bit more people that love proofs, but okay. So, there is no correct answer, right? So, I would be — well, depending on where — in between.

Classroom Interaction: Voting on Mathematical Proofs

The lecturer writes a two-column comparison on the blackboard titled **Attitude towards proofs**:

- **Positive Column:** Essential, Illuminating, Useful, the best part.
- **Negative Column:** Unnecessary, Confusing, Mysterious, No clear rules of the game.

Students raise hands to vote on which side better aligns with their personal learning experience.

Mündliches Protokoll [00:26:22 - 00:28:15]

So, why am I asking this? Does it have some correlation with physicists and mathematicians, or not? Sometimes. What do you think? It could make sense; not always, there are physicists that... but it could make sense.

So, I think that about this discussion of proofs I wanted to use 10 minutes today to tell you two things: first, why I believe — many people can have many answers, but what I believe — that proofs are important; and what we will test in this course, right? Because it's a bit correlated, of course.

The first thing is that in algebra, many times you have a theorem that is very beautiful, very clean, and you can apply it in many situations without modifying it, right? In analysis, as we will see, sometimes the theorem that you learned doesn't apply exactly to the situation you need. Then you need to know how to modify the theorem a little bit. And the only way to do that is to know the proof, and then modify some step here and some step there.

Also, many times the proof contains the ideas on why the theorem is true, right? The theorem is a bunch of symbols, as we wrote there, sometimes, or statements, but it's not very illuminating. It's okay to know that a fact is true, but when you know a bit why, it's more useful many times, right?

Mündliches Protokoll [00:28:15 - 00:30:15]

So, two things: being able to export and to use in situations that are new, and also understanding better — meaning being also sometimes able to remember the theorem. Because if you get quick at remembering proofs and arguments, sometimes when you don't remember all the assumptions of a theorem, you can figure them out just by repeating the proof in your mind, right?

This doesn't mean that you need to learn the proof in all the details all the time. For the proof to be useful and illuminating, it's very important that when you have a super long proof, you extract the main ideas of the proof. So, you need to have several levels in your head. For a proof that is that long, you can remember 3 sentences — like hints — that are the key ingredients of the proof for you. And it's a very good exercise to do that. Then you expand! You learn how to expand these summaries.

And how do you learn how to expand? You learn how to expand hints by doing exercises. That's why the exercises are important. If you become very good at solving exercises, you will be able to compress the information very well, because then you don't need to remember all the details. You just remember the highlights of the proof, and then

you fill in the details, because you are an expert in doing that, right? So, becoming an expert in exercises is good to find proofs more interesting and more useful, and not that much of an effort.

Einblick: **Didactic Philosophy: Multi-Tiered Proof Mastery**

Prof. Serra highlights a fundamental learning strategy for theoretical analysis:

(1) **Level 1 (The Core Idea / Hint):** Compress a 3-page proof into 2–3 pivotal mathematical insights (e.g., **"For FTC, rewrite the total change as a telescopic sum over a partition and apply the Mean Value Theorem"**).

(2) **Level 2 (Structural Expansion):** Unfold the core hints into formal claim structures, bounding inequalities, and limit setups.

(3) **Level 3 (Computational Mechanics):** Fill in epsilon-delta estimates, algebra, and algebraic routine through regular problem set practice.

Attempting to memorize proofs line-by-line without Level 1 abstraction leads to rapid cognitive overload.

Mündliches Protokoll [00:30:15 - 00:33:00]

Now, some people here who find proofs confusing, mysterious, or unnecessary feel that way because sometimes the rules of the game, and why the game is interesting, are not that clear. So, why, when you finish a proof, can you say that the proof is finished, right? Because the professor says so, that this is the proof, Q.E.D.? Is it common sense? Most people see that: *„Oh, this looks like an argument that is conclusive“* — most people, but not every time! And there are very subtle arguments.

So, that's why we need to spend sometimes a lot of time on things that seem obvious. Because if I tell you: a continuous function crosses the zero axis at some point, okay, of course it crosses! The Intermediate Value Theorem, right? A continuous function crosses the zero line... of course! But you need to prove it, right? And the proof cannot be just the intuition of: I make a drawing, I have the zero line, I have two points — one here, one here —, I have a continuous function, so if I cannot lift the pen from the paper, I need to cross the line at some point. This is not a mathematical proof, but it's a correct intuition.

Now, what can help with this? I think something that can help with the rules of what is a proof and what is not a proof... *[switches on the projector and opens browser]*

Mündliches Protokoll [00:33:00 - 00:35:40]

If someone is interested, you can Google **Natural Number Game**. This is serious stuff — I'm not... this is very serious stuff, it's difficult! I mean, I know that there are some cartoons, but this is a serious, difficult game.

And now seriously, in the light of artificial intelligence and all that is going on, one argument that is very interesting about proofs is that — maybe some of you didn't realize — a proof is **computable**. So, whether a proof is correct or not can be computed! That's why it's not an opinion, right?

There are several programming languages for proofs — one of the most famous is Lean — where you can write the proof in the language. It's very tedious to write the proof in the language, but now there are tools that help you. And now it's like a computer program: if the program compiles, the proof is correct up to the axioms! You provide the axioms, you provide the reasoning, and if the program compiles, the proof is correct. And if it doesn't compile, it doesn't mean that it's not correct; it means that you need to put more details. Maybe it's not correct, maybe yes, but it's not detailed enough, right?

So, the first argument is that proofs are computable. This means that this ends all discussion: if I write a proof and it's true, that's it, end of the discussion, okay? Which is interesting — so it doesn't matter if Donald Trump says that the fundamental theorem... I mean, no! He can say whatever he wants, but the theorem will be true, okay? Nowadays, this is an interesting aspect.

Mündliches Protokoll [00:35:40 - 00:38:15]

This is also an interesting aspect for these guys, right? *[opens ChatGPT 5.2 on the browser screen]* So, I had on purpose ChatGPT or Gemini or whatever... so, you are using those, right? You are using these — I know that some people say *„don't use"*, well, I know you are using them. And I think you know much better how to use them than people think.

But you see, these guys still — especially with more advanced proofs — hallucinate and say things sometimes that are wrong! And then, knowing how to spot this is another motivation. Knowing how to spot wrong arguments, and being careful with correct and incorrect arguments, is a good motivation to learn how to do correct proofs, okay?

And you can train by doing exercises. If you are not sure what the rules of the game are — what is a proof, what is not a proof — try to go to the Natural Number Game and play a bit, and you will see what a fully rigorous proof up to the end is.

Mündliches Protokoll [00:38:15 - 00:40:10]

And why don't we do this all the time? Because the next thing I wanted to tell you is that in Analysis 2, the pace will be quicker than in Analysis 1. We'll start roughly at the same pace doing many details, and then we will go a bit faster. What does faster mean? That I will not be as detailed, you see? Some things will be: *„Because of this and that, then this follows."* And then, if you don't see that this follows — if many don't see — you ask a question: *„Wow, why does this follow? We don't see it."* Then we can provide details. But we will not be able to stop all the time to provide details, because some of the arguments you already know how to do — the Analysis 1 arguments. I will say: *„Okay, this is like in Analysis 1"*, right?

So, the pace is quicker and quicker, and the more you advance in your studies, the quicker it will be. This doesn't mean that you can be okay with not understanding, right? If there is some statement you don't know how to prove, you can try by yourself, you can ask your peers, you can unfold everything. And this is a very good exercise.

So, you need to do your exercises, and you need to try to understand all the steps of the proof — meaning that ideally, with time and effort, you should be able to go back to the axioms, right? You don't need to do it every time; you just need to know that you are able to. And then you are okay. So, if you can do this with most proofs, it means that you are following the course very well.

Is there something else I wanted to tell you before we start with the lecture? I don't think so. So, we have roughly 10 minutes before the pause. I will start with the proper lecture in the second half. And then let me ask you: do you have some questions about what I said in this first part, a bit the plan of the course, how to be successful in the course, what do we expect from you?

If there are no questions, let me summarize in 3 points what the main learning goals are, okay?

1. 1. Point number 1, which is very relevant to both physicists and mathematicians, is **being able to do computations**. Computations mean advanced computations: compute the solution of some ODE, several variables stuff, compute some integral (why the area of the sphere of radius 1 is 4π, you need to be able to compute these integrals with a concrete numerical value).
2. 2. You need to be able to **follow proofs and follow the logic**. This you train with exercises (some computational, some theoretical) and with the lecture notes.
3. 3. The third leg that I will introduce a bit is **assessing if a proof is correct**. This is something we didn't test in the past, but now it's becoming more and more important: giving you some proof with a flawed argument (sometimes I make mistakes, so then it's good because you can practice this!), and spotting subtleties where an AI like ChatGPT made an hallucination.

Mündliches Protokoll [00:40:10 - 00:40:20]

[A student in the audience raises a hand to ask a question] Ah, there? Is there a question? Yeah, okay. Thank you, sorry, I cannot see very well... Okay, now this will be tricky. *[throws the catch-box microphone to the student]* Oof, sorry!

Student Question on Proof Rigor in Exams

I wanted to ask: how rigorous do the proofs have to be in the test? Do we have to do them like the Analysis 1 test, where we don't have to write down the axioms every time or we just mention the most important theorem, or do we have to do very rigorous proofs?

Mündliches Protokoll [00:41:23 - 00:44:34]

Let me put the question a bit differently. The proofs are always **very rigorous**. The only thing is that they don't need to be as **detailed**! You see?

You can make some claims, and you don't need to justify to the smallest detail every single claim. In Analysis 1, you see, the difference of the theorem is clear. So, what

would be a proof... how would you summarize in a sentence a proof of this equation over here?

[points at the 1D Fundamental Theorem of Calculus $\int_a^b f'(x)\,dx = f(b) - f(a)$ on the board]

Fundamental Theorem of Calculus. So, can someone tell me 1 sentence that kind of explains the idea of the proof of that?

Student Response via Catch-Box

We can show that the derivative of the primitive converges to the actual value of the function at, I think it was a or no, b.

Mündliches Protokoll [continued]

The derivative of the... yeah, so the derivative with respect to this value, right? Yeah, this is a good answer.

So, everyone has... this is what I was telling you about before. When you see something like this, the first time you see the whole proof, and then there needs to be something that you remember — some summary of the proof that, for each person, might be a bit different. There are also several ways to do the proof. And then with this, you are able to reconstruct the details.

You want to know what I remember when I see this? I always remember the **telescopic sum**. You know what a telescopic sum is, right? So, when I see this, I see that!

[draws interval partition and telescopic sum equations on the lower board]

The proof in 2 minutes, right? You take your interval $[a, b]$, and you divide it into tiny, smaller intervals. This is x_0, this is x_N (capital N), and this is x_i. And then you write $f(b) - f(a) = f(x_N) - f(x_0)$ equals the sum of increments. You unfold the telescope!

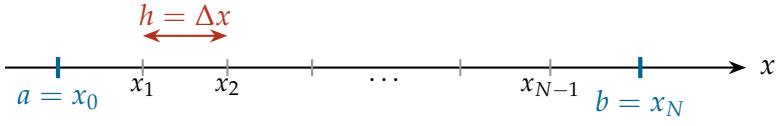
If you call the distance between two consecutive points h, you multiply and divide by h, and each of these pieces is approximately the derivative. End of the proof, okay?

I will not go at this speed, but this is a way to remember the proof, and you need to find your own way in each proof. Then it's not that hard to remember them.

Okay, so it's $9:00$. We do the 15-minute pause, and then we start with the proper lecture.

Telescopic Sum Representation of FTC

To illustrate how a complex 1D proof is compressed into a single memorable cognitive anchor, consider the Fundamental Theorem of Calculus (Theorem 1.1).



Proof Idea via Telescopic Partitioning

Let $a = x_0 < x_1 < x_2 < \cdots < x_N = b$ be a uniform partition of $[a, b]$ with mesh width $h = \frac{b-a}{N}$. The total difference $f(b) - f(a)$ decomposes as a *„telescopic sum"*:

$$\begin{aligned} f(b) - f(a) &= f(x_N) - f(x_0) \\ &= \sum_{i=1}^{N} (f(x_i) - f(x_{i-1})) \\ &= \sum_{i=1}^{N} \frac{f(x_i) - f(x_{i-1})}{x_i - x_{i-1}} \cdot (x_i - x_{i-1}). \end{aligned} \tag{1.4}$$

By the Mean Value Theorem (MVT), on each sub-interval $[x_{i-1}, x_i]$ there exists $\xi_i \in (x_{i-1}, x_i)$ such that $\frac{f(x_i)-f(x_{i-1})}{x_i - x_{i-1}} = f'(\xi_i)$. Substituting this into (1.4) yields a Riemann sum:

$$f(b) - f(a) = \sum_{i=1}^{N} f'(\xi_i)\, \Delta x \xrightarrow{N\to\infty} \int_a^b f'(x)\, dx.$$

Q.E.D.

15-Minute Lecture Break

The lecturer announces a 15-minute mid-lecture break at 09:00. The video recording pauses during the break and resumes at timestamp 00:45:00.

$\mathbb{R}^n$ as Euclidean Space

Mündliches Protokoll [00:45:00 - 00:47:19]

So, I will do my best to respect the schedule, okay, and the times. But please, I would ask you to also do your best to respect the 15-minute break and be here 30 seconds before the bell rings, because otherwise we lose 2 minutes that, of course, are not subtracted from the exam — it's just that I rush some proof, right? So, it's in your interest to be on time.

So, $\mathbb{R}^n$, as we said before, is the set of n-tuples of real numbers, right?

$$\mathbb{R}^n = \{x = (x_1, \ldots, x_n) \mid x_i \in \mathbb{R}\}.$$

Here $n \in \mathbb{N}$ is the dimension of the space.

Can you see from up there? Yeah? You have very good eyes, because I couldn't see anything. Okay.

Now, $\mathbb{R}^n$ by itself — well, just vectors of real numbers — is not that interesting; we will need to add some extra structure. So, what is the Euclidean structure? You already know, I think, but we will say it again.

So, in $\mathbb{R}^n$, what can we do? We can measure... we can, well, sum vectors:

$$x + y = (x_1 + y_1, \ldots, x_n + y_n).$$

If $\lambda \in \mathbb{R}$, we can multiply by a scalar, a real number:

$$\lambda x = (\lambda x_1, \ldots, \lambda x_n).$$

You know that from linear algebra very well.

Definition 1.3 (*Euclidean n-Space*): For any $n \in \mathbb{N}$ ($n \geq 1$), the *„Euclidean n-space“* (or n-dimensional real space) $\mathbb{R}^n$ is defined as the Cartesian product of n copies of the real line $\mathbb{R}$:

$$\mathbb{R}^n := \underbrace{\mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R}}_{n \text{ times}} = \{x = (x_1, x_2, \ldots, x_n) \mid x_i \in \mathbb{R} \text{ for all } i \in \{1, \ldots, n\}\}. \quad (1.5)$$

An element $x \in \mathbb{R}^n$ is called a *„point“* or a *„vector“* in $\mathbb{R}^n$, and $x_i \in \mathbb{R}$ is referred to as the i-th *„coordinate“* (or component) of x. The integer $n \in \mathbb{N}$ specifies the *„dimension“* of the space.

Vector Addition and Scalar Multiplication in $\mathbb{R}^n$

The set $\mathbb{R}^n$ possesses a canonical real vector space structure:

$$x + y := (x_1 + y_1, x_2 + y_2, \ldots, x_n + y_n) \quad \text{(Vector Addition)} \quad (1.6)$$

$$\lambda x := (\lambda x_1, \lambda x_2, \ldots, \lambda x_n) \quad \text{(Scalar Multiplication, } \lambda \in \mathbb{R}) \quad (1.7)$$

Mündliches Protokoll [00:47:19 - 00:50:09]

And then the Euclidean structure means that we can also measure distances between points, right?

So, we have the norm — the Euclidean norm — and distance. First, the norm: the norm of a vector x is the square root of the sum of the squares of the components:

$$\|x\| = \sqrt{x_1^2 + \cdots + x_n^2}.$$

This is the length of the vector.

In the plane, let's say this is x_1 and this is x_2, and this is the point x. The length of this vector x is $\sqrt{x_1^2 + x_2^2}$ — Pythagoras, right? Pythagorean theorem, and the same applies to every dimension.

And we can also measure scalar products or inner products that have to do with angles, right? The scalar products. So, we will use these notations: $x \cdot y$ or sometimes $\langle x, y \rangle$, both are fine:

$$x \cdot y = \langle x, y \rangle := \sum_{i=1}^{n} x_i y_i.$$

This shouldn't be too surprising: scalar product.

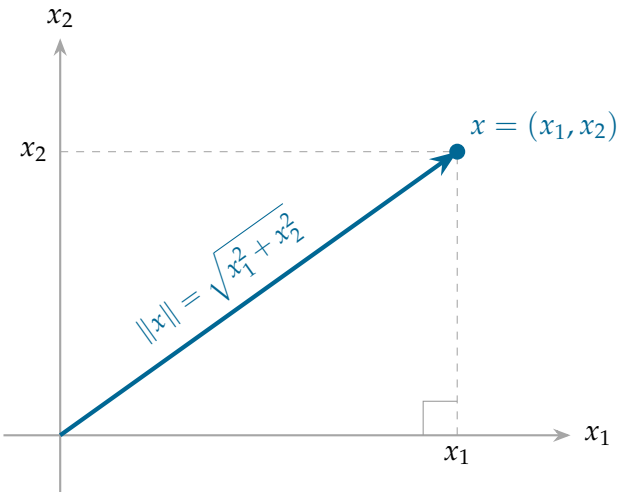
Definition 1.4 (*Euclidean Norm and Scalar Product*): For any $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$ in $\mathbb{R}^n$:

(1) The *„Euclidean norm“* (or length) of x is given by:

$$\|x\| := \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2} = \sqrt{\sum_{i=1}^{n} x_i^2}. \quad (1.8)$$

(2) The *„Euclidean scalar product"* (or dot product / inner product) of x and y is given by:

$$x \cdot y = \langle x, y \rangle := \sum_{i=1}^{n} x_i y_i. \tag{1.9}$$



Mündliches Protokoll [00:50:09 - 00:50:59]

And then, I said the norm of a vector is the length of a vector. And what is the distance between points?

If I take two points here — one point x and one point y over here — the distance between the two points is the length of the vector that joins them, right? The distance between x and y is the length of the vector that joins them:

$$d(x, y) := \|y - x\|.$$

This is a definition. This is the Euclidean distance.

Definition 1.5 (*Euclidean Distance*): The *„Euclidean distance"* $d(x, y)$ between two points $x, y \in \mathbb{R}^n$ is defined as the norm of their difference vector:

$$d(x, y) := \|y - x\| = \sqrt{\sum_{i=1}^{n} (y_i - x_i)^2}. \tag{1.10}$$

The Triangle Inequality and Cauchy–Schwarz Inequality

Mündliches Protokoll [00:50:59 - 00:53:34]

Now, the first lemma I will prove about Euclidean distance is the triangle inequality.

Lemma: Triangle inequality.

Do you know... do some of you know the triangle inequality? Have you seen that? How many people have not seen the triangle inequality? Ah, you all know the triangle inequality. Okay, then I will be quick on that.

So, the triangle inequality states this: for all $x, y, z \in \mathbb{R}^n$,

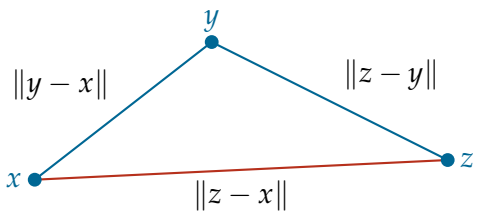
$$\|z - x\| \leq \|y - x\| + \|z - y\|.$$

Maybe you have seen the triangle inequality with the absolute value, but you have also seen that one with the norm, with the length, right?

What this says is that whenever you have three points x, y, z in space in any position, the distance between x and z is always smaller than or equal to the sum of these two distances, right? And actually, it is equal when the point is aligned with the two, right? But I'm not saying this yet. So, let's prove that.

Lemma 1.6 (*Triangle Inequality*): For all vectors $x, y, z \in \mathbb{R}^n$, the Euclidean norm satisfies:

$$\|z - x\| \leq \|y - x\| + \|z - y\|. \tag{1.11}$$



Mündliches Protokoll [00:53:34 - 00:55:34]

Equivalently, I'm saying that:

$$\|x + y\| \leq \|x\| + \|y\| \quad \forall x, y \in \mathbb{R}^n. \tag{$*$}$$

Why is this equivalent to that? Well, equivalent means that if you know how to prove this, you easily prove that, right? Yeah...

Student Question on Vector Equivalence

If we look at the vector representation of x and y... if we draw them as vectors, and then if we put z as the origin, we have the same... the idea is actually the same.

Mündliches Protokoll [00:55:50 - 00:57:09]

Yeah! You put one of the points as the origin; that's correct. You can put z as the origin, and then you have the two vectors. Or another way to do this — it's the same, it's equivalent — is you can do a change of variables, right?

You can call $a := y - x$, and $b := z - y$. And then automatically, if you sum $a + b$, it is $z - x$, right? So, if you want to go from here to there, you just make this substitution, and it works both ways, right? Because if you do this, you will get $\|a + b\| \leq \|a\| + \|b\|$.

Now, this is as detailed as I will be — again, a bit of the difference between Analysis 1 and 2. If this equivalence poses some trouble — it shouldn't pose too much — you

can work a bit by yourself; it's a nice exercise. And if you cannot, ask later.

Okay, so since we convinced ourselves that the two statements are equivalent and this one is shorter, I will prove this one. So, now the goal is to prove that inequality (∗).

Proof of the Triangle Inequality

Mündliches Protokoll [00:57:09 - 01:01:21]

So, I start writing... let me call this inequality star (∗).

So, what I want to show is equivalent after squaring, right? I square and I say:

$$\|x+y\|^2 \leq \big(\|x\|+\|y\|\big)^2 = \|x\|^2+\|y\|^2+2\|x\|\|y\|.$$

That inequality is equivalent to this one because it's an inequality between non-negative numbers, and squaring is a monotone function. This is the part that I say in words, but I will not write, because we are in Analysis 2, okay? This you learned in Analysis 1.

Now, once I write it in this form, I can expand what the norm squared is. This is equal by definition to:

$$\sum_{i=1}^{n}(x_i+y_i)^2.$$

Now, these are real numbers, so I can expand the squares:

$$\sum_{i=1}^{n}(x_i^2+y_i^2+2x_iy_i) = \|x\|^2+\|y\|^2+2(x\cdot y).$$

So, what I'm left with... I know it's a bit like a snake, but these are equalities... what I'm left with is proving that this thing is less than or equal to this thing! And you see, here I can simplify these, right? I can simplify because they are on both sides.

So, what I want to show is that $2(x\cdot y)\leq 2\|x\|\|y\|$, or equivalently:

$$x\cdot y \leq \|x\|\|y\|. \tag{⋆}$$

And this is an inequality that is famous. Now I will prove it in case you don't know it, but this is the Cauchy–Schwarz inequality.

Reduction of Triangle Inequality to Cauchy–Schwarz

To establish (∗), squaring both sides yields the equivalent inequality:

$$\|x+y\|^2 \leq \|x\|^2+\|y\|^2+2\|x\|\|y\|. \tag{1.12}$$

Expanding the left-hand side using the definition of the Euclidean norm:

$$\begin{aligned}\|x+y\|^2 &= \sum_{i=1}^{n}(x_i+y_i)^2\\ &= \sum_{i=1}^{n}(x_i^2+y_i^2+2x_iy_i)\\ &= \sum_{i=1}^{n}x_i^2+\sum_{i=1}^{n}y_i^2+2\sum_{i=1}^{n}x_iy_i\\ &= \|x\|^2+\|y\|^2+2(x\cdot y). \end{aligned}\tag{1.13}$$

Comparing (1.13) with the right-hand side of (1.12), canceling $\|x\|^2+\|y\|^2$, and dividing

by 2 reduces the claim to proving:

$$x \cdot y \leq \|x\|\|y\|. \tag{1.14}$$

Mündliches Protokoll [01:01:21 - 01:04:54]

And this is true because it's the Cauchy–Schwarz inequality.

And now, just to warm up and in case you didn't remember or you never saw the proof of the Cauchy–Schwarz inequality, I will prove the Cauchy–Schwarz inequality, right?

So, if I can prove this inequality, you see that the proof of the triangle inequality is concluded. I'm just left with proving this inequality that, as I told you, is called the Cauchy–Schwarz inequality.

So, let me do this... I will put a square that means end of the proof, but up to this ingredient that I will prove next.

Lemma: Cauchy–Schwarz inequality. This holds for all $x, y \in \mathbb{R}^n$.

Proof: In the notes I have put two proofs — there are many ways to prove this. I will give one, okay?

The first observation is that if either $x = 0$ or $y = 0$, the inequality is $0 \leq 0$, which is trivially true.

By this observation, I can assume without loss of generality that both vectors are non-zero. So, *„assume w.l.o.g."* both $x, y \neq 0$.

I guess you have seen this *„assume w.l.o.g."* phrasing many times now that you are in the second semester.

Lemma 1.7 (*Cauchy–Schwarz Inequality*): For all vectors $x, y \in \mathbb{R}^n$, the inner product is bounded by the product of their norms:

$$x \cdot y \leq \|x\|\|y\|. \tag{1.15}$$

Furthermore, $|x \cdot y| \leq \|x\|\|y\|$.

Mündliches Protokoll [01:04:54 - 01:08:19]

Okay, so both vectors are non-zero. Now, what I will do... I want to write what the scalar product is:

$$x \cdot y = \sum_{i=1}^{n} x_i y_i.$$

And now I will do a trick: for every $\lambda > 0$, I have the right to divide and multiply by λ. I will put λ here and λ over here:

$$x \cdot y = \sum_{i=1}^{n} (\lambda x_i) \left(\frac{y_i}{\lambda}\right).$$

I did nothing, because I multiplied and divided by λ.

And now I will use n times an inequality of real numbers. You know that $2ab \leq a^2 + b^2$, which is the same as $(a - b)^2 \geq 0$.

So, this inequality of real numbers I will apply n times to each of these pairs: this will be the a and this will be the b in each case:

$$2(x \cdot y) = 2\sum_{i=1}^{n}(\lambda x_i)\left(\frac{y_i}{\lambda}\right) \leq \sum_{i=1}^{n}\left(\lambda^2 x_i^2 + \frac{y_i^2}{\lambda^2}\right) = \lambda^2\|x\|^2 + \frac{1}{\lambda^2}\|y\|^2.$$

And this holds for every $\lambda > 0$.

Now is where I use that both are non-zero, so I can take:

$$\lambda^2 = \frac{\|y\|}{\|x\|}.$$

We take that λ, and you get:

$$\lambda^2\|x\|^2 + \frac{1}{\lambda^2}\|y\|^2 = \frac{\|y\|}{\|x\|}\|x\|^2 + \frac{\|x\|}{\|y\|}\|y\|^2 = 2\|x\|\|y\|.$$

You see, this simplifies to what you want, and you get $2\|x\|\|y\|$.

This is the proof of Cauchy–Schwarz, and this ends the proof of the triangle inequality.

Algebraic Proof of Cauchy–Schwarz

If $x = 0$ or $y = 0$, then $x \cdot y = 0$ and $\|x\|\|y\| = 0$, so $0 \leq 0$ holds trivially. Assume w.l.o.g. $x \neq 0$ and $y \neq 0$. For any $\lambda > 0$, apply the real number inequality $2ab \leq a^2 + b^2$ (derived from $(a-b)^2 \geq 0$) with $a_i = \lambda x_i$ and $b_i = \frac{y_i}{\lambda}$:

$$\begin{aligned} 2(x \cdot y) = 2\sum_{i=1}^{n} x_i y_i &= 2\sum_{i=1}^{n}(\lambda x_i)\left(\frac{y_i}{\lambda}\right) \\ &\leq \sum_{i=1}^{n}\left(\lambda^2 x_i^2 + \frac{y_i^2}{\lambda^2}\right) \\ &= \lambda^2\sum_{i=1}^{n} x_i^2 + \frac{1}{\lambda^2}\sum_{i=1}^{n} y_i^2 \\ &= \lambda^2\|x\|^2 + \frac{1}{\lambda^2}\|y\|^2. \end{aligned} \tag{1.16}$$

Since $x \neq 0$ and $y \neq 0$, choose the optimal scaling parameter $\lambda > 0$ such that $\lambda^2 := \frac{\|y\|}{\|x\|}$. Substituting λ^2 into (1.16):

$$2(x \cdot y) \leq \left(\frac{\|y\|}{\|x\|}\right)\|x\|^2 + \left(\frac{\|x\|}{\|y\|}\right)\|y\|^2 = \|x\|\|y\| + \|x\|\|y\| = 2\|x\|\|y\|.$$

Dividing by 2 proves $x \cdot y \leq \|x\|\|y\|$. Replacing x by $-x$ yields $-(x \cdot y) \leq \|x\|\|y\|$, establishing $|x \cdot y| \leq \|x\|\|y\|$.

Mündliches Protokoll [01:08:19 - 01:08:53]

Is there any question about this proof?

Student Correction on Index Sign

Shouldn't it be $x + y$ at the beginning of the proof?

Mündliches Protokoll [01:08:54 - 01:09:24]

Ah, yes! Actually it's the same, but yeah, now that you say it, it's better. So, you can put $x+y$ or $x-y$, but to match this inequality... Thank you! Thank you for the... nice!

Q.E.D.

Metric Spaces

Mündliches Protokoll [01:09:24 - 01:12:04]

So, this — $\mathbb{R}^n$ with this structure of the scalar product and length — is the main structure we will use in proofs to do analysis.

But now we need to understand convergent sequences in $\mathbb{R}^n$, Cauchy sequences, and all these things that you understood in $\mathbb{R}$.

To make it more efficient, we will learn all these concepts from a more general viewpoint, that is, from the viewpoint of metric spaces. So that later, when you go to other courses, or later in this course when we have different metric spaces, I don't need to redefine all the time what convergence means, and so on.

Metric spaces.

Now, I need to warn you — especially the physicists — now it will get a bit more abstract! But then we will come back... this will be for a couple of weeks or so. Later we will come back to $\mathbb{R}^n$ — everything in $\mathbb{R}^n$.

But metric spaces and topology are more axiomatic, more abstract. It's good to practice proofs. Because here you need to use axioms; you cannot use your intuition all the time. You need to go back to the axioms of what a metric space is, which I will tell you now, and do the proofs.

Sometimes working, like mathematicians, from a more abstract viewpoint with fewer axioms is also easier. Because if you have fewer axioms... in $\mathbb{R}^n$ we have many axioms: we have the axiom of the supremum for each coordinate, you have many things.

Now, in a metric space, I will introduce 3 axioms. Just 3! And then all the proofs need to use these 3 axioms. So, you don't need to think too much... it's like playing chess — as I sometimes say — like playing chess only with the rooks!

If you play chess only with the rooks, it's not as difficult as playing with many pieces, right? But then you need a bit more abstraction, because it's not what you are used to.

So, we are going to play chess with the rooks, and we will define: a metric space is a pair (X,d)...

***Einblick:* Analogy: Playing Chess with Only Rooks**

Prof. Serra compares abstract metric space theory to playing chess with only rooks. In $\mathbb{R}^n$, one possesses a vast array of algebraic, geometric, and order properties. In a general metric space (X,d), all structures are stripped away except for a single distance function $d\colon X\times X\to[0,\infty)$ satisfying three simple axioms. This constraint forces strict

adherence to first-principles axiomatic reasoning without relying on Euclidean visual intuition.

Mündliches Protokoll [01:12:04 - 01:16:09]

...where X is a non-empty set, and d is a map from $X \times X$ to $[0, \infty)$, which will be called the distance.

Definition of a metric space.

And d is a map from $X \times X$ to the non-negative real numbers $[0, \infty)$ that will be called the distance.

And this distance satisfies three axioms:

Axiom **(1)**: Definiteness. For every pair of points x and y in X, $d(x, y) = 0 \iff x = y$. The distance between a point and itself is zero, of course, because what sort of distance would be non-zero from a point to itself? And if the points are different, the distance will not be zero, because otherwise this distance would not be able to distinguish points, right?

Axiom **(2)**: Symmetry. For every $x, y \in X$, $d(x, y) = d(y, x)$. This axiom is also very reasonable for a distance. In applications, this is not always true if you think of a distance... like the distance of flying with a plane between two cities. You know that when you fly to the west, it usually takes less time than flying to the east, due to the wind, right? To atmospheric drift. So, it's not always true in every situation that the distance is symmetric, but in many situations it is true.

And the third axiom, Axiom **(3)**: The triangle inequality. For all $x, y, z \in X$,

$$d(x, z) \leq d(x, y) + d(y, z).$$

These are the axioms of a metric space.

Definition 1.8 (*Metric Space*): A *„metric space"* is an ordered pair (X, d), where X is a non-empty set and $d \colon X \times X \to [0, \infty)$ is a function (called the *„metric"* or *„distance"*) satisfying the following three axioms for all $x, y, z \in X$:

(1) Definiteness (Positivity):

$$d(x, y) = 0 \iff x = y. \tag{1.17}$$

(2) Symmetry:

$$d(x, y) = d(y, x). \tag{1.18}$$

(3) Triangle Inequality:

$$d(x, z) \leq d(x, y) + d(y, z). \tag{1.19}$$

Mündliches Protokoll [01:16:09 - 01:19:28]

First example of a metric space: the Euclidean space $\mathbb{R}^n$ with the Euclidean distance $d_{\text{Euclidean}}(x, y) = \|y - x\|$, the length of the vector. It is very easy to see that it satisfies the three axioms.

Examples:

1. $(\mathbb{R}^n, d_{\text{Euclidean}})$.

Why do we specify both the set and the distance? Because sometimes the same set has 2 or more interesting distances, right?

For instance, in $\mathbb{R}^2$, I can put the Euclidean distance, or I can put a different distance that is called the New York distance or Manhattan distance:

$$d_{\text{NY}}(x,y) = |x_1 - y_1| + |x_2 - y_2|.$$

Why? Because this is the sort of distance when you live in one of these modern American cities where the streets go north to south and east to west, and you want to go from point x to point y, right? Then you cannot go diagonally because you hit the buildings, and so you need to go like that! And then the distance between two points is this thing. You can check very easily that this still satisfies all the axioms of distance, right?

A **3rd** example, a bit more abstract: when you have any metric space (X,d) and $Y \subset X$ is a subset of it, you can just restrict this distance to Y. The distance $d|_{Y\times Y}$ restricted to Y, together with Y, is another metric space! If you take a subset of your given metric space and restrict the distance, it will always be a metric space.

Example: if this is $\mathbb{R}^3$ with the Euclidean distance, and I take a subset that is the two-dimensional sphere:

$$S^2 := \left\{ x \in \mathbb{R}^3 \mid \|x\| = 1 \right\}.$$

Then (S^2, d) is a metric space with the Euclidean distance. The distance between two points of the sphere — say this one and this one — is just: you take the straight segment and measure the length. And this satisfies the axioms of distance.

You can say: *„Okay, but this distance maybe is not the best, because if you travel by plane, this is not the distance between points."* The distance between points when you travel by plane is the geodesic distance, that is, the length of the arc, the shortest arc on the surface of the Earth. This is another distance!

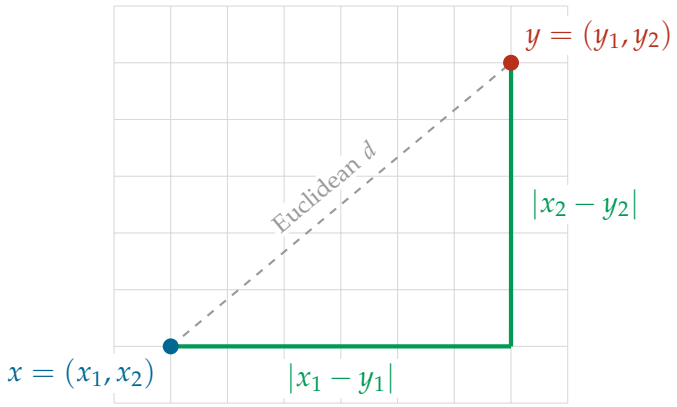
The same set can have multiple meaningful distances depending on the context.

Examples of Metric Spaces

Beispiel 1.9 (*Euclidean Space*): The space $(\mathbb{R}^n, d_{\text{Euclidean}})$ with $d(x,y) = \|y - x\|_2$ is a metric space.

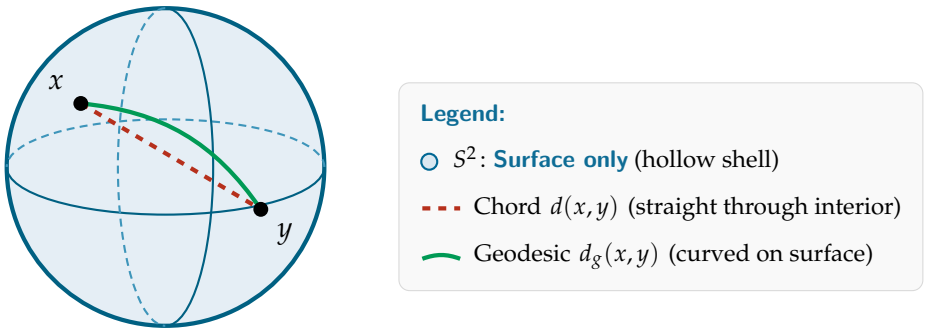
Beispiel 1.10 (*Manhattan / New York Metric on* $\mathbb{R}^2$):** In $\mathbb{R}^2$, the *„Manhattan metric"* (or L^1-metric / taxicab metric) d_{NY} is defined by:

$$d_{\text{NY}}(x,y) := |x_1 - y_1| + |x_2 - y_2|. \tag{1.20}$$



Beispiel 1.11 (*Metric Subspaces and the 2-Sphere S^2*)**:** Let (X, d) be a metric space and $Y \subseteq X$ a non-empty subset. The restriction $d_Y := d|_{Y \times Y}$ defines an *„induced metric“* (or subspace metric) on Y.

For $X = \mathbb{R}^3$ with $d_{\text{Euclidean}}$ and $Y = S^2 := \{x \in \mathbb{R}^3 \mid \|x\| = 1\}$, the pair $(S^2, d_{\text{Euclidean}})$ is a metric subspace.



Mündliches Protokoll [01:19:28 - 01:24:54]

So, one thing that can be useful when you try to do a proof in this setup of metric spaces, which is a bit abstract: remember, you have very few axioms you can use, but then you have less intuition!

So, how can you overcome your lack of intuition? What you do is you think of a specific example. Don't think of a general metric space. A general metric space means a billion things at the same time, right? It is an abstract model of many things at the same time. Pick a concrete example! Pick Euclidean space. Do your proof in Euclidean space using the intuition that you have from that space. And then try to see which axioms you are using. If you... maybe you use a wrong proof and you are not using just those axioms, but then try to modify your proof using your intuition until you can do it with those axioms, okay? This can be useful.

So, whenever things are abstract, don't think always in the abstract; just go to concrete examples. And this applies always, because I have seen many times in more advanced courses during examinations that people know the abstract theory very well, and then when you ask in a concrete case: *„What is this in Euclidean space?“*, people sometimes don't know how to answer. And this is a problem. So, abstract theory is good, but you need to know very well and spend time thinking about concrete exam-

ples.

So, let me give you some more interesting examples of metric spaces, so that you can see things start to be interesting very soon.

Now we saw spaces of points. We can also consider spaces of functions!

You learned what a continuous function is. Let's say $a, b \in \mathbb{R}$ with $a < b$, and you can take X as the space of continuous functions from the closed interval $[a, b]$ to $\mathbb{R}$:

$$X = \{f \colon [a,b] \to \mathbb{R} \mid f \text{ continuous}\}.$$

This is a space of functions.

One point in this space would look something like this: an entire curve! This is one point in this space.

How can we define a distance between functions? Well, in many ways. Something that is used often is the supremum distance, or the maximum distance:

$$d(f,g) := \max_{x \in [a,b]} |f(x) - g(x)|.$$

So, you take two different curves here, and then you look at the maximal oscillation — the maximum difference between the two functions. This is a distance between functions.

If this is 0, are the two functions equal? Yes! Only if? Yes, they are continuous, so you can do that. Exercise!

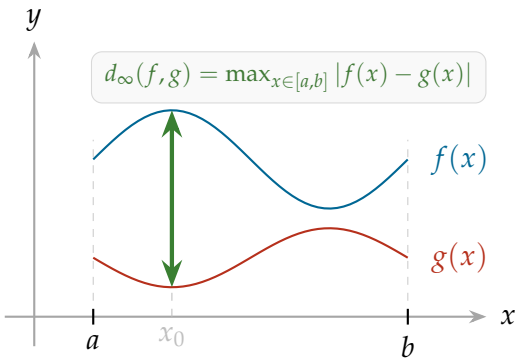
Does it satisfy the triangle inequality? Prove it! Prove that it satisfies the triangle inequality. The symmetry is very easy, because if you swap f and g it's the same, because here you have the absolute value. But the triangle inequality you can prove using the triangle inequality for the absolute value and playing with the maximum.

Beispiel 1.12 (*Function Space $C([a,b])$ with Supremum Metric*)**:** Let $a, b \in \mathbb{R}$ with $a < b$. The space of continuous real-valued functions on $[a, b]$ is denoted by:

$$X = C([a,b], \mathbb{R}) := \{f \colon [a,b] \to \mathbb{R} \mid f \text{ is continuous on } [a,b]\}. \tag{1.21}$$

The *„supremum metric"* (or uniform metric / L^∞-metric) d_∞ on $C([a,b], \mathbb{R})$ is defined by:

$$d_\infty(f,g) := \max_{x \in [a,b]} |f(x) - g(x)|. \tag{1.22}$$



Mündliches Protokoll [01:24:54 - 01:27:09]

Another distance between functions:

$$d_2(f,g) := \left(\int_a^b \big(f(x) - g(x)\big)^2 \, dx \right)^{1/2}.$$

You measure the difference, you square, you integrate, and you take the square root.

This is another distance, a very important one: the L^2 distance! We will not use this in this course, but you will see that you will use this later on many, many times.

So, we will continue next time from here. Thank you! *[applause from students; lecturer packs up equipment]*

Beispiel 1.13 (L^2-Metric on $C([a,b])$): On $X = C([a,b],\mathbb{R})$, another standard metric is the „L^2-metric“ (or root-mean-square integral metric) d_2:

$$d_2(f,g) := \left(\int_a^b \big(f(x) - g(x)\big)^2 \, dx \right)^{1/2}. \tag{1.23}$$

Summary of Key Concepts

- **High-Level Course Vision:** Analysis II elevates single-variable calculus to multivariable Euclidean space $\mathbb{R}^n$, culminating in integral theorems on surfaces (Gauss–Green–Stokes Divergence Theorem).
- **Structure of $\mathbb{R}^n$:** Formalized as a real vector space equipped with the Euclidean scalar product $x \cdot y = \sum x_i y_i$, norm $\|x\| = \sqrt{\sum x_i^2}$, and metric distance $d(x,y) = \|y - x\|$.
- **Fundamental Inequalities:** Rigorously established that the Triangle Inequality $\|x + y\| \leq \|x\| + \|y\|$ in $\mathbb{R}^n$ reduces directly to the Cauchy–Schwarz Inequality $|x \cdot y| \leq \|x\|\|y\|$.
- **Metric Space Axioms:** Introduced the general abstraction (X,d) defined by non-negativity/definiteness ($d(x,y) = 0 \iff x = y$), symmetry ($d(x,y) = d(y,x)$), and the triangle inequality ($d(x,z) \leq d(x,y) + d(y,z)$).
- **Metric Hierarchy & Function Spaces:** Surveyed diverse metrics including the Euclidean metric on $\mathbb{R}^n$, Manhattan taxicab metric d_{NY} on $\mathbb{R}^2$, subspace metrics on S^2, and metrics on continuous function spaces $C([a,b])$ (L^∞ supremum metric d_∞ vs. L^2 integral metric d_2).

CHAPTER 1

Wednesday 18 February 2026

Metric Spaces: Sequences, Completeness, and Euclidean Space $\mathbb{R}^n$

Course Content Overview

In this lecture, we study sequences and convergence in general metric spaces (X, d), proving the uniqueness of limits and introducing subsequences and accumulation points. We examine subtleties in characterizing convergence via subsequences, including a counterexample in $\mathbb{R}$ and the sub-subsequence criterion. Finally, we introduce Cauchy sequences and completeness, analyze complete and incomplete subspaces of $\mathbb{R}$, and prove that n-dimensional Euclidean space $(\mathbb{R}^n, d_{\text{Eucl}})$ is complete via componentwise convergence.

Sequences in Metric Spaces

Definition and Notation

Mündliches Protokoll [00:00:00 - 00:01:45]

Hopefully by the second week, it will only take us two seconds to go from chatting to full silence.

We will continue with metric spaces, provided my voice allows. Remember, last time we introduced this pair *[writes (X, d) on the blackboard]*: X is a set, and d was a map from $X \times X$ to positive numbers, or non-negative numbers.

And now, for today you can forget about the rest of the math you learned, because we are going to use only this and the three axioms. So, you just need to remember three pieces of information: the three axioms.

Now, we will learn again concepts you saw in Analysis I, and the first concept is that of sequence. *[writes Def'n (sequence) on the blackboard]* You saw sequences of real numbers. If X is a set *[writes X set]*, we call a sequence in X a map *[writes we call sequence in X a map]* that we write like this: small x from the natural numbers to X *[writes $x \colon \mathbb{N} \to X$]*. This is small x, this is big X, okay?

Mündliches Protokoll [00:01:45 - 00:02:55]

And, as in Analysis I, instead of denoting this map $x(n)$ *[writes $x(n)$]* as we do with functions, we will use this notation: x_n *[writes x_n]*, right? Sub-index.

This is just one element, the n-th element of the sequence *[writes $\longleftarrow$ n-th element]* that belongs to X *[writes $\in X$]* — belongs to the set X.

When we want to denote the full sequence, we will say: let (x_n) like this be a sequence. Then we can write this *[writes $(x_n)_{n\geq 0}$]*, or we can write this *[writes $(x_n)_{n=0}^{\infty}$]* — all of these are equivalent notations — or we can write *[writes $(x_n)_{n\in\mathbb{N}}$]*, things like that. So, this is our notation for sequences, right? *[draws a horizontal dividing line across the blackboard]*

Definition 1.14 (*Sequence in a Set*): Let X be a non-empty set. A *„sequence“* in X is a mapping:

$$x\colon \mathbb{N} \to X, \quad n \mapsto x(n).$$

Notation 1.15 (*Element and Sequence Notations*): Instead of standard functional notation $x(n)$, the value of the sequence at index $n \in \mathbb{N}$ is denoted using a subscript:

$$x_n := x(n) \in X \quad \text{(the } n\text{-th element of the sequence)}.$$

The entire sequence as a unified mathematical object is denoted interchangeably by:

$$(x_n)_{n\geq 0}, \quad (x_n)_{n=0}^{\infty}, \quad \text{or} \quad (x_n)_{n\in\mathbb{N}}.$$

Functions vs. Index Notation: Formally, a sequence is simply a function whose domain is the set of natural numbers $\mathbb{N} = \{0, 1, 2, \dots\}$ (or starting from 1, depending on the convention). Rather than writing $x(0), x(1), x(2)$, mathematicians use indexed family notation $x_0, x_1, x_2, \dots$ to emphasize the discrete, sequential progression of points in the target space X.

Convergence and Limits

Mündliches Protokoll [00:02:55 - 00:04:33]

Now we are in a metric space, so we can also define convergent sequences and limits.

Definition number two will be convergent sequence and limit *[writes Def'n (convergent seq'ce and limit) on the upper board]*. We say (x_n)... *[writes We say $(x_n)_{n\geq 0}$]*

This is for a metric space: (X, d) is a metric space *[writes (X, d) metric space]*, d is the distance, and (x_n) has limit x in X *[writes has limit $x \in X$]* if...

Well, we can put it in many ways, but essentially what we will say is that if we have a sequence of points in our space, it converges to our given limit point if the distance converges to zero. We have the distance; we need to phrase everything in terms of the distance.

Mündliches Protokoll [00:04:33 - 00:06:22]

So, if the distance between x_n and the limit point x converges to zero as real numbers *[writes if $d(x_n, x) \to 0$ as real numbers]*, right?

These are points in your abstract space, but the distance between the two points is a number. So, I can say: if this distance goes to zero, this means that this point is the limit of that sequence.

Equivalently — if you want something very similar, equivalently *[writes Equivalently]* —: for all ε positive, there exists capital N that depends on ε, such that the distance between x_n and x is less than ε whenever n is bigger than capital N *[writes $\forall \varepsilon > 0$ $\exists N$ such that $d(x_n, x) < \varepsilon$ whenever $n \geq N$]*, right? This is more analogous to what you have seen in Analysis I for sequences.

Now, this is super general, because we saw last time that metric spaces could be a space of functions, could be anything. Now you know how to do convergent sequences in a very general setup.

Definition 1.16 (*Convergent Sequence and Limit*)**:** Let (X, d) be a metric space. A sequence $(x_n)_{n\geq 0}$ in X is said to *„converge“* to a point $x \in X$ (or to have *„limit“* x) if:

$$\lim_{n\to\infty} d(x_n, x) = 0 \quad \text{(as a sequence of real numbers in } \mathbb{R}\text{)}. \tag{1.24}$$

Equivalently, expressed in ε-N language:

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\colon \quad d(x_n, x) < \varepsilon. \tag{1.25}$$

Reduction to Real Sequences: Notice the elegance of Definition 1.16:

(1) The elements x_n and x live in an arbitrary, abstract set X (which could be a set of matrices, a space of continuous functions $C([0,1])$, or a Riemannian manifold).

(2) However, the distance $d(x_n, x)$ is a non-negative **real number**.

(3) Therefore, the sequence of distances $\big(d(x_n, x)\big)_{n\in\mathbb{N}}$ is an ordinary sequence in $\mathbb{R}_{\geq 0}$, allowing us to reduce convergence in any abstract metric space directly to standard single-variable convergence in $\mathbb{R}$.

Mündliches Protokoll [00:06:22 - 00:07:35]

Well, let me once and for all give you the notations we will use. *[draws a horizontal dividing line]*

When the metric space you are using is clear — most of the time in this course will be $\mathbb{R}^n$ with the Euclidean distance, or maybe the complex numbers, that is $\mathbb{R}^2$, right? — then, to say that x_n has limit x, we will use these notations: the limit in n, or when n goes to infinity, of x_n is equal to x *[writes $\lim_{n\to\infty} x_n = x$]*. This we can write when everything is clear from the context, right? When the space X and the distance are clear.

Or we can also put it even more shortly: x_n converges to x *[writes $x_n \to x$]*. This I like a lot; it is very compact, right? As in here — I used it in here *[points at $d(x_n, x) \to 0$]*.

Notation 1.17 (*Limit of a Sequence*): When the underlying metric space (X,d) is clear from the context, the statement that $(x_n)_{n\in\mathbb{N}}$ converges to x is denoted by:

$$\lim_{n\to\infty} x_n = x \quad \text{or simply} \quad x_n \to x.$$

Uniqueness of the Limit

Mündliches Protokoll [00:07:35 - 00:09:20]

Now, given a convergent sequence, we defined what its limit is.

First question: is the limit unique? Let's start from the very basic, fundamental questions.

First lemma... *[notices a smiley face chalk drawing on the board]* Oh, nice! What is this? *[laughs]*

Lemma *[writes Lemma]*: The limit of a sequence is unique. In a metric space (X,d) *[writes (X,d) metric space]*, take a sequence *[writes $(x_n)_{n\geq 0}$]*, and assume that x_n converges to x, and you know also that x_n converges to y *[writes $x_n \to x, x_n \to y$]*, for two given points of X *[writes $x,y \in X$]*.

So, this is a sequence, and this is an assumption: assume that $x_n \to x$ and $x_n \to y$. Then $x = y$ *[writes then $x = y$ and frames it in a box]*, right? *[draws a dividing line under the lemma statement]*

How do we prove this?

Lemma 1.18 (*Uniqueness of the Limit*): Let (X,d) be a metric space. If a sequence $(x_n)_{n\geq 0} \subseteq X$ converges to $x \in X$ and also converges to $y \in X$, then:

$$x = y. \tag{1.26}$$

Proof of Lemma 1.18

Mündliches Protokoll [00:09:20 - 00:11:04]

[draws two points on the board] These are two potential limit points, x and y.

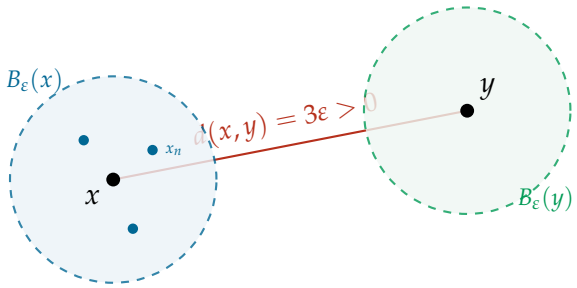
Remember that the first axiom of the distance says that the distance distinguishes points, so the distance between two distinct points is positive. Let me call the distance between x and y two times ε, which is a positive number *[writes $d(x,y) = 2\varepsilon > 0$ in red chalk]*, right? Or, to make it more dramatic, let me put three times ε *[changes 2ε to 3ε]* — and also more symmetric, because the three and the ε are symmetric. Okay.

So, this is 3ε. Then the definition of convergence tells us that after a certain point — for very, very large n — all the elements of the sequence will fall close to the limit x. But at the same time, all the elements of the sequence will fall close to that other limit y. And this should contradict another axiom, which is the triangle inequality, okay?

Here we see the intuition: all the elements cannot be at the same time here and here, right? So, this is what we are going to do. And now we write this in symbols using the

definition, right? *[slides the blackboards upward to reveal fresh blackboard space below]*

Geometric Intuition of the Uniqueness Proof



The Geometric Contradiction: If $x \neq y$, the first metric axiom guarantees $d(x,y) > 0$. By choosing $\varepsilon := \frac{1}{3}d(x,y) > 0$, the open balls $B_\varepsilon(x)$ and $B_\varepsilon(y)$ are strictly disjoint. For sufficiently large n, convergence to x requires $x_n \in B_\varepsilon(x)$, while convergence to y requires $x_n \in B_\varepsilon(y)$. A single point x_n cannot reside in two disjoint sets simultaneously.

Mündliches Protokoll [00:11:04 - 00:12:40]

By definition of the limit, given that ε over there, there exists N — let's say N_x — such that *[writes $\exists N_x \in \mathbb{N}$ st]* the distance between x_n and x is less than ε for all n bigger than N_x *[writes $d(x_n,x) < \varepsilon \; \forall n \geq N_x$]*.

Right? And the same with y: there also exists some N_y such that the distance between x_n and y is less than ε for all n bigger than or equal to N_y *[writes $\exists N_y$ st $d(x_n,y) < \varepsilon \; \forall n \geq N_y$]*.

And then, if I take capital N to be the maximum of these two numbers *[writes $N = \max(N_x, N_y)$]*, what happens? Then for all $n \geq N$ *[writes $\forall n \geq N$]*, since N is bigger than both, I will have: the distance between x and y — this was what? By definition, this was three times ε *[writes $3\varepsilon = d(x,y)$]*, right?

Mündliches Protokoll [00:12:40 - 00:14:15]

Now, I said I would use the triangle inequality, right? So, the distance between x and y should be less than or equal to the distance between x and x_n plus the distance between x_n and y *[writes $\leq d(x,x_n) + d(x_n,y)$ and labels it „Triangle" with circled 3 in red chalk]*. I'm using the triangle inequality with these three points: x, y, and x_n. This is the triangle axiom.

Now, I want to have this, right? So, I need to switch these two arguments. I'm using the second axiom here: symmetry. Axiom number three is the triangle inequality; by axiom number two, I can switch the points in the distance *[writes circled 2 and red arrows switching x and x_n]*.

And then I get that this is strictly less than $\varepsilon + \varepsilon$ *[writes $< \varepsilon + \varepsilon$]*: 3ε is less than 2ε,

contradiction!

The contradiction comes from the fact that I assumed that the points are different and the distance is positive, okay? If the points are the same and the distance is zero, there is no contradiction, of course. So, this finishes the proof of the uniqueness of the limit. *[draws a boundary line around the proof]*

Derivation of the Contradiction

Beweis

Let $(x_n)_{n\geq 0} \subseteq X$ be a sequence such that $x_n \to x$ and $x_n \to y$. We proceed by contradiction: assume that $x \neq y$. By the first metric axiom (**Positivity/Separation**), the distance between distinct points is strictly positive. We define:

$$d(x,y) =: 3\varepsilon > 0 \implies \varepsilon := \frac{1}{3}d(x,y) > 0. \tag{1.27}$$

By the definition of convergence, there exist indices $N_x, N_y \in \mathbb{N}$ such that:

$$d(x_n,x) < \varepsilon \qquad \forall n \geq N_x, \tag{1.28}$$

$$d(x_n,y) < \varepsilon \qquad \forall n \geq N_y. \tag{1.29}$$

Define $N := \max(N_x, N_y) \in \mathbb{N}$. Then for all $n \geq N$, both inequalities (1.28) and (1.29) hold simultaneously. Using the metric axioms, we evaluate the distance $d(x,y)$ for any $n \geq N$:

$$\begin{aligned} 3\varepsilon &= d(x,y) \\ &\leq d(x,x_n) + d(x_n,y) && \text{(by Axiom 3: Triangle Inequality)} \\ &= d(x_n,x) + d(x_n,y) && \text{(by Axiom 2: Symmetry)} \\ &< \varepsilon + \varepsilon = 2\varepsilon && \text{(by (1.28) and (1.29)).} \end{aligned} \tag{1.30}$$

This yields:

$$3\varepsilon < 2\varepsilon \implies 3 < 2,$$

which is an absurd contradiction since $\varepsilon > 0$. Hence, the assumption $x \neq y$ is false, concluding that $x = y$.

Q.E.D.

Q.E.D.

Mündliches Protokoll [00:14:15 - 00:16:35]

Can you tell me now: how is this pace? Is it very slow? Is it very quick? Is it okay?

It's important — I tell you because I have the experience from two years ago: there are some people who, whatever I do, always find it super slow, right? And these are the people who sometimes are more active looking at me and giving feedback. So, it's very important that if people start to be confused, I know before it's too late.

In this direction, today before the break, there will be the election of the class speakers. This is very useful, and I ask you to volunteer, because then I have a more fluid communication. Maybe some people don't come to complain to me that I'm going too fast or too slow, but you can go to your speakers and they will give feedback. Like this,

we will adjust as we go.

[walks back to the upper blackboard] The next thing that you learned in Analysis I is the definition — it will be the same — of a subsequence *[writes def'n (subsequence)]*, right?

If $(x_n)_{n\geq 0}$ is a sequence in a set X *[writes $(x_n)_{n\geq 0}$ seq in X]*, we define... *[notices confused look in audience]* Is there some problem, or do you have some... yeah? No?

Is there some confusion with this proof? I can spend more time; we have time. Yeah, good. *[walks to the front desk and picks up the blue Catchbox throwable microphone]* Now, here's the tricky part, right? *[throws the Catchbox into the audience]* Almost.

Student Question

Um, I do not quite understand where the contradiction is, so why this proof actually works. Um, I do not get the last line you wrote there.

Mündliches Protokoll [00:17:01 - 00:17:08]

This last line? *[points at the inequality line $3\varepsilon \leq d(x_n, x) + d(x_n, y) < 2\varepsilon$]*

Student Question continued

Yes. Like, why we can...

Mündliches Protokoll [00:17:08 - 00:18:25]

Is that okay? This is the definition of 3ε.

[Student nods]

Now, the triangle inequality says that when you have three points, the distance between two of them is shorter than going from one to the third, and from the third to the second, right? This is the triangle; this is less than or equal to.

Now, if we switch these two *[points at $d(x, x_n)$]*, and we use that and that *[points at $d(x_n, x) < \varepsilon$ and $d(x_n, y) < \varepsilon$]* — the switching is by the second axiom: I can switch the two points — I get strictly less than $\varepsilon + \varepsilon$. So, I am getting that $3\varepsilon < 2\varepsilon$, which is a contradiction, right?

[Student responds: „Okay"]

The only place where the contradiction came from is because I assumed that the points are different and ε is positive. If ε is zero, you cannot do that, okay?

[Student responds: „Okay"]

[Student tosses the Catchbox back to the front]

Almost! Thank you.

[Lecturer takes the Catchbox and plugs it into the charger on the desk] Where's the charger? I think it should be somewhere here. Ah, here.

Mündliches Protokoll [00:18:25 - 00:18:57]

Okay, so a subsequence, yeah? *[notices another student raising their hand]* Oh, good.

Student Question

Um, so can we use the terminology of limit if x isn't in the set X? So, for example, if we have like $\mathbb{R}$ but without 0 and like x_n is defined as $1/n$, it converges to 0, but 0 isn't in X.

Mündliches Protokoll [00:19:19 - 00:20:15]

No, in this context, no.

[Student acknowledges: „Okay"]

Here you can use it because you are artificially removing part of your set, but you can measure the distance between 0 and the positive numbers, because you are in $\mathbb{R}$, and $\mathbb{R}$ has its distance.

When you look at the definition, all the things that appear in the definition need to have a meaning. When you have that the distance of x_n and x goes to zero, so that this real number is well defined, you need both points to be in your space. Otherwise, I don't know what the distance is between something that lives in one space and something that lives in another galaxy, right?

[catches the Catchbox, drops it accidentally and picks it up] Okay, oops.

***Einblick:* Ambient Space Requirement for Limits**

In a metric space (X, d), a sequence $(x_n)_{n\in\mathbb{N}} \subseteq X$ can only converge to a limit point x if $x \in X$. For instance, in the metric space $X = (0, \infty)$ equipped with the standard Euclidean distance, the sequence $x_n = 1/n$ does **not** converge in X, because its candidate limit $0 \notin X$. The metric d is a function $d\colon X \times X \to \mathbb{R}_{\geq 0}$; the expression $d(x_n, 0)$ is undefined within the space X.

Subsequences

Mündliches Protokoll [00:20:15 - 00:22:38]

[walks back to the blackboard] Okay, subsequence.

A subsequence is any sequence of the form... *[writes we define subsequence as any sequence of the form $(x_{f(n)})_{n\geq 0}$ where $f\colon \mathbb{N} \to \mathbb{N}$ is some increasing function]* where $f\colon \mathbb{N} \to \mathbb{N}$ is some increasing function, an increasing map.

You know what increasing is, right? When I say increasing, it means: $f(n) < f(m)$ if and only if $n < m$ *[writes $f(n) < f(m) \iff n < m$]*. It's with a strict inequality; this is increasing. When you get this inequality *[writes $\leq$]* for a map, we say that this is non-decreasing *[writes non-decreasing in red chalk]*. Increasing is strict, okay?

And these subsequences, as in Analysis I, it's customary not to denote them this way, but something like this: x_{n_k} *[writes $(x_{n_k})_{k\geq 0}$]*. So, instead of putting $f(n)$ or $f(k)$ — let's say $f(k)$, it's the same — we will write n_k, because n was the original index, and when we are running this index at a faster speed, we write n_k.

Same notation, same concept as in Analysis I, just now that the elements live in some abstract space. *[draws a stepped dividing line on the blackboard]*

Definition 1.19 (*Subsequence*): Let $(x_n)_{n\geq 0}$ be a sequence in a set X. A *„subsequence"* of (x_n) is any sequence of the form:

$$(x_{f(k)})_{k\geq 0}, \tag{1.31}$$

where $f\colon \mathbb{N} \to \mathbb{N}$ is a strictly increasing function, meaning:

$$f(k) < f(\ell) \iff k < \ell. \tag{1.32}$$

Notation 1.20 (*Standard Subsequence Notation*): Instead of writing $f(k)$, it is customary to write $n_k := f(k)$, denoting the subsequence as:

$$(x_{n_k})_{k\geq 0} \quad \text{with} \quad n_0 < n_1 < n_2 < \cdots < n_k < \dots$$

Here, k acts as the new running index that samples the original sequence at strictly advancing positions n_k.

Bemerkung 1.21 (*Increasing vs. Non-Decreasing Maps*): A mapping $f\colon \mathbb{N} \to \mathbb{N}$ is:

- **Strictly Increasing:** $k < \ell \implies f(k) < f(\ell)$.
- **Non-Decreasing (Weakly Increasing):** $k \leq \ell \implies f(k) \leq f(\ell)$.

The definition of a subsequence strictly requires f to be **strictly increasing**, which guarantees that $n_k \geq k \to \infty$ as $k \to \infty$ and prevents the sequence from stalling or repeating elements.

Accumulation Points

Mündliches Protokoll [00:22:37 - 00:24:12]

Okay. Something you already learned in $\mathbb{R}$ also: accumulation points. You know what an accumulation point is. Let's put in here: definition of accumulation points *[writes „def'n (accumulation points)" on the blackboard]*. We will give two things, two definitions.

I will not write it every time, but unless stated otherwise, (X, d) is a metric space *[writes (X, d)]*.

And Y is a subset of X, right? *[writes „$\bullet$ $Y \subset X$ subset"]* We say that $y \in X$ *[writes „We say that $y \in X$"]* is an accumulation point *[writes „is an acc. pt."]* of Y *[writes „of Y"]* if there exists a sequence (y_n) contained in Y *[writes „if $\exists (y_n)_{n\geq 0} \subset Y$"]* — so a sequence of points in Y — such that y_n converges to y *[writes „such that $y_n \to y$"]*.

Bullet point number one.

Mündliches Protokoll [00:24:12 - 00:26:03]

And bullet point number two *[writes •]* is for sequences. If $(x_n)_{n\geq 0}$ is a sequence in X *[writes „$(x_n)_{n\geq 0}$ sequence of X“]*, we say that x is an accumulation point *[writes „we say the x is an acc. pt.“]* of this sequence if there exists a subsequence that converges to x — if there exists an increasing f *[writes „if $\exists f$ increasing st“]* such that $x_{f(n)}$ converges to x *[writes „$x_{f(n)} \to x$“]*.

Or, if you prefer, with the notation you are more used to *[draws a boxed region on the right board]*: if there exists some subsequence *[writes „$\exists x_{n_k}$ subsequence“]* — it's the same, I'm just repeating — a subsequence $(x_{n_k})_{k\geq 0}$ such that, when $k \to \infty$ *[writes „$k \to \infty$“]*, x_{n_k} converges to x *[writes „$x_{n_k} \to x$“]*.

Let (X, d) be a metric space.

Definition 1.22 (*Accumulation Point of a Subset*): Let $Y \subseteq X$ be a subset. A point $y \in X$ is called an *„accumulation point“* of Y if there exists a sequence $(y_n)_{n\geq 0} \subseteq Y$ such that:

$$\lim_{n\to\infty} y_n = y \quad (y_n \to y). \tag{1.33}$$

Definition 1.23 (*Accumulation Point of a Sequence*): Let $(x_n)_{n\geq 0} \subseteq X$ be a sequence. A point $x \in X$ is called an *„accumulation point“* of the sequence $(x_n)_{n\geq 0}$ if there exists a convergent subsequence $(x_{n_k})_{k\geq 0}$ converging to x:

$$\lim_{k\to\infty} x_{n_k} = x \quad \big(\text{or equivalently, } x_{f(n)} \to x \text{ for some strictly increasing } f\colon \mathbb{N} \to \mathbb{N}\big). \tag{1.34}$$

Key Distinctions in Accumulation Points:

(1) **Accumulation Point of a Set vs. Sequence:** An accumulation point of a set Y is any point that can be reached as the limit of elements in Y. Note that y does not need to belong to Y; it can lie on the boundary of Y. In contrast, an accumulation point of a sequence is a point that is visited infinitely often or approached along an infinite sub-index chain n_k.

(2) **Multiple Accumulation Points:** A sequence can have zero accumulation points (e.g., $x_n = n$ in $\mathbb{R}$), exactly one accumulation point (e.g., $x_n = \frac{1}{n}$ in $\mathbb{R}$), or multiple accumulation points (e.g., $x_n = (-1)^n$, which has accumulation points -1 and $+1$).

Discussion on Pacing and Definitions

Mündliches Protokoll [00:26:03 - 00:27:29]

All of these are definitions of concepts you already know in a more general framework. And now we will start to do more interesting things with these definitions.

You will notice that I'm going very slowly — speaking slowly, writing slowly — but I'm not writing very much on the blackboard. I'm giving you a lot of time to see if what I write makes sense, but I'm not writing a lot, because when I start later to increase

the speed and write like this *[gestures broadly towards the board]*, you will need to follow, right?

So, if something is not clear because I need to write more... I say everything speaking, right? But I don't write everything on the blackboard. If someone is confused: raise your hand, stop me, we have time, okay?

Mündliches Protokoll [00:27:29 - 00:28:00]

This will be very useful: for instance, I use abbreviations or things like that, so later, when I want to go slightly quicker, it will be very useful that we are all used to that.

Yeah. *[takes the Catchbox microphone from the table]* Oops.

Student Question

Um, is there a reason why we often use for definitions something we kind of learned in Analysis I as corollaries or lemmas? For example, for the, um, accumulation point, we usually saw the definition with this epsilon-delta thing, and that whenever you have this delta neighborhood around your point and find the **Schnittmenge** *[laughs]* with Y, for example, the set is not empty. Is there a reason why we avoid epsilon-delta, or is this just...

Mündliches Protokoll [00:28:00 - 00:29:05]

With the limit, no. The ε-δ is more usually for continuity, right? In continuity, we will see the sequence formulation and the ε-δ formulation, and we will see that everything is equivalent.

Here, you cannot do that much ε-δ. You can use epsilons, or you can use limits, and in the notion of a limit the epsilon is implicit, because when I gave the definition of limit, there was an epsilon: I gave it without the epsilon and with the epsilon. But no, it should be exactly equivalent to Analysis I.

When there is something like this, where you have more than one possibility — as when we define the continuity of maps, of functions — I will give all the equivalent definitions systematically. But here, there are no epsilons and deltas that I can see yet. *[erases the middle upper blackboard to make space]*

***Einblick:* Sequential Characterization versus Neighborhood Characterization**

In single-variable calculus, an accumulation point y of a set $Y \subseteq \mathbb{R}$ is often introduced via open metric balls:

$$\forall \varepsilon > 0: \quad B_\varepsilon(y) \cap Y \neq \emptyset \quad (\text{or } B_\varepsilon(y) \cap (Y \setminus \{y\}) \neq \emptyset).$$

In any metric space (X, d), this metric ball condition is completely equivalent to the existence of a sequence $(y_n)_{n \geq 0} \subseteq Y$ converging to y:

- If $y_n \to y$, then for any $\varepsilon > 0$, almost all terms y_n lie in $B_\varepsilon(y)$, so $B_\varepsilon(y) \cap Y \neq \emptyset$.

- Conversely, if $B_\varepsilon(y) \cap Y \neq \emptyset$ for every $\varepsilon > 0$, choosing $\varepsilon_n = \frac{1}{n+1}$ allows one to pick points $y_n \in B_{1/(n+1)}(y) \cap Y$, yielding a sequence with $d(y_n, y) < \frac{1}{n+1} \to 0$.

The lecturer purposefully builds on sequences because limits in metric spaces have already been established.

Convergence Characterized by Accumulation Points

Mündliches Protokoll [00:29:05 - 00:30:50]

Let me prove something: an easy lemma.

A sequence in a metric space (X, d) *[writes „Lemma Sequence $(x_n)_{n\geq 0} \subset X$, (X,d) metric space“]* converges to x *[writes „converges to x“]* if and only if *[writes $\iff$]* x is its sole accumulation point — its only accumulation point *[writes „x is its only acc. pt.“]*. *[draws a dividing line under the lemma statement]*

This is to emphasize also that a sequence, in general, is not convergent, so it will have several accumulation points — many. A sequence being convergent is exactly when the sequence has only one accumulation point. This is something you know for real numbers, and now we see that it's the same in a metric space.

Let's prove that. Proof: *[writes „proof“]*

Lemma 1.24 (*Convergence and Unique Accumulation Point (Initial Statement)*)**:** Let (X, d) be a metric space and $(x_n)_{n\geq 0} \subseteq X$ a sequence. Then $(x_n)_{n\geq 0}$ converges to $x \in X$ if and only if x is its only accumulation point:

$$\lim_{n\to\infty} x_n = x \iff x \text{ is the sole accumulation point of } (x_n)_{n\geq 0}. \tag{1.35}$$

Mündliches Protokoll [00:30:50 - 00:32:51]

What we need to show is that x_n converges to x... *[writes $x_n \to x \iff$]* I will put it in this phrasing: x_n converges to x is equivalent to: for every convergent subsequence converging to y for some y, we have $y = x$ *[writes $(x_{n_k} \to y \implies y = x)$]*, right? An accumulation point is a limit of a subsequence. I'm rephrasing this; I'm saying these two things are equivalent.

Now, there should be one implication that is easy, right? *[writes $\impliedby$ in a box]*

I need to do the two arrows. This implication is easy, because I can take x_{n_k} to be x_n *[writes $(x_{n_k}) = (x_n)$]*, right? The full sequence is always a subsequence: it's the case where f is the identity, $f(n) = n$. Taking this, I find that x_n converges to x *[writes $\implies x_n \to x$]*.

So, this proves this implication *[draws a small square □ closing the $\impliedby$ step]*, just taking a particular case of a subsequence.

Proof Setup and the Leftward Implication

Proof of Lemma 1.24

We rephrase the assertion in terms of convergent subsequences:

$$x_n \to x \iff \big(x_{n_k} \to y \implies y = x\big). \tag{1.36}$$

Direction $\Longleftarrow$ (as presented on the board):
Assume that any convergent subsequence of $(x_n)_{n\geq 0}$ converges to x. Consider the full sequence $(x_n)_{n\geq 0}$ itself. This corresponds to the subsequence with the identity index map:

$$f(n) = n \quad (n_k = k).$$

The lecturer writes:

$$(x_{n_k}) = (x_n) \implies x_n \to x.$$

Q.E.D.

Mündliches Protokoll [00:32:51 - 00:35:20]

[writes $\implies$ in a box] Okay.

Now, assume that x_n converges to x, and I take a subsequence. For convenience, I will write the subsequence as: let $f\colon \mathbb{N} \to \mathbb{N}$ be increasing *[writes „let $f\colon \mathbb{N} \to \mathbb{N}$ increasing“]*. The goal is to show that $x_{f(n)}$ converges to x *[writes „Goal show $x_{f(n)} \to x$“]*.

This is an assumption now: I know that x_n converges to x, and I want to prove that $x_{f(n)}$ converges to x for every increasing f. Now I need to use the definition of limit.

What does it mean that x_n converges to x? It is the same as: given $\varepsilon > 0$, there exists N depending on ε, such that the distance between x_n and x is less than ε when $n \geq N$ *[writes $x_n \to x \iff \forall \varepsilon > 0\ \exists N$ st $d(x_n, x) < \varepsilon$ if $n \geq N$]*.

Now, since f is increasing, $f(n) \geq n$, right? *[writes $f(n) \geq n$]* That f is increasing implies that $f(n) \geq n$ *[boxes the implication f increasing $\implies f(n) \geq n$]*. Using this, I get that the distance between $x_{f(n)}$ and x is less than ε if $n \geq N$ *[writes $\implies d(x_{f(n)}, x) < \varepsilon$ if $n \geq N$]*.

Given an ε, the same capital N works for this sequence or for any subsequence. So, I proved that $x_{f(n)}$ converges to x *[writes $\implies x_{f(n)} \to x$ and draws $\square$]*.

Okay? This was the first one.

Proof of the Rightward Implication

Continuation of Proof: Direction $\Longrightarrow$

Assume that $\lim_{n\to\infty} x_n = x$. Let $(x_{f(n)})_{n\geq 0}$ be an arbitrary subsequence, where $f\colon \mathbb{N} \to \mathbb{N}$ is strictly increasing.

Goal: Show that $\lim_{n\to\infty} x_{f(n)} = x$.

By the definition of the limit:

$$x_n \to x \iff \forall \varepsilon > 0\ \exists N \in \mathbb{N}\colon \quad d(x_n, x) < \varepsilon \quad \text{whenever } n \geq N. \tag{1.37}$$

Since $f : \mathbb{N} \to \mathbb{N}$ is strictly increasing, an immediate induction on n yields:

$$f(n) \geq n \quad \text{for all } n \in \mathbb{N}. \tag{1.38}$$

Consequently, whenever $n \geq N$, it follows from (1.38) that $f(n) \geq n \geq N$. Applying (1.37) to the index $f(n)$ yields:

$$d(x_{f(n)}, x) < \varepsilon \quad \text{for all } n \geq N.$$

Since $\varepsilon > 0$ was arbitrary, this proves that:

$$\lim_{n\to\infty} x_{f(n)} = x.$$

Q.E.D.

Subsequences of a Convergent Sequence: This step rigorously establishes a standard theorem in analysis: **Every subsequence of a convergent sequence converges to the same limit.** Because the re-indexing function satisfies $f(n) \geq n$, any tail bound $n \geq N$ is automatically satisfied by $f(n)$, meaning the exact same cutoff index $N(\varepsilon)$ chosen for the parent sequence works simultaneously for every subsequence.

Student Interjections and the Counterexample in $\mathbb{R}$

Mündliches Protokoll [00:35:20 - 00:35:37]

Where is the cube? Can you throw it there? *[tosses the Catchbox microphone into the auditorium]*

Student Question

Um, in the first, like, in the first implication we showed to the left, how do we know x_n is convergent? Because we because we assumed that any convergent subsequence converges to x, but we do not know that x_n is convergent a priori.

Mündliches Protokoll [00:35:56 - 00:37:10]

You say: any convergent subsequence converges to x... Ah, how did I phrase the lemma? Now you are asking me... let me think again.

A sequence x_n converges to x... yes, I said that it converges to x if it is the only accumulation point, yeah. And now you... No, this is a good question!

You're telling me: if I take a subsequence that converges to y, I know that $y = x$. Good. But you are telling me: why should the full sequence be a convergent subsequence?

[Student confirms: „Yes"]

Good! Good question, well spotted!

Yes, I had in mind a slightly weaker result, and then I stated... This is also true, but then this implication is a bit harder, okay? Very good, very good!

Mündliches Protokoll [00:37:13 - 00:37:22]

Can you throw it there? Yeah, let's ask all the questions and then we...

Student Question

Um, as far as I understand, why can't we let n_k be equal to n, or in other words $f(n) = n$? It's still increasing, and it's still a sub-... so, in other words, a sequence is a subsequence of itself.

Mündliches Protokoll [00:37:41 - 00:38:50]

Yes, this is correct. But I stated a slightly stronger statement — and he's right: I stated that whenever I pick an accumulation point, then it must be x.

You see, what is written here is: **if** there exists a convergent subsequence, then the limit must be x. But he's right in saying: okay, if I take the sequence (x_n), it is a subsequence, but I don't know that it is converging.

So, this *„if"* is not satisfied. I cannot yet conclude that the limit is x with the statement I have written here. It's a subtlety, but it's an important subtlety, and in some proofs we will use this, so we need to fix that.

Is it clear? Yeah, there.

Student Question: The Counterexample

So, what would happen if you take a sequence that for even n is... well, it's just n, and for odd n is $1/n$? Because then you have an accumulation point at zero, but the entire sequence has no limit. What happens then?

Mündliches Protokoll [00:39:30 - 00:40:45]

Let me ask this question out loud, okay?

[writes the sequence on the rightmost blackboard] You say: okay, we have $\mathbb{R}$, and you take this sequence where, let's say, $x_{2n} = n$ for $n \geq 0$, and $x_{2n+1} = 1/n$ for $n > 0$. Is that the question?

And then you say: this sequence satisfies that the only accumulation point is zero, right? Because the even terms are unbounded. *[writes 0 below the definition on the board]* The only accumulation point seems to be zero, while this goes to infinity. And then every subsequence seems to satisfy this, yeah?

Good question!

The Student's Counterexample in $\mathbb{R}$

Beispiel 1.25 (*A Divergent Sequence with a Unique Accumulation Point*)**:** Consider the

real line $\mathbb{R}$ equipped with the standard Euclidean metric. We define a sequence $(x_n)_{n\geq 0}$ piecewise by splitting even and odd indices:

$$\begin{cases} x_{2n} = n & \text{for } n \geq 0, \\ x_{2n+1} = \dfrac{1}{n} & \text{for } n > 0. \end{cases} \tag{1.39}$$

The sequence alternates between unbounded growth and terms approaching zero:

$$(x_n)_{n\geq 0} = \left(0, 1, 1, \frac{1}{2}, 2, \frac{1}{3}, 3, \frac{1}{4}, 4, \dots\right).$$

Analysis of the Counterexample:

- **Accumulation Points:** The odd subsequence converges:
$$x_{2n+1} = \frac{1}{n} \xrightarrow{n\to\infty} 0,$$
so 0 is an accumulation point. Any subsequence containing infinitely many even indices $x_{2n_k} = n_k$ is unbounded and diverges to $+\infty$, which has no limit in $\mathbb{R}$. Any subsequence with only finitely many even indices eventually consists purely of odd terms and must converge to 0. Hence, 0 **is the unique accumulation point** of $(x_n)_{n\geq 0}$ in $\mathbb{R}$.
- **Divergence of the Full Sequence:** Because the even terms diverge to infinity, the full sequence $(x_n)_{n\geq 0}$ does not converge.
- **Conclusion on the Lemma:** This concrete example demonstrates that the statement „$x_n \to x \iff x$ *is the only accumulation point of* (x_n)“ is **false in non-compact metric spaces**! For the backward implication ($\Longleftarrow$) to hold, one either requires:
  - **(a)** The ambient space (X, d) to be compact (guaranteeing that every subsequence has a convergent sub-subsequence), or
  - **(b)** The stronger assumption: **Every subsequence of (x_n) converges to x** (as stated in the course notes).

Mündliches Protokoll [00:40:45 - 00:41:05]

Good. Any other questions? We will try to solve all of them, okay? It's a very good discussion. Yeah.

Student Question

I mean in the in the if you look in the script, the assumption is slightly different: we assume that every subsequence converges to x. We don't say that x is the only accumulation point. Which would be correct.

Mündliches Protokoll [00:41:06 - 00:42:38]

Yeah, this is the stronger statement I was talking about. What script are you looking at? At the one from two years ago?

[Student confirms: „Yes“]

Yeah, this is much easier, and actually it is true: every subsequence converges to x.

Let me think what we will do. The problem is that, for the result we have in the script, I know that in the last lecture I will need a stronger version. So, I wanted to put the stronger version that you will need in the last lecture to do some proof exercise, but then I was not very fortunate in choosing it.

Your comments were right, and what we will do is: I will put... I mean, anyway, this result in the lecture notes is not very interesting. I will put a new version that is slightly stronger than this, but not saying that this is false, right? And then we will prove this. We can do this after the break if you want, because we have a lot of time today.

Mid-Lecture Break

The lecturer announces a short break. During the break, class representatives are elected, and the lecturer writes the corrected lemmas on the blackboard. The video resumes after the break at timestamp 00:04:27.

Corrected Characterizations of Convergence (Lemmas 2 and 3)

Mündliches Protokoll [00:42:38 - 00:44:28]

This is probably the most stupid lemma in the lecture, and it's the one I always have problems with, because... *[laughs]*

Also, it's the first day... Last time I taught, two years ago, I was also sick the first week, and this time again! So, it's like my enemy, this lemma, okay? *[laughs]*

But it's very interesting, because we could see a discussion where, from what you know in $\mathbb{R}$ — in this abstract setup, the claim I made seemed more or less reasonable, but it was false, and it was a bit subtle that it was false. Some of your peers, thinking about what you know of $\mathbb{R}$, of the real line, said: okay, here is a counterexample. There is a problem in the fact that you could have sequences where some subsequences do not converge. Then it's not true that if zero is the only accumulation point, then the sequence converges, right?

So, Lemma 1 that I wrote and the proof had a problem: this is false, and this is the example. So, this is false *[writes „FALSE“ in a box under Lemma 1]*. Why? Because this is the example *[points at the counterexample on the right]*. This is false *[erases the FALSE mark and draws a new box around FALSE]*.

Yeah?

Student Question

Isn't this also, like, the strongest possible thing that you could prove, because if there exists any subsequence which does not converge, then you can't prove that the entire sequence does not converge... so this is the strongest possible thing you could get anyway?

Mündliches Protokoll [00:44:44 - 00:45:46]

This one?

[Student clarifies: „No, the second one."]

Ah, the second. No, there is something stronger. Now I will tell you: this is Lemma 2 that is in the notes, for those who read the notes. What I proved is weaker than that, but it is true: a sequence converges if and only if every subsequence converges to the same limit point, right?

And for this, the implication I gave that is written here was correct, while the other implication is the one that is tricky — well, that is false for the first lemma. For this second, corrected lemma, that implication is trivial, as I said. It is trivial because here it says *„for every subsequence"*, not *„for every convergent subsequence"*. For every subsequence, in particular, I can take $x_{n_k} = x_n$. Now it's correct.

But what I wanted to put...

Lemma 1.26 (*Convergence Characterized by All Subsequences*): Let (X, d) be a metric space and $(x_n)_{n\geq 0} \subseteq X$ a sequence. Then:

$$x_n \to x \iff \forall (x_{n_k})_{k\geq 0}: \quad x_{n_k} \to x. \tag{1.40}$$

Beweis

Direction $\Longrightarrow$:
If $\lim_{n\to\infty} x_n = x$, then for any $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $d(x_n, x) < \varepsilon$ for all $n \geq N$. Since an index map $f\colon \mathbb{N} \to \mathbb{N}$ defining a subsequence $(x_{n_k})_{k\geq 0} = (x_{f(k)})_{k\geq 0}$ is strictly increasing, it satisfies $n_k = f(k) \geq k$. Hence, for all $k \geq N$, we have $n_k \geq N$, which implies:

$$d(x_{n_k}, x) < \varepsilon \implies \lim_{k\to\infty} x_{n_k} = x.$$

Direction $\Longleftarrow$:
Conversely, assume that every subsequence of $(x_n)_{n\geq 0}$ converges to x. Since the sequence $(x_n)_{n\geq 0}$ is a subsequence of itself (corresponding to the identity re-indexing map $f(k) = k$), it follows immediately that:

$$\lim_{n\to\infty} x_n = x.$$

Q.E.D.

Mündliches Protokoll [00:45:46 - 00:47:26]

Lemma 3 *[writes „Lemma 3" on the blackboard]*, which is a bit more interesting, is that... well, the same assumptions as above *[writes „same assumptions as above"]*.

Then x_n converges to x *[writes $x_n \to x \iff$]* if and only if every subsequence (x_{n_k}) *[writes „Every subsequence (x_{n_k})"]* has a sub-subsequence *[writes „has a sub-subsequence"]* — a subsequence of the subsequence. You can say: what the... Well, let me finish the sentence and then I will tell you why.

So: it has a sub-subsequence such that $x_{n_{k_\ell}}$ converges to x *[writes „st $x_{n_{k_\ell}} \to x$"]*.

Mündliches Protokoll [00:47:26 - 00:49:06]

Whenever I choose a subsequence, maybe this subsequence does not converge. But then, we will see next time the definition of compact spaces, as you know in $\mathbb{R}$. In compact spaces, like here in your example, you realized that you could construct this counterexample because the space is unbounded — $\mathbb{R}$ is unbounded, or is not compact, right? If you restrict to a compact interval of $\mathbb{R}$, what I wrote here was true! So, we see that it's not completely false; we just need some extra assumptions. In a compact space, this will be true, but we need some extra assumptions.

Instead, this thing is always true: if every subsequence has a sub-subsequence that converges to x, then the limit is x, right?

Since today I need to do other things, and this is a bit difficult but a good exercise, I will put this statement as an exercise. I hope it is clear; otherwise you will have it in the notes. Try to solve it, and then next time or in a couple of days, when many people have tried, I will solve it here and we will discuss it again. Like this, you can practice a bit. It's a good exercise — it's not trivial, that one. Lemma 2 is easier.

So, this is an exercise.

Mündliches Protokoll [00:49:06 - 00:49:34]

[writes „exercise" in a box around Lemma 3] You have the solution in the notes, but don't look at the solution! Try to do it, right? And then we will solve it here.

[coughs, takes a wiper and starts erasing the right blackboard] Okay.

Lemma 1.27 (*Sub-Subsequence Criterion for Convergence*): Let (X,d) be a metric space and $(x_n)_{n\geq 0} \subseteq X$ a sequence. Then:

$$\lim_{n\to\infty} x_n = x \iff \text{Every subsequence } (x_{n_k})_{k\geq 0} \text{ has a sub-subsequence } (x_{n_{k_\ell}})_{\ell\geq 0} \text{ such that } \lim_{\ell\to\infty} x_{n_{k_\ell}} = x. \tag{1.41}$$

Übung 1.28 (*Proof of the Sub-Subsequence Criterion*): Prove Lemma 1.27.

Why the Sub-Subsequence Criterion is Powerful:

(1) **Forward Direction ($\Longrightarrow$):** If $x_n \to x$, then every subsequence (x_{n_k}) converges to x. Thus, any sub-subsequence $(x_{n_{k_\ell}})$ also converges to x trivially.

(2) **Backward Direction ($\Longleftarrow$):** Suppose $x_n \not\to x$. Then there exists $\varepsilon_0 > 0$ and a subsequence (x_{n_k}) such that $d(x_{n_k}, x) \geq \varepsilon_0$ for all $k \geq 0$. This subsequence cannot have any sub-subsequence converging to x, contradicting the hypothesis.

(3) **Compact Spaces:** In a compact metric space, every sequence has a convergent subsequence. Therefore, in compact spaces, the condition that x is the unique accumulation point does indeed guarantee convergence! The failure of Lemma 1 occurred purely because the counterexample sequence escaped to infinity.

Cauchy Sequences and Complete Metric Spaces

Definitions of Cauchy Sequences and Completeness

Mündliches Protokoll [00:49:59 - 00:51:39]

Actually, this false lemma is very appropriate because it will be a very good example to introduce two concepts. The next concept I'm going to introduce is the concept of a complete space, right?

Complete space. We know that $\mathbb{R}$ is complete — we proved this in Analysis I. Now we are going to see the definition of a complete space at this abstract level, and we will prove that $\mathbb{R}^n$ is complete.

First, we need to define Cauchy sequences. *[writes „Def'n (cauchy sequence)" on the blackboard]* It's the same definition as in $\mathbb{R}$, right?

$(x_n)_{n\geq 0}$ is Cauchy *[writes $(x_n)_{n\geq 0}$ is (cauchy) if]* if, for all $\varepsilon > 0$, there exists capital N *[writes $\forall\varepsilon\ \exists N$ st]* such that the distance between x_n and x_m is less than ε *[writes $d(x_n, x_m) < \varepsilon$]* whenever $n, m \geq N$ *[writes whenever $n, m \geq N$]*.

Mündliches Protokoll [00:51:39 - 00:52:59]

Right? If you wait long enough — if both indices are sufficiently large — then any two points of the sequence become close, right? This is the same as in $\mathbb{R}$.

[draws a dividing line] Now, once we know Cauchy sequences, we can define a complete metric space *[writes „Def'n (complete metric space)" on the blackboard]*. We say that (X, d) is complete *[writes (X, d) is complete]* if every Cauchy sequence *[writes if every (cauchy) sequence]* converges to some limit in the space *[writes converges to some limit in X]*, right?

Mündliches Protokoll [00:52:59 - 00:53:56]

The fact that it converges to some limit in the space I can put between parentheses, because in my definition of convergence, we say that a sequence converges if there is a point of the space towards which it is converging. Whenever I say that a sequence converges, it is to some point in the space; we cannot remove points from the space.

If we start to remove points — as your peer was asking before: what happens if I take the positive numbers? — okay, then I'm taking a complete space, which is $\mathbb{R}$, and I'm making it incomplete by removing points, right?

[underlines „complete"] So, examples... Yeah? Questions?

Definition 1.29 (*Cauchy Sequence*): Let (X, d) be a metric space. A sequence $(x_n)_{n\geq 0} \subseteq X$ is called a *„Cauchy sequence"* if:

$$\forall\varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n, m \geq N\colon \quad d(x_n, x_m) < \varepsilon. \tag{1.42}$$

Definition 1.30 (*Complete Metric Space*): A metric space (X, d) is called *„complete"* if every Cauchy sequence in X converges to a limit point in X:

$$(x_n)_{n\geq 0} \text{ is Cauchy in } X \implies \exists x \in X\colon \lim_{n\to\infty} x_n = x. \tag{1.43}$$

Intuition of Completeness: A Cauchy sequence is one whose terms become arbitrarily close to each other as the indices grow. Completeness means there are no *„missing points“* or *„holes“* in the space: whenever a sequence clusters together, the point it clusters towards is guaranteed to exist within the space itself.

Examples of Complete and Incomplete Metric Subspaces of $\mathbb{R}$

Student Question: Non-Convergent Cauchy Sequences

So here we get that if the Cauchy sequence doesn't converge, then the metric space is not complete?

Mündliches Protokoll [00:54:08 - 00:54:09]

Exactly.

Student Question continued

But my question is: what does it mean for a Cauchy sequence to not converge?

Mündliches Protokoll [00:54:15 - 00:55:45]

I can give you some examples. Let's give some examples of non-complete spaces that you know very well, okay?

As an example of a complete metric space, we have $\mathbb{R}$, right? And then let's take subsets of $\mathbb{R}$. A subset of $\mathbb{R}$ is a metric space, because we have seen that whenever you have a metric space, a subset is a metric space.

If you take a closed subset of a complete space *[walks to the leftmost blackboard and starts cleaning it]* — you don't know what a closed subset of a complete space is yet — but let me give some examples in $\mathbb{R}$, where you know everything.

Mündliches Protokoll [00:55:45 - 00:57:31]

Example: *[writes $\mathbb{R}, d_{Euclidean}$]* $\mathbb{R}$ with the Euclidean distance. What is the Euclidean distance in $\mathbb{R}$? It's just the absolute value of the difference between two numbers, right? *[writes $d(x,y) = |x-y|$]*

Now, we take a subset $A \subset \mathbb{R}$ *[writes $A \subset \mathbb{R}$]*. Then we can restrict the distance to A: the absolute value of the difference.

Case number one: $A = (0,\infty)$ *[writes • $A = (0,\infty)$]*. Is this complete?

What is the problem here? This is not complete because you are missing zero. This is open, right? *[sketches an open interval $(0,\infty)$ on the board]* And you can have a Cauchy sequence: the sequence $x_n = 1/n$ *[writes $x_n = \frac{1}{n}$]* is Cauchy, and the limit would be

zero, but you are removing the limit from your space. So, this space is not complete, right?

Instead, if you take a closed set — let's say $[a,b]$ *[writes* $\bullet\ A = [a,b]$*]*, a closed interval — then, as a metric space with the distance of the absolute value, this is a complete metric space. When you take a Cauchy sequence, it converges in $\mathbb{R}$, so the limit belongs to $\mathbb{R}$; but since this is a closed interval, the limit will belong to your space, right?

Mündliches Protokoll [00:57:31 - 00:58:51]

So, this is not complete *[writes „not complete" next to* $(0,\infty)$*]*, and this is complete *[writes „complete" next to* $[a,b]$*]*.

A more interesting example: *[writes* $\bullet\ \mathbb{Q} \subset \mathbb{R}$*]* $A = \mathbb{Q}$. The rational numbers are not complete. That's because you can take a sequence of rational numbers converging to $\sqrt{2}$, right? *[writes* $x_n \to \sqrt{2}$*]*

You can take a sequence of rational numbers that, as real numbers, converges to $\sqrt{2}$. If your space were the rational numbers — just fractions of integers — and you hadn't invented the reals yet, you would have a problem, because you don't have this completeness property. And this is the whole point of inventing the reals, right? To make the rationals complete.

Consider the real line equipped with the standard Euclidean metric:

$$(\mathbb{R}, d_{\text{Euclidean}}), \quad d(x,y) = |x-y|.$$

For any non-empty subset $A \subseteq \mathbb{R}$, $(A, d|_A)$ forms a metric subspace.

Beispiele 1.31 (*Subspaces of $\mathbb{R}$ and Completeness*):

(1) Open Half-Line $A = (0,\infty)$ (Not Complete):
The sequence $x_n = \frac{1}{n}$ $(n \geq 1)$ lies entirely in $A = (0,\infty)$. It is a Cauchy sequence because:

$$|x_n - x_m| = \left|\frac{1}{n} - \frac{1}{m}\right| \xrightarrow{n,m\to\infty} 0.$$

As a sequence of real numbers, $\lim_{n\to\infty} x_n = 0$. However, $0 \notin A$. Because the limit point does not belong to the space A, the sequence $(x_n)_{n\geq 1}$ does **not** converge in $(A, d|_A)$. Therefore, $(0,\infty)$ is **not complete**.

(2) Closed Bounded Interval $A = [a,b]$ (Complete):
Let $(x_n)_{n\geq 0} \subseteq [a,b]$ be any Cauchy sequence. Since $\mathbb{R}$ is complete, there exists $x \in \mathbb{R}$ such that $x_n \to x$. Since $a \leq x_n \leq b$ for all $n \in \mathbb{N}$, taking the limit preserves the weak inequalities:

$$a \leq \lim_{n\to\infty} x_n = x \leq b \implies x \in [a,b].$$

Thus, the limit point belongs to the space $[a,b]$, proving that $[a,b]$ is **complete**.

(3) The Rational Numbers $A = \mathbb{Q}$ (Not Complete):
Consider a sequence of rational numbers $(x_n)_{n\geq 0} \subseteq \mathbb{Q}$ approximating $\sqrt{2}$ (for example, by truncated decimal expansions $1.4, 1.41, 1.414, \dots$). The sequence is Cauchy in $\mathbb{Q}$, but its unique limit in $\mathbb{R}$ is $\sqrt{2} \notin \mathbb{Q}$. Hence, $(x_n)_{n\geq 0}$ does not converge in $(\mathbb{Q}, d|_{\mathbb{Q}})$, proving that $\mathbb{Q}$ is **not complete**.

The Essential Purpose of the Real Numbers: The historical motivation for constructing $\mathbb{R}$ from $\mathbb{Q}$ (via Dedekind cuts or equivalence classes of rational Cauchy sequences) was precisely to achieve **metric completeness**: filling in all the *„holes“* such as $\sqrt{2}, \pi, e$ where rational Cauchy sequences want to converge.

Mündliches Protokoll [00:58:51 - 00:59:16]

Any questions about this completeness? Yeah.

Student Question: Constant Sequence Outside the Subspace

Um, if you take your second example...

Mündliches Protokoll [00:59:25 - 00:59:26]

Yeah.

Student Question continued

...and we consider the function x_n is equal to the constant function... to the constant value $a-2$...

Mündliches Protokoll [00:59:37 - 00:59:38]

In this example? *[points to $[a,b]$]*

Student Question continued

Yes.

Mündliches Protokoll [00:59:40 - 00:59:41]

$x_n = \dots$

Student Question continued

$a-2$, or $a-3$ to be more dramatic, yes.

Mündliches Protokoll [00:59:47 - 00:59:48]

$a-3$? *[writes $x_n = a-3$ on the board with an arrow pointing to $[a,b]$]*

Student Question continued

Then the sequence converges to some value that is not in A. So it's not...

Mündliches Protokoll [01:00:04 - 01:00:41]

Yeah, but the sequence needs to be in my metric space! You are taking a sequence that is outside of...

[Student responds: „Oh, yeah, okay, because in the third we are taking in $\mathbb{Q}$. Okay. Thank you."]

Is it clear?

[Student confirms: „Yes, clear."]

Yes, of course, one needs to be careful going back to the definitions. This sequence is not a valid sequence because it does not belong to my space, right? My space is $[a, b]$. Okay, good.

Bemerkung 1.32 (*Sequences Must Lie in the Metric Space*): When testing whether a metric space (A, d_A) is complete, all candidate Cauchy sequences $(x_n)_{n\geq 0}$ must by definition consist entirely of elements belonging to A:

$$x_n \in A \quad \forall n \in \mathbb{N}.$$

A sequence like $x_n = a - 3$ for $A = [a, b]$ does not lie in A and is therefore not a valid sequence in the metric space (A, d_A).

Lecture Transition and Overview of Upcoming Topics

Mündliches Protokoll [01:00:57 - 01:02:42]

[sets down the chalk and paces near the podium] Okay, so now I can do two things. I don't see more questions, right?

We can start to introduce open and closed sets in a metric space, okay? And then, in the last five minutes, I will discuss a bit... Because this year you already saw with Professor Figalli the construction of the reals with Dedekind cuts, I think, right? So, you have a construction of the real numbers. But one can also construct them as a completion, as I said here: if you make $\mathbb{Q}$ complete, there is an abstract way to make any metric space complete. We can discuss this very quickly for those interested; this procedure applies to $\mathbb{Q}$ and gives you $\mathbb{R}$. This is another way to construct a model of $\mathbb{R}$. But since this is not strictly needed now — it's just for extra information — I will spend just the last five minutes on that.

Now let's advance and introduce open and closed sets. You know what an open and closed interval is in $\mathbb{R}$, and now... *[erases the middle chalkboard with a damp cloth]*

Or maybe... No, I think that... Sorry, maybe now it's better that I prove that $\mathbb{R}^n$ is complete; it's a good time. So, let's state this theorem. Sorry.

Completeness of Euclidean Space $\mathbb{R}^n$

Theorem Formulation and Coordinate Notation

Mündliches Protokoll [01:02:42 - 01:04:02]

[writes „Thm $(\mathbb{R}^n, d_{Eucl})$ is complete" on the upper board] This is a theorem you have for $\mathbb{R}$. To prove the theorem for $\mathbb{R}^n$, I will use the result in $\mathbb{R}$: I will use that $\mathbb{R}$ is complete. This is a highly non-trivial fact; it's almost axiomatic — you define $\mathbb{R}$ axiomatically at the beginning to make it complete. So, this is hard. But once I know that $\mathbb{R}$ is complete, it's very easy to show that $\mathbb{R}^n$ is complete, right?

I will prove this theorem, but to prove this theorem I will need a lemma that will be useful for you many, many times, which says the following: *[draws a dividing line under the theorem and writes „Lemma"]*

If x_m... Ah, now we have the usual notational problem, so I will write x_m *[writes $(x_m)_{m\geq 0} \subset \mathbb{R}^n$]*, a sequence in $\mathbb{R}^n$. I use the sub-index m here because when I'm in $\mathbb{R}^n$, n will be the dimension in this case, so I cannot use n here, okay?

Satz 1.33 (*Completeness of Euclidean Space*): The n-dimensional Euclidean space $(\mathbb{R}^n, d_{\text{Eucl}})$ equipped with the standard Euclidean metric is complete:

$$(x_m)_{m\geq 0} \text{ is a Cauchy sequence in } \mathbb{R}^n \implies \exists x \in \mathbb{R}^n \colon \lim_{m\to\infty} x_m = x. \tag{1.44}$$

Mündliches Protokoll [01:04:02 - 01:05:47]

When you have a sequence of points like this, x_m, remember that for a point in $\mathbb{R}^n$... This is a bit of notation, but it's important that we clarify it right now.

In $\mathbb{R}^n$, we were denoting points as $x = (x_1, \dots, x_n)$ *[writes $x = (x_1, \dots, x_n)$]*, and these coordinates are n real numbers. But now, if I have a sequence of points, we will denote the coordinates like this: x_m is a point in the sequence, and we will denote the coordinates as $x_{m,1}, x_{m,2}, \dots, x_{m,n}$ *[writes $x_m = (x_{m,1}, x_{m,2}, \dots, x_{m,n})$]*.

You see: the first sub-index is which element of the sequence, and the second sub-index is which coordinate, right? It must be like that.

If you want, there is a small notational conflict, because when you see x_m, you are not sure — well, you will be sure from the context, but if I only tell you x_m, what is this? Is it the m-th coordinate of the point x, or the m-th element of a sequence? You will always know by looking at the context. But this is a small notational conflict that I pointed out in the notes as well, and you need to resolve it by context.

Mündliches Protokoll [01:05:47 - 01:06:45]

So, lemma: (x_m) is a sequence in $\mathbb{R}^n$. Then x_m converges to x *[writes $x_m \to x$]* if and only if *[writes $\iff$]* every coordinate converges *[writes $x_{m,i} \to x_i$]* to the corresponding coordinate, for all i from 1 up to n *[writes $\forall i = 1, \dots, n$]*.

The way to find whether a sequence converges in $\mathbb{R}^n$ is just to look at each coordinate; if each coordinate converges to the corresponding coordinate of the limit — these are the coordinates of that, right? *[points to $x = (x_1, \ldots, x_n)$]* — then you have convergence, and this is an *„if and only if"*.

We will first prove the lemma, and from this lemma we will get immediately that $\mathbb{R}^n$ is complete.

Notation 1.34 (*Coordinates of Sequence Elements in $\mathbb{R}^n$*): For a point $x \in \mathbb{R}^n$, we write its coordinates as:

$$x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n, \quad \text{where } x_i \in \mathbb{R} \text{ for each } i \in \{1, \ldots, n\}.$$

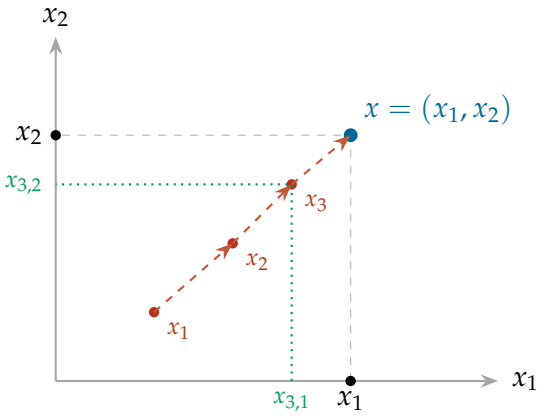
For a sequence of vectors $(x_m)_{m\geq 0} \subseteq \mathbb{R}^n$, the index $m \in \mathbb{N}$ denotes the sequence term, while the index $i \in \{1, \ldots, n\}$ denotes the coordinate:

$$x_m = (x_{m,1}, x_{m,2}, \ldots, x_{m,n}) \in \mathbb{R}^n.$$

Lemma 1.35 (*Componentwise Convergence in $\mathbb{R}^n$*): Let $(x_m)_{m\geq 0} \subseteq \mathbb{R}^n$ be a sequence and let $x \in \mathbb{R}^n$. Then:

$$x_m \xrightarrow{m\to\infty} x \text{ in } (\mathbb{R}^n, d_{\text{Eucl}}) \iff x_{m,i} \xrightarrow{m\to\infty} x_i \text{ in } (\mathbb{R}, |\cdot|) \quad \forall i \in \{1, \ldots, n\}. \tag{1.45}$$

Geometrical Meaning of Componentwise Convergence: In $\mathbb{R}^n$, convergence in the Euclidean metric is equivalent to convergence in each coordinate independently:



The Euclidean distance $\|x_m - x\|$ controls, and is controlled by, the individual coordinate distances $|x_{m,i} - x_i|$.

Proof of Lemma 1.35: Componentwise Convergence

Proof of Lemma 1.35

Forward Direction: Vector Convergence Implies Coordinate Convergence

Mündliches Protokoll [01:06:45 - 01:07:47]

[slides the blackboards upward to reveal clean space below] Okay, let's do the implication to the right. *[writes $\Longrightarrow$ on the board]*

The starting point is: I know that x_m converges to x. *[writes $x_m \to x$]* What does it mean? *[writes $\Longleftrightarrow$ $\forall \varepsilon > 0$ $\exists N$ such that]* For all $\varepsilon > 0$, there exists capital N such that the norm $\|x_n - x\| < \varepsilon$ *[writes $\|x_n - x\| < \varepsilon$]* if $n \geq N$ *[writes if $n \geq N$]*.

This is the definition of convergence in a metric space, and I'm using that in my metric space the distance is the norm of the difference of points. So, this is the same as the distance between x_n and x *[writes $d(x_n, x)$]*: it's the definition of the Euclidean distance, right? This is by definition of convergence.

Mündliches Protokoll [01:07:47 - 01:09:17]

Now I want to prove that this implies that each coordinate converges.

You see that this implies that... let me put it here: *[writes $\Longrightarrow$ $|x_{n,i} - x_i|$]* for a given coordinate i *[writes Given $i \in \{1, \ldots, n\}$]*, what is $|x_{n,i} - x_i|$?

This is less than or equal to the square root of the sum for $j = 1$ to n of $(x_{n,j} - x_j)^2$ *[writes $\leq \sqrt{\sum_{j=1}^{n} |x_{n,j} - x_j|^2}$]*, right? Given some coordinate i, $|x_{n,i} - x_i|$ is less than the sum of the squares for all coordinates, and then you take the square root.

By definition, this is the norm *[writes $= \|x_n - x\|$]*, and I'm saying that this will be less than ε *[writes $< \varepsilon$]*, right?

Mündliches Protokoll [01:09:17 - 01:10:16]

And this is what I wanted to show, right?

Given ε, there exists some N — here it is the same capital N — such that, when $m \geq N$... This is an m, sorry: m, m *[corrects the sequence index n to m across the board equations: x_m, $\|x_m - x\|$, and $m \geq N$]*. This is n, this is m, okay?

So, the i-th coordinate difference $|x_{m,i} - x_i|$ will be less than ε. This just follows from this inequality: the comparison between the difference of coordinates and the square root. This implication is rather easy, and is done here. *[draws a small square $\square$ to complete the $\Longrightarrow$ direction]*

Forward Implication: Vector to Coordinates

Direction ($\Longrightarrow$):
Assume that $x_m \to x$ in $(\mathbb{R}^n, d_{\text{Eucl}})$. By definition of convergence in a metric space:

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall m \geq N: \quad d_{\text{Eucl}}(x_m, x) = \|x_m - x\| < \varepsilon. \tag{1.46}$$

Let $i \in \{1, \ldots, n\}$ be an arbitrary coordinate. For any $m \in \mathbb{N}$, we compare the i-th coordinate difference to the full Euclidean norm:

$$|x_{m,i} - x_i| = \sqrt{(x_{m,i} - x_i)^2} \leq \sqrt{\sum_{j=1}^{n} (x_{m,j} - x_j)^2} = \|x_m - x\|. \tag{1.47}$$

Combining (1.47) with (1.46), for all $m \geq N$:

$$|x_{m,i} - x_i| \leq \|x_m - x\| < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, this establishes that $\lim_{m\to\infty} x_{m,i} = x_i$ in $\mathbb{R}$ for each $i \in \{1, \ldots, n\}$.

Backward Direction: Coordinate Convergence Implies Vector Convergence

Mündliches Protokoll [01:10:16 - 01:12:27]

The other implication is almost the same, but slightly harder. *[writes $\Longleftarrow$ in a circle]*

Now I start at the other point. Suppose that... Ah, this is an m, sorry, you needed to tell me: these are m's *[corrects another index to m]*.

My starting point is that for each coordinate i, $x_{m,i} \to x_i$. So, for all $\varepsilon > 0$, there exists an N_i such that $|x_{m,i} - x_i| < \varepsilon$ if $m \geq N_i$ *[writes $\forall \varepsilon > 0\ \exists N_i$ st $|x_{m,i} - x_i| < \varepsilon$ if $m \geq N_i$]*.

I know that these kinds of things are always the most confusing, but I'm just starting on the right-hand side. For each coordinate i, $x_{m,i}$ converges to x_i, and I'm writing the definition: given $i \in \{1, \ldots, n\}$ *[writes Given $i \in \{1, \ldots, n\}$]*, for all $\varepsilon > 0$ there exists some N that depends on i — that's why I call it N_i — such that this holds. Good?

Now, if I can do this, I can take ε divided by n *[modifies ε to ε/n in the inequality: $|x_{m,i} - x_i| < \frac{\varepsilon}{n}$]*, right? You are just taking a different ε. This is true.

Mündliches Protokoll [01:12:27 - 01:14:32]

What you do now is: if this happens, then the norm $\|x_m - x\|$ *[writes $\|x_m - x\| = \sqrt{\sum_{i=1}^{n} |x_{m,i} - x_i|^2}$]*... What is this? This is the square root of the sum for $i = 1$ to n of $|x_{m,i} - x_i|^2$.

Now, I will take m bigger than or equal to the maximum of these numbers for i between 1 and n *[writes $m \geq \max_{1\leq i\leq n} N_i =: N$]*. I take the maximum of all these numbers, and I call this capital N. If I take this N that is the maximum of all these numbers, and m is larger, it will satisfy all these conditions at the same time.

Then I will get that this is strictly less than the square root of n times ε^2/n^2 *[writes $\leq \sqrt{n\frac{\varepsilon^2}{n^2}}$]*. Each term is less than ε/n, so when squared it is ε^2/n^2. And this is equal to $\varepsilon/\sqrt{n}$ *[writes $= \frac{\varepsilon}{\sqrt{n}}$]*, which is less than or equal to ε *[writes $< \varepsilon$]*.

You see, this establishes that implication *[draws a small square $\square$ to complete the $\Longleftarrow$ direction]*. I assumed that the coordinates converge; I needed to take a smaller epsilon, divided by n, but then the vector converges, right?

Backward Implication: Coordinates to Vector

Direction ($\Longleftarrow$):
Assume that for every coordinate $i \in \{1, \dots, n\}$, we have $x_{m,i} \to x_i$ in $\mathbb{R}$. Let $\varepsilon > 0$. We apply the definition of convergence to each coordinate with the tighter tolerance $\frac{\varepsilon}{n} > 0$. For each $i \in \{1, \dots, n\}$, there exists an index $N_i \in \mathbb{N}$ such that:

$$|x_{m,i} - x_i| < \frac{\varepsilon}{n} \quad \text{whenever } m \geq N_i. \tag{1.48}$$

Define the unified cutoff index:

$$N := \max_{1 \leq i \leq n} N_i \in \mathbb{N}. \tag{1.49}$$

For all $m \geq N$, the bound (1.48) holds simultaneously for all n coordinates. We compute the Euclidean norm of the difference vector:

$$\begin{aligned} \|x_m - x\| &= \sqrt{\sum_{i=1}^{n} |x_{m,i} - x_i|^2} \\ &< \sqrt{\sum_{i=1}^{n} \left(\frac{\varepsilon}{n}\right)^2} = \sqrt{n \cdot \frac{\varepsilon^2}{n^2}} \\ &= \frac{\varepsilon}{\sqrt{n}} \leq \varepsilon. \end{aligned} \tag{1.50}$$

Thus, $\|x_m - x\| < \varepsilon$ for all $m \geq N$, proving that $x_m \to x$ in $(\mathbb{R}^n, d_{\text{Eucl}})$.

Q.E.D.

Proof of the Completeness of $\mathbb{R}^n$

Proof of Theorem 1.33

Mündliches Protokoll [01:14:32 - 01:16:07]

Now, with exactly the same trick, exactly the same setup, I can prove the theorem: the completeness of $\mathbb{R}^n$. $\mathbb{R}^n$ is complete! *[writes „proof of Thm ($\mathbb{R}^n$ complete)“ on the board]*

How do you prove that something is complete? You give me a Cauchy sequence, and I must show that the Cauchy sequence converges.

Given a Cauchy sequence *[writes „Given $(x_n)_{n\geq 0}$ Cauchy sequence“]*, let us show that there exists some limit x such that the sequence converges to x *[writes „Let us show $\exists x\colon x_n \to x$“]*.

How will we find the limit? Remember, I told you: we need to use that $\mathbb{R}$ is complete.

When we have a sequence... Ah, sorry, this is in $\mathbb{R}^n$, so I need to use m or something else: m *[corrects index n to m: $(x_m)_{m\geq 0}$, $x_m \to x$]*. m here, n there.

Mündliches Protokoll [01:16:07 - 01:17:55]

We need to use that $\mathbb{R}$ is complete, so what we will do is pass to the components. For each index $i = 1, \dots, n$, $(x_{m,i})_{m\geq 0}$ is a Cauchy sequence in $\mathbb{R}$ *[writes* $\forall i = 1, \dots, n \quad (x_{m,i})_{m\geq 0}$ *is Cauchy in* $\mathbb{R}$*]*.

Why is this true? By the same inequality! When you have a Cauchy sequence, by the definition with m, we said that the component differences $|x_i - y_i|$ for two points are bounded by the norm $\|x - y\|$ *[writes* $|x_i - y_i| \leq \|x - y\|$*]*.

And then we also had the opposite inequality: that the minimum — sorry, the minimum in i of $|x_i - y_i|$ bounds the norm *[writes* $\|x - y\| \leq \min_i |x_i - y_i|\sqrt{n}$*]*, but we need to pay a factor of $\sqrt{n}$ here. That is what we did here, right?

Mündliches Protokoll [01:17:55 - 01:19:37]

Between these two quantities — the difference of the components or the norm of the vector — you can pass from one to the other. Just when you go in one direction, you need to pay a factor of $\sqrt{n}$.

This shows that this sequence is Cauchy. Being a Cauchy sequence in $\mathbb{R}$, it converges, so $x_{m,i} \to x_i$ *[writes* $x_{m,i} \to x_i$*]*. This is my definition of x_i.

Then I put all these real numbers together, and I form the point x: I define $x := (x_1, \dots, x_n)$ *[writes* $\rightsquigarrow x = (x_1, \dots, x_n)$*]*. And the original sequence of vectors converges to this limit, again by the same reasoning I used here: each component converges to the corresponding component implies that the vectors converge, right? I'm using the lemma — this implication.

It's almost time. So, in one sentence, because it's time: the proof of the completeness of $\mathbb{R}^n$ is just pass to the components. The components are Cauchy sequences, so they converge, and then you put together the components to make the point, proving the existence of the limit.

Yeah. Hm?

Proof of Completeness of $\mathbb{R}^n$

Beweis

Let $(x_m)_{m\geq 0} \subseteq \mathbb{R}^n$ be an arbitrary Cauchy sequence in $(\mathbb{R}^n, d_{\text{Eucl}})$. **Goal:** Show that there exists $x \in \mathbb{R}^n$ such that $\lim_{m\to\infty} x_m = x$.

Step 1 (Components are Cauchy in $\mathbb{R}$):
By definition of a Cauchy sequence in $\mathbb{R}^n$:

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall m, k \geq N: \quad \|x_m - x_k\| < \varepsilon.$$

For each coordinate $i \in \{1, \dots, n\}$, applying inequality (1.47) yields:

$$|x_{m,i} - x_{k,i}| \leq \|x_m - x_k\| < \varepsilon \quad \text{for all } m, k \geq N.$$

Thus, for every $i \in \{1, \dots, n\}$, the real sequence $(x_{m,i})_{m\geq 0}$ is a Cauchy sequence in $(\mathbb{R}, |\cdot|)$.

Step 2 (Convergence in $\mathbb{R}$ by Completeness of $\mathbb{R}$):
Since $\mathbb{R}$ is complete, every Cauchy sequence in $\mathbb{R}$ converges. Therefore, for each coor-

dinate $i \in \{1, \dots, n\}$, there exists a real limit:

$$x_i := \lim_{m \to \infty} x_{m,i} \in \mathbb{R}.$$

Step 3 (Reconstruction of the Vector Limit):
We assemble these n coordinate limits into a candidate vector:

$$x := (x_1, x_2, \dots, x_n) \in \mathbb{R}^n.$$

By Lemma 1.35 (Direction $\Longleftarrow$), since $x_{m,i} \to x_i$ for all $i \in \{1, \dots, n\}$, it follows that:

$$\lim_{m \to \infty} x_m = x \quad \text{in } (\mathbb{R}^n, d_{\text{Eucl}}).$$

Hence, every Cauchy sequence in $\mathbb{R}^n$ converges, establishing that $(\mathbb{R}^n, d_{\text{Eucl}})$ is complete.

Q.E.D.

Q.E.D.

Student Question on the Norm Equivalence Bound

Mündliches Protokoll [01:19:37 - 01:19:44]

[tosses the Catchbox throwable microphone to a student in the front rows]

Student Question: Minimum versus Maximum in Norm Bound

Could you please elaborate why you used the minimum with the distances there, instead of for instance the maximum?

Mündliches Protokoll [01:19:52 - 01:19:55]

It's the maximum.

Student Question continued

Okay. Thank you.

Mündliches Protokoll [01:19:57 - 01:20:22]

Thank you. *[walks over to the board, erases „min" and corrects it to „max" in the inequality $\|x - y\| \leq \max_i |x_i - y_i| \sqrt{n}$, then sets down the chalk] [lecturer packs up his materials at the podium as students pack up their bags; audio ends]*

Bemerkung 1.36 (*Two-Sided Bounds Relating Coordinates and Norms*): For any two vectors $x, y \in \mathbb{R}^n$, the individual coordinate distances and the Euclidean norm satisfy

the two-sided bounds:

$$\max_{1\le i\le n} |x_i - y_i| \le \|x - y\| \le \sqrt{n} \max_{1\le i\le n} |x_i - y_i|. \tag{1.51}$$

- **Lower Bound on the Norm:** Since $\|x-y\| = \sqrt{\sum_{j=1}^n (x_j - y_j)^2} \ge \sqrt{(x_i - y_i)^2} = |x_i - y_i|$ for every i, taking the maximum over all i yields the left inequality.
- **Upper Bound on the Norm:** Since $|x_i - y_i| \le \max_{1\le j\le n} |x_j - y_j|$ for each i:

$$\|x-y\|^2 = \sum_{i=1}^n |x_i - y_i|^2 \le \sum_{i=1}^n \left(\max_{1\le j\le n} |x_j - y_j|\right)^2 = n \left(\max_{1\le j\le n} |x_j - y_j|\right)^2.$$

Taking the square root yields the factor $\sqrt{n}$. The lecturer originally wrote min by a slip of the pen and corrected it to max following the student's question.

Summary of Key Concepts

- **Sequences and Limits in Metric Spaces:** A sequence $(x_n)_{n\ge 0}$ in (X,d) converges to $x \in X$ if and only if $d(x_n, x) \to 0$ in $\mathbb{R}$. The limit is unique by the triangle inequality and separation axiom ($x \ne y \implies d(x,y) > 0$).
- **Subsequences and Accumulation Points:** A subsequence is formed by a strictly advancing index map $f\colon \mathbb{N} \to \mathbb{N}$ ($n_k \ge k$). An accumulation point of a sequence is the limit of a convergent subsequence.
- **Subsequence Criteria for Convergence:** In general metric spaces, $x_n \to x$ if and only if every subsequence converges to x, or equivalently, if every subsequence contains a sub-subsequence converging to x. Having a unique accumulation point does not suffice in non-compact spaces (e.g., $x_{2n} = n, x_{2n+1} = 1/n$ in $\mathbb{R}$).
- **Cauchy Sequences and Completeness:** A metric space (X,d) is complete if every Cauchy sequence in X converges to a limit in X. Subspaces like $(0,\infty)$ and $\mathbb{Q}$ are incomplete because their Cauchy sequences lack limits in the space, whereas closed intervals $[a,b]$ are complete.
- **Completeness of $(\mathbb{R}^n, d_{\text{Eucl}})$:** A sequence in $\mathbb{R}^n$ converges (or is Cauchy) if and only if it converges (or is Cauchy) componentwise in each coordinate. The metric completeness of $\mathbb{R}^n$ follows directly from the completeness of $\mathbb{R}$ via coordinate-wise reconstruction.

CHAPTER 1

Thursday 19 February 2026

Open and Closed Sets, Topology, and Continuity in Metric Spaces

Course Content Overview: Topology and Continuity in Metric Spaces

In this lecture, we transition from sequential notions in metric spaces (convergence, Cauchy sequences, and completeness) to the topological foundations of multi-variable analysis:

- **Open Balls and Open Sets:** Defining open balls $B(x,r)$ and open subsets $U \subseteq X$ via local ball containment.
- **Closed Sets and Topology:** Defining closed sets via complements and introducing the metric topology $\mathcal{T}$ associated to a distance d.
- **Stability Properties:** Proving that arbitrary unions and finite intersections of open sets are open, and employing De Morgan's laws to establish dual properties for closed sets.
- **Interior, Closure, and Boundary:** Formalizing $\text{int}(\Omega)$, $\overline{\Omega}$, and $\partial\Omega$, with concrete geometric examples in Euclidean space $\mathbb{R}^2$.
- **Sequential Characterization:** Characterizing openness and closedness via convergent sequences and the concept of eventual containment.
- **Continuity in Metric Spaces:** Formulating the three equivalent characterizations of continuity (ε-δ, sequential, and topological pre-images of open sets).

Review of Metric Spaces and Motivation

Mündliches Protokoll [00:00:00 - 00:01:50]

Good afternoon. We can continue where we left off last time.

Just a quick summary of what we saw in the last lectures: The first lecture was mostly an introduction, but then we saw metric spaces. And I told you — because people already asked me, even though I addressed it the first day — why we study metric spaces. Whenever it gets too abstract, the model you need to have in mind all the time for this (X,d) metric space is $\mathbb{R}^n$ with the Euclidean distance, okay? So, whenever you have doubts, just plug $\mathbb{R}^n$ where there is an X — a capital X — and d is the usual distance, the modulus of the difference of the points.

And I said that I wanted to do this in this more general framework because later, in

the last part of the course, we will use other spaces, okay? And for good reason: we will use the space of continuous functions to prove some theorems in the course. But if you want — this is not something that belongs only to mathematicians, right? Let me say very quickly for the physicists: in classical mechanics, the point in space is a vector in $\mathbb{R}^n$, right? But when you go to quantum mechanics, what is a point? What is the position of a particle in space? It's the wave function; it's a function. So, the wave functions are naturally points in your space, right?

This is what we are saying here: We are trying to state things more generally to be able to understand the things that come naturally in physics, for instance.

Mündliches Protokoll [00:01:50 - 00:02:24]

Okay. Now, in this framework of metric spaces, we saw what a sequence means, what it means that a sequence converges, what a Cauchy sequence is, and what a complete metric space is. And we proved that $\mathbb{R}^n$ is always a complete metric space, using the fact that $\mathbb{R}$ is complete, last time.

And today we will see the definition of open and closed sets in a metric space, and the notion of continuous functions in the setup of metric spaces. So, let me start with the definition of...

Summary of Previous Lectures and Canonical Models

Foundational Concepts in Metric Spaces (X, d):

- **Canonical Finite-Dimensional Model:** The primary intuitive model for any metric space (X, d) is Euclidean space $\mathbb{R}^n$ equipped with the standard Euclidean distance:
$$d_{\text{Euc}}(x, y) := \|x - y\|_2 = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}.$$
- **Sequences and Convergence:** A sequence $(x_n)_{n\geq 0} \subseteq X$ converges to a limit $x \in X$ (written $x_n \to x$) if:
$$\lim_{n\to\infty} d(x_n, x) = 0 \iff \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \geq N \colon d(x_n, x) < \varepsilon.$$
- **Cauchy Sequences and Completeness:** A sequence $(x_n)_{n\geq 0}$ is a *„Cauchy sequence"* if $\lim_{n,m\to\infty} d(x_n, x_m) = 0$. A metric space (X, d) is *„complete"* if every Cauchy sequence in X converges to a limit in X. In particular, $(\mathbb{R}^n, d_{\text{Euc}})$ is complete.

Physical Motivation: From Classical Points to Function Spaces: In classical mechanics, the state of a point particle is described by a position vector $x \in \mathbb{R}^3$. In quantum mechanics, however, the state of a particle is given by a complex-valued wave function $\psi \colon \mathbb{R}^3 \to \mathbb{C}$ living in an infinite-dimensional function space (such as $L^2(\mathbb{R}^3)$). Developing analysis in general metric spaces allows us to treat such function spaces rigorously using the exact same geometric intuition as $\mathbb{R}^n$.

Open Balls and Open Sets

Open Balls

Mündliches Protokoll [00:02:24 - 00:03:54]

...definition of open set... So, let me start maybe with the definition of an open ball. Open ball. *[writes „defn (open ball)“]*

We define the open ball of radius r positive, *[writes „Define open ball of radius $r > 0$“]* centered at a given point x in my metric space, *[writes „centered at $x \in X$ as“]* as — and we will denote it by this: the ball centered at x with radius r *[writes $B(x,r)$]*; sometimes you will see the equivalent notation of the ball of radius r centered at x *[writes $B_r(x) = B(x,r)$]*, which is also in many notes and books, it's the same, okay? — this is by definition the set of points in my space such that the distance... Ah, sorry, the points will be y, such that the distance between y and x is less than r, right? *[writes $= \{y \in X \mid d(y,x) < r\}$]*

Definition 1.37 (*Open Ball*): Let (X,d) be a metric space. For any point $x \in X$ and any real number $r > 0$, the *„open ball“* of radius r centered at x is defined as the set:

$$B(x,r) := \{y \in X \mid d(y,x) < r\}. \tag{1.52}$$

Notation 1.38 (*Alternative Notation for Open Balls*): In mathematical literature and lecture notes, the open ball is frequently denoted by $B_r(x)$ interchangeably with $B(x,r)$:

$$B_r(x) \equiv B(x,r).$$

Key Structural Features of Open Balls:

- **Strict Inequality:** The condition defining the ball is strictly $d(y,x) < r$, excluding points at distance exactly equal to r.
- **Symmetry:** Since any metric satisfies $d(y,x) = d(x,y)$, the definition can equivalently be written as $d(x,y) < r$.
- **Center Inclusion:** Since $d(x,x) = 0 < r$ for any $r > 0$, the center point x is always an element of $B(x,r)$.

Mündliches Protokoll [00:03:54 - 00:04:52]

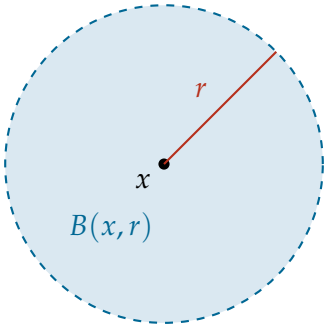
So, if you want a drawing of that, in Euclidean space — in a general metric space I cannot draw anything because it's too general, but in Euclidean space, let's say in $\mathbb{R}^2$ *[writes $\mathbb{R}^2$]* — the metric ball would be the ball.

The open ball would be this, right? *[draws open disk with center x, radius r, and dashed boundary]* What is inside of the ball. And it's open because you don't include... If you want, I can put the boundary of the ball, which we will introduce today also in a general topological space, what the boundary means. This surface of the ball is not included, because there is a strict inequality, okay?

The points that are at distance exactly r, I don't take them. This is the open ball.

Geometric Representation of an Open Ball in $\mathbb{R}^2$

In the Euclidean plane $(\mathbb{R}^2, d_{\text{Euc}})$, an open ball $B(x,r)$ corresponds geometrically to an open circular disk whose boundary circle is excluded.



Dashed boundary:
$\{y \mid d(y,x) = r\} \notin B(x,r)$

Strict Boundary Exclusion: The dashed circumference represents the sphere $S(x,r) := \{y \in X \mid d(y,x) = r\}$. Because the defining condition is strictly $d(y,x) < r$, points residing on this boundary circumference do not belong to the open ball $B(x,r)$.

Definition of Open Sets

Mündliches Protokoll [00:04:52 - 00:05:52]

Now, the next definition *[writes „defn (open & closed sets)“]* is open and closed sets. Open and closed sets.

We say that a set U — a set U of my metric space X — is open *[writes „• $U \subset X$ is open“]* if for every x in U, there exists some positive radius r, possibly depending on x, such that the ball of radius r centered at x is contained in U. *[writes „if $\forall x \in U\ \exists r > 0$ st $B(x,r) \subset U$“]*

Definition 1.39 (*Open Set*): Let (X,d) be a metric space. A subset $U \subseteq X$ is called an *„open set“* if around every point of U, there exists an open ball entirely contained within U:

$$\forall x \in U\ \exists r > 0: \quad B(x,r) \subseteq U. \tag{1.53}$$

Dependence of the Radius on the Point: The radius r in (1.53) depends on the chosen point $x \in U$ (i.e., $r = r(x)$). As x moves closer to the edge or boundary of U, the permissible radius r will generally need to become strictly smaller to prevent the ball from spilling outside U.

Mündliches Protokoll [00:05:52 - 00:07:12]

In $\mathbb{R}^2$, an open set is essentially any set... This is heuristically the idea, right? Then we will see concrete open sets. But it's a set that does not include its contour — its

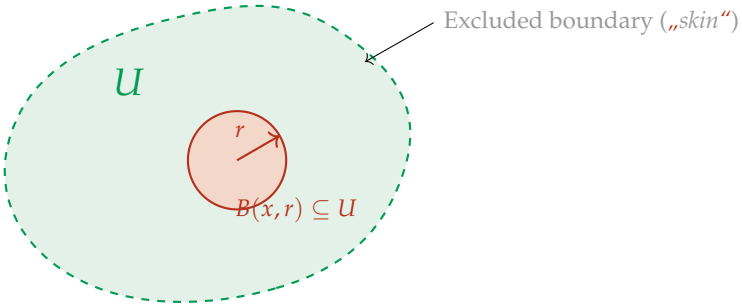
boundary.

Again, I didn't define the boundary yet, but now you have some common-sense knowledge of what the boundary of this set is, which is this part: the skin of the set, right? If the set is the inside without the skin, it is open.

Why? Because for any point here, you can take a ball. Depending on the distance to the boundary, right? Depending on the distance to the skin, you make the ball super, super small, and it will be contained in the set. Like I said, any point in the interior will be at a positive distance from the skin, right?

Geometric Intuition: The „*Skin*“ of an Open Set

Geometrically, a subset of Euclidean space $\mathbb{R}^2$ is open if it does not contain its „*skin*“ (its boundary contour). Every point in the interior of the set sits at a strictly positive distance from the boundary, allowing a sufficiently small ball to fit entirely inside.



The „*No Skin*“ Principle: For any interior point $x \in U$, let $\delta := \inf_{y \in \partial U} d(x, y)$ denote the distance from x to the boundary of U. Since the boundary is not part of U, we have $\delta > 0$. By selecting any radius $0 < r < \delta$, the entire open ball $B(x, r)$ is guaranteed to lie strictly inside U.

Example: The Open Unit Square in $\mathbb{R}^2$

Mündliches Protokoll [00:07:12 - 00:08:35]

If you want a more concrete example, take the set $(0,1)\times(0,1)$ *[writes $(0,1)\times(0,1)$]*, the open square, subset $\mathbb{R}^2$ *[writes $\subset\mathbb{R}^2$ open]*. Prove that it is open: exercise.

You take this square *[draws dashed square]*, and you take a point here. A point in the square will have two coordinates. And now, what is the definition of open? If you take the point (x_1,x_2) *[writes (x_1,x_2)]*, you need to construct a positive radius r such that the whole ball centered at this given point belongs to the square.

In the case of the square, what is the formula for the radius? Well, you can take r equal to, I don't know, something like the minimum between this distance, this distance, this distance, and this distance. So, it will be something like the minimum between x_1, x_2, $1-x_1$, and $1-x_2$ *[writes $r=\min\{x_1,x_2,1-x_1,1-x_2\}$]*.

You can check that this radius works — the minimum of these four numbers, right?

Beispiel 1.40 (*Open Unit Square in $\mathbb{R}^2$*): The Cartesian product of open intervals:

$$U:=(0,1)\times(0,1)=\{(x_1,x_2)\in\mathbb{R}^2\mid 0<x_1<1 \text{ and } 0<x_2<1\}\subset\mathbb{R}^2 \tag{1.54}$$

is an open subset of $(\mathbb{R}^2,d_{\text{Euc}})$.

Verification of Openness

Let $x=(x_1,x_2)\in(0,1)\times(0,1)$. We seek a radius $r>0$ such that $B(x,r)\subseteq(0,1)\times(0,1)$. The distances from x to the four boundary edges of the square are:

- Distance to the left edge ($x_1=0$): x_1,
- Distance to the bottom edge ($x_2=0$): x_2,
- Distance to the right edge ($x_1=1$): $1-x_1$,
- Distance to the top edge ($x_2=1$): $1-x_2$.

Since $0<x_1<1$ and $0<x_2<1$, each of these four quantities is strictly positive. We define:

$$r:=\min\{x_1,x_2,1-x_1,1-x_2\}>0. \tag{1.55}$$

Now let $y=(y_1,y_2)\in B(x,r)$. By definition of the Euclidean metric:

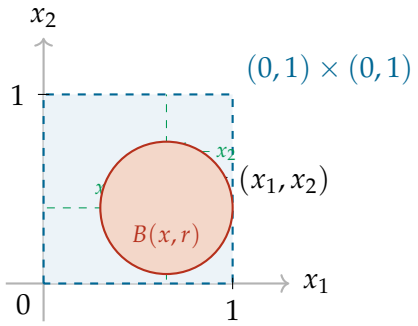
$$|y_1-x_1|=\sqrt{(y_1-x_1)^2}\le\sqrt{(y_1-x_1)^2+(y_2-x_2)^2}=d_{\text{Euc}}(y,x)<r\le x_1,$$

which implies $y_1>x_1-x_1=0$. Similarly:

$$y_1-x_1\le|y_1-x_1|<r\le 1-x_1 \implies y_1<1.$$

Applying the identical estimate to the second coordinate gives $0<y_2<1$. Hence, $y\in(0,1)\times(0,1)$, confirming $B(x,r)\subseteq(0,1)\times(0,1)$.

Q.E.D.



Closed Sets and Topology

Closed Sets

Mündliches Protokoll [00:08:35 - 00:09:15]

Okay, the second point in the definition is what a closed set is.

A subset $A \subseteq X$ is called closed if the complement is open: $X \setminus A$ is open. *[writes „• $A \subset X$ is closed if $X \setminus A$ is open“]*

Definition 1.41 (*Closed Set*): Let (X, d) be a metric space. A subset $A \subseteq X$ is called a *„closed set“* if its complement in X is open:

$$A \text{ is closed} \iff X \setminus A \text{ is open.} \tag{1.56}$$

Complementary Duality: Closedness is defined directly through openness. A set is not merely *„not open“*; rather, a set is closed precisely when its complement contains an open ball around each of its points.

Mündliches Protokoll [00:09:15 - 00:10:15]

This is closed; this is open.

Now, this set is not closed, as you can show. Instead, this set, for instance, is closed: $[0,1] \times [0,1]$ *[writes $[0,1] \times [0,1]$]*.

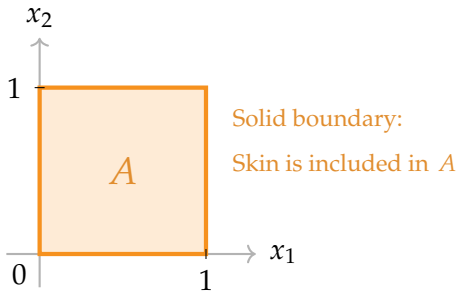
Now it needs to contain the skin to be closed, because when you pass to the complement, which is by definition open, it must not contain the boundary, right?

So, the closed sets, we draw them with their skin. This is closed, A. *[draws solid square labeled A]*

Beispiel 1.42 (*Closed Unit Square in $\mathbb{R}^2$*): The closed unit square:

$$A := [0,1] \times [0,1] = \{(x_1, x_2) \in \mathbb{R}^2 \mid 0 \leq x_1 \leq 1 \text{ and } 0 \leq x_2 \leq 1\}$$

is a closed subset of $\mathbb{R}^2$.



Why the Closed Square Has an Open Complement: The complement $\mathbb{R}^2 \setminus ([0,1] \times [0,1])$ is the union of four unbounded open half-planes:

$$\{x_1 < 0\} \cup \{x_1 > 1\} \cup \{x_2 < 0\} \cup \{x_2 > 1\}.$$

Since each of these half-planes is open, their union is open, proving that the square $[0,1] \times [0,1]$ is closed.

The Topology Associated to a Metric

Mündliches Protokoll [00:10:15 - 00:11:35]

And maybe the last part of this definition, which we will not actually use, but it is a common term that you will find in books and it's good that you know, is what a topology is. *[writes „Topology associated to d"]*

You will see in the notes the word *„topology"* many times. You are not yet studying topology, but just for your information, the topology associated to the distance d is — we can put a $\mathcal{T}$ like this, a sort of calligraphic $\mathcal{T}$ — the collection of all open sets: *[writes $\mathcal{T} = \{U \subset X \mid U \text{ open}\}$]*

This is the topology. So, when we say that something is a topological notion, it has to do with this collection of open sets.

Definition 1.43 (*Metric Topology*): Let (X,d) be a metric space. The *„topology associated to d"* (or metric topology) is the collection $\mathcal{T}$ of all open subsets of X:

$$\mathcal{T} := \{U \subseteq X \mid U \text{ is open in } (X,d)\}. \tag{1.57}$$

Topological Properties: Any mathematical property that can be defined purely in terms of the collection $\mathcal{T}$ (without directly referring to the numerical distances $d(x,y)$) is called a *„topological property"*. These include openness, closedness, continuity, compactness, and connectedness.

Properties of Open and Closed Sets

Unions and Intersections of Open Sets

Mündliches Protokoll [00:11:35 - 00:12:55]

Now, because this is still a bit abstract, to see a bit more practically how we identify open and closed sets, we are going to do it through sequences, okay?

Well, let me first do a lemma that we will use many times. Lemma: unions and intersections of open sets. *[writes „Lemma ($\cup, \cap$ of open sets)“]*

This is one of the basic properties. It's very simple, but since we will use it many, many times, I will prove it now. It says that — I will put it in words first, and then we will prove it with symbols — arbitrary unions of open sets are open. *[writes „• Arbitrary unions of open sets are open“]*

Mündliches Protokoll [00:12:55 - 00:13:17]

I'm sorry, there is a bit of discussion. Would someone tell me what is going on? Who? Who's brave enough? Ah, good.

Student Remark: Exam Results

I think we're waiting for the results. *[laughter]*

Mündliches Protokoll [00:13:21 - 00:13:24]

For results of what?

Student Remark, continued

From exams... exams...

Mündliches Protokoll [00:13:26 - 00:13:27]

Ah, of some exam?

Student Remark, continued

Yeah.

Mündliches Protokoll [00:13:28 - 00:14:09]

Ah, okay! So, it's not that you are not following... *[laughter from the classroom]* Okay, good.

So, this means that... So, today is the result of the first exams of the first semester?

Okay, I can see that you are stressed, then. *[laughter]*

Could we try to be stressed in a lower voice? *[laughter]* Because otherwise, it will be a bit difficult.

Mündliches Protokoll [00:14:09 - 00:15:45]

So, arbitrary unions of open sets are open. This means that if you have some collection $\{U_i\}_{i\in I}$ *[writes $\{U_i\}_{i\in I}$]* of open subsets $U_i \subset X$ *[writes $U_i \subset X$ open]*, then the union of this collection is also open, right? *[writes $\bigcup_{i\in I} U_i$ is open]*

And also, finite intersections of open sets are open, right? *[writes „• Finite intersections of open sets are open"]* This means that if I is a finite set — let's put it like this: the intersection of U_i for i from 1 to capital N, a bounded number — this is an open set. *[writes $\bigcap_{i=1}^{N} U_i$ open]*

Lemma 1.44 (*Unions and Intersections of Open Sets*): Let (X,d) be a metric space.

(i) Arbitrary Unions: If $\{U_i\}_{i\in I}$ is any arbitrary (finite or infinite) collection of open sets in X, then their union is open:

$$\bigcup_{i\in I} U_i \text{ is open.} \tag{1.58}$$

(ii) Finite Intersections: If $U_1,\dots,U_N$ is a finite collection of open sets in X, then their intersection is open:

$$\bigcap_{i=1}^{N} U_i \text{ is open.} \tag{1.59}$$

Proof of Lemma 1.44

Mündliches Protokoll [00:15:45 - 00:17:40]

The proof of this is just using the definition. *[writes „proof"]*

If I want to check that a union is open, I take a point x in this set that is the union: *[writes „take $x \in \bigcup_{i\in I} U_i$"]* Then, in particular, x belongs to one of them, because it's a union. I call it i_0 — it's one of the indices i, any of them. *[writes $\implies x \in U_{i_0}$]* And then, since this set is open by the statement, there exists some radius $r > 0$ *[writes $\implies \exists r > 0$]* such that the ball of radius r centered at x is contained in U_{i_0}. But this is contained in the union of the U_i's: *[writes st $B(x,r) \subset U_{i_0} \subset \bigcup_{i\in I} U_i$]* which proves that this union is open, right?

Proof of Part (i): Arbitrary Unions

Let $x \in \bigcup_{i\in I} U_i$. By definition of the union, there exists an index $i_0 \in I$ such that:

$$x \in U_{i_0}.$$

Since U_{i_0} is open, there exists a radius $r > 0$ such that:

$$B(x, r) \subseteq U_{i_0}.$$

Since $U_{i_0} \subseteq \bigcup_{i \in I} U_i$, it follows immediately that:

$$B(x, r) \subseteq U_{i_0} \subseteq \bigcup_{i \in I} U_i.$$

Since x was arbitrary, this proves that $\bigcup_{i \in I} U_i$ is open.

Mündliches Protokoll [00:17:40 - 00:18:04]

With the finite intersection, you can do it as an exercise. But I will tell you how to do it, okay? Now I will explain the proof with... Yeah? *[throws catch-box microphone to a student]*

Student Question: Clopen Sets

Hello? Yeah. My question was: except for the empty set and the set X itself, can you give us some examples of both closed and open sets?

Mündliches Protokoll [00:18:16 - 00:19:06]

We will discuss this when we discuss connected spaces, okay? This is a good question.

The whole space and the empty set are always open and closed, and I will emphasize this when we see connected spaces. So, you need to wait a bit more. I could give an example, but it's not very interesting — it's just a disconnected space, like the disjoint union of two intervals or something like this, where each interval is open and closed.

But this would not be very interesting. You will see the interest of this question when we see connected spaces, okay? But it's a fair question.

Mündliches Protokoll [00:19:06 - 00:20:34]

So, I am saying that when you take open sets... *[catches microphone back]* When you take finitely many intersections of open sets, you get something open. Why? I will say it in words, and then you try to write it down, okay?

You have finitely many open sets. And now you take a point that is in the intersection of these finitely many, right? This means that the point is in all of them, right? And now, for each of the sets, you pick a radius — which is different every time — but there are finitely many radii such that you can put your ball inside the open set. If you take the minimum of all these radii, it will be good: with the minimum of the finitely many radii, you will fall inside the intersection.

Okay? You don't need to understand what I just said right away; please try to do the exercise. If there are problems doing this, you have it in the notes. But I think it is very good to try these first proofs of the theory, which are very short, one-line proofs. It's

a very good opportunity to do it as an exercise to see if we understood the definitions, because it's one line, so you don't need to think too much. Okay.

Proof of Part (ii): Finite Intersections

Let $U_1, \ldots, U_N$ be open sets in X, and let $x \in \bigcap_{i=1}^{N} U_i$. By definition of the intersection:

$$x \in U_i \quad \text{for every } i \in \{1, \ldots, N\}.$$

Since each U_i is open, there exist radii $r_1, \ldots, r_N > 0$ such that:

$$B(x, r_i) \subseteq U_i \quad \text{for each } i \in \{1, \ldots, N\}.$$

We define the radius:

$$r := \min\{r_1, r_2, \ldots, r_N\}. \tag{1.60}$$

Since the set $\{r_1, \ldots, r_N\}$ is finite and each $r_i > 0$, its minimum is strictly positive: $r > 0$. Now, since $r \leq r_i$ for all $i \in \{1, \ldots, N\}$, we have:

$$B(x, r) \subseteq B(x, r_i) \subseteq U_i \quad \text{for every } i \in \{1, \ldots, N\}.$$

Consequently, the ball is contained in all U_i simultaneously:

$$B(x, r) \subseteq \bigcap_{i=1}^{N} U_i.$$

This proves that $\bigcap_{i=1}^{N} U_i$ is open.

Why Finiteness is Essential for Intersections: The positivity of $r = \min\{r_1, \ldots, r_N\}$ relies fundamentally on taking the minimum over a **finite** number of strictly positive elements. For an infinite collection of open sets, the infimum of positive radii can equal zero:

$$\inf_{n \in \mathbb{N}} \left\{ \frac{1}{n} \right\} = 0.$$

For example, in $\mathbb{R}$ with open intervals $U_n = \left(-\frac{1}{n}, \frac{1}{n}\right)$, their infinite intersection is a single point:

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n} \right) = \{0\},$$

which is not open.

Q.E.D.

Unions and Intersections of Closed Sets

Mündliches Protokoll [00:20:34 - 00:22:04]

Now, an analogous lemma, but for closed sets: Lemma: unions and intersections of closed sets. *[writes „Lemma ($\cup, \cap$ of closed sets)"]*

And it's the same, but the opposite. It says that finite unions of closed sets are closed *[writes „• Finite unions of closed sets are closed"]*, and arbitrary intersections — it could be finite or infinite — of closed sets are closed, okay? *[writes „• Arbitrary intersections of closed sets are closed"]*

You can repeat the same proof, or a variant of the same proof, or you can reason by passing to the complement. Remember that being closed is defined as the complement being open. If you pass to the complement, complementation converts closed sets into open sets, and open sets into closed sets, right?

So, you can always pass statements about open sets to the complement, and you get a statement about closed sets. If you have some property of open sets and you pass the whole statement to the complements, you get a property of closed sets.

Lemma 1.45 (*Unions and Intersections of Closed Sets*)**:** Let (X, d) be a metric space.

(i) Finite Unions: If $A_1, \ldots, A_N$ is a finite collection of closed sets in X, then their union is closed:

$$\bigcup_{i=1}^{N} A_i \text{ is closed.} \tag{1.61}$$

(ii) Arbitrary Intersections: If $\{A_i\}_{i \in I}$ is any arbitrary (finite or infinite) collection of closed sets in X, then their intersection is closed:

$$\bigcap_{i \in I} A_i \text{ is closed.} \tag{1.62}$$

Mündliches Protokoll [00:22:04 - 00:23:55]

And this is what we have here. Why? Because X minus a union of sets *[writes* $X \setminus \bigcup_{i \in I} A_i$*]* is the same as the intersection of the complements, right? *[writes* $= \bigcap_{i \in I} (X \setminus A_i)$*]* When you take the complement of a union, you get the intersection of the complements.

And the same: the complement of an intersection *[writes* $X \setminus \bigcap_{i \in I} A_i$*]* is equal to the union of the complements. *[writes* $= \bigcup_{i \in I} (X \setminus A_i)$*]*

Then you apply the previous result. You apply this property of how intersections pass to unions when you complement, and vice versa, and then you apply the previous result. Because if these sets are closed, these sets will be open, and the finite intersection of open sets is open, okay?

Proof via De Morgan's Laws

Proof of Lemma 1.45

We use De Morgan's laws to translate the properties of open sets directly to closed sets:

Proof of Part (i) (Finite Unions): Let $A_1, \ldots, A_N$ be closed sets. By definition, each complement $X \setminus A_i$ is open. By De Morgan's law:

$$X \setminus \left(\bigcup_{i=1}^{N} A_i \right) = \bigcap_{i=1}^{N} (X \setminus A_i). \tag{1.63}$$

By Lemma 1.44 part **(ii)**, the finite intersection of open sets $\bigcap_{i=1}^{N} (X \setminus A_i)$ is open. Therefore, the complement $X \setminus \left(\bigcup_{i=1}^{N} A_i \right)$ is open, which proves that $\bigcup_{i=1}^{N} A_i$ is closed.

Proof of Part (ii) (Arbitrary Intersections): Let $\{A_i\}_{i\in I}$ be an arbitrary collection of closed sets. Then each $X \setminus A_i$ is open. By De Morgan's law:

$$X \setminus \left(\bigcap_{i\in I} A_i \right) = \bigcup_{i\in I} (X \setminus A_i). \tag{1.64}$$

By Lemma 1.44 part **(i)**, the arbitrary union of open sets $\bigcup_{i\in I}(X \setminus A_i)$ is open. Therefore, the complement $X \setminus (\bigcap_{i\in I} A_i)$ is open, which proves that $\bigcap_{i\in I} A_i$ is closed. Q.E.D.

Duality Summary: Open vs. Closed Sets:

Open Sets $\mathcal{T}$	Closed Sets
Arbitrary unions are open	Finite unions are closed
Finite intersections are open	Arbitrary intersections are closed

De Morgan's laws swap unions and intersections upon complementation, directly exchanging the roles of arbitrary and finite operations.

Importance of Finiteness: Infinite Intersections and Unions

Mündliches Protokoll [00:23:54 - 00:25:17]

Okay. What else...

Let me give an example, very, very easy, showing that this property — where we said finite intersections or finite unions — is important that it is finite. When you take infinite intersections of open sets, you can get a closed set that is not open.

Example: finite is important. *[writes „Example (finite is important)"]*

If you take one of the spaces that you are most familiar with, namely $\mathbb{R}$ with the distance given by the absolute value *[writes $(\mathbb{R}, d_{|\cdot|})$]* — with the Euclidean distance — and you take the sets $U_k = (-1/k, 1/k)$ *[writes $U_k = (-1/k, 1/k)$]*, and you take their intersection: then the intersection for k from 1 to infinity of U_k *[writes $\bigcap_{k=1}^{\infty} U_k$]* gives you the point zero: *[writes $= \{0\}$]*

Mündliches Protokoll [00:25:17 - 00:26:14]

These are open intervals, and since the intervals do not contain their endpoints, each of them is open. But when you take their intersection, you get just the point zero, and the point zero is not open, because you cannot put any ball in it, right? The point zero is closed, but it is not open.

So, these are open, and this is not open. *[writes „open" next to U_k and „not open!" next to $\{0\}$]*

That's why it is important that the intersection is of finitely many sets. And passing to the complement, you get a counterexample for infinite unions of closed sets.

Any questions about this?

Beispiel 1.46 (*Infinite Intersection of Open Sets Need Not Be Open*)**:** In $(\mathbb{R}, d_{\text{Euc}})$, consider the family of open intervals indexed by $k \in \mathbb{N}_{\geq 1}$:

$$U_k := \left(-\frac{1}{k}, \frac{1}{k}\right) \quad \text{(open)}. \tag{1.65}$$

Taking the infinite intersection over all $k \geq 1$ yields:

$$\bigcap_{k=1}^{\infty} U_k = \{0\} \quad \text{(not open!)}. \tag{1.66}$$

Why the Singleton $\{0\}$ is Closed but Not Open:

- $\{0\}$ **is not open:** For any radius $r > 0$, the open ball centered at 0 is the interval $B(0,r) = (-r, r)$. For any $r > 0$, $(-r, r) \not\subseteq \{0\}$ because it contains non-zero points (e.g., $r/2$). Thus, no open ball around 0 is contained in $\{0\}$.
- $\{0\}$ **is closed:** Its complement $\mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$ is an arbitrary union of two open intervals, hence open in $\mathbb{R}$.
- **Dual Counterexample via Complements:** Taking the complements $A_k := \mathbb{R} \setminus U_k = (-\infty, -1/k] \cup [1/k, \infty)$ yields an infinite family of closed sets whose union:

$$\bigcup_{k=1}^{\infty} A_k = \mathbb{R} \setminus \bigcap_{k=1}^{\infty} U_k = \mathbb{R} \setminus \{0\}$$

is open, and hence not closed. This demonstrates that arbitrary unions of closed sets need not be closed.

Interior, Closure, and Boundary in Metric Spaces

Definitions of Interior, Closure, and Boundary

Mündliches Protokoll [00:26:14 - 00:27:44]

Now we will define something interesting, which is what we were saying earlier heuristically: what the interior and the boundary of a set in a topological space are.

In a metric space: *[writes „Defn (Interior, bdry, in metric space)" on the middle board]* Given a set... Let me call the set $Y \subset X$ *[writes „Given $Y \subset X$"]*, (X, d) a metric space *[writes (X, d) metric space]*. Should I call it Y, or let me call it... I don't know, later Y will be something else. Ω, whatever. *[erases Y and writes $\Omega \subset X$]*

Given a set in my metric space,

Mündliches Protokoll [00:27:44 - 00:29:19]

I define the interior of Ω: *[writes „• $\text{int}(\Omega) =$"]* Maybe I will not put parentheses; I will say $\text{int}\,\Omega$, or sometimes you put Ω with a circle like this *[writes Ω°]*, depending on the source. This is what is called the interior of Ω. *[underlines and writes „interior of Ω"]*

This is, by definition, the largest open set contained in Ω. You have your set Ω, you look at all the open sets that are contained in it, and the largest one is the union of all open sets contained in Ω. That is, the union of $U \subset \Omega$ such that U is open: *[writes*

$= \bigcup\{U \subset \Omega \mid U \text{ is open}\}$]

Right? This is the interior.

Definition 1.47 (*Interior of a Set*)**:** Let (X,d) be a metric space and $\Omega \subseteq X$ an arbitrary subset. The „*interior*" of Ω, denoted by $\text{int}(\Omega)$ or Ω°, is defined as the union of all open subsets of X contained in Ω:

$$\text{int}(\Omega) = \Omega^\circ := \bigcup\{U \subseteq \Omega \mid U \text{ is open in } (X,d)\}. \tag{1.67}$$

Characterization of the Interior:

- **Openness:** Since Ω° is an arbitrary union of open sets, it is itself open by Lemma 1.44.
- **Largest Open Subset:** If $V \subseteq \Omega$ is any open set, then V is one of the sets in the union (1.67), so $V \subseteq \Omega^\circ$. Thus, Ω° is the unique maximal open subset of Ω.
- **Openness Criterion:** A set Ω is open if and only if $\Omega = \Omega^\circ$.

Mündliches Protokoll [00:29:19 - 00:30:44]

Now, I also define the closure, which you know from $\mathbb{R}$, right? The closure of some set is the set of points $x \in X$ such that there exists some sequence (x_n) contained inside the set Ω, and the sequence converges to x: *[writes* $\bullet\ \overline{\Omega} = \{x \in X \mid \exists (x_n)_{n\geq 0} \subset \Omega,\ x_n \to x\}$ *]*

Definition 1.48 (*Closure of a Set via Sequences*)**:** Let (X,d) be a metric space and $\Omega \subseteq X$. The „*closure*" of Ω, denoted by $\overline{\Omega}$, is defined as the set of all limit points of sequences in Ω:

$$\overline{\Omega} := \{x \in X \mid \exists (x_n)_{n\geq 0} \subseteq \Omega \text{ such that } x_n \to x\}. \tag{1.68}$$

Mündliches Protokoll [00:30:44 - 00:32:34]

Here I can do a drawing, right? Let's look at an example. *[moves to the right chalkboard and writes „Example* $[0,1) \times [0,1) = \Omega$*"]* Let's take this example: say, $[0,1) \times [0,1)$ — closed-open, closed-open.

How do you draw this example? It would be something like this, right? *[draws a square with solid bottom and left edges, dashed top and right edges, and hatched interior]* It is the inside of this square, but you are missing this part of the boundary and this part of the boundary.

Now, can you see that these points satisfy this definition here? If I take a point x here on the boundary — sorry, on the closure — because I can take a sequence of points from the inside, from the red part (i.e., the hatched interior), that approaches x, right? But these points that are not in my set, because I am excluding them, are also in the closure. This point x over here is also in the closure. *[draws point* x *on the dashed top edge with sequence dots approaching it, labeled* $\cdot \to x \in \overline{\Omega}$*]* This belongs to the closure because I can

take a sequence from here that approaches my point, right?

You can also define this — and actually we will prove this — as the intersection of all closed sets that are bigger than Ω: *[writes $= \bigcap\{A \supset \Omega \mid A \text{ closed}\}$ on the left board]* So, the closure is the smallest closed set that contains Ω.

Definition 1.49 (*Closure as Smallest Closed Superset*): Equivalently, the closure $\overline{\Omega}$ of $\Omega \subseteq X$ is defined as the intersection of all closed subsets of X containing Ω:

$$\overline{\Omega} = \bigcap\{A \subseteq X \mid \Omega \subseteq A \text{ and } A \text{ is closed in } (X,d)\}. \tag{1.69}$$

Equivalence of the Two Definitions of Closure:

- **Smallest Closed Set:** By Lemma 1.45, an arbitrary intersection of closed sets is closed. Hence, $\overline{\Omega}$ defined via (1.69) is closed and is the unique minimal closed set containing Ω.
- **Closedness Criterion:** A set Ω is closed if and only if $\Omega = \overline{\Omega}$.
- **Equivalence with Sequences:** We will formally prove in Lemma 1.52 that the sequential definition (1.68) and the topological definition (1.69) coincide in any metric space.

Mündliches Protokoll [00:32:34 - 00:33:29]

And their difference — this minus that — is called the boundary, and is denoted with this symbol, the del symbol: *[writes $\bullet\ \overline{\Omega} \setminus \Omega^\circ =: \partial\Omega \longleftarrow$ boundary of Ω]* This is called the boundary of Ω.

And this boundary of Ω is the skin. *[writes „„skin““]* Right? If you do any example in the plane, you will see what this is doing: the closure is taking the set with the skin, including the inside, and then for the boundary you are removing the inside.

Definition 1.50 (*Boundary of a Set*): Let (X,d) be a metric space and $\Omega \subseteq X$. The *„boundary“* of Ω, denoted by $\partial\Omega$, is defined as the set difference between the closure and the interior:

$$\partial\Omega := \overline{\Omega} \setminus \Omega^\circ. \tag{1.70}$$

Heuristic and Topological Role of the Boundary: Heuristically, the boundary $\partial\Omega$ is the *„skin“* of the set. Points in $\partial\Omega$ have the property that every open neighborhood intersects both Ω and its complement $X \setminus \Omega$.

Mündliches Protokoll [00:33:29 - 00:35:04]

For this set over here, the interior would be this: *[draws a square with all 4 dashed edges and hatched interior]* This would be Ω°, the interior $(0,1)^2$. *[labels Ω° and writes $(0,1)^2$]*

Then, when you take the closure, you also get the points that can be approximated by a sequence — or if you take the intersection of closed sets, you get the closed square

$[0,1]^2$: *[draws a square with all 4 solid edges and hatched interior, labels $[0,1]^2$]*

And if you want the boundary, it's the difference between the two, which is just this part, without the inside: *[draws a hollow square with 4 solid edges and empty interior, labels $\partial\Omega$]* Right?

Again, all of these are perfectly common-sense notions, but we can formalize them and make them perfectly rigorous just using these few axioms. And then... *[pauses and pushes the chalkboards upward]*

Beispiel 1.51 (*Decomposition of the Half-Open Unit Square in $\mathbb{R}^2$*)**:** Let $\Omega = [0,1) \times [0,1) \subset \mathbb{R}^2$. Then:

- **Interior:** $\Omega^\circ = (0,1) \times (0,1) = (0,1)^2$,
- **Closure:** $\overline{\Omega} = [0,1] \times [0,1] = [0,1]^2$,
- **Boundary:** $\partial\Omega = \overline{\Omega} \setminus \Omega^\circ = ([0,1] \times [0,1]) \setminus ((0,1) \times (0,1))$.



$\Omega = [0,1) \times [0,1)$ (Original Set) | $\Omega^\circ = (0,1)^2$ (Interior) | $\overline{\Omega} = [0,1]^2$ (Closure) | $\partial\Omega = \overline{\Omega} \setminus \Omega^\circ$ (Boundary / „*Skin*“)

Sequential Characterization of Open and Closed Sets

Statement of the Lemma

Mündliches Protokoll [00:35:04 - 00:36:04]

Now, I gave two different definitions of the closure: one through sequences — points that can be approximated by a sequence from the set — and one that is the intersection of all closed sets that contain my set, right? And I said these are the same.

I'm going to give a lemma that actually shows that these are the same. After this lemma, you can take whichever of the two definitions is more convenient each time.

This lemma is called „*open and closed sets through sequences*“: how we determine if a set is open or closed using convergent sequences. *[writes „Lemma (open & closed through sequences)“ on the lower left chalkboard]*

Mündliches Protokoll [00:36:04 - 00:37:39]

It says that U — I don't write it every time, but throughout all this chapter, every time it should say (X,d) is a metric space, right? *[writes (X,d) ms]*

A subset $U \subseteq X$ is open if and only if, whenever I have a sequence x_n in my space and x_n converges to a point $x \in U$, then x_n belongs to U for all n up to finitely many exceptions: *[writes „(1) $U \subset X$ open $\iff$ [whenever $(x_n)_{n\geq 0} \subset X$, $x_n \to x \in U \implies$*

$x_n \in U$ for all n up to finitely many exceptions]"]

Lemma 1.52 (*Sequential Characterization of Open and Closed Sets*): Let (X, d) be a metric space.

(1) Open Sets: A subset $U \subseteq X$ is open if and only if for every sequence $(x_n)_{n\geq 0} \subseteq X$ converging to a point $x \in U$, the sequence belongs to U for all n up to finitely many exceptions:

$$U \subseteq X \text{ is open} \iff \left[\begin{array}{l}\text{whenever } (x_n)_{n\geq 0} \subseteq X \text{ with } x_n \to x \in U, \\ \quad \implies x_n \in U \text{ for all } n \text{ up to finitely many exceptions}\end{array}\right]. \tag{1.71}$$

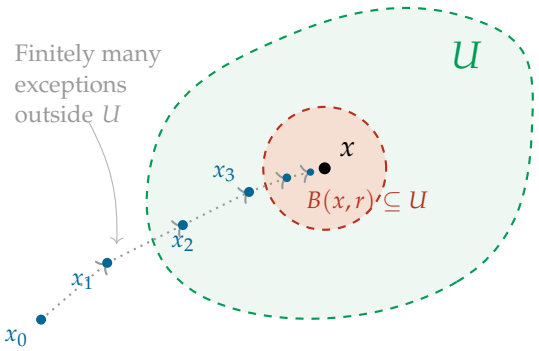
Mündliches Protokoll [00:37:39 - 00:38:59]

You see, I'm going to do the drawing. *[draws a sketch of an open set U with dashed boundary and an incoming sequence x_n converging to $x \in U$]* This is my drawing of an open set U. I have some sequence in the space that is going to converge to this point. It could start anywhere, but it is going to converge to this point. So, after finitely many exceptions, you will be inside the open set.

Why? Because the open set always contains a ball, and the definition of convergence means that at some point you will be inside this ball, whatever the radius of the ball is. This is exactly the definition of convergence.

And this will be the proof, which I will write down, right? This is almost a proof with a picture. You need to translate this picture into symbols, but that is the proof; there is nothing else in the proof than that.

Geometric Intuition: Sequences Converging in Open Sets



Core Mechanism of the Forward Implication: Since U is open and $x \in U$, there exists an open ball $B(x, r) \subseteq U$. By definition of convergence $x_n \to x$, for this radius $r > 0$, there exists an index $N \in \mathbb{N}$ such that $x_n \in B(x, r)$ for all $n \geq N$. Since $B(x, r) \subseteq U$, it follows that $x_n \in U$ for all $n \geq N$. The only terms that can possibly lie outside U belong to the finite initial set $\{x_0, x_1, \ldots, x_{N-1}\}$.

The Notion of „*Eventually*“

Mündliches Protokoll [00:38:59 - 00:39:59]

This concept of „*up to finitely many exceptions*“ is very useful, and it is very nice to say things concisely, so we give a name to this. You have a more formal definition in the notes.

When you have a sequence, and some property — like, for instance, being in U — is true up to finitely many exceptions, we say that this property happens **eventually**. *[writes $\iff$ eventually]*

This is a word I will use many times, okay? Whenever I have a sequence and I say that this happens eventually, it means that maybe at the beginning of the sequence it is not happening, but if I wait long enough, for n large enough, it will happen, and then it never stops happening. Yeah? *[taps the board]*

Student Question: Formal Meaning of „*Eventually*“

Is this eventually equivalent to saying there exists a big N such that for all n bigger than big N, this holds?

Mündliches Protokoll [00:40:08 - 00:40:54]

Exactly. Up to finitely many exceptions means that — if you want, this is equivalent by definition to „*eventually*“, which is equivalent to: *[writes $\iff \exists N \text{ st } \forall n \geq N$]* there exists some N such that for all $n \geq N$, the property holds. And this N just counts how many exceptions you have; it is an upper bound for the number of exceptions.

All of these mean the same thing. You will see that it is very useful to say things concisely with this „*eventually*“. That's why I introduced it, and it's very standard as well.

Definition 1.53 (*Eventually*): Let $(x_n)_{n\geq 0}$ be a sequence and $P(x_n)$ a mathematical predicate. We say that $P(x_n)$ holds „*eventually*“ (or for all n up to finitely many exceptions) if:

$$\exists N \in \mathbb{N}\ \forall n \geq N: \quad P(x_n) \text{ is true.} \tag{1.72}$$

In particular, $x_n \in U$ eventually if and only if $\exists N \in \mathbb{N}$ such that $x_n \in U$ for all $n \geq N$.

Sequential Characterization of Closed Sets

Mündliches Protokoll [00:40:54 - 00:42:14]

And the second part: *[writes „(2) $A \subset X$ closed $\iff$ [whenever $(x_n)_{n\geq 0} \subset A$, $x_n \to x \implies x \in A$]“]* A subset A is closed if and only if, whenever I have a sequence contained in A and the sequence converges to some limit point in X, then the limit belongs to A.

That is what I said earlier, more or less, right? Being closed.

So, here, by closed... *[bell rings in the hallway, students begin talking]* Well, let's take a break, yeah.

Lemma 1.54 (*Sequential Characterization of Closed Sets*): Let (X,d) be a metric space.

(2) Closed Sets: A subset $A \subseteq X$ is closed if and only if it is closed under taking limits of sequences in A:

$$A \subseteq X \text{ is closed} \iff \left[\begin{array}{l}\text{whenever } (x_n)_{n\geq 0} \subseteq A \text{ and } x_n \to x \in X, \\ \implies x \in A\end{array}\right]. \tag{1.73}$$

Equivalence with Sequential Closure: Recall that the closure $\overline{A}$ was defined in (1.68) as the set of all limits of sequences in A. Lemma 1.54 asserts that A is closed if and only if $A = \overline{A}$, establishing complete consistency with the topological definition.

Full Statement: Sequential Characterization of Open and Closed Sets

Lemma 1.55 (*Sequential Characterization of Open and Closed Sets (Complete Statement)*): Let (X,d) be a metric space.

(1) Open Sets: A subset $U \subseteq X$ is open if and only if for every sequence $(x_n)_{n\geq 0} \subseteq X$ converging to a point $x \in U$, the sequence belongs to U eventually:

$$U \subseteq X \text{ is open} \iff \Big(\forall (x_n)_{n\geq 0} \subseteq X \colon x_n \to x \in U \implies \exists N \in \mathbb{N}\, \forall n \geq N \colon x_n \in U\Big). \tag{1.74}$$

(2) Closed Sets: A subset $A \subseteq X$ is closed if and only if it contains all limits of sequences in A:

$$A \subseteq X \text{ is closed} \iff \Big(\forall (x_n)_{n\geq 0} \subseteq A \colon x_n \to x \in X \implies x \in A\Big). \tag{1.75}$$

Clarification on Quantifiers and Commencement of the Proof

Mündliches Protokoll [00:42:14 - 00:43:04]

[classroom noise and chatter as students talk loudly] Please... Please... Okay, I will start if you don't listen... *[laughter]*

May you forgive me for the stereotype, but when I studied mathematics in Barcelona — which is a Mediterranean country — people started the lecture on time, so they were silent. We are in Switzerland: you should be like the champions of starting on time! And I'm seeing the contrary; I'm very surprised. Okay. *[wipes the lower right chalkboard with a sponge]*

And you know this is recorded, so the proof of the misbehavior will be there for history. Okay.

Mündliches Protokoll [00:43:05 - 00:43:55]

We said we are going to prove **(1)**.

So, I said that U open is equivalent to this property. Let me clarify, because this is a terminology that is very useful, but maybe you are not familiar with it: when I say *„whenever“*, it means whenever. That is, for all sequences... *[writes ∀ seq above „whenever“]* For every sequence satisfying this, then that happens, right? *„Whenever“* is like a *„for all“*.

So, *„whenever“* essentially means *„for all“*.

Logical Parsing of the Statement

In the formulation of Lemma 1.52:

$$\text{„whenever “}(x_n)_{n\geq 0} \subseteq X \text{ with } x_n \to x \in U \implies x_n \in U \text{ eventually}$$

the term *„whenever“* is a formal universal quantifier over all sequences in X:

$$\forall (x_n)_{n\geq 0} \subseteq X\colon \quad \big(x_n \to x \text{ with } x \in U \implies \exists N \in \mathbb{N}\, \forall n \geq N\colon x_n \in U\big).$$

Proof of the Sequential Characterization of Open Sets

Proof of Lemma 1.52, Part (1)

Forward Implication ($\implies$)

Mündliches Protokoll [00:43:55 - 00:45:06]

Proof of **(1)**. *[writes „proof (1) $\implies$ “ on the lower right chalkboard]* I need to do two implications. First, I will do this direction, which is the one I've been drawing here.

Assume that U is open. Now, I'm just going to put in symbols the drawing that I did here. Take $x \in U$ *[writes „take $x \in U$“]*, any point. By definition of an open set, since U is open, there exists some $r > 0$ such that the ball of radius r centered at x is contained in U, right? *[writes $\implies \exists r > 0$ st $B(x,r) \subset U$ and notes „given U open“]*

Forward Implication: Openness Implies Eventual Containment

Goal ($\implies$): Assume $U \subseteq X$ is open. Let $(x_n)_{n\geq 0} \subseteq X$ be any sequence converging to a point $x \in U$. We must show that $x_n \in U$ eventually.

Let $x \in U$. Since U is open by assumption, there exists a radius $r > 0$ such that:

$$B(x,r) \subseteq U. \tag{1.76}$$

Mündliches Protokoll [00:45:06 - 00:46:58]

And now, since by assumption x_n converges to x *[writes „Since $x_n \to x$“]*... What was the definition of convergence? Remember: given any ε — in the definition I called this ε *[points at $r > 0$]*, right? Given any ε, now it is r, but it is the same — for n large enough, bigger than a certain capital N, my sequence will belong to the ball of radius r.

So, since x_n converges to x, there exists some capital N, depending on r, such that *[writes $\exists N$ st]* x_n belongs to the ball of radius r centered at x for all $n \geq N$: *[writes $x_n \in B(x,r)\ \forall n \geq N$]*

Now, in the definition of convergence, maybe there was not this ball, but there was something equivalent: distance. This is the same: saying that $x_n \in B(x,r)$ is the same as saying that the distance between x_n and x is less than r. *[boxes $x_n \in B(x,r)$ and writes $\iff d(x_n,x) < r$]* Because when does a point belong to the ball? When the distance to the center is strictly less than r. This is if and only if, right?

And this is what I'm saying here: end of the proof. *[draws $\square$]* Because up to finitely many exceptions — that is this capital N — all subsequent points of the sequence belong to the ball, and the ball is contained in U, so they belong to U, right? So, this proves this implication.

Conclusion of Forward Implication

Since $x_n \to x$, by the definition of convergence with $\varepsilon = r > 0$, there exists an index threshold $N = N(r) \in \mathbb{N}$ such that:

$$d(x_n, x) < r \quad \forall n \geq N. \tag{1.77}$$

By definition of the open ball, this condition is identical to:

$$x_n \in B(x,r) \quad \forall n \geq N. \tag{1.78}$$

Since $B(x,r) \subseteq U$ from (1.76), we conclude:

$$x_n \in U \quad \forall n \geq N.$$

Thus, $x_n \in U$ eventually (for all n up to finitely many exceptions). This completes the proof of the forward implication $\implies$.

Reverse Implication ($\Leftarrow$) by Contraposition

Mündliches Protokoll [00:46:58 - 00:48:45]

Now, still for **(1)**, I will prove the second implication — the one this way: *[writes $\Leftarrow$ in a box]*

And I will prove this by contraposition, right? $A \implies B$ is equivalent to $\neg B \implies \neg A$, right? We know that. To prove this, I will prove that not-left gives not-right: *[writes $\neg$left $\implies$ $\neg$right and „contraposition“]* This is a proof by contraposition.

Now, let's do a bit of logic. What is the negation of U open? Remember the definition of U open: for every point in my set U, there exists some radius such that the ball is contained in U, right?

And now we take the negation. The negation of *„for every point“* is *„there exists a point“*. So, the negation — U is not open — is the same as: there exists $x \in U$ *[writes*

„U *not open* $\iff$ $\exists x \in U$“*]* such that for every radius $r > 0$ *[writes* $\forall r > 0$*]*, the ball centered at x with radius r is not contained in U: *[writes* $B(x,r) \not\subset U$*]*

Logical Setup of Contraposition

Goal ($\Leftarrow$): We wish to prove:

$$\big(\text{whenever } x_n \to x \in U \implies x_n \in U \text{ eventually}\big) \implies U \text{ is open.}$$

We prove this by contraposition ($\neg$left $\implies$ $\neg$right):

$$U \text{ is not open} \implies \neg\big(\forall (x_n) \subseteq X\colon x_n \to x \in U \implies x_n \in U \text{ eventually}\big).$$

Negating Openness: Recall the definition of openness:

$$U \text{ is open} \iff \forall x \in U\ \exists r > 0\colon B(x,r) \subseteq U.$$

Negating this statement yields:

$$\begin{aligned} U \text{ is not open} &\iff \neg\big(\forall x \in U\ \exists r > 0\colon B(x,r) \subseteq U\big) \\ &\iff \exists x \in U\ \forall r > 0\colon B(x,r) \not\subseteq U. \end{aligned} \tag{1.79}$$

Mündliches Protokoll [00:48:45 - 00:50:46]

And that the ball is not contained in U is the same as saying that there exists some point — let's call it x_r, because it depends on r — that belongs to the complement: *[writes* $\iff \exists x_r \in B(x,r) \cap (X \setminus U)$*]*

What am I saying here? Now I will do a drawing of that.

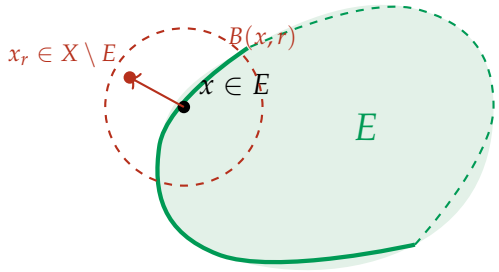
I take a set that is not open — let me call it E: *[draws a sketch of set E on the middle-left chalkboard with a solid boundary segment]* Heuristically, if the skin were not contained, it would be open. I draw it like this because there is a piece of skin, so it is not open, okay? This is just to make the drawing.

Now, I'm saying: there exists a point x on the skin *[marks x on the solid boundary of E]* such that for every radius that I choose here, I can find a point outside my set. This point is what I called x_r, right? *[marks x_r outside E in the circle around x]* In other words, there is some point in the set that can be approximated by a sequence of points from outside the set.

Geometric Construction of Approximating Sequence

The condition $B(x,r) \not\subseteq U$ means that for every $r > 0$, the ball contains at least one point outside U:

$$B(x,r) \not\subseteq U \iff \exists x_r \in B(x,r) \cap (X \setminus U). \tag{1.80}$$



Heuristic Idea: Approximating a Boundary Point from Outside: If E contains a portion of its boundary (*„skin“*), any point x on this boundary cannot be insulated by an open ball. Every ball $B(x,r)$, no matter how small, must leak into the exterior $X \setminus E$, providing an escape point $x_r \notin E$.

Mündliches Protokoll [00:50:46 - 00:52:28]

Taking $r = 1/n$ for $n \geq 1$ *[writes „taking $r = 1/n, n \geq 1$“]*, I produce a sequence of points. These balls are smaller and smaller, and these points x_r — I define $x_n := x_{r=1/n}$ *[writes $x_n := x_{r=1/n}$]*. These points are in balls closer and closer to x, so they converge to the point x, right? *[writes $x_n \to x$]*

I found a sequence that converges to the point x, and that comes from the outside.

But then, this is false. Because the statement says that up to finitely many exceptions, the sequence should be contained in U, whereas all the elements of this sequence are outside of U! So, this contradicts that x_n belongs to U eventually: *[writes „contradicts $x_n \in U$ eventually“]*

This contradicts that $x_n \in U$ eventually, and that ends the proof of this implication. *[draws □]*

I proved that not-left implies not-right, right? Except for the spelling. *[laughter]*

Contradiction and Conclusion of Reverse Implication

We now construct an explicit escaping sequence. For each $n \in \mathbb{N}_{\geq 1}$, we choose the radius:

$$r_n := \frac{1}{n} > 0. \tag{1.81}$$

By (1.80), for each $n \geq 1$ there exists a point $x_n := x_{1/n}$ such that:

$$x_n \in B\left(x, \frac{1}{n}\right) \cap (X \setminus U). \tag{1.82}$$

Because $x_n \in B(x, 1/n)$, the distance satisfies:

$$d(x_n, x) < \frac{1}{n} \xrightarrow{n\to\infty} 0,$$

which proves that $x_n \to x \in U$.

However, by construction, every term of the sequence lies in the complement of U:

$$x_n \in X \setminus U \iff x_n \notin U \quad \forall n \geq 1.$$

Therefore, x_n can never belong to U, which directly contradicts the assertion that $x_n \in U$ eventually!

Thus, assuming U is not open implies the existence of a sequence $x_n \to x \in U$ that is not eventually in U. By contraposition, this completes the proof of the reverse implication $\Leftarrow$.

Q.E.D.

Student Interaction: Notation of Subset Symbols

Mündliches Protokoll [00:52:28 - 00:53:26]

Is this okay, this type of proof?

This is the usual problem when I see people who look very serious: I don't know if they are completely lost, if they find it super boring, or what the problem is. So, if someone has questions and wants to participate, we can slow down and ask questions. Otherwise, we continue with other stuff.

We still need to do this one, right? *[points at Part **(2)** of the Lemma on the left board]* Good.

Ah, there is a question... Sorry? Up there, I don't see? Ah, there! Oh, nice. Thank you. *[tosses catch-box microphone up to student]*

Student Question: Notation for Subsets

You write the... you write the subset symbol without the equality sign at the bottom. Does it have a meaning, or like that it cannot be equal, or is that just notation? Like, subset without the extra line at the bottom.

Mündliches Protokoll [00:53:38 - 00:54:06]

Ah, you mean this? *[points to $\subset$ on the board]* It's the same. You mean this? *[points to $\subset$ again]* Well, it's not the same symbol, but it's the same meaning.

Some people use the equality line. Since it means the same, I save one line, right? But this is very standard; many people just use this. So, I will always use this *„contained“* symbol without the equality sign.

Bemerkung 1.56 (*Standing Convention: Subset Notation $\subset$ vs. $\subseteq$*)**:** Throughout this course, the symbol $\subset$ is used interchangeably with $\subseteq$ to denote non-strict set inclusion:

$$A \subset B \iff A \subseteq B \iff \forall x \in A\colon x \in B.$$

In particular, $A \subset B$ permits the possibility that $A = B$. If a strict (proper) subset is intended, it will be stated explicitly.

Exercises: Sequential Closedness and Metric Subspaces

Mündliches Protokoll [00:54:06 - 00:55:23]

Any others? Thanks. *[catches microphone back]*

Let's do this, because maybe you will find continuous functions more exciting. Since this proof over here is super similar to the previous one — and again, you see that these proofs are one-line proofs, just using the definitions and manipulating them around — I leave this proof as an exercise. Try to do it. If you understood the definitions well and can draw a picture of what's going on, this shouldn't be too difficult.

It can be a bit difficult at the beginning because there are a lot of new definitions and you need to learn them. But the best way is to practice with these exercises and try to do them by yourself. Because if I do them, it looks like a bunch of symbols that all work out, but you don't internalize the definitions. The best way is to try by yourself.

Übung 1.57 (*Sequential Characterization of Closed Sets*): Prove Part (2) of Lemma 1.52: Let (X,d) be a metric space. A subset $A \subseteq X$ is closed if and only if:

$$\text{whenever } (x_n)_{n\geq 0} \subseteq A \text{ and } x_n \to x \in X \implies x \in A.$$

Proof Strategy for the Exercise:

- **Direction $\implies$:** Assume A is closed, so $X \setminus A$ is open. If $(x_n) \subseteq A$ converges to $x \in X \setminus A$, then by Part (1), $x_n \in X \setminus A$ eventually, contradicting that all $x_n \in A$. Hence, $x \in A$.
- **Direction $\impliedby$:** Assume A is sequentially closed. To show $X \setminus A$ is open, apply Part (1): if $x_n \to x \in X \setminus A$ but $x_n \in A$ for infinitely many n, a subsequence in A converges to $x \in A$, a contradiction.

Mündliches Protokoll [00:55:23 - 00:56:54]

Now I will do something more exciting, which is continuity. Or maybe I can put another exercise — you have more in the exercise list: *[writes „exercise" on the board]*

Exercise before continuity: Now that you have all of this, you can prove that if (X,d) is a complete metric space *[writes (X,d) complete]* and $A \subset X$ is a closed subset *[writes $A \subset X$ closed]*, then A with the same distance restricted is complete: *[writes (A,d) complete]*

Example: we know that $\mathbb{R}^n$ is a complete metric space. Then any closed subset of $\mathbb{R}^n$ will be a complete metric space when you restrict the distance.

Now you have all the ingredients to prove that. It's not too difficult; it's a good exercise.

Übung 1.58 (*Closed Subsets of Complete Metric Spaces Are Complete*): Let (X,d) be a complete metric space and $A \subseteq X$ a closed subset. Prove that the subspace $(A, d|_{A\times A})$ is a complete metric space.

Beispiel 1.59 (*Complete Subspaces of $\mathbb{R}^n$*): Since $(\mathbb{R}^n, d_{\text{Euc}})$ is complete, every closed subset $A \subseteq \mathbb{R}^n$ (such as closed balls, closed cubes, or submanifolds without boundary) equipped with the Euclidean distance is a complete metric space.

Guided Derivation for Übung 1.58:

(1) Let $(x_n)_{n\geq 0} \subseteq A$ be a Cauchy sequence in $(A, d|_{A\times A})$.
(2) Since the metric on A is the restriction of d, $(x_n)_{n\geq 0}$ is also a Cauchy sequence in the ambient space (X, d).
(3) Since (X, d) is complete, there exists a limit $x \in X$ such that $x_n \to x$ in (X, d).
(4) Because A is closed, by Lemma 1.54 the limit point must lie in A ($x \in A$).
(5) Thus, the Cauchy sequence converges to a limit inside A, proving that $(A, d|_{A\times A})$ is complete.

Continuity in Metric Spaces

Definition of Continuous Maps Between Metric Spaces

Mündliches Protokoll [00:56:54 - 00:58:23]

I will switch to something that is a bit more fun, which is continuity, because you already know...

Now we start to have problems with the shadows. Someone — I don't remember who — complained about shadows. Now I can write here, right? It's just when the other blackboard is annoying, okay.

Now we'll see continuity. *[writes „Continuity" as header on the left board]* For continuity, we will define continuous maps. We will be in this setup: f will be a map between two sets X and Y *[writes $f: X \to Y$]*, and they are both metric spaces. So, we will have X with its distance d_X, and Y with its distance d_Y: *[writes $(X, d_X), (Y, d_Y)$ metric spaces]*

Setup for Continuity

Setting for Continuous Mappings: Let (X, d_X) and (Y, d_Y) be two metric spaces, and let:

$$f: X \to Y$$

be a mapping from X into Y.

Mündliches Protokoll [00:58:23 - 00:59:43]

Many times for us, these sets will be the Euclidean space of some dimension and the Euclidean space of some other dimension, right? But they can be whatever. Both are metric spaces — different or the same.

Now, the definition: *[writes „Def'n"]* We say that a map f is continuous *[writes „f is continuous"]* if one of the following three equivalent properties holds true: *[writes „if one*

of the following 3 equivalent properties hold true:"]

I will state the three definitions of continuity, because you are already very familiar with at least two of them from Analysis I, right? But I will state them now by postulating that they are equivalent, and later we will prove immediately that they are indeed equivalent, right?

Definition 1.60 (*Continuous Mapping between Metric Spaces*): Let (X, d_X) and (Y, d_Y) be metric spaces. A mapping $f: X \to Y$ is called *„continuous"* if any one of the following three equivalent properties holds true:

(1) ε-δ Continuity (Local/Metric Form): For every point $x \in X$, small variations in x produce arbitrarily small variations in $f(x)$.

(2) Sequential Continuity: f preserves convergent sequences: whenever $x_n \to x$ in X, then $f(x_n) \to f(x)$ in Y.

(3) Topological Continuity (Pre-images of Open Sets): For every open subset $V \subseteq Y$, the pre-image $f^{-1}(V)$ is open in X.

Definition (1): ε-δ Continuity

Mündliches Protokoll [00:59:43 - 01:00:23]

Definition **(1)** of continuity is the ε-δ continuity that you are familiar with. *[writes „(1) ε-δ" in a box]*

You know this very well for functions of one real variable. Now we need to abstract the same notion to metric spaces. How was the definition with ε-δ? Can someone remind me: on the real line, what is the definition of continuity with ε-δ? Does someone remember? *[holds up catch-box microphone and looks at the class]*

Yeah, good. *[throws microphone to student]*

Student Answer: Definition of ε-δ Continuity

Hello? Hello. Is it working? Okay, so I think it was... a function is continuous at x_0 if for every ε larger than zero, there exists a δ such that for all x minus, that's $x - x_0$ smaller than δ, $f(x) - f(x_0)$ is smaller than ε, and if that holds, yeah.

Mündliches Protokoll [01:00:52 - 01:02:43]

Up to here is very good. You said what is continuous at a point. For a continuous function, it is continuous at all points — you were going to say that. But I will use directly continuous at every point.

I will start exactly as you said. First I will add — since I want it to be continuous at every point — *„for every point, for all $x \in X$"*: *[writes $\forall x \in X$]*

Next is what you said: for every x, given — *„given"* is *„for all $\varepsilon > 0$"* — there exists $\delta > 0$: *[writes $\forall \varepsilon > 0\ \exists \delta > 0$]*

And now the rest of your definition used intervals and stuff that are not available in metric spaces, so how do we translate? The notion will be that if the distance between x and x' is less than δ, then the distance between $f(x)$ and $f(x')$ is less than ε: *[writes* $[d(x,x') < \delta \implies d(f(x),f(x')) < \varepsilon]$*]*

And what is this distance? This distance is the distance in the space here, X, and these guys belong to Y, so this is the distance in Y. *[adds subscripts* d_X *and* d_Y*]* Right? What else could it be? It cannot be anything else. If you think about this, it's the only way to generalize.

Another way to say this: for all $x' \in X$ that satisfies this, it also satisfies that, okay? *[writes* $(\forall x' \in X)$ *above]*

Definition 1.61 (ε-δ *Continuity*): Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f\colon X \to Y$ is *„ε-δ continuous“* if for every point $x \in X$ and every tolerance $\varepsilon > 0$, there exists a radius $\delta > 0$ such that all points within distance δ from x map to points within distance ε from $f(x)$:

$$\forall x \in X\ \forall \varepsilon > 0\ \exists \delta > 0\colon \quad \Big[\forall x' \in X\colon d_X(x,x') < \delta \implies d_Y(f(x),f(x')) < \varepsilon\Big]. \tag{1.83}$$

From $\mathbb{R}$ to Metric Spaces: In standard real analysis on $\mathbb{R}$, distance is measured via the absolute value $|x - x'| < \delta$ and $|f(x) - f(x')| < \varepsilon$. In arbitrary metric spaces, these absolute values are replaced by the corresponding metric distance functions $d_X(x,x')$ and $d_Y(f(x),f(x'))$.

Reformulation via Open Balls

Mündliches Protokoll [01:02:43 - 01:03:44]

Another way to say this — because this is a bit annoying to write, it's a bit long — is to say that this thing here, the red bracket, is exactly the same as saying that when I compute f, the image of the ball of radius δ centered at x will be contained in the ball of radius ε centered at $f(x)$: *[writes in red chalk* $\iff [f(B_\delta(x)) \subset B_\varepsilon(f(x))]$*]*

Mündliches Protokoll [01:03:44 - 01:04:19]

You see this? What does f of this ball mean? f of the ball is the image of all the points in the ball. And I'm saying that for every point in the ball of radius δ here, the image will be here, in the other ball, right?

This thing is a compact way of writing that condition, and I will use this bottom part.

So, this is the ε-δ definition.

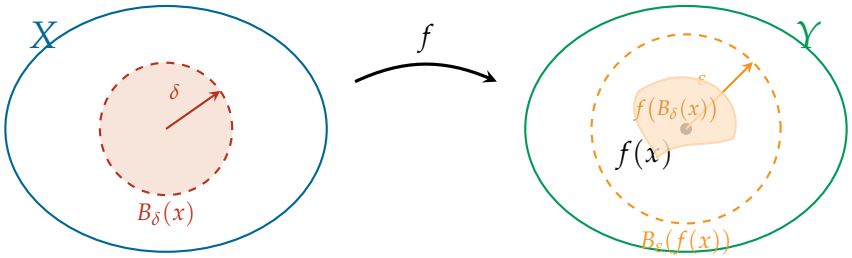
Reformulation of Continuity via Open Balls

The bracketed implication in (1.83) can be formulated compactly using open balls:

$$\forall x' \in X\colon \Big(d_X(x,x') < \delta \implies d_Y(f(x),f(x')) < \varepsilon\Big) \iff f\big(B_\delta(x)\big) \subseteq B_\varepsilon\big(f(x)\big). \tag{1.84}$$

Thus, the condition of ε-δ continuity becomes:

$$\forall x \in X\, \forall \varepsilon > 0\, \exists \delta > 0\colon \quad f\big(B_\delta(x)\big) \subseteq B_\varepsilon\big(f(x)\big). \tag{1.85}$$



Geometric Interpretation of Ball Containment: The condition $f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$ encapsulates the geometric essence of continuity: the mapping f contracts a sufficiently small neighborhood $B_\delta(x)$ in the domain into the prescribed target tolerance ball $B_\varepsilon(f(x))$ in the codomain.

Definition (2): Sequential Continuity

Mündliches Protokoll [01:04:19 - 01:05:40]

The second definition is with sequences, right? Continuity with sequences. *[writes „**(2)**“]*

This is essentially the same as in $\mathbb{R}$. If (x_n) is a sequence converging to a point x, then the sequence of images, $(f(x_n))_{n\geq 0}$, is convergent *[writes $(x_n)_{n\geq 0} \subset X$ & $x_n \to x \implies (f(x_n))_{n\geq 0}$]* in the space Y with limit $f(x)$. *[writes „is convergent in Y with limit $f(x)$“]*

Mündliches Protokoll [01:05:40 - 01:06:11]

A compact way to write this, if you want, is that $f(x_n)$ converges as $n \to \infty$ to $f(x)$: *[writes $\iff [f(x_n) \xrightarrow{n\to\infty} f(x)]$]*

For this, you essentially don't even need anything new: it's really the same definition as in $\mathbb{R}$; you plug it in here and it makes sense.

Definition 1.62 (*Sequential Continuity*): Let (X,d_X) and (Y,d_Y) be metric spaces. A mapping $f\colon X \to Y$ is called *„sequentially continuous“* if it preserves limits of convergent sequences:

$$\forall (x_n)_{n\geq 0} \subseteq X \text{ with } x_n \xrightarrow{n\to\infty} x \text{ in } X \implies f(x_n) \xrightarrow{n\to\infty} f(x) \text{ in } Y. \tag{1.86}$$

Sequential Preservation: Sequential continuity states that the operations of apply-

ing the function f and taking sequential limits commute:

$$\lim_{n\to\infty} f(x_n) = f\left(\lim_{n\to\infty} x_n\right).$$

Definition (3): Topological Continuity

Mündliches Protokoll [01:06:11 - 01:07:32]

And then there is a new... *[a student sneezes: „bless you“]* ...a new notion of continuity, which is the topological notion of continuity. Number **(3)**: this one is new, but it will be very easy to prove that it is the same. *[moves to the chalkboard on the right and writes „**(3)**“]*

We will use this all the time, which is why I am giving it. Topological: *[writes „topological“]* it says that for every open $V \subset Y$ *[writes $\forall V \subset Y$ open]*, the preimage $f^{-1}(V)$ *[writes $f^{-1}(V)$]*, which is a subset of X *[writes $\subset X$]*, is open. *[writes „open“]*

This is the third definition of continuity. A map, a function is continuous whenever you take an open set in the target, and you look back at it — you take f^{-1} — you get an open set here in the domain.

Definition 1.63 (*Topological Continuity*): Let (X, d_X) and (Y, d_Y) be metric spaces. A mapping $f\colon X \to Y$ is called *„topologically continuous“* if the pre-image of every open subset of Y is open in X:

$$\forall V \subseteq Y \text{ open in } (Y, d_Y) \implies f^{-1}(V) \subseteq X \text{ is open in } (X, d_X). \tag{1.87}$$

Pre-images vs. Forward Images: A crucial asymmetry in topology is that continuity is defined via **pre-images** ($f^{-1}(V)$), not forward images ($f(U)$). The forward image of an open set under a continuous map need not be open. For instance, the constant map $f\colon \mathbb{R} \to \mathbb{R}$ given by $f(x) = 0$ is continuous, but the image of the open set $U = (0,1)$ is the singleton $\{0\}$, which is not open in $\mathbb{R}$.

Equivalence of the Three Definitions of Continuity

Mündliches Protokoll [01:07:32 - 01:07:57]

Now, maybe it's not obvious that the three are equivalent, but we will show that.

Proposition: *[writes „Prop“ and **(1)** $\iff$ **(2)** $\iff$ **(3)** on the board]*

Proposition 1.64 (*Equivalence of Definitions of Continuity*): Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f\colon X \to Y$ be a mapping. The following three statements are equivalent:

(1) f is ε-δ continuous (Definition 1.61).
(2) f is sequentially continuous (Definition 1.62).

(3) f is topologically continuous (Definition 1.63).

End of Video Recording and Remaining Board State

The recording concludes with the formulation of the equivalence of the three definitions of continuity (Proposition 1.64). On the final blackboard, the lecturer presented:

- The proof that **(1)** $\Longrightarrow$ **(2)** using the definition of convergence and open ball containment.
- The proof that **(2)** $\Longrightarrow$ **(3)** by contraposition via the sequential characterization of open sets (Lemma 1.52).
- The closing implication **(3)** $\Longrightarrow$ **(1)** left as a direct verification for the students.

Summary of Key Concepts

- **Open Balls and Sets:** An open ball is $B(x,r) := \{y \in X \mid d(y,x) < r\}$. A set $U \subseteq X$ is open if $\forall x \in U\ \exists r > 0 \colon B(x,r) \subseteq U$.
- **Closed Sets and Topology:** A set $A \subseteq X$ is closed if its complement $X \setminus A$ is open. The metric topology $\mathcal{T}$ is the collection of all open subsets of (X,d).
- **Stability Properties:** Arbitrary unions and finite intersections of open sets are open. Dually, finite unions and arbitrary intersections of closed sets are closed.
- **Interior, Closure, and Boundary:**
  - Interior: $\operatorname{int}(\Omega) = \Omega^\circ := \bigcup\{U \subseteq \Omega \mid U \text{ open}\}$ (the unique largest open subset of Ω).
  - Closure: $\overline{\Omega} := \bigcap\{A \supseteq \Omega \mid A \text{ closed}\} = \{x \in X \mid \exists (x_n) \subseteq \Omega \colon x_n \to x\}$ (the unique smallest closed superset of Ω).
  - Boundary: $\partial\Omega := \overline{\Omega} \setminus \Omega^\circ$.
- **Sequential Characterizations:**
  - U is open $\iff$ whenever $x_n \to x \in U$, then $x_n \in U$ eventually (for all $n \geq N$).
  - A is closed $\iff$ whenever $(x_n) \subseteq A$ and $x_n \to x \in X$, then $x \in A$.
- **Equivalent Characterizations of Continuity:** For a map $f \colon (X,d_X) \to (Y,d_Y)$, the following are equivalent:
  (1) ε-δ continuity: $\forall x \in X\ \forall \varepsilon > 0\ \exists \delta > 0 \colon f(B_\delta(x)) \subseteq B_\varepsilon(f(x))$.
  (2) Sequential continuity: $\forall (x_n) \subseteq X \colon x_n \to x \implies f(x_n) \to f(x)$.
  (3) Topological continuity: $\forall V \subseteq Y \text{ open} \implies f^{-1}(V) \subseteq X$ is open.

CHAPTER 2

Monday 23 February 2026

Continuity, the Banach Fixed Point Theorem, and Compactness

Course Content Overview

This lecture completes the proof of the equivalence between the three characterizations of continuity (ε-δ, sequential, and topological) by demonstrating that topological continuity implies ε-δ continuity. We then introduce the stronger notions of uniform continuity and Lipschitz continuity, prove that the distance to a fixed point is 1-Lipschitz continuous via the reverse triangle inequality, and state and prove the fundamental Banach Fixed Point Theorem for contraction mappings on complete metric spaces along with its connections to ordinary differential equations. Finally, we turn to compactness in metric spaces, defining covers, subcovers, sequential compactness, and topological compactness, and prove that sequential compactness implies topological compactness via an inscribed-radius contradiction argument.

Equivalence of Definitions of Continuity

Review of Continuity Definitions

Mündliches Protokoll [00:00:00 - 00:01:57]

Okay, so, good morning.

I remind you that on Thursday we were talking about continuity. And we gave three definitions of continuity: number (1), number (2), and number (3). The first one was the ε-δ continuity, the second one was the sequential continuity, and the third one was the topological continuity.

On Thursday we proved that... I think you missed about the first two minutes, or one minute and a half of recap, but it's for you that I do the recap. So, if you want to skip it, this is like the series in Netflix, isn't it? You can skip the introduction.

We gave these three equivalent definitions of continuity. We proved that the ε-δ continuity implied the sequential continuity. Then the sequential continuity implied the topological continuity. That maybe was the one that was a bit new, the topological continuity: it was this thing that the preimage of an open set is an open set.

And now it remained to be shown that (3) implies (1). So, topological continuity implies ε-δ continuity. This will be actually the easiest of all the three implications,

right? We will do it now. But maybe it is also the newest.

Review: Equivalent Characterizations of Continuity

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f\colon X \to Y$ be a mapping. The three definitions of continuity introduced in the previous lecture are:

(1) ε-δ Continuity:

$$\forall x \in X,\ \forall \varepsilon > 0,\ \exists \delta > 0 \quad \text{such that} \quad f(B_\delta(x)) \subseteq B_\varepsilon(f(x)).$$

(2) Sequential Continuity:

$$\forall (x_n)_{n\geq 0} \subseteq X \quad \text{with} \quad x_n \xrightarrow{n\to\infty} x \implies f(x_n) \xrightarrow{n\to\infty} f(x).$$

(3) Topological Continuity:

$$\forall V \subseteq Y \text{ open} \implies f^{-1}(V) \subseteq X \text{ is open.}$$

Status of Equivalence from Last Lecture:

(1) $\implies$ **(2)**	(proven)
(2) $\implies$ **(3)**	(proven)
(3) $\implies$ **(1)**	(to be done)

Proof: Topological Continuity Implies ε-δ Continuity

Mündliches Protokoll [00:01:57 - 00:03:55]

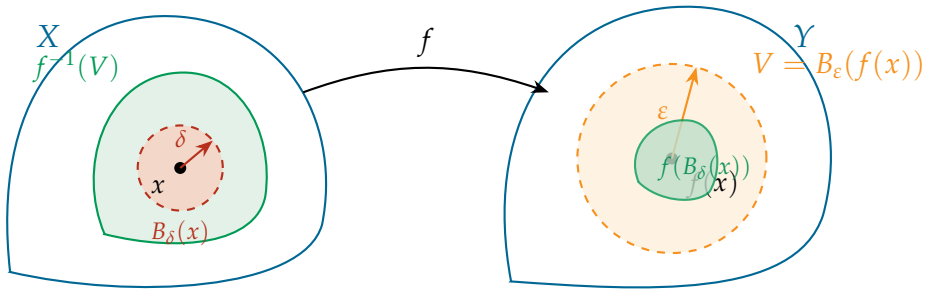
Here you have a small picture of what we try to do when we prove ε-δ continuity. We are given an ε — you give me the ε positive —, and what I will try to find... I have this ball of radius ε, and I need to find here some ball of radius δ *[draws dashed circle around x]* such that when I compute the image of this ball — that maybe is not a ball, because the continuous map can distort shapes —, the image of this ball will be some sort of potato over here, and this potato needs to be contained in the yellow, in the ball of radius ε. So, this is the ball of radius δ, and this is the ball of radius ε.

This is what I need to show. And my assumption is the topological continuity. What I need to use is that the preimage of an open set is an open set. What is the open set that I can take here that can be helpful? Well, the open set can be precisely the ball V, the ball of radius ε centered at $f(x)$ *[writes $V = B_\varepsilon(f(x))$]*. This will be my open set.

Then I know that the preimage of an open set is an open set *[draws green boundary in X and writes $f^{-1}(V)$]*. This would be $f^{-1}(V)$. By the topological continuity, this preimage is an open set. And now x is a point that belongs to this open set, so I can always find a ball of radius δ inside *[draws $B_\delta(x)$ inside $f^{-1}(V)$]*. And this δ works! So, this is the proof, right? This shows...

Should I put it with symbols? Now I will put what I just described here with symbols.

Geometric Idea: Preimage of an Open Ball



Logic of the Topological Argument: To establish ε-δ continuity at an arbitrary point $x \in X$:

(1) Choose the Target Open Set: Given $\varepsilon > 0$, construct the open target neighborhood $V := B_\varepsilon(f(x)) \subseteq Y$.

(2) Pull Back via Continuity: By topological continuity, the preimage $f^{-1}(V)$ is open in X.

(3) Point Membership: Because $f(x) \in V$, the point x necessarily belongs to $f^{-1}(V)$.

(4) Fit an Open Ball: Since $f^{-1}(V)$ is open in X and contains x, the definition of an open set guarantees the existence of a radius $\delta > 0$ such that $B_\delta(x) \subseteq f^{-1}(V)$.

(5) Push Forward to Conclude: Applying f to both sides yields $f(B_\delta(x)) \subseteq f(f^{-1}(V)) \subseteq V = B_\varepsilon(f(x))$.

Mündliches Protokoll [00:03:55 - 00:05:25]

So, take V, the ball of radius ε centered at $f(x)$ *[writes take $V = B_\varepsilon(f(x))$]*. I missed the parentheses here, right?

Then, by topological continuity — so, since f is topologically continuous *[writes $\Longrightarrow$ (f top. cont.)]* —, the set $f^{-1}(V)$ is open *[writes $f^{-1}(V)$ open]*. And since this contains $f(x)$, the point x belongs here, right? *[writes $\ni x$]*

And then, by definition of open, there exists $\delta > 0$ *[writes $\Longrightarrow$ $\exists \delta > 0$]* such that the ball of radius δ centered at the point x is contained in this open set *[writes st $B_\delta(x) \subset f^{-1}(V)$]*.

And then one checks that the image under f of the ball of radius δ is contained in the ball of radius ε centered at $f(x)$ *[writes $\Longrightarrow$ $f(B_\delta(x)) \subset B_\varepsilon(f(x))$]*. That is what we want. Okay? *[draws a shaded square box for Q.E.D.]*

So, this finishes the third implication, all right?

Formal Proof of $\Longrightarrow$

Proposition 2.1 (*Topological Continuity Implies ε-δ Continuity*)**:** Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f\colon X \to Y$ be a map. If f is topologically continuous, then f is ε-δ continuous.

Beweis

Let $x \in X$ and let $\varepsilon > 0$ be given. We define the target set as the open ball in Y:

$$V := B_\varepsilon(f(x)). \tag{2.1}$$

Since every open ball in a metric space is an open set, $V \subseteq Y$ is open. By the assumption of topological continuity:

$$f \text{ topologically continuous } \implies f^{-1}(V) \subseteq X \text{ is open.} \tag{2.2}$$

Furthermore, since $f(x) \in V$, we have $x \in f^{-1}(V)$.

Because $f^{-1}(V)$ is an open subset of X containing x, there exists, by definition of openness in a metric space, a radius $\delta > 0$ such that:

$$B_\delta(x) \subseteq f^{-1}(V). \tag{2.3}$$

Applying the forward mapping f to both sides of (2.3) gives:

$$f(B_\delta(x)) \subseteq f\big(f^{-1}(V)\big) \subseteq V = B_\varepsilon(f(x)). \tag{2.4}$$

This establishes that f is ε-δ continuous at x. Since $x \in X$ was arbitrary, f is ε-δ continuous on X.

Q.E.D.

Uniform and Lipschitz Continuity

Concepts and Definitions

Mündliches Protokoll [00:05:25 - 00:07:38]

Now you've got three equivalent definitions of continuity that you can use throughout the lecture, and even in the setup of general metric spaces. As I said, most of the time, if not all of the time after a couple of weeks more, we will use $\mathbb{R}^n$ and $\mathbb{R}^m$, most of the time. Just at the very end of the lecture we will see again spaces of functions for ordinary differential equations, that you have already seen when you have seen Cauchy–Lipschitz. Well, I will tell you something later about Cauchy–Lipschitz.

Now let me introduce some notion that you are also familiar with when you work in $\mathbb{R}$, that is uniform continuity. Remember uniform continuity.

So, definition... *[starts writing on the board]* Same setup as before: we have a map f from X to Y *[writes $f\colon X \to Y$]*, where X and Y are metric spaces — a map between metric spaces *[writes (map between metric spaces)]*.

Now, we say that f is uniformly continuous *[writes $\circ$ f is unif. cont.]*. So, uniformly continuous, as in $\mathbb{R}$, is the same as ε-δ continuous — almost the same.

Mündliches Protokoll [00:07:38 - 00:09:32]

The only difference is that this *„for all x"*, you put it here *[points at the board]*, which is very different, because, you know, in the quantifiers in math, the order of the *„exists"*, *„for all"*, and so on matters a lot. It's not the same to have the *„for all x"* here and the *„for all x"* at the beginning. This is what makes it uniform.

Let's do that. For every ε — you first give me the ε *[writes $\forall \varepsilon > 0$]* —, now you give me the ε, I need to give you the δ: there exists $\delta > 0$ *[writes $\exists \delta > 0$]*, and this δ needs to be good for all x. That's why this is more difficult. Such that for all x in my space *[writes st $\forall x \in X$]*, the ball of radius δ centered at x is mapped to the ball of radius ε centered at $f(x)$ *[writes $f(B_\delta(x)) \subset B_\varepsilon(f(x))$]*.

Again, exactly as in Analysis I, the key difference is that the δ that I give you will be independent of the point x. Because you didn't tell me the point yet. You just told me the ε, and I need to tell you the δ, and it needs to work for every x. Instead, there, I insist: first you tell me the point, then you tell me the ε, and then knowing the ε and the point I need to give you the δ. Very different.

Definition of Uniform Continuity

Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f\colon X \to Y$ be a map.

Definition 2.2 (*Uniform Continuity*): The map $f\colon X \to Y$ is called *„uniformly continuous"* if:

$$\forall \varepsilon > 0,\ \exists \delta > 0 \quad \text{such that} \quad \forall x \in X, \quad f\big(B_\delta(x)\big) \subseteq B_\varepsilon\big(f(x)\big). \tag{2.5}$$

Equivalently, in terms of distance functions:

$$\forall \varepsilon > 0,\ \exists \delta > 0 \quad \text{such that} \quad \forall x, y \in X, \quad d_X(x,y) < \delta \implies d_Y\big(f(x), f(y)\big) < \varepsilon. \tag{2.6}$$

Quantifier Ordering: Pointwise vs. Uniform Continuity: The critical conceptual distinction between ordinary continuity and uniform continuity lies entirely in the order of the logical quantifiers:

- **Pointwise Continuity:**

  $$\forall x \in X,\ \forall \varepsilon > 0,\ \exists \delta > 0 \quad \text{such that} \quad d_X(x,y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon.$$

  Here, the choice of $\delta = \delta(x, \varepsilon)$ depends on **both** the precision ε and the specific baseline point x. Near regions where the function steepens, δ can become arbitrarily small.
- **Uniform Continuity:**

  $$\forall \varepsilon > 0,\ \exists \delta > 0 \quad \text{such that} \quad \forall x \in X, \quad d_X(x,y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon.$$

  Here, $\delta = \delta(\varepsilon)$ is chosen knowing only ε. It must serve simultaneously as a valid modulus of continuity across the **entire space** X, regardless of the point x.

Mündliches Protokoll [00:09:32 - 00:11:08]

And a particular case of uniform continuity that will be very important, and we will use many, many times, is that of Lipschitz continuity. Have you seen this in $\mathbb{R}$? Lipschitz continuity.

Now, in a metric space it is essentially the same. We call it λ-Lipschitz *[writes $\circ$ f is λ-Lipschitz]*. λ is the Lipschitz constant, or L — let me call it capital L *[overwrites λ with L and puts a green box around L]*. This L will be what is called the Lipschitz constant; it is a positive number *[writes L Lipschitz ctt]*, that is, a number > 0 *[writes > 0 number]*, right?

And the definition is that the distance in Y between $f(x)$ and $f(y)$ *[writes $d_Y(f(x), f(y))$]* is less than or equal to the Lipschitz constant — now it is L, right? — L times the distance in X between x and y *[writes $\leq L\, d_X(x,y)$]*. And this is true for every pair of points, for all x and y in X *[writes $\forall x, y \in X$]*.

Mündliches Protokoll [00:11:08 - 00:12:15]

So, if you want, an exercise that should be quite immediate is: show that a Lipschitz function is uniformly continuous *[writes on the board: „Show Lipschitz fctn is uniformly continuous." accompanied by an exercise icon]*.

For a Lipschitz function, the relation between the ε and the δ is very explicit, right? Because since the map is multiplying the distance by L at most, if you want to make sure that the ball of radius δ is mapped to the ball of radius ε, you just need to take $\delta = \varepsilon/L$, and then it will work. But you can write this down. Okay.

Definition of Lipschitz Continuity and Exercise

Definition 2.3 (*Lipschitz Continuity*): Let (X, d_X) and (Y, d_Y) be metric spaces. A map $f\colon X \to Y$ is called *„L-Lipschitz"* (or *„Lipschitz continuous"*) if there exists a real constant $L > 0$ such that:

$$d_Y\big(f(x), f(y)\big) \leq L \cdot d_X(x,y) \quad \forall x, y \in X. \tag{2.7}$$

The constant L is called the *„Lipschitz constant"* of f.

Übung 2.4 (*Lipschitz Implies Uniform Continuity*): Show that any Lipschitz continuous function $f\colon X \to Y$ between metric spaces is uniformly continuous.

Explicit Selection of δ: Let $L > 0$ be the Lipschitz constant of f. To verify uniform continuity:

(1) Let $\varepsilon > 0$ be given.

(2) Define $\delta := \frac{\varepsilon}{L} > 0$. Notice that δ depends strictly on ε and L, and is completely independent of any base point $x \in X$.

(3) For any points $x, y \in X$ satisfying $d_X(x,y) < \delta$, the Lipschitz bound (2.7) gives:

$$d_Y\big(f(x), f(y)\big) \leq L \cdot d_X(x,y) < L \cdot \delta = L \cdot \frac{\varepsilon}{L} = \varepsilon.$$

Thus, f is uniformly continuous on X.

Example: Distance Functions Are 1-Lipschitz

Mündliches Protokoll [00:12:15 - 00:13:30]

Maybe I give some example of something that is interesting... *[slides the blackboards up and pulls down a fresh board]*

Example: *[writes „Example" on the board]* As always, (X, d) is a metric space *[writes (X,d) metric space]*, and fix a point $x_0 \in X$ *[writes $x_0 \in X$]*. And now we define the

function $f(x)$ as the distance from x to the point x_0 *[writes $f(x) := \operatorname{dist}(x, x_0)$]*.

Why do I put *„dist“*? Okay, this is d *[erases dist and writes $d(x, x_0)$]*. The distance, but we call it d, right? Then f is 1-Lipschitz *[writes then f is 1-Lipschitz]*, so Lipschitz with constant 1, right?

Mündliches Protokoll [00:13:30 - 00:14:31]

Let's prove this, okay? We need to compute the distance between $f(x)$ and $f(y)$... *[writes $d(f(x), f(y))$]*

The first thing that I need to identify is: what is the target space here? This map f goes from X to what? What is Y? The question is: what is Y? *[walks over to retrieve the catch-box microphone]*

Student Answer

The positive real numbers?

Mündliches Protokoll [00:14:33 - 00:16:22]

Yeah, the non-negative real numbers, right? *[writes $Y = [0, \infty)$ on the board]* Exactly. Thank you. *[catches the microphone back]*

So, Y is that one. And when we don't explicitly specify, if our Y is a subset of $\mathbb{R}$, the distance will be the Euclidean distance, the one given by the absolute value, right?

So, this, by definition of distance, is $|f(x) - f(y)|$ *[writes $\overset{d_Y}{=} |f(x) - f(y)|$]*.

Now, I copy the definition of f: this is equal to the distance between x and x_0 minus the distance between y and x_0, in absolute value *[writes $\overset{def}{=} |d(x, x_0) - d(y, x_0)|$]*.

And now my goal is to show that the map is 1-Lipschitz, so we want to show that this is less than or equal to the distance between x and y *[writes $\leq d(x, y)$ with an arrow labeled „goal“]*.

And this would be the λ, or the L — sorry, I always call it... So, the L in this case is equal to 1, right? *[writes 1 and 1 under the coefficient]*

Mündliches Protokoll [00:16:22 - 00:18:22]

And this inequality, I'm saying this is equivalent to the triangle inequality. This is precisely the triangle inequality *[writes „triangle inequality“ with an arrow pointing to the inequality]*.

Why? Because if you remember, the triangle inequality says that the distance between A and C is less than or equal to the distance between A and B plus the distance between B and C *[writes $d(A, C) \leq d(A, B) + d(B, C)$]*, right?

And then this is the same as... you can pass this one to the left: so, the distance between A and C minus the distance between A and B is less than or equal to the

distance between B and C *[writes* $\implies d(A,C) - d(A,B) \leq d(B,C)$*]*, right?

And then, since you can reorder the points and do all possible combinations, you get actually the positive and the minus version of that, right? So, you get this *[draws absolute value bars* $|d(A,C) - d(A,B)| \leq d(B,C)$ *and encloses the derivation in brackets]*. Right?

So, this is just one example. In every metric space, when you compute the distance to a point, something that is interesting from here is that the distance is continuous in particular. It's Lipschitz continuous, so it is uniformly continuous, and then it is continuous. The distance with respect to each of its arguments is a continuous map.

Distance to a Fixed Point is 1-Lipschitz

Beispiel 2.5 (*Distance to a Fixed Point is 1-Lipschitz*): Let (X,d) be a metric space and let $x_0 \in X$ be a fixed reference point. Consider the distance function to x_0:

$$f\colon X \to [0,\infty), \quad f(x) := d(x,x_0). \tag{2.8}$$

Then f is 1-Lipschitz continuous with respect to the standard Euclidean metric on the codomain $Y = [0,\infty) \subset \mathbb{R}$.

Beweis

Let $x,y \in X$. Since the codomain is $Y = [0,\infty) \subset \mathbb{R}$, the metric on Y is the standard absolute value metric $d_Y(a,b) = |a-b|$. Using the definition of f, we have:

$$\begin{aligned} d_Y\big(f(x),f(y)\big) &= |f(x) - f(y)| \\ &= \big|d(x,x_0) - d(y,x_0)\big|. \end{aligned} \tag{2.9}$$

We claim that for any $x,y,x_0 \in X$:

$$\big|d(x,x_0) - d(y,x_0)\big| \leq 1 \cdot d(x,y). \tag{2.10}$$

To prove (2.10), apply the standard triangle inequality for (X,d) with points x,y,x_0:

$$d(x,x_0) \leq d(x,y) + d(y,x_0) \implies d(x,x_0) - d(y,x_0) \leq d(x,y).$$

By symmetry, interchanging the roles of x and y and using $d(y,x) = d(x,y)$:

$$d(y,x_0) \leq d(y,x) + d(x,x_0) \implies -\big(d(x,x_0) - d(y,x_0)\big) \leq d(x,y).$$

Combining both inequalities yields the reverse triangle inequality:

$$\big|d(x,x_0) - d(y,x_0)\big| \leq d(x,y). \tag{2.11}$$

Substituting (2.11) into (2.9) gives:

$$d_Y\big(f(x),f(y)\big) \leq 1 \cdot d(x,y) \quad \forall x,y \in X.$$

This proves that f is 1-Lipschitz continuous (with Lipschitz constant $L = 1$). Q.E.D.

Consequences of the 1-Lipschitz Property: From the hierarchy of continuity conditions:

$$\text{Lipschitz Continuous} \implies \text{Uniformly Continuous} \implies \text{Continuous},$$

we immediately obtain:

- The function $x \mapsto d(x,x_0)$ is continuous on (X,d).
- More generally, the metric $d\colon X \times X \to [0,\infty)$ is jointly continuous in both arguments with respect to the product metric on $X \times X$.

The Banach Fixed Point Theorem

Statement of the Theorem

Mündliches Protokoll [00:18:22 - 00:20:15]

And now we will show some nice application, some very important property of Lipschitz maps from a space onto itself that are λ-Lipschitz with $\lambda < 1$. This is a theorem that is called the Banach Fixed Point Theorem *[writes „Thm (Banach fixed pt)" on the board]*.

The first thing is the assumption on the space: (X, d) is a complete metric space *[writes (X, d) complete m.s.]*. We have a complete metric space, and you are given a map T from X to X *[writes Suppose $T\colon X \to X$]* that is λ-Lipschitz *[writes map λ-Lipschitz]*.

Suppose that we have a map like this that is λ-Lipschitz for $\lambda \in (0,1)$ *[writes $\lambda \in (0,1)$]*. There are two very important things in this theorem: it's very, very general, but you need a complete space — and that's actually one of the reasons why complete metric spaces are so important.

And then you need what is called a contraction mapping. So, you need a λ-Lipschitz map for $\lambda < 1$.

Mündliches Protokoll [00:20:15 - 00:21:58]

When this happens, you will see sometimes in books and so on that it is said that T is a contraction *[underlines in green and writes T is a contraction]*.

Contraction means that it shortens the distance. You take two points, you compute the images of these two points, and they will always be closer than before.

And the conclusion is: then T has a unique fixed point *[writes Then T has a unique fixed pt]*. Fixed point means a point such that... there exists a unique $x \in X$ such that $T(x) = x$ *[writes $(\exists!\, x \in X \text{ st } T(x) = x)$]*.

A fixed point is a point that does not move. The map is sending x to itself. And the theorem is saying that there is always one of those points, and just one of them. *[underlines the statement and draws a dividing line]*

The Banach Fixed Point Theorem

Satz 2.6 (*Banach Fixed Point Theorem / Contraction Mapping Principle*): Let (X, d) be a complete metric space, and let $T\colon X \to X$ be a contraction mapping, that is, a λ-Lipschitz map with contraction factor $\lambda \in (0,1)$:

$$d\big(T(x), T(y)\big) \leq \lambda \cdot d(x, y) \quad \forall x, y \in X. \tag{2.12}$$

Then T possesses a unique fixed point $x^* \in X$:

$$\exists!\, x^* \in X \quad \text{such that} \quad T(x^*) = x^*. \tag{2.13}$$

Core Hypotheses and Architecture of the Theorem: The Banach Fixed Point Theorem relies on three critical mathematical components:

- **Metric Completeness of (X,d):** The metric space must have no *„holes“*. Every Cauchy sequence must converge to a point within X. (Without completeness, iterates could converge toward a missing boundary point, e.g., $f(x) = x/2$ on $(0,1]$ has no fixed point in $(0,1]$).
- **Strict Contraction ($\lambda < 1$):** It is not sufficient that $d(T(x),T(y)) < d(x,y)$ for $x \neq y$; there must exist a uniform factor $\lambda \in (0,1)$ strictly bounding the shrinkage rate.
- **Uniqueness of the Fixed Point:** Uniqueness is immediate from the contraction property: if x^* and y^* are two fixed points, then:
$$d(x^*,y^*) = d\big(T(x^*),T(y^*)\big) \leq \lambda \cdot d(x^*,y^*).$$
Since $(1-\lambda) > 0$, this requires $(1-\lambda)d(x^*,y^*) \leq 0$, forcing $d(x^*,y^*) = 0 \implies x^* = y^*$.

Heuristic Geometric Sketch and Iteration Dynamics

Mündliches Protokoll [00:21:58 - 00:23:28]

Before writing the proof, let me do a small sketch and tell you why it is reasonable to expect such a theorem, okay?

Now we have our set X *[draws a large closed curve representing X on the right board]*. The map T goes from X onto itself *[draws a loop arrow labeled T from X to X]*.

So, if I take a point x, the image of my point, $T(x)$, is perhaps here *[marks x and $T(x)$]*. And if I take a point y here *[marks y]*, the image of the point y is, I don't know, maybe here *[marks $T(y)$]*.

But I can be very, very sure that the distance between these two points that are the images is shorter than the distance between these two. This is the contraction: I take any two points, I compute the images, and the distance is shorter, right?

Mündliches Protokoll [00:23:28 - 00:25:05]

And now, what we will do is this: we will take a single point, say x_0 *[marks x_0 in green chalk near the center of X]*. We will compute its image, $T(x_0)$ *[draws arrow to $T(x_0)$]*. And now we will think of this point as the y.

Then I will compute its image again, let's say $T^2(x_0)$ *[writes $T^2(x_0)$ and draws arrow]*, the image $T(T(x_0))$. This is just a notation: T^2 means I apply the map T twice, right?

I'm just looking at where this point is sent: x_0 is sent to $T(x_0)$, this is sent to $T(T(x_0))$, then $T(T(T(x_0)))$ *[writes $T^3(x_0)$ and draws arrow]*, right?

And what happens? These were my first point, the second point... and you see, these two are the images: this is sent to this, and this is sent to this, right?

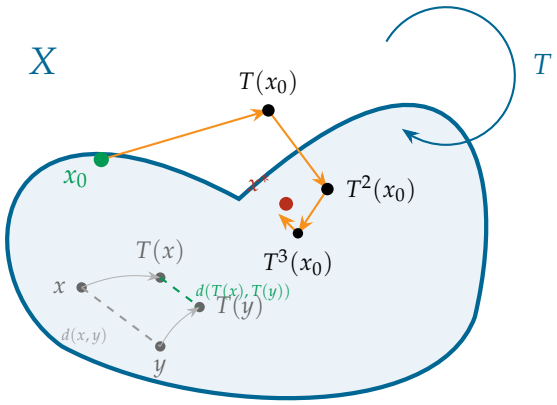
So, the distance here is bigger than the distance here; the distance here is bigger than the distance here. And it's not only bigger: it is geometrically bigger, the distances form a geometric sequence. It's contracting with a geometric factor.

So, every time the distances will become smaller and smaller in a geometric fashion.

And since we are in a complete metric space, we will have a Cauchy sequence, which will force the convergence of the sequence, okay?

This is the plan of the proof, and now we write it.

Heuristic Visualization of the Picard Iteration



The Picard Iteration Mechanism: The heuristic idea of the constructive proof consists in setting up the *„Picard iteration"*:

(1) Pick an arbitrary initial point $x_0 \in X$ and define recursively:

$$x_1 := T(x_0), \quad x_2 := T(x_1) = T^2(x_0), \quad \dots, \quad x_{n+1} := T(x_n) = T^{n+1}(x_0).$$

(2) By the contraction property (2.12):

$$d(x_{n+1}, x_n) = d\big(T(x_n), T(x_{n-1})\big) \leq \lambda \cdot d(x_n, x_{n-1}) \leq \cdots \leq \lambda^n \cdot d(x_1, x_0).$$

(3) Since $\lambda < 1$, the terms λ^n form a convergent geometric series. For any $m > n$:

$$d(x_m, x_n) \leq \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \leq d(x_1, x_0) \sum_{k=n}^{m-1} \lambda^k \leq d(x_1, x_0) \cdot \frac{\lambda^n}{1-\lambda}.$$

As $n \to \infty$, the right-hand side tends to 0, proving that $(x_n)_{n\geq 0}$ is a Cauchy sequence.

(4) Because (X, d) is complete, the Cauchy sequence converges to a limit $x^* \in X$, which is the unique fixed point.

Proof of the Banach Fixed Point Theorem

Proof of the Banach Fixed Point Theorem 2.6

Construction of the Sequence and Contraction Decay

Mündliches Protokoll [00:25:05 - 00:26:36]

[writes „Proof" on the upper right board] What did I say? Fix a point. A bit the strength and the magic about this theorem is that it doesn't matter. You want to find a solution of this equation, right?

This is a non-linear map, so this is any map. It's not linear algebra, it's any map. Finding a solution to an equation with a non-linear map is typically very, very hard, right?

You can start with any initial point. Start with any x_0.

[writes fix $x_0 \in X$] And now you start to bootstrap. So, you start to say: okay, let me define x_1 equal to $T(x_0)$ *[writes $x_1 = T(x_0)$]*. Then x_2 is equal to $T(x_1)$ *[writes $x_2 = T(x_1)$]*, that is, $T(T(x_0))$. And so on: x_{n+1} is equal to $T(x_n)$ *[writes $x_{n+1} = T(x_n)$]*, for all $n \geq 0$ *[writes $\forall n \geq 0$]*. Right?

This is the recurrence. *[draws a frame around $x_{n+1} = T(x_n)$, $\forall n \geq 0$]*

Iterative Sequence Construction

Step 1: Picard Iteration Scheme.
Fix an arbitrary point $x_0 \in X$. We define the sequence of iterates $(x_n)_{n=0}^{\infty} \subseteq X$ recursively by:

$$\begin{aligned} x_1 &:= T(x_0), \\ x_2 &:= T(x_1) = T\big(T(x_0)\big) = T^2(x_0), \\ &\vdots \\ x_{n+1} &:= T(x_n) \quad \forall n \geq 0. \end{aligned} \tag{2.14}$$

Mündliches Protokoll [00:26:36 - 00:28:27]

And now I use that it is a contraction, right? I say: okay, the distance between x_{n+1} and x_n *[writes $d(x_{n+1}, x_n)$]*...

What is x_{n+1}? This is the distance between $T(x_n)$ and $T(x_{n-1})$ *[writes $= d(T(x_n), T(x_{n-1}))$]*, right? Each point is, by definition, T applied to the previous point in the sequence. So, the distance between these points is the same, because this is this, and this is that.

And now I use that T is a contraction. So, this distance is less than or equal to λ times the distance between x_n and x_{n-1} *[writes $\leq \lambda\, d(x_n, x_{n-1})$]*. And $\lambda < 1$, right?

[draws a green box around the inequality $d(x_{n+1}, x_n) = d(T(x_n), T(x_{n-1})) \leq \lambda\, d(x_n, x_{n-1})$] And this is the key property I will use to prove that my sequence is Cauchy. *[writes $(*)$ in green chalk underneath the boxed inequality]*

The Fundamental Step Estimate

Step 2: Consecutive Distance Decay.
Using the recursive definition (2.14) and the contraction property (2.12), we compute the distance between consecutive iterates for any $n \geq 1$:

$$d(x_{n+1}, x_n) = d\big(T(x_n), T(x_{n-1})\big) \leq \lambda \cdot d(x_n, x_{n-1}). \tag{$*$}$$

The relation $(*)$ is our key structural estimate: each step is strictly contracted by the constant factor $\lambda \in (0,1)$.

Proof of the Cauchy Property

Mündliches Protokoll [00:28:27 - 00:30:11]

Now I claim that $(*)$ *[points at $(*)$]* will give me that the sequence (x_n) is Cauchy *[writes $(*) \implies (x_n)$ is Cauchy on the board]*.

Well, first I need to compound property $(*)$, right? I need to say: okay, a first consequence of $(*)$ is that the distance between x_{n+1} and x_n is less than or equal to λ times the distance between x_n and x_{n-1} *[writes $d(x_{n+1}, x_n) \leq \lambda\, d(x_n, x_{n-1})$]*.

And I can compare to the previous step, right? This is less than or equal to $\lambda \cdot \lambda$ times the distance between x_{n-1} and x_{n-2} *[writes $\leq \lambda\lambda\, d(x_{n-1}, x_{n-2})$]*.

And then you can prove by induction — if you want to write a full, rigorous proof — that after n steps you arrive at λ^n times the distance between x_1 and x_0 *[writes $\leq \cdots \leq \lambda^n\, d(x_1, x_0)$]*, right? Okay.

Compounding the Contraction Estimate

Step 3: Iterated Contraction Bound.
Applying the fundamental estimate $(*)$ iteratively n times yields:

$$\begin{aligned} d(x_{n+1}, x_n) &\leq \lambda \cdot d(x_n, x_{n-1}) \\ &\leq \lambda^2 \cdot d(x_{n-1}, x_{n-2}) \\ &\leq \dots \\ &\leq \lambda^n \cdot d(x_1, x_0) \quad \forall n \geq 0. \end{aligned} \tag{2.15}$$

Mündliches Protokoll [00:30:11 - 00:31:57]

And now we will use the triangle inequality.

I want to prove that the sequence is Cauchy, right? So, I want to control the distance between x_n and x_m *[writes $d(x_n, x_m)$]*.

Let me assume, because the distance is symmetric, that $m < n$ *[writes $m < n$ in a small box]*. This is the large one. I can assume this without loss of generality, because the distance is symmetric in its arguments.

Then I want to control that, and what I can do is use the triangle inequality *[writes $\leq$ triangle inequality]*. This is bounded by the sum for $k = m$ to $n-1$ *[writes $\sum_{k=m}^{n-1}$]*, and

now I need to take the distance between x_{k+1} and x_k *[writes $d(x_{k+1}, x_k)$]*. So...

Mündliches Protokoll [00:31:57 - 00:33:27]

Let's go step by step. What I'm saying over here is that this is a triangle inequality. Actually, it is a compounded triangle inequality: I'm using the triangle inequality many times, right?

You will see this very clearly in a picture. What I'm saying is: I have my point x_m. Now comes the point x_{m+1}, x_{m+2}, x_{m+3}, and so on, up to the point x_n, right? Every time they are closer.

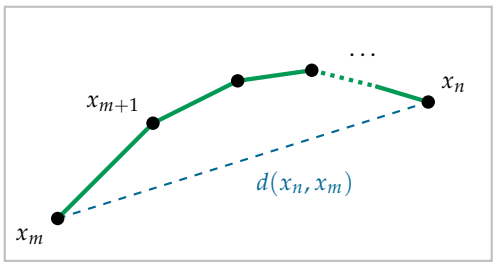
And what I'm saying is that the distance between these two points *[draws a chord connecting x_m and x_n directly]* is less than or equal to the sum of the distances between this and this, this and this, this and this, this and this, and so on *[draws segments connecting the intermediate points forming a polygonal path]*. Right?

Now, if you want a fully rigorous, fully detailed proof that this form of the triangle inequality is true — intuitively it's clear, but if you want to write the full proof — you need to prove it by induction on the number of segments, right? You start with three points: that's the standard triangle inequality. Then you add one, step by step, and you get the proof of that.

The Polygonal Triangle Inequality

Step 4: Compounded Triangle Inequality.
Let $n, m \in \mathbb{N}$ with $m < n$ (assumed without loss of generality since $d(x_n, x_m) = d(x_m, x_n)$). By repeatedly applying the triangle inequality across intermediate points $x_m, x_{m+1}, \ldots, x_n$:

$$d(x_n, x_m) \leq \sum_{k=m}^{n-1} d(x_{k+1}, x_k). \tag{2.16}$$



Inductive Justification of the Polygonal Inequality: For two points, the inequality is trivial ($d(x, x) = 0$). For three points, it is the standard triangle inequality $d(x_3, x_1) \leq d(x_3, x_2) + d(x_2, x_1)$. Assuming the claim holds for p segments:

$$d(x_{m+p}, x_m) \leq d(x_{m+p}, x_{m+p-1}) + d(x_{m+p-1}, x_m) \leq d(x_{m+p}, x_{m+p-1}) + \sum_{k=m}^{m+p-2} d(x_{k+1}, x_k) = \sum_{k=m}^{m+p-1} d(x_{k+1}, x_k$$

which proves (2.16) by mathematical induction for all $n > m$.

Mündliches Protokoll [00:33:27 - 00:35:09]

And now, when I have this, I use the property that I showed here *[points at (2.15)]*.

This implies that the distance between x_n and x_m *[writes* $\Longrightarrow\ d(x_n, x_m)$*]* is less than or equal to the sum from $k = m$ to $n-1$ *[writes* $\leq \sum_{k=m}^{n-1}$*]*. And now, remember that the distance between x_{k+1} and x_k was λ^k times the distance between x_1 and x_0 *[writes* $\lambda^k\, d(x_1, x_0)$*]*.

And this is now a common factor *[underlines $d(x_1, x_0)$ in green]*, so I can take it out: this is the distance between x_1 and x_0 *[writes* $\leq d(x_1, x_0)$*]* times the sum of a geometric series. Remember that I said before that there will be a geometric series? So, here you have your geometric series.

When you sum this geometric series, you get λ^m *[writes* λ^m*]*, right? And then something that is less than $1/(1-\lambda)$ *[writes* $\frac{1}{1-\lambda}$*]*.

This $1/(1-\lambda)$ is the sum of the geometric series up to infinity. When I truncate, it's less than that, right? Now you know how to sum geometric series: in Analysis I you did a lot of that.

Mündliches Protokoll [00:35:09 - 00:35:47]

Okay, this is just a constant, this is just a constant, and what is important is this *[draws a green box around λ^m]*: since when m goes to infinity, this λ^m goes to zero, my sequence is Cauchy.

So, as $\lambda^m \to 0$ as $m \to \infty$ *[writes* $\Longrightarrow\ \lambda^m \to 0$ *as* $m \to \infty$*]*, the sequence is Cauchy *[writes sequence is as Cauchy and underlines Cauchy in green]*, right?

Geometric Series Summation and the Cauchy Condition

Step 5: Summing the Geometric Series.
Substituting the iterated bound (2.15) into the polygonal triangle inequality (2.16):

$$\begin{aligned} d(x_n, x_m) &\leq \sum_{k=m}^{n-1} d(x_{k+1}, x_k) \\ &\leq \sum_{k=m}^{n-1} \lambda^k \cdot d(x_1, x_0) \\ &= d(x_1, x_0) \sum_{k=m}^{n-1} \lambda^k. \end{aligned} \tag{2.17}$$

Factoring out λ^m from the summation:

$$\sum_{k=m}^{n-1} \lambda^k = \lambda^m \sum_{j=0}^{n-1-m} \lambda^j < \lambda^m \sum_{j=0}^{\infty} \lambda^j = \lambda^m \cdot \frac{1}{1-\lambda}. \tag{2.18}$$

Therefore, combining (2.17) and (2.18):

$$d(x_n, x_m) \leq d(x_1, x_0) \cdot \frac{\lambda^m}{1-\lambda} \quad \forall n > m \geq 0. \tag{2.19}$$

Since $\lambda \in (0, 1)$, we have:

$$\lim_{m \to \infty} \lambda^m = 0. \tag{2.20}$$

Hence, for any $\varepsilon > 0$, choosing $M \in \mathbb{N}$ sufficiently large such that:

$$d(x_1, x_0) \cdot \frac{\lambda^M}{1-\lambda} < \varepsilon$$

guarantees that $d(x_n, x_m) < \varepsilon$ for all $n > m \geq M$. This proves that $(x_n)_{n=0}^{\infty}$ is a **Cauchy sequence** in (X, d).

Convergence and Existence of the Fixed Point

Mündliches Protokoll [00:35:47 - 00:36:41]

Now that I know that the sequence is Cauchy, I use my assumption that the metric space is complete, and this sequence has a limit in my space.

So then, since X is complete *[writes X is complete]*, there exists $x = \lim_{n\to\infty} x_n$ *[writes* $\exists\, x = \lim_{n\to\infty} x_n$*]*, right? In my drawing...

[points at the Picard sketch on the right board] I was taking the sequence of points: every time they are closer and closer, and in the end, I get a point that is so close to itself that it doesn't move, right? So, this will be the fixed point.

Mündliches Protokoll [00:36:41 - 00:37:50]

Now, why is this a fixed point? Now it's very easy, because you see, if I compute $T(x)$ *[writes* Tx*]*... Well, the map is continuous, right? The map is Lipschitz, so it is continuous *[writes T continuous above the equals sign]*.

Since T is continuous, this is the limit as $n \to \infty$ of $T(x_n)$ *[writes* $= \lim_{n\to\infty} Tx_n$*]*, right?

But this is the limit as $n \to \infty$ of x_{n+1} *[writes* $= \lim_{n\to\infty} x_{n+1}$*]*. And (x_{n+1}) is just the same sequence, right? It is a subsequence, just shifting by one. So, this has the same limit, x *[writes* $= x$*]*.

So, my point x is a fixed point. *[draws a green box around* $Tx = \lim_{n\to\infty} Tx_n = \lim_{n\to\infty} x_{n+1} = x$*]*

Convergence to a Fixed Point

Step 6: Existence of the Limit and Fixed Point Property.
Because (X, d) is assumed to be a **complete** metric space, every Cauchy sequence converges to a limit in X:

$$\exists\, x^* \in X \quad \text{such that} \quad x^* = \lim_{n\to\infty} x_n. \tag{2.21}$$

We verify that x^* is indeed a fixed point of T. Since T is λ-Lipschitz, it is in particular continuous. Passing the limit inside T:

$$\begin{aligned} T(x^*) &= T\left(\lim_{n\to\infty} x_n\right) \\ &= \lim_{n\to\infty} T(x_n) && \text{(since } T \text{ is continuous)} \\ &= \lim_{n\to\infty} x_{n+1} && \text{(by definition of the sequence)} \\ &= x^* && \text{(shifting indices preserves the limit).} \end{aligned} \tag{2.22}$$

Thus, $T(x^*) = x^*$, establishing the existence of a fixed point.

Uniqueness of the Fixed Point

Mündliches Protokoll [00:37:50 - 00:38:44]

And I'm almost done. I just need to check something: now I showed the existence of a fixed point.

What I need to show to close the proof of the theorem is that this fixed point is unique, that there cannot be two fixed points. But this is very easy because of the contraction property, right?

If x and y are two fixed points *[writes If x, y are 2 fixed pts]*, because they are fixed points, $x = T(x)$ *[writes $x = T(x)$]* and $y = T(y)$ *[writes $y = T(y)$]*.

And now I use the contracting property: $d(x,y) = d(T(x),T(y))$ *[writes $d(x,y) = d(T(x),T(y))$]*.

Mündliches Protokoll [00:38:44 - 00:39:49]

Then this is less than or equal to λ times the distance between x and y *[writes $\leq \lambda\, d(x,y)$]*. And $\lambda < 1$.

So, you can never have a positive number less than or equal to λ times itself, with $\lambda < 1$, unless this number is zero.

This implies that $d(x,y) = 0$ *[writes $\implies d(x,y) = 0$]*, and then by one of the axioms of the metric, this implies $x = y$ *[writes $\implies x = y$]*.

And this ends the proof of the theorem. *[draws a shaded box on the board for Q.E.D.]*

Uniqueness of the Fixed Point

Step 7: Uniqueness.
Suppose $x, y \in X$ are both fixed points of T, so that $T(x) = x$ and $T(y) = y$. Applying the contraction property (2.12):

$$\begin{aligned} d(x,y) &= d\big(T(x),T(y)\big) \\ &\leq \lambda \cdot d(x,y). \end{aligned} \tag{2.23}$$

Subtracting $\lambda \cdot d(x,y)$ from both sides gives:

$$(1-\lambda)\cdot d(x,y) \leq 0. \tag{2.24}$$

Since $\lambda \in (0,1)$, the coefficient $1-\lambda$ is strictly positive ($1-\lambda > 0$). Because distances in a metric space are non-negative ($d(x,y) \geq 0$), inequality (2.24) forces:

$$d(x,y) = 0. \tag{2.25}$$

By the definiteness axiom of a metric:

$$d(x,y) = 0 \implies x = y.$$

Therefore, the fixed point is unique, completing the proof of the Banach Fixed Point Theorem.

Q.E.D.

Connection to Ordinary Differential Equations and the Break

The Cauchy–Lipschitz Theorem and Fixed Point Methods

Mündliches Protokoll [00:39:49 - 00:40:57]

Now I'm going to wash my hands and erase the blackboard, and then you think if you have questions about the theorem, okay? *[walks over to the sink to wash hands, then wipes the lower blackboard with a wet sponge and squeegee]*

Blackboard Maintenance

The lecturer washes his hands at the sink and thoroughly cleans the lower blackboard panels with a sponge and squeegee while the students review the proof.

Mündliches Protokoll [00:40:57 - 00:41:52]

Since it is almost the time of the break, and you can also think about questions during the break, let me tell you something to connect this theorem with something you already know: that is Cauchy–Lipschitz.

In Analysis I, you should have proven the Cauchy–Lipschitz theorem for ordinary differential equations, haven't you? *[throws the orange catch-box microphone to a student in the audience]* Sorry.

Student Remark

We did it at the end, but it was more like, how to say it, freiwilliger Zusatzstoff. He said that we're going to do it in Analysis II.

Mündliches Protokoll [00:42:01 - 00:42:03]

Could you speak a bit louder? Yeah.

Student Remark, Continued

Yes. Um, I'm not sure if I'm the right spokesperson for this, but Professor Figalli said we did it at the end, but it was more like Zusatzmaterial. He said we will do it more profoundly in Analysis II.

Mündliches Protokoll [00:42:14 - 00:44:24]

This is correct. Cauchy–Lipschitz we will do at the end of this course, in all detail in Analysis II. That's why we bother about spaces of functions.

When you want to prove the existence of a solution of an ODE, you have some equation — it is a non-linear equation —, but the space where your equation lives is a space of functions: it will be the trajectories of your ODE. Every point in your space is a trajectory. This is how you do it.

And then the key idea of this proof — the Picard proof — is to set up a contraction map. Since it is a contraction, it will easily give you the existence of a fixed point.

Whenever you want to solve a complicated equation, there is a general strategy that does not always work, but works in many cases and is very effective: to rewrite your equation as a fixed point of some contraction. And then you just run the machine, you compose your map with itself, and you end up with a fixed point. *[wipes down the remainder of the lower board with a sponge]*

We will also use this in this course to prove a very fundamental and important theorem in the course: that is the Inverse Function Theorem. This is one of the big theorems in the course that is very useful, and the proof is not trivial; it will be again based on a variant of this Banach Fixed Point Theorem.

***Einblick:* The Contraction Mapping Principle as a Universal Existence Machine**

The Banach Fixed Point Theorem is one of the most powerful foundational tools in modern analysis because it transforms an otherwise intractable **non-linear equation solving problem**:

$$F(x) = 0$$

into a **fixed point problem**:

$$T(x) = x \quad \text{(for instance by setting } T(x) := x - \alpha F(x)\text{)}.$$

When T is arranged to be a contraction on a complete space:

- **Constructive Nature:** The proof does not merely assert existence abstractly; it provides an explicit algorithm (Picard iteration $x_{n+1} = T(x_n)$) that generates approximations with quantifiable geometric error bounds:
$$d(x_n, x^*) \leq \frac{\lambda^n}{1-\lambda} \cdot d(x_1, x_0).$$
- **Infinite-Dimensional Function Spaces:** When applied to spaces of continuous trajectories $C([0,T], \mathbb{R}^n)$ equipped with the supremum metric $\|u - v\|_\infty$, the theorem yields the Picard–Lindelöf (Cauchy–Lipschitz) Theorem for existence and uniqueness of solutions to initial value problems $\dot{x}(t) = f(t, x(t))$.
- **Multivariable Calculus:** In later chapters of this course, an adapted form of this

principle will serve as the engine driving the proof of the Inverse Function Theorem and the Implicit Function Theorem in $\mathbb{R}^n$.

Mid-Lecture Break and Q&A

Mündliches Protokoll [00:44:24 - 00:44:32]

Is there any question before the break, or shall we just take the break? Okay, then let's take the break. *[puts down microphone and announces break]*

Mid-Lecture Break

The lecturer announces a short break. Students use the time to think about questions regarding the Banach Fixed Point Theorem.

Mündliches Protokoll [00:44:33 - 00:46:20]

Again, is there some question about the Banach Fixed Point Theorem? No?

Does anyone have any question about the theorem? I left it on the blackboard to discuss it if you would like to. I don't see any question.

Now let's try to be honest, so that I can adjust the pace: you have no questions because everything is so clear — option number one —, or you have no questions because you are completely lost? Option number one: who has no questions because it's so clear? *[some students raise hands]*

Okay.

And who has no questions because they are a bit lost? *[other students raise hands]*

Well. The advantage is that many people... Thanks for the honesty. It's very important, because then I can see better.

Since many people got something, or at least they say so... The problem is that if you don't ask me questions, it's very difficult for me to reply to questions. *[laughter]*

What you can do when you think more about your questions: you can go to the TAs and to the peers that already understood the theorem, or some parts of it, and then we can continue the discussion another day.

Compactness

Covers, Subcovers, and Open Covers

Mündliches Protokoll [00:46:20 - 00:48:31]

We can then switch to another topic, that is compactness. This starts to be... *[walks to the left blackboard and writes „Compactness"]*

This starts to be a bit more difficult today, okay? I warn you. We will give the first

proof that is a bit more difficult: it is not long, but it is more difficult.

First, let me define what a cover is. Given a set X... Let me give a definition. *[writes „Def'n (cover)" on the board]* Definition of cover.

Given a set X *[writes X set]* and $E \subseteq X$ *[writes $E \subset X$]*, we say that *[writes we say that]* this calligraphic letter $\mathcal{U}$ *[writes $\mathcal{U}$]* — that is a collection of sets *[writes $= \{U_i\}_{i\in I}$]* — is a cover of E *[writes is a cover of E]* if the union of the sets covers my set E. So, if E is contained in the union of this family, capital $\mathcal{U}$ *[writes if $E \subset \bigcup \mathcal{U}$]*, which just means the union for $i \in I$ of the U_i *[writes $= \bigcup_{i\in I} U_i$]*, right?

Mündliches Protokoll [00:48:31 - 00:49:50]

And whenever you have a cover, if I give you $\mathcal{V} \subseteq \mathcal{U}$ that still covers *[writes $\mathcal{V} \subset \mathcal{U}$] [writes still covers]*, this is still a cover, we call it a subcover *[writes we call subcover]*, right?

And maybe the last point: when all the sets in my cover are open *[draws a box around the first part and writes above right]*... If $\mathcal{U}$ is a collection of open sets that covers, as before *[writes if $\mathcal{U}$ is a collection of open sets]*, we call it an open cover *[writes we call it open cover]*, right?

Definition of Cover, Subcover, and Open Cover

Let X be a set and let $E \subseteq X$ be a subset.

Definition 2.7 (*Cover, Subcover, and Open Cover*): Let $\mathcal{U} = \{U_i\}_{i\in I}$ be a collection of subsets of X indexed by a set I.

(1) Cover: The collection $\mathcal{U}$ is called a *„cover"* of E if E is contained in the union of the sets in $\mathcal{U}$:

$$E \subseteq \bigcup \mathcal{U} = \bigcup_{i\in I} U_i. \tag{2.26}$$

(2) Subcover: If $\mathcal{V} \subseteq \mathcal{U}$ is a subcollection of $\mathcal{U}$ that still covers E, that is:

$$E \subseteq \bigcup \mathcal{V} = \bigcup_{V\in\mathcal{V}} V, \tag{2.27}$$

then $\mathcal{V}$ is called a *„subcover"* of E. If $\mathcal{V}$ contains only finitely many sets, it is called a *„finite subcover"*.

(3) Open Cover: If (X, d) is a metric space and every set $U_i \in \mathcal{U}$ is an open subset of X, then $\mathcal{U}$ is called an *„open cover"* of E.

Notation and Structure of Covers:

- **Index Set I:** The index set I can be arbitrary: finite, countably infinite, or uncountably infinite.
- **The Operator $\bigcup \mathcal{U}$:** In set theory, $\bigcup \mathcal{U}$ denotes the union of all elements of the family $\mathcal{U}$, meaning $x \in \bigcup \mathcal{U} \iff \exists U \in \mathcal{U}$ such that $x \in U$.
- **Subcover vs. Subcollection:** A subcover $\mathcal{V} \subseteq \mathcal{U}$ consists of a selection of sets from $\mathcal{U}$. Crucially, $\mathcal{V}$ must still cover the entire set E, so no points of E are left uncovered when discarding the remaining sets $\mathcal{U} \setminus \mathcal{V}$.

Geometric Example of a Cover in $\mathbb{R}^2$

Mündliches Protokoll [00:49:50 - 00:50:33]

Sometimes the set E will be the whole space. Now, when I discuss compactness, the set E will be the whole space, so it can be everything. But in general it could be — and we will use many times in this course — a subset.

It is something like this. Let's say that $X = \mathbb{R}^2$ *[writes $X = \mathbb{R}^2$]* to make a drawing, and then I have my set... This is my set E, the square *[draws a square and writes $E = [0,1]^2$]*:

Mündliches Protokoll [00:50:33 - 00:51:43]

Then I can draw a cover of E that could be this thing *[draws rectangles covering E]*, right? A cover by three sets. Or let's put four: these three, and then we can have another one that is, I don't know, this one *[draws dashed rectangle with internal dots]*.

This is a cover, because every point is contained in one of the sets. Every point in my original square is contained in at least one of the rectangles I drew, isn't it? This point is contained actually in the yellow and in the dashed rectangle, right? The property to be a cover is that every point should be contained in at least one. It can be contained in many, but at least one.

And then a subcover: if I want a subcover of this, I can take just the colored ones. So, I erase the dashed one, and it is still a cover, because every point will still be contained in at least one.

Mündliches Protokoll [00:51:43 - 00:52:13]

And that's why we put it like this: we will also use infinite families of sets that cover my given space.

Yeah, there is a question... *[throws catch-box microphone to a student in the audience]*

Student Question

So, um, should $\mathcal{V}$ as a subcover now be a subset of the family or of the union of the family?

Mündliches Protokoll [00:52:21 - 00:52:34]

A cover is a collection of sets, and a subcover is a subcollection that still covers.

Student Answer

Okay.

Mündliches Protokoll [00:52:35 - 00:52:51]

Right? Yes.

Okay, this is a definition. Now we will see how to use that for compactness.

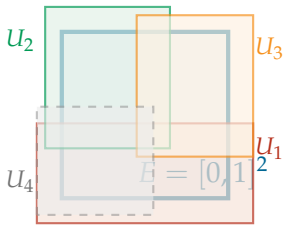
Geometric Example of a Cover and Subcover in $\mathbb{R}^2$

Let $X = \mathbb{R}^2$ and consider the unit square $E = [0,1]^2 \subset \mathbb{R}^2$.

Beispiel 2.8 (*Cover and Subcover of a Square*)**:** Consider four open rectangles $U_1, U_2, U_3, U_4 \subset \mathbb{R}^2$ covering E:

$$U_1 := (-0.15, 1.15) \times (-0.15, 0.45), \quad \text{(lower horizontal rectangle)}$$
$$U_2 := (-0.1, 0.65) \times (0.3, 1.15), \quad \text{(upper-left rectangle)}$$
$$U_3 := (0.45, 1.15) \times (0.25, 1.1), \quad \text{(upper-right rectangle)}$$
$$U_4 := (-0.15, 0.55) \times (-0.1, 0.55). \quad \text{(redundant dashed rectangle)}$$

Then $\mathcal{U} := \{U_1, U_2, U_3, U_4\}$ is an open cover of E. The subcollection $\mathcal{V} := \{U_1, U_2, U_3\} \subset \mathcal{U}$ already satisfies $E \subseteq U_1 \cup U_2 \cup U_3$, so $\mathcal{V}$ is a valid subcover of E.



Reading the Diagram:

- **Cover Condition:** Every point $(x, y) \in [0,1]^2$ belongs to at least one of the rectangles: points near the bottom lie in U_1, points on the upper-left lie in U_2, and points on the upper-right lie in U_3. Thus, $\mathcal{U} = \{U_1, U_2, U_3, U_4\}$ covers E.
- **Subcover Extraction:** The dashed rectangle U_4 covers points that are already covered by U_1 and U_2. Discarding U_4 leaves $\mathcal{V} = \{U_1, U_2, U_3\}$, which still covers E, forming a subcover.
- **Student Clarification:** As the professor clarified in the lecture, a subcover is a **subcollection of sets** $\mathcal{V} \subseteq \mathcal{U}$ (not a geometric sub-region or subset of the union) that retains the covering property $E \subseteq \bigcup \mathcal{V}$.

Sequential Compactness

Mündliches Protokoll [00:52:52 - 00:53:52]

Now, definition... *[writes (X, d) metric space]* Let (X, d) be a metric space.

Let $K \subseteq X$ be a set *[writes $K \subset X$ set is called]*. A set K is called...

As for continuity, we will have two equivalent definitions of compactness. Now I will say them: this is compactness, this is compactness. Later I will prove that they are

really equivalent, right?

Now I give different names. The first one is called sequential compactness:

Mündliches Protokoll [00:53:52 - 00:56:11]

[writes (1) sequential compactness] Sequential compactness.

We say that K is sequentially compact... *[writes K is seq cpt if]*

You know that sometimes I use abbreviations: they are useful for the blackboard, and also for you to take notes. The standard abbreviation of compactness is *„CPT“*, compact. Sequential I put as *„seq“*. Sequential compactness.

K is sequentially compact if for every sequence $(x_n)_{n\geq 0} \subseteq K$ *[writes* $\forall (x_n)_{n\geq 0} \subset K$*]* contained in my set K, I can find a convergent subsequence *[writes* $\exists (x_{n_k})$ *subsequence s.t.]*: there exists a subsequence (x_{n_k}) such that $\lim_{k\to\infty} x_{n_k}$ *[writes* $\lim_{k\to\infty} x_{n_k}$*]* exists *[writes exists]* and belongs to K *[writes and belongs to K]*.

So, the first definition of compactness: a set is compact if whenever I take any sequence in the set, I can extract a subsequence that is convergent, and the limit is inside the set, okay?

Definition of Sequential Compactness

Let (X,d) be a metric space and let $K \subseteq X$ be a subset.

Definition 2.9 (*Sequential Compactness*): The set K is called *„sequentially compact“* (abbreviated *„seq. cpt“*) if every sequence in K possesses a subsequence that converges to a limit in K:

$$\forall (x_n)_{n\geq 0} \subseteq K, \quad \exists (x_{n_k})_{k\geq 0} \subseteq (x_n) \quad \text{such that} \quad \lim_{k\to\infty} x_{n_k} \text{ exists and } \lim_{k\to\infty} x_{n_k} \in K. \tag{2.28}$$

Two Indispensable Requirements of the Definition: The condition of sequential compactness imposes two simultaneous constraints:

(1) Existence of a Subsequential Limit (Boundedness/No Escape to Infinity): For any sequence $(x_n) \subseteq K$, the points cannot escape to infinity or oscillate without accumulation points; there must exist at least one cluster point x^*.

(2) Containment of the Limit (Closedness/No Escape Across Boundaries): The limit point x^* must remain strictly inside K ($x^* \in K$). If points could accumulate on a missing boundary point outside K, the condition would fail.

Examples: Bolzano–Weierstrass and Non-Compactness of $[0,1]\cap\mathbb{Q}$

Mündliches Protokoll [00:56:12 - 00:57:26]

Let me give an example from Analysis I *[writes „Example (Analysis 1)“ on the board]*.

Probably you have seen it under this name: the Bolzano–Weierstrass theorem *[writes „Bolzano-Weierstrass“]*. Bolzano–Weierstrass says that if you take a closed interval

$[a,b] \subset \mathbb{R}$ *[writes* $[a,b] \subset \mathbb{R}$*]*, it is always sequentially compact *[writes is seq compact]*, right? A closed interval with finite endpoints.

Where was the question? Here? *[throws catch-box microphone to a student in the audience]* Oops, sorry, I fell short.

Student Question

Um... Isn't this just our definition of completeness, with the Cauchy sequence, but with a subsequence?

Mündliches Protokoll [00:57:36 - 00:57:37]

Any sequence.

Student Question, Continued

With, yeah, with any sequence and any subsequence. Is not the the same definition?

Mündliches Protokoll [00:57:42 - 00:57:46]

For any sequence there exists some subsequence.

Student Question, Continued

Okay.

Mündliches Protokoll [00:57:47 - 00:57:51]

Given a sequence, there is a subsequence. It's very strong, this.

Student Question, Continued

Okay. So it is like our completeness definition with the Cauchy sequence? 'Cause our completeness definition with the Cauchy sequence is that every, uh, sequence converges to a point inside the complete interval.

Mündliches Protokoll [00:58:09 - 00:58:19]

What you are asking is if compact implies complete, right? *[student nods „Yes"]* Yes. *[catches microphone back]* Yes.

Mündliches Protokoll [00:58:20 - 00:59:47]

Compact will be stronger than complete. Complete in $\mathbb{R}$ is closed sets.

In $\mathbb{R}$ or in $\mathbb{R}^n$ — we will see, not in a general metric space, but in $\mathbb{R}^n$ — we will see a theorem that says that compact in $\mathbb{R}$ is closed and bounded. And the same in $\mathbb{R}^n$. So, you are adding boundedness.

Another example *[writes „Example" on the board]* of something that is bounded and is not compact — you will be missing completeness —: if you take the closed interval $[0,1]$ intersected with $\mathbb{Q}$, the rational numbers *[writes* $[0,1]\cap\mathbb{Q}$*]*, right? Here, if you take any sequence, it will have a convergent subsequence, but in $\mathbb{R}$, not in $\mathbb{Q}$, right?

So, this is not compact *[writes is not cpt and underlines „not cpt"]*.

And now there is the definition... *[pulls down the middle blackboard panel]*

Examples: Bolzano–Weierstrass and Non-Compactness of Rational Subsets

Beispiel 2.10 (*The Bolzano–Weierstrass Theorem in* $\mathbb{R}$): In $\mathbb{R}$ equipped with the standard Euclidean metric, any closed and bounded interval $[a,b]\subset\mathbb{R}$ (with $-\infty < a \le b < \infty$) is sequentially compact:

$$[a,b]\subset\mathbb{R} \quad \text{is sequentially compact.}$$

Every sequence $(x_n)\subseteq[a,b]$ has a subsequence converging to a point $x^*\in[a,b]$.

Beispiel 2.11 (*A Bounded Non-Compact Subset:* $[0,1]\cap\mathbb{Q}$): Consider the set of rational numbers in the unit interval:

$$K := [0,1]\cap\mathbb{Q}\subset\mathbb{Q}.$$

Although K is bounded, it is **not** sequentially compact:

- Choose an irrational number in the interval, for instance $\frac{1}{\sqrt{2}}\in(0,1)\setminus\mathbb{Q}$.
- Construct a sequence of rational numbers $(q_n)_{n\ge0}\subseteq[0,1]\cap\mathbb{Q}$ converging to $\frac{1}{\sqrt{2}}$ in $\mathbb{R}$ (e.g., via decimal truncations).
- Any subsequence (q_{n_k}) must also converge to $\frac{1}{\sqrt{2}}$ in $\mathbb{R}$.
- However, $\frac{1}{\sqrt{2}}\notin\mathbb{Q}$, so (q_{n_k}) has no limit in $[0,1]\cap\mathbb{Q}$. Thus, no subsequence converges in K.

***Einblick:* Compactness vs. Completeness**

The student's question highlights a fundamental distinction in metric geometry:

- **Completeness:** Requires that every **Cauchy** sequence converges in the space. It ensures there are no *„missing puncture points"* or *„holes"*.
- **Sequential Compactness:** Requires that **every sequence whatsoever** (even sequences that wander wildly, like $(-1)^n$) possesses a convergent subsequence.
- **Relation:** Compactness is strictly stronger than completeness:

$$\text{Compact} \implies \text{Complete and Bounded.}$$

In $\mathbb{R}$ (and $\mathbb{R}^n$), the Heine–Borel theorem establishes that a subset is compact if and only if it is closed and bounded. In $[0,1]\cap\mathbb{Q}$, boundedness is satisfied, but completeness fails because of the gaps formed by irrational numbers.

Topological Compactness

Mündliches Protokoll [00:59:57 - 01:01:19]

This was definition number one of compactness.

Now I continue... *[writes „Compactness“ at the top left of the lower middle board]*

Definition number two: this one is new. But you will see that we will use it many, many times; it's very useful.

A set K is called topologically compact if for every open cover $\mathcal{U}$ of K *[writes $\forall \mathcal{U}$ open cover of K]*, there exists a finite subcover *[writes exists a finite subcover]*. *[draws a frame around this definition]*

Okay?

Definition of Topological Compactness

Let (X, d) be a metric space.

Definition 2.12 (*Topological Compactness*): A subset $K \subseteq X$ is called *„topologically compact“* if every open cover of K contains a finite subcover:

$$\forall \mathcal{U} \text{ open cover of } K, \quad \exists \{U_1, \dots, U_N\} \subseteq \mathcal{U} \quad \text{such that} \quad K \subseteq \bigcup_{j=1}^{N} U_j. \tag{2.29}$$

The Finite Subcover Condition: Topological compactness allows one to reduce a potentially intractable, uncountably infinite collection of local approximations to a finite collection. In analysis, finiteness allows one to take maximums, minimums, and finite sums, making global estimates possible.

Counterexample: Infinite Open Cover of $(0,1) \cap \mathbb{Q}$ without Finite Subcover

Mündliches Protokoll [01:01:19 - 01:02:26]

Here it is very important that the open cover can be infinite, right?

Let me give you an example. Again, this set $X = (0,1) \cap \mathbb{Q}$ *[writes $X = (0,1) \cap \mathbb{Q}$]*, right? You learned in Analysis I that the rationals are countable.

So, you can enumerate the rationals, right? *[writes $\mathbb{Q} \subset \{x_n \mid n > 0\}$]* They are contained in the image of a sequence.

We can take a sequence... I think I saw in your notes of Analysis I even the nice drawing of how you do this sequence. You remember that, right? *[sketches snake diagonal pattern in the air]* That sequence that is like a snake, and enumerates the rationals, right? You know what I'm talking about, yeah?

There is a sequence that will enumerate, or that will cover, all the rationals.

Mündliches Protokoll [01:02:26 - 01:03:58]

Now, for every element of the sequence, you can take a ball, let's say the ball of radius 2^{-n} centered at x_n *[writes $B_{2^{-n}}(x_n)$]*, right?

If you start with 2^{-n}, maybe it covers everything, right? But you can take $2^{-n-1000}$ *[modifies exponent to $2^{-n-1000}$]*, right?

2^{-1000} is a ridiculously small number, but it's positive, right?

What we are doing here is: for every rational number, we put on top of it a tiny interval *[draws a point with a tiny interval on the board]*. But we have infinitely many of them, right?

Of course, this covers all the rationals. This set *[writes $\bigcup_n B_{2^{-n-1000}}(x_n)$ on the board]*... These balls are intervals, right? In this case, a ball in $\mathbb{R}$ is just the interval of this radius.

These sets cover my whole interval. There are infinitely many of them, but I can never extract a finite subcover.

Mündliches Protokoll [01:03:58 - 01:05:35]

We will see more about this, but if you want some strong heuristic of why you cannot cover this: it is because this thing is geometric. If you sum the lengths of all these intervals, it's a geometric series. You will see that the length of all these intervals is about $2 \cdot 2^{-10^3}$ *[writes $2 \cdot 2^{-10^3}$ on the board]*.

Two times a ridiculous number.

I have infinitely many intervals, but when I sum all of their lengths — actually it's four times, but who cares *[changes 2 to 4: $4 \cdot 2^{-10^3}$]* —, it's four times this.

Of course, I cannot cover an interval of length 1 by tiny, tiny intervals whose total length is ridiculously small, right?

We will see more about that. *[draws a square on the board]* You can also prove in other ways that it doesn't cover. *[writes an arrow from $4 \cdot 2^{-10^3}$ to „total length"]* This is the total length.

What I would like to prove today is that these two definitions — sequential compactness and topological compactness — are equivalent: **(1)** implies **(2)**, and **(2)** implies **(1)**, okay?

And that's what we are going to do now.

Equivalence of Compactness Definitions and Board Reset

Mündliches Protokoll [01:05:35 - 01:06:06]

In this way we will also understand the definition a bit better. *[walks over to the board controls]*

Let me put it there... *[slides boards and operates motorized shades]*

Now we start with the shades... This gets complicated. *[shades settle into position]*

Yeah, I hope you can see. I will try not to use that part of the blackboard.

Blackboard Transition

The lecturer operates the motorized controls to reposition the blackboard panels and shades, bringing down a large clean black surface on the lower level while preserving the previous summary of compactness definitions on the upper boards.

Proof of Equivalence: Sequential Implies Topological Compactness

Equivalence Theorem

Satz 2.13 (*Equivalence of Compactness Notions in Metric Spaces*)**:** Let (X, d) be a metric space, and let $K \subseteq X$ be a subset. Then the following statements are equivalent:

(1) Sequential Compactness: Every sequence $(x_n)_{n\geq 0} \subseteq K$ has a subsequence converging to a limit in K.

(2) Topological Compactness: Every open cover $\mathcal{U}$ of K possesses a finite subcover $\mathcal{V} \subseteq \mathcal{U}$.

Proof of (1) $\Longrightarrow$ (2): Sequential Compactness Implies Topological Compactness

Proof Strategy and Open Cover Characterization

Mündliches Protokoll [01:06:06 - 01:07:41]

I told you: this proof is a bit hard.

It's not long, but it's one of these proofs that... if you didn't know the trick of how to start, it's hard to do it, which is an indication that there is something deep going on in this proof. So, let's be careful with this proof, okay?

The other implication is also delicate.

Our assumption now is that we have a subset... Assume $K \subseteq X$ is sequentially compact *[writes* ***(1)*** $\Longrightarrow$ ***(2)*** *and „Assume $K \subset X$ is seq. cpt."]*.

Our goal is to show *[writes „Goal show"]* that for every open cover $\mathcal{U} = \{U_i\}_{i\in I}$ *[writes $\forall \mathcal{U} = \{U_i\}_{i\in I}$ open cover]*, I can extract a finite subcover: there exists $\mathcal{V} \subseteq \mathcal{U}$ finite subcover *[writes $\exists \mathcal{V} \subset \mathcal{U}$ finite subcover]*.

This is the same as **(1)** implies **(2)**, right? But I repeat everything again so that we are all on the same page to start this proof.

Setup and Objective of the Implication

Implication: (1) $\Longrightarrow$ (2).

- **Assumption:** $K \subseteq X$ is *„sequentially compact"*:

$$\forall (x_n)_{n\geq 0} \subseteq K, \quad \exists (x_{n_\ell})_{\ell \geq 0} \quad \text{such that} \quad \lim_{\ell \to \infty} x_{n_\ell} = x \in K.$$

- **Goal:** Prove that K is *„topologically compact“*:
  $\forall \mathcal{U} = \{U_i\}_{i\in I}$ open cover of K, $\exists \mathcal{V} \subseteq \mathcal{U}$ finite subcover such that $K \subseteq \bigcup_{V\in\mathcal{V}} V$.

Mündliches Protokoll [01:07:41 - 01:09:11]

Okay.

This is the goal, I want to show that. You give me the cover, $\mathcal{U}$, and I will need to provide you with a finite subcover. This is my task: someone gives me the cover, and I need to give you the finite subcover.

Let $\{U_i\}$ be an open cover *[writes „Let $\{U_i\}$ be a open cover $\iff$ “]*.

This is the same as: for every $x \in K$, there exists at least one U_i such that $x \in U_i$ *[writes $x \in K \implies \exists U_i$ open st $x \in U_i$]*. These guys are always open.

Pointwise Characterization of an Open Cover

Let $\mathcal{U} = \{U_i\}_{i\in I}$ be an arbitrary open cover of K. By definition of a cover:

$$K \subseteq \bigcup_{i\in I} U_i \iff \forall x \in K,\ \exists i \in I \text{ such that } U_i \text{ is open and } x \in U_i. \tag{2.30}$$

Because each U_i is an open subset of (X,d), every point $x \in K$ possesses an open neighborhood within the cover.

Construction of the Inscribed Radius Function

Mündliches Protokoll [01:09:11 - 01:10:46]

If you give me many, many sets that cover my original compact set, you are giving me more freedom. But actually it's also tricky, because some of the sets can be very bad to cover.

I need to identify which sets here see a big chunk of my set, so that I can pick them.

The way to do this is:

Given $x \in K$ *[writes „Given $x \in K$ let“]*, I will define $r(x)$, radius, that is the supremum of $r > 0$ such that the ball of radius r centered at x is contained in U_i for some i *[writes $r(x) := \sup\{r > 0 \mid B_r(x) \subset U_i,\ U_i \in \mathcal{U}\}$]*, where U_i is some element in the cover.

This is U_i, and this is the cover.

Mündliches Protokoll [01:10:46 - 01:12:43]

Let me draw this, because otherwise it's complicated.

Here we have our set K *[draws a smooth closed curve representing K on the board]*. This is K.

And now you give me some cover *[draws overlapping closed curves in white and red chalk]* across K.

Now I have another one here. I put different colors because otherwise they are very hard to see: maybe one is this, and now is that. Now every point in my set is contained in one of the sets of the cover, right?

And now I pick a point here, this one.

There is a biggest possible radius such that the ball of radius r centered at my point — this is x — is contained in the set of my cover, in the blue set *[draws a circle centered at x touching the boundary of the covering set]*.

Just pick the point and enlarge the radius until the ball touches the boundary of my set: this is the greatest radius.

It could be that some of the open sets in my cover are really, really large, and then potentially the radius could be infinity. To rule this out, I will just put the minimum between this number and 1.

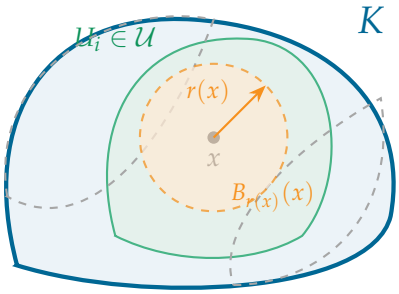
Let me redefine $r(x)$ as the minimum between this number and the number 1 *[modifies definition to $r(x) := \min\{\sup\{\dots\}, 1\}$]*.

Okay? This is just the definition.

The Inscribed Radius Function $r(x)$

Definition 2.14 (*Inscribed Radius Function*): Let $\mathcal{U} = \{U_i\}_{i\in I}$ be an open cover of K. For each point $x \in K$, we define the *„inscribed radius“* $r(x)$ by:

$$r(x) := \min\left\{\sup\{r > 0 \mid \exists i \in I \text{ such that } B_r(x) \subseteq U_i\},\, 1\right\}. \tag{2.31}$$



Properties of the Radius Function:

- **Strict Positivity ($r(x) > 0$):** Because $\mathcal{U}$ is an open cover of K, for any $x \in K$ there is some index $i \in I$ such that $x \in U_i$. Since U_i is open, there exists an open ball $B_\varepsilon(x) \subseteq U_i$ for some $\varepsilon > 0$. Hence the set of valid radii contains $\varepsilon > 0$, guaranteeing:
$$r(x) > 0 \quad \forall x \in K.$$
- **Capping by 1:** If $U_i = X$ is unbounded or infinitely large, the supremum could be $+\infty$. Capping the definition with $\min\{\dots, 1\}$ ensures $0 < r(x) \leq 1$ for every $x \in K$.

Selection of Efficient Open Sets

Mündliches Protokoll [01:12:43 - 01:14:45]

Now, given $x \in K$, select U_i in the cover such that the ball of radius $r(x)/2$ centered at x is contained in U_i *[writes Given $x \in K$ select U_i st $B_{\frac{r(x)}{2}}(x) \subset U_i$]*.

Select U_i where i depends on x *[writes $i = i(x)$ and $U_{i(x)}$]*.

For every point, this supremum is $r(x)$. But you know that the supremum maybe is not attained.

However, half of the supremum is always attained, right?

If you want to put the supremum here, maybe it's just attained in the limit, along a sequence of U_i; but if I put half of the supremum, there is always one set that gives me half of the supremum.

That's why I take this $r(x)/2$, just to take care of this supremum.

So, for every point, there exists some U_i that is efficient covering the point in this sense: it's quite large.

Selection of Covering Sets with Half-Radius Balls

For each $x \in K$, we select an index $i(x) \in I$ and an associated open set $U_{i(x)} \in \mathcal{U}$ satisfying:

$$B_{\frac{r(x)}{2}}(x) \subseteq U_{i(x)}. \tag{2.32}$$

Why Halving the Radius is Necessary: Because $r(x)$ is defined as a supremum, an inscribed ball of the exact radius $r(x)$ might not be contained in any single element $U_i \in \mathcal{U}$ (the supremum could be a supremum of radii over an infinite sequence of distinct covering sets). However, by definition of the supremum, any strictly smaller radius—specifically:

$$\frac{r(x)}{2} < r(x)$$

—is strictly exceeded by at least one valid radius in the set. Therefore, there is guaranteed to exist at least one set $U_{i(x)} \in \mathcal{U}$ containing the ball $B_{r(x)/2}(x)$.

Iterative Construction of the Sequence of Points

Mündliches Protokoll [01:14:45 - 01:16:31]

Now you will see how it goes.

Pick any point — you can choose it — $x_0 \in K$ *[writes „Pick any $x_0 \in K$"]*, right?

And now I will start to construct the finite cover for you.

Pick any point. The finite subcover I call $\mathcal{V}$, right?

The first element of the subcover will be this U_i associated to my first point, x_0 *[writes $\mathcal{V} := \{U_{i(x_0)}, \dots\}$]*. Right?

But this is not covering everything. So, what I do is... I will draw it here, okay?

I have my set *[draws a new sketch of set K on the right board]*: this is the set K.

I start with any point, x_0.

And then I pick what I will call $U_{i(x_0)}$ — let me call this for brevity U_0 *[writes U_0 above $U_{i(x_0)}$]*.

U_0 will be this one. And this one is good because at least I have some ball here: the ball of radius $r(x_0)/2$, right?

This is my first set of the cover.

Mündliches Protokoll [01:16:31 - 01:18:11]

Now I will take a second. All the points that are covered by this first set are already taken care of. So, let me take any point outside of this.

I will take x_1 here *[marks x_1 outside U_0]*.

And this point, because it's a cover, belongs to one of these U: this will be U_1.

Let me define $U_{i(x_1)}$, which I call U_1 *[writes $U_1 in \mathcal{V}$]*, where x_1 will be any point in $K \setminus U_0$ *[writes $x_1 \in K \setminus U_0$]*.

And I continue like this.

Now, if these two sets cover the whole K, I'm happy, because I gave you a finite subcover of two sets.

So, I can assume that these two don't cover, and I have a point x_2 *[marks x_2 in blue chalk outside $U_0 \cup U_1$]*.

Now in blue here: x_2. And this x_2 is contained in some U_2, that is $U_{i(x_2)}$ *[writes $U_2 in \mathcal{V}$]*.

And so on and so forth.

Greedy Extraction Algorithm for the Subcover

We attempt to construct a finite subcover $\mathcal{V} \subseteq \mathcal{U}$ iteratively:

(1) Base Step: Choose an arbitrary point $x_0 \in K$. Select $U_0 := U_{i(x_0)} \in \mathcal{U}$ satisfying $B_{r(x_0)/2}(x_0) \subseteq U_0$.
- If $K \subseteq U_0$, then $\mathcal{V} := \{U_0\}$ is a finite subcover, and the proof terminates.

(2) First Iteration: If $K \not\subseteq U_0$, choose a point $x_1 \in K \setminus U_0$. Select $U_1 := U_{i(x_1)} \in \mathcal{U}$ satisfying $B_{r(x_1)/2}(x_1) \subseteq U_1$.
- If $K \subseteq U_0 \cup U_1$, then $\mathcal{V} := \{U_0, U_1\}$ is a finite subcover, and the proof terminates.

(3) Second Iteration: If $K \not\subseteq U_0 \cup U_1$, choose a point $x_2 \in K \setminus (U_0 \cup U_1)$. Select $U_2 := U_{i(x_2)} \in \mathcal{U}$ satisfying $B_{r(x_2)/2}(x_2) \subseteq U_2$.

Infinite Sequence Generation and Sequential Compactness

Mündliches Protokoll [01:18:11 - 01:19:21]

Now, if this process... I would pick a point x_3 here, right? Now it's the last color I have, so it's the last example: x_3 here, and it's covered by its set. The important point is that $x_2 \in K \setminus (U_0 \cup U_1)$ *[writes $x_2 \in K \setminus (U_0 \cup U_1)$]*, and so on and so forth.

If eventually I get finitely many sets that cover, I'm happy.

Otherwise, this sequence that I'm constructing will go on forever: I will have infinitely many elements of this sequence.

And now I plan to reach a contradiction, using my assumption that K is sequentially compact.

You see? I did all of this construction to construct a sequence.

Now I will use my assumption on this sequence: the subset is sequentially compact, this sequence will have a convergent subsequence, and I will reach some sort of contradiction.

Let's do that.

Mündliches Protokoll [01:19:21 - 01:21:33]

Doing so, unless I find a finite subcover *[writes „Doing so, unless I find finite subcover,"]*, I'll produce a sequence *[writes „I'll produce a sequence"]* (x_n) such that $x_n \in K \setminus \bigcup_{k=0}^{n-1} U_{i(x_k)}$ for $n > 0$ *[writes $x_n \in K \setminus \bigcup_{k=0}^{n-1} U_{i(x_k)}$, $n > 0$]*, right?

And this sequence is infinite.

By sequential compactness *[writes „By seq. compactness,"]*, this sequence has a convergent subsequence $(x_{n_\ell})_{\ell \geq 0}$ *[writes (x_n) has converging subsequence $(x_{n_\ell})_{\ell \geq 0}$]*.

Let me define $x := \lim_{\ell \to \infty} x_{n_\ell}$ *[writes $x := \lim_{\ell \to \infty} x_{n_\ell}$]*.

This limit exists by assumption *[writes $\exists$ by assumption]*, and belongs to K *[writes $\in K$]*.

Remember that $r(x) > 0$, right? *[writes $r(x) > 0$]*

We have three minutes, let's try to make it as clear as possible.

The Infinite Sequence and Subsequential Convergence

The Infinite Escape Sequence:

Suppose for the sake of contradiction that no finite subcollection of $\mathcal{U}$ covers K. Then this iterative process never terminates, producing an infinite sequence $(x_n)_{n=0}^{\infty} \subseteq K$ satisfying:

$$x_n \in K \setminus \bigcup_{k=0}^{n-1} U_{i(x_k)} \quad \forall n \geq 1. \tag{2.33}$$

In particular, for any two indices $n > m \geq 0$:

$$x_n \notin U_{i(x_m)}. \tag{2.34}$$

Application of Sequential Compactness:

By hypothesis **(1)**, the set K is sequentially compact. Therefore, the infinite sequence $(x_n)_{n=0}^{\infty} \subseteq K$ admits a convergent subsequence $(x_{n_\ell})_{\ell \geq 0}$ whose limit lies in K:

$$\lim_{\ell \to \infty} x_{n_\ell} = x \in K. \tag{2.35}$$

Since $x \in K$, the inscribed radius at the limit point is strictly positive:

$$r(x) > 0. \tag{2.36}$$

Geometric Contradiction and Completion of the Proof

Mündliches Protokoll [01:21:33 - 01:23:04]

Right, we have our set K. We have U_0, then x_0, then x_1 is covered by someone, x_2 is covered by someone. But now these converge to x, right?

That a subsequence converges to x means that eventually I will have infinitely many elements that are here, right?

So, I will have...

Mündliches Protokoll [01:23:04 - 01:24:25]

Remember that this point is always outside, right? Every element of the sequence is outside the previous sets.

By construction of the sequence, now the radius $r(x_n)$ needs to converge to zero *[writes „by construction $r(x_n) \to 0$“]*.

Because otherwise the sequence could not converge to x. These radii of the balls need to go to zero, right?

Mündliches Protokoll [01:24:25 - 01:26:37]

On the other hand, the limit point x is in one of the sets of the cover: there is $U_{i(x)}$, an efficient set that covers my point x. This means that the ball of radius $r(x)/2$ *[writes $B_{\frac{r(x)}{2}}(x)$]* is contained in $U_{i(x)}$ *[writes $\subset U_{i(x)}$]*.

You see now the contradiction: eventually all these points x_{n_ℓ} will be very, very close to x because they are convergent.

The points x_{n_ℓ} will be super close to x. And I'm picking a radius $r(x_{n_\ell})$ that is going to zero, yet I could have chosen a radius that is much larger!

$r(x)/4$ would work for me.

This will be the contradiction. Let me just finish this sentence:

On the one hand, $r(x_{n_\ell}) \to 0$ *[writes $r(x_{n_\ell}) \to 0$]*.

On the other hand, for ℓ large, $r(x)/4$ is a good radius here: the ball of radius $r(x)/4$ centered at x_{n_ℓ} is contained in $U_{i(x)}$ *[writes for ℓ large $B_{\frac{r(x)}{4}}(x_{n_\ell}) \subset U_{i(x)}$]*. And this is a contradiction! *[draws a box around the contradiction statement]*

Okay, we will revisit next time. *[students applaud, lecturer concludes]*

Derivation of the Contradiction

1. The Radii Must Shrink to Zero ($r(x_{n_\ell}) \to 0$):
By construction, every iterate x_{n_k} lies outside all previous sets:

$$x_{n_k} \notin U_{i(x_{n_j})} \quad \forall k > j.$$

Since $B_{\frac{r(x_{n_j})}{2}}(x_{n_j}) \subseteq U_{i(x_{n_j})}$, this non-inclusion implies:

$$d(x_{n_k}, x_{n_j}) \geq \frac{r(x_{n_j})}{2} \quad \forall k > j. \tag{2.37}$$

Because the subsequence $(x_{n_\ell})_{\ell \geq 0}$ converges to x, it is a Cauchy sequence, so $d(x_{n_k}, x_{n_j}) \to 0$ as $j, k \to \infty$. In particular, taking $k = j + 1 \to \infty$:

$$\lim_{\ell \to \infty} r(x_{n_\ell}) = 0. \tag{2.38}$$

2. The Persistent Ball Around the Limit:
By (2.32), the target set $U_{i(x)}$ contains the ball of radius $\frac{r(x)}{2}$ around the limit x:

$$B_{\frac{r(x)}{2}}(x) \subseteq U_{i(x)}. \tag{2.39}$$

Since $x_{n_\ell} \to x$, there exists $L \in \mathbb{N}$ such that for all $\ell \geq L$:

$$d(x_{n_\ell}, x) < \frac{r(x)}{4}. \tag{2.40}$$

For any such $\ell \geq L$ and any point $z \in B_{\frac{r(x)}{4}}(x_{n_\ell})$, the triangle inequality gives:

$$d(z, x) \leq d(z, x_{n_\ell}) + d(x_{n_\ell}, x) < \frac{r(x)}{4} + \frac{r(x)}{4} = \frac{r(x)}{2}.$$

Consequently:

$$B_{\frac{r(x)}{4}}(x_{n_\ell}) \subseteq B_{\frac{r(x)}{2}}(x) \subseteq U_{i(x)}. \tag{2.41}$$

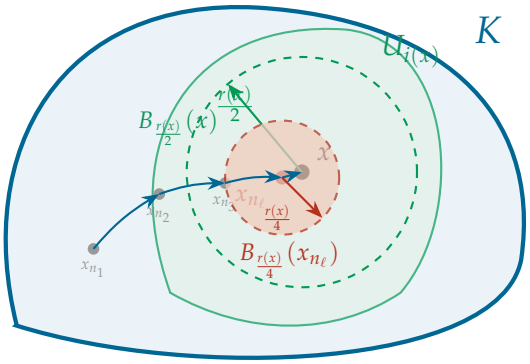
By the definition of the inscribed radius $r(x_{n_\ell})$ as the supremum in (2.31), the inclusion (2.41) implies:

$$r(x_{n_\ell}) \geq \min\left\{\frac{r(x)}{4}, 1\right\} = \frac{r(x)}{4} > 0 \quad \forall \ell \geq L. \tag{2.42}$$

3. The Contradiction:
Inequality (2.42) states that $r(x_{n_\ell}) \geq \frac{r(x)}{4} > 0$ for all large ℓ, which directly contradicts (2.38), where $\lim_{\ell \to \infty} r(x_{n_\ell}) = 0$.

Hence, the assumption that no finite subcover exists is false. Every open cover of K must admit a finite subcover, concluding the proof that sequential compactness implies topological compactness.

Geometric Visualization of the Contradiction



Core Contradiction Mechanism: The visual clarifies the geometric tension that forces the contradiction:

(1) Cauchy Separation: Every point x_{n_ℓ} was chosen strictly outside the preceding covering sets, forcing the distance between iterates to be at least $\frac{r(x_{n_k})}{2}$. Since the subsequence converges to x, the points cluster together, which forces the radii $r(x_{n_\ell})$ to shrink to zero.

(2) Room in the Target Neighborhood: At the limit x, the set $U_{i(x)}$ is open and contains a fixed ball of radius $\frac{r(x)}{2} > 0$. As soon as x_{n_ℓ} comes within $\frac{r(x)}{4}$ of x, a ball of radius $\frac{r(x)}{4}$ around x_{n_ℓ} fits completely inside $U_{i(x)}$.

(3) Contradiction: This guarantees that $r(x_{n_\ell}) \geq \frac{r(x)}{4} > 0$ for all sufficiently large ℓ, which directly contradicts the requirement that $r(x_{n_\ell}) \to 0$.

Q.E.D.

Summary of Key Concepts

- **Equivalence of Continuity Definitions:** Sequential continuity, ε-δ continuity, and topological continuity ($f^{-1}(V)$ is open for every open V) are equivalent in general metric spaces. We proved the final implication **(3)** $\implies$ **(1)** using the preimage of the open ball $B_\varepsilon(f(x))$.
- **Uniform and Lipschitz Continuity:** Uniform continuity requires a single modulus $\delta(\varepsilon) > 0$ valid for all $x \in X$. Lipschitz continuity ($d_Y(f(x), f(y)) \leq L \cdot d_X(x, y)$) implies uniform continuity with $\delta = \varepsilon / L$. The distance function to any fixed reference point $x \mapsto d(x, x_0)$ is 1-Lipschitz continuous by the reverse triangle inequality.
- **Banach Fixed Point Theorem:** Every contraction mapping $T\colon X \to X$ with contraction factor $\lambda \in (0,1)$ on a complete metric space (X, d) possesses a unique fixed point $x^* \in X$. The Picard iterates $x_{n+1} = T(x_n)$ form a Cauchy sequence by geometric decay of step lengths, converging to $x^* = T(x^*)$ with error bound $d(x_n, x^*) \leq \frac{\lambda^n}{1-\lambda} d(x_1, x_0)$.
- **Compactness in Metric Spaces:**
  - **Sequential Compactness:** Every sequence in K admits a subsequence converging to a point in K.
  - **Topological Compactness:** Every open cover $\mathcal{U}$ of K admits a finite subcover

$\mathcal{V} \subseteq \mathcal{U}$.

Sequential compactness implies topological compactness via an inscribed-radius construction $r(x)$ where assuming non-existence of a finite subcover produces an infinite sequence with $r(x_{n_\ell}) \to 0$, contradicting the persistent containment $B_{r(x)/4}(x_{n_\ell}) \subseteq U_{i(x)}$.

CHAPTER 2

Wednesday 25 February 2026

Compactness in Metric Spaces, the Heine–Borel Theorem, and Introduction to Connectedness

Course Content Overview

In this lecture, we continue and expand our investigation of compactness in metric spaces before transitioning into the topological study of connectedness:

- **Equivalence of Compactness Notions:** We prove both directions of the fundamental equivalence between sequential compactness and topological compactness (the open-cover property).
- **Fundamental Properties of Compact Sets:** We establish that compact sets are closed and complete, and that closed subsets of compact sets remain compact.
- **Continuous Maps and Optimization:** We show that the continuous image of a compact set is compact and prove the Extreme Value Theorem (Weierstrass Theorem) for real-valued continuous functions on compact metric spaces.
- **The Heine–Borel Theorem in $\mathbb{R}^n$:** We characterize compactness in finite-dimensional Euclidean space as being closed and bounded, proving sufficiency via reduction to closed cubes and componentwise extraction using the Bolzano–Weierstrass theorem.
- **Introduction to Connectedness:** We introduce clopen sets, motivate the partition problem on surfaces like S^2 and intervals in $\mathbb{R}$, and formalize the definitions of disconnected and connected sets.

Equivalence of Compactness Notions

Mündliches Protokoll [00:00:00 - 00:01:45]

Okay. Good morning.

We will continue where we left it last lecture. So, I remind you that during the previous lecture we introduced compactness, right? We introduced compactness and we gave two equivalent notions or two equivalent definitions of compactness: given a subset of our metric space, we say that it is sequentially compact if every sequence in the set has a subsequence converging to a limit in the set, okay? Number 1.

Number 2 was the topological compactness that says that for every open cover — this is an open cover, so a collection of open sets that cover my given set —, I can extract some finite subcover, right? And we were in the process to prove that both are

equivalent.

Some peers of you asked questions about this definition, and something I want to emphasize is that... *[points at the board]* it is very important, so that the notion of compactness is really strong, that there is a *„for every"* here, right? So, it is: for every sequence there exists a subsequence; for every open cover there exists a subcover, okay?

Definitions of Compactness in Metric Spaces

Let (X, d) be a metric space and $K \subseteq X$ a subset.

Definition 2.15 (*Sequential Compactness*): A subset $K \subseteq X$ is *„sequentially compact"* if every sequence $(x_n)_{n\geq 0} \subseteq K$ has a convergent subsequence $(x_{n_k})_{k\geq 0}$ whose limit belongs to K:

$$\forall (x_n)_{n\geq 0} \subseteq K \quad \exists (x_{n_k})_{k\geq 0} \text{ such that } x_{n_k} \xrightarrow{k\to\infty} x \in K.$$

Definition 2.16 (*Topological Compactness*): A subset $K \subseteq X$ is *„topologically compact"* if every open cover has a finite subcover: for every family $\mathcal{U} = \{U_i\}_{i\in I}$ of open sets in X satisfying $K \subseteq \bigcup_{i\in I} U_i$, there exist finitely many indices $i_1, \ldots, i_N \in I$ such that:

$$K \subseteq \bigcup_{j=1}^{N} U_{i_j}.$$

Satz 2.17 (*Equivalence of Compactness in Metric Spaces*): Let (X, d) be a metric space and $K \subseteq X$. Then K is sequentially compact if and only if K is topologically compact:

$$K \text{ is sequentially compact} \iff K \text{ is topologically compact.}$$

The Strength of the Universal Quantifiers: As emphasized by the lecturer, compactness is an exceptionally rigid and powerful topological property precisely because of the leading universal quantifiers:

- In Definition 2.15, the condition must hold for **every** sequence in K, no matter how wildly it oscillates.
- In Definition 2.16, the condition must hold for **every** open cover $\mathcal{U}$, even covers consisting of uncountably many tiny open neighborhoods.

Proof of Theorem 2.17

Sequential Implies Topological Compactness ($(1) \implies (2)$)

Mündliches Protokoll [00:01:45 - 00:03:00]

I had to rush a little bit, so I am going to repeat the argument, because it was the end of the lecture: how **(1)** implies **(2)**. And here was a delicate proof — not very long, but delicate.

And we said that, given a cover — so, I need to prove that for every cover something happens, so you give me the cover $\mathcal{U}$ —, and then, given a point x in K, we defined this radius $r(x)$, this positive number. And the important piece was this one: you take the supremum of all the $r > 0$ such that the ball of radius r centered at x is contained

in some of the open sets in your cover, right?

And then, just for technical reasons, we were putting the minimum between this and 1, because we don't want this $r(x)$ to be infinite. But this is not important; this is the key part.

Construction of the Local Lebesgue Radius $r(x)$

Let $\mathcal{U} = \{U_i\}_{i\in I}$ be an arbitrary open cover of K. For each $x \in K$, we define the local radius function:

$$r(x) := \min\left\{\sup\left\{r > 0 \;\middle|\; \exists U \in \mathcal{U} \text{ such that } B_r(x) \subseteq U\right\}, 1\right\}. \tag{2.43}$$

Why the Radius Is Strictly Positive: Since $\mathcal{U}$ covers K, for any $x \in K$ there exists some $U \in \mathcal{U}$ with $x \in U$. Because U is open, there exists an open ball $B_\varepsilon(x) \subseteq U$ for some $\varepsilon > 0$. Hence, the set of admissible radii is non-empty, which guarantees:

$$r(x) > 0 \quad \text{for all } x \in K.$$

The capping by 1 is merely a technical normalization to prevent $r(x) = \infty$ if an open set $U \in \mathcal{U}$ happens to contain arbitrarily large balls.

Mündliches Protokoll [00:03:00 - 00:04:47]

And then, using this function $r(x)$, we defined an assignment: to each x, we pick an open set in our cover such that the ball of radius $r(x)/2$ is contained in the open set. Which we can do because the supremum is not attained, but it is almost attained, right? So, that's why I put the $r/2$.

And then, we constructed a sequence. Remember that we need to use our assumption that every sequence has a convergent subsequence. So, we constructed the sequence as follows: we picked any point — you can choose it x_0 —, and then we pick the associated open set $U(x_0)$ here in the cover, and we say: is there a point in the complement of $U(x_0)$? Wait, if there is not such a point, we found a cover that is finite, by one open set, right? So, we can always assume that there is a next point x_1 in $K \setminus U(x_0)$, then an x_2 in $K \setminus (U(x_0) \cup U(x_1))$, and so on and so forth.

Now, x_{n+1}, the $(n+1)$-th element of the sequence, would be in the complement of all the open sets associated to the previous points. And if this process ends at some moment, that means that we cannot find the next point — it means that this difference is empty —, but this means that our set was covered by finitely many open sets in our cover. So, we are happy! If the sequence is finite and ends at a certain point, we are happy, right?

Otherwise, the sequence is infinite, and we will derive a contradiction. And this is the argument I did very quickly, and I will repeat now.

Assignment of Neighborhoods and Iterative Sequence Construction

By definition of the supremum in (2.43), for each $x \in K$, there exists an open set $U(x) \in$

$\mathcal{U}$ containing the halved ball:

$$B_{r(x)/2}(x) \subseteq U(x). \tag{2.44}$$

Iterative Choice of Points: We choose points in K sequentially:

- Pick an arbitrary initial point $x_0 \in K$.
- If $K \subseteq U(x_0)$, the single open set $\{U(x_0)\}$ already forms a finite subcover.
- Otherwise, pick $x_1 \in K \setminus U(x_0)$.
- If $K \subseteq U(x_0) \cup U(x_1)$, we have a finite subcover. Otherwise, pick $x_2 \in K \setminus \big(U(x_0) \cup U(x_1)\big)$.
- In general, having chosen $x_0, x_1, \ldots, x_n$, if $K \subseteq \bigcup_{j=0}^{n} U(x_j)$, we stop. Otherwise, pick:

$$x_{n+1} \in K \setminus \bigcup_{j=0}^{n} U(x_j). \tag{2.45}$$

Dichotomy:

(1) The process terminates at step N: Then $K \subseteq \bigcup_{j=0}^{N} U(x_j)$, providing a finite subcover $\{U(x_0), \ldots, U(x_N)\} \subseteq \mathcal{U}$. In this case, topological compactness is established directly.

(2) The process never terminates: Then we obtain an infinite sequence $(x_n)_{n=0}^{\infty} \subseteq K$. Since K is assumed to be sequentially compact, there exists a subsequence $(x_{n_k})_{k=0}^{\infty}$ converging to some limit $x \in K$:

$$x_{n_k} \xrightarrow{k\to\infty} x \in K.$$

Mündliches Protokoll [00:04:47 - 00:06:15]

So, now we have our infinite sequence. This sequence is in K, right? So, because K is sequentially compact, there exists a subsequence converging to a limit in K.

And now... notice that by construction, right? If m is bigger than n, *[writes $m > n$]* you always take the next points of the sequence in the complement of the previous ones. So, you know that x_m is going to belong to the complement of all the previous open sets: *[writes $x_m \in K \setminus (U(x_0) \cup \cdots \cup U(x_n))$]* of $U(x_0)$ union... $U(x_n)$, right? n, sorry.

And since this is an open set, right, this property... we can pass to the limit. And if you want the limit of this, this is the limit of a subsequence of that, so this limit is also contained in this. Hence, also x, the limit, belongs to the complement of all of this for every n, right?

Exactly. And then, this implies — well, on the one hand, remember that... — so, this means that x does not belong to the ball of radius $r(x_n)/2$... sorry, let me put not x_n, let me put x_{n_k}, $r(x_{n_k})$ divided by 2, right? *[writes $x \notin B_{r(x_{n_k})/2}(x_{n_k})$]* x does not belong here.

Separation of the Limit Point from the Sequence Balls

By construction of the sequence in (2.45), for any pair of indices with $m > n$, we have:

$$x_m \in K \setminus \bigcup_{j=0}^{n} U(x_j) \implies x_m \notin U(x_n). \tag{2.46}$$

Since $U(x_n)$ is an open neighborhood containing $B_{r(x_n)/2}(x_n)$, the closed complement $X \setminus U(x_n)$ contains all subsequent terms x_m for $m > n$.

Taking the limit along the subsequence (x_{n_k}) as $k \to \infty$, the closedness of $X \setminus U(x_n)$ ensures that the limit point $x = \lim_{k\to\infty} x_{n_k}$ must also lie in the complement:

$$x \in K \setminus \bigcup_{j=0}^{n} U(x_j) \quad \text{for every } n \geq 0.$$

In particular, for every element x_{n_k} of the subsequence, $x \notin U(x_{n_k})$, and therefore:

$$x \notin B_{r(x_{n_k})/2}(x_{n_k}) \quad \text{for all } k \geq 0. \tag{2.47}$$

Mündliches Protokoll [00:06:15 - 00:07:35]

x does not belong here, but at the same time, the distance...

So, we have x, the limit point, and then this is the point x_{n_k}. *[draws a diagram on the board showing x, points x_{n_k}, and a ball around x_{n_k}]* We have our point in the sequence, this is x_{n_k}. These points are approaching this one, this is converging to that, and we have the ball of radius $r(x_{n_k})/2$; this point is outside.

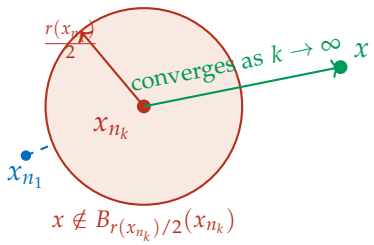
But since these points are converging to here, this radius needs to go to zero, right? So, then this $r(x_{n_k})$ divided by 2 — well, or not divided by 2 — converges to zero, *[writes $\implies r(x_{n_k}) \to 0$]* right? Because if we had a ball of positive radius uniform along the sequence, this could never converge to this point. You can use the triangle inequality and the definition of convergence, just that.

And now that we know that this... this is as k goes to infinity. *[writes $k \to \infty$]*

Vanishing Radii along the Convergent Subsequence

From (2.47), the limit point x lies outside the ball $B_{r(x_{n_k})/2}(x_{n_k})$, which means:

$$d(x, x_{n_k}) \geq \frac{r(x_{n_k})}{2} \quad \text{for all } k \geq 0. \tag{2.48}$$



Since $x_{n_k} \xrightarrow{k\to\infty} x$, the distance $d(x, x_{n_k})$ converges to 0. By the squeeze principle applied to (2.48):

$$0 \leq \frac{r(x_{n_k})}{2} \leq d(x, x_{n_k}) \xrightarrow{k\to\infty} 0 \implies \lim_{k\to\infty} r(x_{n_k}) = 0. \tag{2.49}$$

Mündliches Protokoll [00:07:35 - 00:10:05]

And now that we know that this r goes to zero, we will find easily a contradiction, because whatever is the r, r of the limit point is a positive number, right? *[writes $r(x) > 0$]*

And then, since x_{n_k} again is converging to x, *[writes $x_{n_k} \to x$]* eventually the ball of radius $r(x)/4$, centered at one of these points, x_{n_k}, is contained in the ball of radius $r(x)/2$, centered at x. *[writes $B_{r(x)/4}(x_{n_k}) \subset B_{r(x)/2}(x) \subset U(x)$]*

You see? The points are converging to here, and they are very, very close, and in this ball I have a lot of space: that is $r/4$ versus $r/2$. So, eventually — you have the ball of radius $r/2$ is like this, and then I'm considering a point that is almost this one, this is the x_{n_k}, and the ball of radius $r/4$ is smaller —, eventually this ball is inside of my bigger ball, right? Even if the center is slightly shifted.

And this, by definition, is contained in one of the open sets of the cover, right? But then this means that my choice of $r(x_{n_k})$ was not optimal; it was not respecting this, right? It was not the supremum. Because I'm saying that the ball of radius... remember that by optimality, or almost optimality, of $r(x_{n_k})$, *[writes 'by optimality']* we should be able to take this $r(x_{n_k})$ as big as this number, right? As big as this number, yet we are saying that this goes to zero.

So, this is a contradiction, right? *[writes 'is a contradiction' and draws a square]*

This was the first implication, and I did it again because this is a hard proof. Not very long, but it's hard.

Deriving the Contradiction via Optimality of the Radius

Since $x \in K$, Eq. (2.43) guarantees that the limit point has a strictly positive radius:

$$r(x) > 0.$$

By convergence $x_{n_k} \xrightarrow{k\to\infty} x$, there exists an index $K_0 \in \mathbb{N}$ such that for all $k \geq K_0$:

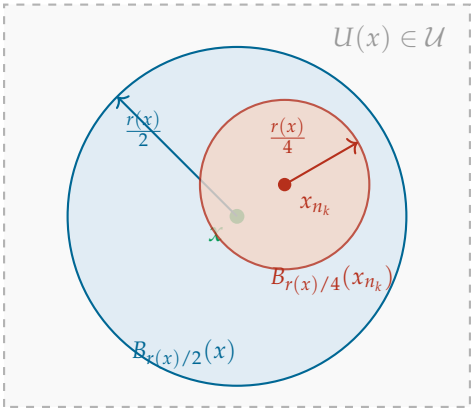
$$d(x_{n_k}, x) < \frac{r(x)}{4}. \tag{2.50}$$

Geometric Nesting of Balls: For any point $y \in B_{r(x)/4}(x_{n_k})$, the triangle inequality gives:

$$d(y, x) \leq d(y, x_{n_k}) + d(x_{n_k}, x) < \frac{r(x)}{4} + \frac{r(x)}{4} = \frac{r(x)}{2}.$$

Therefore, the ball $B_{r(x)/4}(x_{n_k})$ is completely nested within $B_{r(x)/2}(x)$, which itself is contained in the assigned open set $U(x) \in \mathcal{U}$:

$$B_{r(x)/4}(x_{n_k}) \subseteq B_{r(x)/2}(x) \subseteq U(x). \tag{2.51}$$



Contradiction with Radius Vanishing: Since $B_{r(x)/4}(x_{n_k})$ is contained in $U(x) \in \mathcal{U}$, the definition of the supremum for $r(x_{n_k})$ implies:

$$r(x_{n_k}) \geq \min\left\{\frac{r(x)}{4}, 1\right\} > 0 \quad \text{for all } k \geq K_0.$$

In particular, $r(x_{n_k})$ is bounded below by the strictly positive constant $\min\{r(x)/4, 1\}$, which directly contradicts Eq. (2.49):

$$\lim_{k\to\infty} r(x_{n_k}) = 0.$$

This contradiction demonstrates that the sequence cannot be infinite; hence, the inductive construction must terminate after finitely many steps, yielding a finite subcover. Thus, sequential compactness implies topological compactness.

Topological Compactness Implies Sequential Compactness ($(2) \implies (1)$)

Mündliches Protokoll [00:10:05 - 00:12:05]

The second one... *[erases a line on the board and prepares the lower board]*

Okay. The second one, **(2)** implies **(1)**, is slightly easier; it's a bit easier, right?

To show **(2)** implies **(1)**, we need to find... so, given... *[writes 'Given $(x_n)_{n\geq 0} \subset K$ sequence']* given a sequence, our goal... *[writes 'goal: show $\exists$ subsequence (x_{n_k}) convergent with limit in K']* our goal is to show that there exists a subsequence, that will be x_{n_k}, convergent, with limit in K, right?

And we will prove this again by contradiction. Assume that this does not happen. So, by contradiction... *[writes 'by contradiction']*

Assume that this does not happen. Another way to put that is that for every x in K, *[writes '$\forall x \in K$, x is not an accum. pt of (x_n)']* x is not an accumulation point of my sequence.

Is this not an accumulation point? Remember that the definition of accumulation point is precisely that there is a subsequence that converges to x, right?

Proof Setup: Negation of Sequential Compactness

We now prove the converse implication: assuming K is topologically compact, we show

that K is sequentially compact.

Setup: Let $(x_n)_{n\geq 0} \subseteq K$ be an arbitrary sequence in K.

Goal: Show that there exists a subsequence $(x_{n_k})_{k\geq 0}$ converging to some $x \in K$.

Proof by Contradiction: Assume the contrary: no subsequence of (x_n) converges to any point in K. That is:

$$\forall x \in K, \quad x \text{ is not an accumulation point of } (x_n)_{n\geq 0}. \tag{2.52}$$

Mündliches Protokoll [00:12:05 - 00:14:00]

And now I will put it again in another form that is more suitable. So, what does it mean that x is not an accumulation point?

I will claim, *[writes 'claim:']* I will claim that this implies that for every x in K, *[writes '$\forall x \in K$']* there exists some $\varepsilon(x)$ positive *[writes '$\exists \varepsilon(x) > 0$ such that']* such that my sequence... let me put it like this: my sequence x_n visits *[writes 'x_n visits $B_{\varepsilon(x)}(x)$ only finitely many times']* the ball of radius $\varepsilon(x)$ centered at x only finitely many times.

The ball visits... well, let me put this in... this phrasing is easier to get intuition from, but what I mean mathematically is that x_n belongs to the complement of this ball... so x_n is always an element of... so, this is the same, right? *[writes '$\iff x_n \in K \setminus B_{\varepsilon(x)}(x)\ \forall n \geq N(x)$']*

This is a way to say, if you want, that x_n belongs to the complement of the ball for all n bigger than or equal to a certain number that depends on x. Or eventually, all of the elements of the sequence will be in the complement of the ball. I'm saying the same in many equivalent ways. Is it clear?

Characterization via Finite Visits of Neighborhoods

Behauptung 2.18: If $x \in K$ is not an accumulation point of $(x_n)_{n\geq 0}$, then there exists a radius $\varepsilon(x) > 0$ such that the sequence (x_n) visits the open ball $B_{\varepsilon(x)}(x)$ only finitely many times. That is:

$$\exists \varepsilon(x) > 0,\ \exists N(x) \in \mathbb{N} \quad \text{such that} \quad x_n \in K \setminus B_{\varepsilon(x)}(x) \quad \text{for all } n \geq N(x). \tag{2.53}$$

Mündliches Protokoll [00:14:00 - 00:16:25]

So, why does x not being an accumulation point imply this?

Well, because if... let's say, parenthesis, proof of claim: *[writes '(proof of claim)']* if for every... let's say for $\varepsilon = 1/k$, so if for every ε I can take $\varepsilon = 1/k$, and $k = 1$ to infinity, *[writes 'if for $\varepsilon = 1/k$, $k = 1, \ldots, \infty$']* x_n visits infinitely many times the ball of radius $1/k$ centered at x, *[writes 'x_n visits ∞-many times $B_{1/k}(x)$']* right?

Then I can pick a subsequence. I can say: okay, let me take n_0; I define it as 0. *[writes '$n_0 = 0$']* And then, since I always have infinitely many values, I will define n_k always bigger than the previous one, or n_{k+1} bigger than n_k, so that it is a subsequence, and such that $x_{n_{k+1}}$ is contained... let me put it like this to make it nicer: *[writes '$n_k > n_{k-1}$]*

and $x_{n_k} \in B_{1/k}(x)$'] n_k bigger than n_{k-1} and x_{n_k} is contained in the ball of radius $1/k$ centered at x.

You see? If you have infinitely many values, you can always pick one where the index is larger than the previous one, and it is going back to this ball. And this sequence would be a subsequence that converges to x. So, x_{n_k} would converge to x as k goes to infinity. *[writes 'So $x_{n_k} \to x$ as $k \to \infty$']*

Right? So, this is the proof of my claim. If I am assuming that no subsequence converges to x, no subsequence converging to x necessarily implies that there is some ε such that this ball is only visited finitely many times, okay?

Proof of the Finite-Visits Claim

Proof of Claim 2.18

We argue by contraposition. Suppose that for every $\varepsilon > 0$, the sequence $(x_n)_{n\geq 0}$ visits $B_\varepsilon(x)$ infinitely many times.

Taking $\varepsilon_k = \frac{1}{k}$ for each $k \in \{1, 2, 3, \dots\}$, the set of indices $\{n \in \mathbb{N} \mid x_n \in B_{1/k}(x)\}$ is infinite. We can therefore construct a strictly increasing sequence of indices inductively:

- Set $n_0 := 0$.
- For $k = 1$, pick $n_1 > n_0$ such that $x_{n_1} \in B_1(x)$.
- Having chosen n_{k-1}, since the ball $B_{1/k}(x)$ contains infinitely many terms of the sequence, pick an index $n_k > n_{k-1}$ such that:

$$x_{n_k} \in B_{1/k}(x).$$

This defines a valid subsequence $(x_{n_k})_{k\geq 1}$. Furthermore, for every $k \geq 1$:

$$d(x_{n_k}, x) < \frac{1}{k} \xrightarrow{k\to\infty} 0 \implies x_{n_k} \xrightarrow{k\to\infty} x.$$

Thus, (x_{n_k}) is a subsequence of (x_n) converging to $x \in K$, which means that x is an accumulation point of (x_n), contradicting our hypothesis (2.52). This proves the claim. Q.E.D.

Mündliches Protokoll [00:16:25 - 00:19:00]

And now with this, I can conclude, right? Because now I take the cover — remember that now I want to use somewhere that my set is topologically compact, so I need to define a cover.

I define the cover $\mathcal{U}$... *[writes '$\mathcal{U} = \{B_{\varepsilon(x)}(x)\}_{x\in K}$']* You see that these are open balls, so they are in particular open sets, right? There are many of them, because K could be very, very infinite... so in our examples in $\mathbb{R}$, this will be uncountable, many, many times. But okay, we have an arbitrary collection of balls, and now we can extract a finite subcover.

Then we can extract... since K is topologically compact, *[writes 'K top. cpt.']* there exists a finite subcover, so we can put $K \subset \bigcup_{j=1}^{N} B_{\varepsilon(y_j)}(y_j)$... *[writes '$K \subset \bigcup_{j=1}^{N} B_{\varepsilon(y_j)}(y_j)$']*

Let me call the centers y_j, right? And now this is a contradiction. You see why?

Because any sequence has infinitely many terms. And I'm saying that my sequence would be visiting each of these balls finitely many times. But there are finitely many balls. So, at least one of the balls has to be visited infinitely many times. Contradiction!

So, this would imply that the sequence has finitely many terms, *[writes 'this would imply seq has finitely many terms']* okay? Contradiction. *[writes 'Contradiction □']*

Deriving the Contradiction via Topological Compactness

Using the radii $\varepsilon(x) > 0$ provided by Claim 2.18 for each $x \in K$, we construct the open cover:

$$\mathcal{U} := \{B_{\varepsilon(x)}(x) \mid x \in K\}. \tag{2.54}$$

Since each $B_{\varepsilon(x)}(x)$ is an open set containing x, $\mathcal{U}$ is indeed an open cover of K.

Extracting a Finite Subcover: By the topological compactness of K, there exist finitely many points $y_1, \ldots, y_N \in K$ such that:

$$K \subseteq \bigcup_{j=1}^{N} B_{\varepsilon(y_j)}(y_j). \tag{2.55}$$

Contradiction via the Pigeonhole Principle:

- By Claim 2.18, for each $j \in \{1, \ldots, N\}$, the sequence $(x_n)_{n\geq 0}$ visits $B_{\varepsilon(y_j)}(y_j)$ only finitely many times. That is, the set of indices:
$$I_j := \{n \in \mathbb{N} \mid x_n \in B_{\varepsilon(y_j)}(y_j)\}$$
is finite for each $j \in \{1, \ldots, N\}$.
- Since every term of the sequence lies in K and $K \subseteq \bigcup_{j=1}^{N} B_{\varepsilon(y_j)}(y_j)$, we must have:
$$\mathbb{N} = \bigcup_{j=1}^{N} I_j.$$
- A finite union of finite sets is finite, which implies that $\mathbb{N}$ is finite — an immediate contradiction!

Therefore, at least one ball $B_{\varepsilon(y_j)}(y_j)$ must contain infinitely many terms of the sequence, contradicting Claim 2.18.

Thus, the initial assumption (2.52) was false. Every sequence in K possesses an accumulation point $x \in K$, which implies the existence of a convergent subsequence in K. Hence, K is sequentially compact.

Q.E.D.

Mündliches Protokoll [00:19:00 - 00:20:50]

Okay. This is a difficult argument, I know. Now you have seen it. Maybe you have some questions about this, maybe not yet. We can also ask questions later on when you have them.

And then you can read it again... Last time, I also tried to draw pictures of the open sets and so on, so that you gain a bit of intuition, hopefully. And then you can read again the proof in the notes until it is clear. But this is, I know, a bit difficult.

So, the next thing, that will be easier, will be to... *[erases the middle and lower boards and prepares to write consequences of compactness]*

Board Transition

The lecturer concludes the discussion of the equivalence between sequential and topological compactness, takes a brief moment for student questions, and thoroughly erases the lower and middle blackboard panels to introduce the corollaries of compactness.

Corollaries of Compactness

Mündliches Protokoll [00:20:50 - 00:23:12]

Okay, so compact sets are very useful and very important. And now we will see some consequences, and you will start to... So, I think that some people are looking at the proof. Maybe is there some question?

[waits briefly for student questions; none are raised]

Okay, let me give some easier consequences of compactness: consequences, if you want, of the characterization of sequential compactness.

So, easy statements are, if you want, Corollary 1: *[writes 'Cor. 1']* for every metric space, if $K \subseteq X$ is compact, then K is closed. *[writes '$\forall (X,d)$ m.s. if $K \subset X$ cpt $\implies$ K is closed']*

Corollary 2 of sequential compactness: *[writes 'Cor. 2']* if $K \subseteq X$ is compact, then K is complete. *[writes 'if $K \subset X$ cpt $\implies$ K is complete']*

And Corollary 3: *[writes 'Cor. 3']* if $K \subseteq X$ is compact and A is closed, then the intersection of K and A is closed... sorry, closed, of course, but compact. *[writes 'if $K \subset X$ cpt, $A \subset X$ closed $\implies$ $K \cap A$ cpt']*

Fundamental Corollaries of Compactness

Let (X,d) be a metric space.

Korollar 2.19 (*Compact Subsets Are Closed*): If $K \subseteq X$ is compact, then K is closed in X.

Korollar 2.20 (*Compact Subsets Are Complete*): If $K \subseteq X$ is compact, then (K,d) is complete as a metric subspace.

Korollar 2.21 (*Intersection of Compact and Closed Subsets*): If $K \subseteq X$ is compact and $A \subseteq X$ is closed, then $K \cap A$ is compact.

Overview of the Corollaries: These three results provide immediate structural information about compact sets in arbitrary metric spaces:

- **Closedness (Corollary 2.19):** Compact sets automatically contain all their limit points.
- **Completeness (Corollary 2.20):** Every Cauchy sequence in K converges to a point

in K, so compactness guarantees the absence of *„holes“*.

- **Stability under Intersection (Corollary 2.21):** Intersecting a compact set with a closed set retains compactness.

Proof of the Corollaries

Mündliches Protokoll [00:23:12 - 00:25:12]

These are all easy consequences of when you use this definition of compactness, for instance, sequential compactness, right? And what you know about the definitions of complete, closed, and so on and so forth.

So, I will say with words: each of them is a one-line proof using the definitions. But I will explain why this is true, and then it is a good... as always, you have everything in the notes, right? But you can do as an exercise: try to prove them, okay?

First thing: why is a compact set always closed? A compact subset is always closed. Well, you need to remember: compact subset with sequences is: every sequence has a convergent subsequence in the set. And we know also two characterizations of closed sets.

The first definition of a closed set was the complement of an open set. And then we saw also that a set is closed if and only if it contains the limits of convergent sequences. And this is the one we are going to use, right?

So, we will take any sequence in my compact set that is convergent, because we want to check that it is closed, right? Then it has a convergent subsequence, because it's compact, and the limit belongs to the set. And then, since the sequence is convergent, the full sequence converges to the same limit that is in my set, right?

If you understood as I said, it's clear and it's good for you. Otherwise, it's even better, because then it's a good opportunity to try to reproduce and do this proof by yourself.

Proof that Compact Subsets Are Closed

Proof of Corollary 2.19

We use the sequential characterization of closed sets: a subset $K \subseteq X$ is closed if and only if every convergent sequence in K has its limit in K.

Let $(x_n)_{n\geq 0} \subseteq K$ be a sequence such that $x_n \xrightarrow{n\to\infty} x \in X$. Since K is compact (and therefore sequentially compact), there exists a subsequence $(x_{n_k})_{k\geq 0}$ that converges to some limit $y \in K$:

$$x_{n_k} \xrightarrow{k\to\infty} y \in K.$$

Because (x_n) is already convergent to x in X, every subsequence of (x_n) must converge to the same limit. Thus, $y = x$. Since $y \in K$, we obtain $x \in K$. This proves that K contains all its limit points, so K is closed in X.

Q.E.D.

Mündliches Protokoll [00:25:12 - 00:27:12]

Corollary number 2: if K is compact, then it must be complete.

So, you take any Cauchy sequence. A Cauchy sequence is a sequence, so it will have a convergent subsequence in the set. And then, since it is Cauchy, every other subsequence will have the same limit, right? You need to show that; it's not difficult.

Again, Cauchy sequences are always about to converge. A Cauchy sequence is going to converge to some point. Just that sometimes, in some spaces, the limit point is not in my space, and I need to create it, like the rationals. A Cauchy sequence of rational numbers wants to converge; the problem is that maybe it wants to converge to the square root of 2, and then it's not in my space. But compactness warrants that the limit is in the space, so no problem.

Corollary number 3: if you have a compact set and you take a closed subset of my compact set — so, for instance, you take A closed —, then this intersection is closed always, because the compact is closed, and A is closed, and remember that the intersection of closed sets is closed, right? So, this intersection is closed. Then it is also compact. Again, sequential compactness: you just need to take a sequence in the intersection. In particular, the sequence is in K, so it has a convergent subsequence, and the limit is in A because A is closed, right?

So, we have these three consequences.

Proofs of Completeness and Closed Intersections

Proof of Corollary 2.20

Let $(x_n)_{n\geq 0} \subseteq K$ be a Cauchy sequence. Since K is sequentially compact, there exists a subsequence $(x_{n_k})_{k\geq 0}$ converging to some point $x \in K$. A general property of Cauchy sequences in metric spaces states that if a Cauchy sequence possesses a subsequence converging to x, then the entire sequence converges to x:

$$d(x_n, x) \leq d(x_n, x_{n_k}) + d(x_{n_k}, x) \xrightarrow{n,k\to\infty} 0.$$

Therefore, (x_n) converges in K, proving that (K, d) is complete.

Q.E.D.

Proof of Corollary 2.21

Let $(x_n)_{n\geq 0} \subseteq K \cap A$ be an arbitrary sequence. In particular, $(x_n) \subseteq K$. Since K is sequentially compact, there exists a subsequence $(x_{n_k})_{k\geq 0}$ that converges to some limit $x \in K$. At the same time, $(x_{n_k}) \subseteq A$. Since A is closed, it contains all limits of its convergent sequences, whence $x \in A$. Consequently, $x \in K \cap A$, which demonstrates that every sequence in $K \cap A$ has a convergent subsequence whose limit belongs to $K \cap A$. Thus, $K \cap A$ is compact.

Q.E.D.

Continuous Images of Compact Sets

Mündliches Protokoll [00:27:13 - 00:29:41]

Something maybe slightly more interesting is the following proposition, but still easy, still easy. I don't remember if it is a proposition in the notes or a theorem... let me check... *[checks lecture notes]* I don't have it. Okay. Let's call it a proposition.

So, I will state it informally: a continuous image of a compact set is compact. *[writes 'Prop (continuous image of cpt is cpt)']* This is the way you can remember the proposition: the continuous image of a compact set is always compact, right?

And now we phrase it a bit more precisely. So, (X, d_X) is a metric space, as always, *[writes '(X, d_X) m.s.']* $K \subseteq X$ compact... *[writes '$K \subset X$ cpt']* Well, let me put it: (X, d_X) and (Y, d_Y) are metric spaces, both, *[writes '$(X, d_X), (Y, d_Y)$ m.s.']* and f from X to Y is a continuous map, *[writes '$f\colon X \to Y$ continuous']* and then, for every $K \subseteq X$ compact, *[writes '$\forall K \subset X$ cpt']* $f(K)$ is also compact. *[writes '$f(x) \subset Y$ cpt']* So, $f(K)$ is a subset of Y, right? *[writes '$f(K) \subset Y$']*

And again, this is very, very useful; we will use this kind of thing many times. But it is very easy to prove when we have that, because, yeah...

Student Question

You mean $f(K)$?

Mündliches Protokoll [00:29:43 - 00:29:48]

Sorry? Yeah, exactly. *[corrects $f(x)$ to $f(K)$ on the board]* Thank you.

Mündliches Protokoll [00:29:48 - 00:32:15]

So, when you take a compact set and you compute the image by a continuous function, it is also compact.

How can we prove this? We can use either of the two definitions. With the sequential it is very easy; with the other, also. Let me do one of them. Which one? The sequential or the... let's see which one we use, right? Let's take number 2 to change a bit, right?

Okay. So, let's say that I want to show that the image is topologically compact, right? Goal: *[writes 'Goal: show $f(K)$ top. cpt.']* show that $f(K)$ is topologically compact. So, this is the proof. I'm doing the proof of this proposition: Proof. *[writes 'Proof']* My goal is to show that $f(K)$ is topologically compact.

So, you give me some open cover. Given $\mathcal{V}$ an open cover of $f(K)$, *[writes 'given $\mathcal{V}$ open cover of $f(K)$']* there exists a finite subcover; this is what I want to show. I want to show that there exists a finite subcover. But what I can do with this open cover is to send it back by f to an open cover of my set K.

So, given an open cover, I consider $\mathcal{U}$... *[writes '$\mathcal{U} = \{f^{-1}(V) \mid V \in \mathcal{V}\}$']* I hope you can see here... I consider calligraphic $\mathcal{U}$ to be the set of... let me put it like this: $f^{-1}(V)$,

where V belongs to my cover, right? And you see, I have my map f...

Mündliches Protokoll [00:32:15 - 00:34:26]

Here is the space X, *[draws set X on the board]* right? Here is the space Y. *[draws set Y on the board]* My map f goes from here to here. *[draws arrow $f: X \to Y$]* And I have my compact set here, *[draws set K inside X]* K. And this is $f(K)$, *[draws image $f(K)$ inside Y]* and I want to show that that is compact.

So, what I can take is some open cover here, *[draws open cover $\mathcal{V}$ covering $f(K)$]* right? I take my cover; this is $\mathcal{V}$. And remember that a function is continuous if and only if, when I map back open sets, it gives me open sets. And since those cover $f(K)$, when I compute the preimage of all these open sets, they cover K. *[draws preimages covering K]* So, these are the preimages of my sets, right?

Then this thing, being an open cover of K, will have a finite subcover. And now I will take the finitely many sets here, and I map them there; they are a finite subcover. End of the proof, okay?

So, extract a finite subcover, *[writes 'extract finite subcover']* and then map it to the target: *[writes 'and then map it to the target']* map it to the codomain. *[writes 'map it to the codomain']* Put it here. Okay.

With sequential compactness it is very easy also. You use the definition of continuity with sequences, and it will be very easy, one line. Okay.

Continuous Image of Compact Sets and the Topological Proof

Proposition 2.22 (*Continuous Image of a Compact Set*): Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f: X \to Y$ be a continuous map. If $K \subseteq X$ is compact, then its image $f(K) \subseteq Y$ is also compact.

Topological Proof (Using Open Covers)

Goal: Show that $f(K)$ is topologically compact.

Let $\mathcal{V} = \{V_j\}_{j \in J}$ be an arbitrary open cover of $f(K)$ in Y, so that:

$$f(K) \subseteq \bigcup_{j \in J} V_j.$$

For each open set $V_j \in \mathcal{V}$, consider its preimage $f^{-1}(V_j) \subseteq X$. Since f is continuous, the preimage of every open set in Y is open in X.

We define the family of open sets in X:

$$\mathcal{U} := \{f^{-1}(V) \mid V \in \mathcal{V}\}. \tag{2.56}$$

Since $f(K) \subseteq \bigcup_{j \in J} V_j$, taking preimages yields:

$$K \subseteq f^{-1}(f(K)) \subseteq f^{-1}\left(\bigcup_{j \in J} V_j\right) = \bigcup_{j \in J} f^{-1}(V_j).$$

Thus, $\mathcal{U}$ is an open cover of K.

Because K is compact, it possesses a finite subcover: there exist finitely many indices $j_1, \ldots, j_N \in J$ such that:

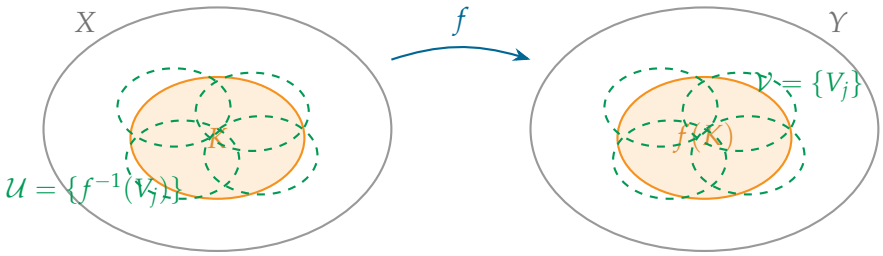
$$K \subseteq \bigcup_{k=1}^{N} f^{-1}(V_{j_k}). \tag{2.57}$$

Applying f to both sides, we obtain:

$$f(K) \subseteq f\left(\bigcup_{k=1}^{N} f^{-1}(V_{j_k})\right) = \bigcup_{k=1}^{N} f\left(f^{-1}(V_{j_k})\right) \subseteq \bigcup_{k=1}^{N} V_{j_k}. \tag{2.58}$$

Therefore, $\{V_{j_1}, \ldots, V_{j_N}\}$ is a finite subcover of $f(K)$ extracted from $\mathcal{V}$. Hence, $f(K)$ is topologically compact.

Q.E.D.



Sequential Alternative Proof: As noted by the lecturer, the proposition also admits a very direct one-line proof using sequential compactness and sequential continuity:

(1) Let $(y_n)_{n\geq 0} \subseteq f(K)$ be an arbitrary sequence. By definition of the image, there exists a sequence $(x_n)_{n\geq 0} \subseteq K$ such that $y_n = f(x_n)$ for all $n \geq 0$.

(2) Since K is sequentially compact, (x_n) has a subsequence $(x_{n_k})_{k\geq 0}$ converging to some $x \in K$.

(3) Since f is continuous, it is sequentially continuous, so:

$$y_{n_k} = f(x_{n_k}) \xrightarrow{k\to\infty} f(x) \in f(K).$$

Thus, (y_n) possesses a subsequence converging to a point in $f(K)$, which proves that $f(K)$ is sequentially compact.

Mündliches Protokoll [00:34:26 - 00:35:43]

Further consequences of compactness, still in the abstract setup, is that continuous functions...

Where is the eraser? Ah, now I lost... *[searches for the eraser]* I lost the eraser, but I put it there, okay.

Many times we will consider real-valued functions on our space, that would be $\mathbb{R}^n$ or some other metric space, but most of the time $\mathbb{R}^n$.

The Extreme Value Theorem (Weierstrass Theorem)

Mündliches Protokoll [00:35:43 - 00:37:43]

So, we will consider maps from X to $\mathbb{R}$ *[writes 'Thm $f: X \to \mathbb{R}$ continuous']* continuous, many times. And then, on compact subsets, this map always attains the supremum and the infimum.

The supremum *[writes '$\sup\{f(x) \mid x \in K\}$']* of $f(x)$ when x in K — so K is compact, *[writes '$K \subset X$ cpt']* — the supremum of $f(x)$ when x in K, and the same with the infimum, *[writes '$\inf$']* are attained. *[writes 'are attained']*

Are attained means that there exists some z in K *[writes '$\exists z \in K$']* — are attained in K, right? *[writes 'attained in K']* — such that $f(z)$ is equal to the supremum, *[writes '$f(z) = \sup$']* right? When we put it this way: the supremum when x in K of $f(x)$. *[writes '$\sup_{x\in K} f(x)$']*

Are you familiar with this notation, supremum? It is the same, right? *[circles $\sup\{f(x) \mid x \in K\}$ and points to $\sup_{x\in K} f(x)$]* I can also put it like this, or with this below the supremum. It's notation.

The Extreme Value Theorem on Compact Metric Spaces

Satz 2.23 (*Extreme Value Theorem / Weierstrass Theorem*)**:** Let (X,d) be a metric space, $K \subseteq X$ a compact subset, and $f: X \to \mathbb{R}$ a continuous function. Then f is bounded on K and attains its supremum and infimum on K. That is, there exist points $z_{\max}, z_{\min} \in K$ such that:

$$f(z_{\max}) = \sup_{x\in K} f(x) \quad \text{and} \quad f(z_{\min}) = \inf_{x\in K} f(x). \tag{2.59}$$

Equivalently:

$$\min_{x\in K} f(x) = f(z_{\min}) \leq f(x) \leq f(z_{\max}) = \max_{x\in K} f(x) \quad \text{for all } x \in K.$$

Mündliches Protokoll [00:37:43 - 00:39:13]

Why is the supremum attained? And the infimum the same?

Well, you can see this in two ways. One way is I can use that... Okay, not yet, because I need to prove something, but let me do the second way, sorry. Later I will do a comment on how you can view this theorem when we have Heine–Borel.

We can prove this very easily using sequential compactness, right? By definition of the supremum, there exists some... if you call it S, the supremum... *[writes 'S']*

I will do it for the supremum, right? But the infimum is the same. If this supremum you call it S, *[boxes $\sup\{f(x) \mid x \in K\}$ as S]* by definition of supremum there exists some sequence *[writes '$\exists (x_n) \subset K$']* x_n such that $f(x_n)$ is converging to S, *[writes '$f(x_n) \to S$']* right?

Mündliches Protokoll [00:39:13 - 00:40:25]

Actually, I can even take the sequence converging like that — I mean, monotonely, right? So, I can even take $f(x_n)$ increasing or non-decreasing, *[writes '($f(x_n)$ non-decreasing)']* if I want.

Maybe the supremum is not attained, but by definition of supremum there have to be points that are closer and closer in the image, right?

And then, this sequence, because K is compact, has a convergent subsequence: *[writes '$\exists x_{n_k} \to z \in K$']* there exists some x_{n_k} converging to some z, right? And this belongs to K. But then, since f is continuous, $f(z)$ is equal to S, *[writes '$f(z) = S$']* so the supremum is attained.

So, this uses that f is continuous, *[writes '(f continuous)']* right? f is sequentially continuous, so f of the limit is equal to the limit of the f's. f of the limit is equal to that.

Mündliches Protokoll [00:40:25 - 00:41:13]

And the same applies to the infimum. The infimum is attained, and we call it minimum.

This you already saw in $\mathbb{R}$, right? In $\mathbb{R}$ you said: if you have a closed, bounded interval, the infimum and the supremum of a function are attained, and we call them minimum and maximum. But this is much more general; this applies to any compact space.

Proof of the Extreme Value Theorem via Sequential Compactness

Proof of Theorem 2.23

We prove that the supremum is attained; the proof for the infimum is completely analogous.

Let $S := \sup_{x \in K} f(x) \in (-\infty, \infty]$. By the definition of the supremum, there exists a sequence of points $(x_n)_{n \geq 0} \subseteq K$ such that:

$$f(x_n) \xrightarrow{n \to \infty} S. \tag{2.60}$$

(Without loss of generality, we can choose (x_n) such that the sequence of values $(f(x_n))_{n \geq 0}$ is monotonically non-decreasing).

Since K is compact, it is sequentially compact. Therefore, (x_n) has a subsequence $(x_{n_k})_{k \geq 0}$ that converges to some limit point $z \in K$:

$$x_{n_k} \xrightarrow{k \to \infty} z \in K.$$

Because f is continuous, it is sequentially continuous. Hence:

$$f(x_{n_k}) \xrightarrow{k \to \infty} f(z).$$

On the other hand, $(f(x_{n_k}))_{k \geq 0}$ is a subsequence of the sequence $(f(x_n))_{n \geq 0}$ which converges to S. By the uniqueness of limits in $\mathbb{R}$:

$$f(z) = S = \sup_{x \in K} f(x).$$

Since $z \in K$, we have $f(z) \in \mathbb{R}$, which in particular implies that $S < \infty$ (so f is bounded from above on K) and that the supremum is attained at $z \in K$.

Applying the exact same argument to $-f$ (or directly to $\inf_{x\in K} f(x)$), there exists a point $z_{\min} \in K$ such that $f(z_{\min}) = \inf_{x\in K} f(x) > -\infty$.

Q.E.D.

The Heine–Borel Theorem in $\mathbb{R}^n$

Mündliches Protokoll [00:41:13 - 00:43:13]

Now, all of this is in this huge level of generality. But we want to understand very, very well for this course the case of $\mathbb{R}^n$. *[writes '$\mathbb{R}^n$']*

So, what happens in $\mathbb{R}^n$? In $\mathbb{R}^n$, we have a theorem that we will prove now, that is called Heine–Borel: *[writes 'Thm (Heine-Borel)']* that says that $K \subseteq \mathbb{R}^n$ is compact *[writes '$K \subset \mathbb{R}^n$ cpt']* if and only if K is bounded and closed. *[writes '$\iff$ K is bdd & closed']* Right?

This equivalence only works for $\mathbb{R}^n$, not general metric spaces; in general metric spaces this is not true. But in $\mathbb{R}^n$, this is the case, and we will see a proof of that.

What does it mean that K is bounded? You can give several equivalent definitions. One thing is that there exists some N large *[writes '$\exists N \in \mathbb{N}$ large']* — you can take it integer — such that the ball of radius N centered at zero *[writes '$B_N(0)$']* — this is a ball in $\mathbb{R}^n$, right? *[writes '$\subset \mathbb{R}^n$']* — contains my set. *[writes '$\supset K$']*

A bounded set you can put inside of a huge ball, right? On the contrary, a set is unbounded if it has a sequence of points... if this cannot exist, it means that it has a sequence of points that diverges to infinity in some direction, right?

The Heine–Borel Theorem and Boundedness in Euclidean Space

Satz 2.24 (*Heine–Borel Theorem in $\mathbb{R}^n$*): Let $K \subseteq \mathbb{R}^n$ be equipped with the standard Euclidean metric. Then:

$$K \text{ is compact} \iff K \text{ is closed and bounded.} \tag{2.61}$$

Definition 2.25 (*Bounded Subset of $\mathbb{R}^n$*): A subset $K \subseteq \mathbb{R}^n$ is *„bounded“* if there exists $N \in \mathbb{N}$ such that K is contained in the open ball of radius N centered at the origin:

$$K \subseteq B_N(0) = \{x \in \mathbb{R}^n \mid \|x\| < N\}. \tag{2.62}$$

Heine–Borel As a Specific Euclidean Property: The lecturer explicitly emphasizes that the equivalence between compactness and being *„closed and bounded“* is a special feature of finite-dimensional Euclidean space $\mathbb{R}^n$:

- In general metric spaces (X, d), compactness always implies closedness and boundedness (Corollary 2.19), but the converse fails dramatically (for example, in infinite-dimensional normed spaces or in discrete metric spaces with infinitely many points, the unit ball is closed and bounded but not compact).

- In $\mathbb{R}^n$, the local geometry allows us to reduce compactness to boundedness and closedness via the Bolzano–Weierstrass theorem.

Metric Characterization of Boundedness

Mündliches Protokoll [00:43:13 - 00:44:52]

Another way to put the same — you can prove that it's the same, and that is a bit more for any metric, whereas this is a bit more for $\mathbb{R}^n$ —: if you want to define an unbounded set in a metric space, you can define that the supremum of the distance between x and x' *[writes '$\sup_{x,x' \in K} d(x,x')$']* when x and x' belong to my set *[writes '$x, x' \in K$']* is finite. Sorry, if it is finite, it is bounded; if it is infinite, it is unbounded.

This applies to any metric space. So, how can you say if a subset is very large in a metric space? You take any pair of points in your set and see if the distance grows more and more and more: the set has a very, very long diameter, right? And then it's unbounded.

After the break, we will prove this theorem. *[writes 'Proof']*

So, proof...

Are you aware of the theorem of conservation of the syllabus, right? Are you aware of this theorem?

It's like the conservation of energy: the syllabus of the course is conserved, and what is not conserved is the pace at which I teach. So, if we lose time, I just will rush theorems, but I will teach the same, so...

Boundedness in General Metric Spaces

Definition 2.26 (*Diameter and Metric Boundedness*): Let (X,d) be a metric space. The *„diameter"* of a non-empty subset $K \subseteq X$ is defined by:

$$\operatorname{diam}(K) := \sup_{x,x' \in K} d(x,x') \in [0,\infty]. \tag{2.63}$$

A subset $K \subseteq X$ is called *„bounded"* if $\operatorname{diam}(K) < \infty$, and *„unbounded"* if $\operatorname{diam}(K) = \infty$.

Equivalence of Boundedness Definitions in $\mathbb{R}^n$: In $\mathbb{R}^n$, the ball-containment condition $K \subseteq B_N(0)$ and the finite-diameter condition $\operatorname{diam}(K) < \infty$ are equivalent:

- If $K \subseteq B_N(0)$, then for all $x, x' \in K$, the triangle inequality gives $d(x,x') \leq \|x\| + \|x'\| < 2N$, so $\operatorname{diam}(K) \leq 2N < \infty$.
- Conversely, if $\operatorname{diam}(K) = D < \infty$, fixing any point $x_0 \in K$ gives $K \subseteq B_{D+1}(x_0)$. By setting $N := D + 1 + \|x_0\|$, we obtain $K \subseteq B_N(0)$.

Einblick:* The *„Conservation of the Syllabus"

The lecturer playfully presents the *„conservation of the syllabus"* in analogy to the conservation of energy in physics: the material to be covered throughout the semester is

constant and invariant. Consequently, if classroom pacing slows down, the velocity of lecturing must accelerate to conserve the total curriculum.

Proof of Theorem 2.24

Necessity: Compact Subsets Are Closed and Bounded ($\Longrightarrow$)

Mündliches Protokoll [00:44:52 - 00:46:48]

Proof: we need to show... there is an easy implication.

If K in $\mathbb{R}^n$ is compact, I want to show that K is... well, that K is closed, I already showed, because this is a general property. So, by one of the corollaries, K is closed, *[writes ' $\Longrightarrow$ 1) K closed ✓']* because every compact set is closed, so this is okay.

And then, I want to show that K is bounded... *[writes '2) K bdd']* Bounded — this means bounded. *[rewrites 'bounded']* That K is bounded.

So, how do I show this? Okay, if K were unbounded... *[writes 'if K were unbounded']* there exists some sequence... *[writes '$\exists x_n$']*

Remember, the definition of bounded is this, right? *[points at $B_N(0) \supset K$]* So, if I say that K is not bounded, it means that there is a sequence of capital N's, if you want, *[writes '$\exists (x_N)_{N\in\mathbb{N}} \subset K$']* such that the modulus of x_N — this is contained in K — is bigger than N *[writes '$|x_N| > N$']* for all N, *[writes '$\forall N$']* right? I'm just using this definition of bounded.

Forward Direction: Closedness and Setup for Boundedness

We prove the forward implication ($\Longrightarrow$): assume $K \subseteq \mathbb{R}^n$ is compact.

1. K is closed: By Corollary 2.19, any compact subset of an arbitrary metric space is closed. Since $(\mathbb{R}^n, d_{\text{Euclidean}})$ is a metric space, K is closed.

2. K is bounded (Proof by Contradiction): Suppose that K were unbounded. By Definition 2.25, $K \not\subseteq B_N(0)$ for any $N \in \mathbb{N}$. Therefore, for each $N \in \mathbb{N}$, there exists an element $x_N \in K$ such that:

$$x_N \in K \quad \text{and} \quad \|x_N\| > N. \tag{2.64}$$

This constructs a sequence $(x_N)_{N\in\mathbb{N}} \subseteq K$ whose norms grow beyond all bounds.

Mündliches Protokoll [00:46:48 - 00:48:31]

So, if it is not bounded, I can have a divergent sequence. And now you know that any subsequence of that cannot converge. So, any subsequence will diverge. *[writes 'any subsequence will diverge']*

Indeed, *[writes 'Indeed']* if I had some x_{n_k} converging to x, *[writes '$x_{n_k} \to x$']* because the modulus... or by the triangle inequality, right? I know that this is equivalent to $x_{n_k} - x$ converging to zero *[writes '$|x_{n_k} - x| \to 0$']* as k goes to infinity. *[writes 'as $k \to \infty$']* But then I would have, by the triangle inequality, that $|x_{n_k}|$ *[writes '$|x_{n_k}| \leq$']* is less

than or equal to this plus this: *[writes '$|x_{n_k} - x| + |x|$']* this is the triangle inequality. Triangle, right? *[writes 'triangle']*

And this is going to zero, and this is a bounded number, so this contradicts that, right? *[points at $|x_N| > N$]*

So, any compact set must be bounded in $\mathbb{R}^n$. *[writes 'This shows $\Longrightarrow$']*

Forward Direction: Deducing Boundedness via Divergence of Subsequences

Deriving the Contradiction: Since K is compact, it is sequentially compact (Theorem 2.17). Therefore, the sequence $(x_N)_{N\in\mathbb{N}} \subseteq K$ must possess a convergent subsequence $(x_{N_k})_{k\in\mathbb{N}}$ converging to some limit point $x \in K$:

$$x_{N_k} \xrightarrow{k\to\infty} x \in K \iff \|x_{N_k} - x\| \xrightarrow{k\to\infty} 0.$$

By the reverse triangle inequality in $\mathbb{R}^n$:

$$\|x_{N_k}\| \leq \|x_{N_k} - x\| + \|x\|. \tag{2.65}$$

As $k \to \infty$, the term $\|x_{N_k} - x\|$ converges to 0, and $\|x\|$ is a fixed constant. Thus, the sequence of norms $(\|x_{N_k}\|)_{k\in\mathbb{N}}$ must be bounded:

$$\limsup_{k\to\infty} \|x_{N_k}\| \leq \|x\| < \infty.$$

However, by construction in (2.64), we have:

$$\|x_{N_k}\| > N_k \geq k \xrightarrow{k\to\infty} \infty,$$

which implies $\|x_{N_k}\| \to \infty$, an immediate contradiction!

Hence, our assumption was false: K cannot be unbounded. Every compact subset $K \subseteq \mathbb{R}^n$ is necessarily bounded and closed. This completes the forward direction ($\Longrightarrow$).

Sufficiency: Closed and Bounded Subsets Are Compact ($\Longleftarrow$)

Mündliches Protokoll [00:48:31 - 00:49:26]

And now we need to prove the hard work, that is the other implication. *[shifts blackboards]*

But... no. Now I'm looking at my notes because I always have the same problem: I remember the theorems, but I never remember the name of the theorem.

And how is this... now you can help me: what is the name of this theorem in $\mathbb{R}$, that when you have a sequence in a closed, bounded interval, there exists a convergent subsequence?

Student Answer

Bolzano-Weierstrass.

Mündliches Protokoll [00:49:28 - 00:50:04]

Ah, this! Thank you. Bolzano–Weierstrass.

Thank you. She said: Bolzano–Weierstrass. Thank you, this was what I was missing.

So, to prove Heine–Borel for $\mathbb{R}^n$, our proof will be based on Bolzano–Weierstrass for $\mathbb{R}$, right? We'll see how to use this property that you proved in Analysis 1 on each of the components of my sequence to extract a convergent subsequence.

Board Transition

The lecturer thoroughly erases the lower blackboard panel in silence to prepare for the proof of the reverse implication ($\Longleftarrow$) of the Heine–Borel theorem.

Reduction to Compactness of Closed Cubes

Mündliches Protokoll [00:50:58 - 00:52:33]

Heine–Borel, the difficult implication: that one. *[writes '$\Longleftarrow$']* Right?

Our goal will be to show that, given N in $\mathbb{N}$... *[writes 'given $N \in \mathbb{N}$']* — let's say, it's not important, given any number — the cube $[-N, N]^n$, *[writes '$[-N, N]^n \subset \mathbb{R}^n$']* the cube in $\mathbb{R}^n$, is compact, right? *[writes 'is cpt']*

Why is this sufficient? This is sufficient *[writes 'This is sufficient']* because then, if $K \subset \mathbb{R}^n$ *[writes 'if $K \subset \mathbb{R}^n$']* is closed and bounded... *[writes 'closed & bdd']* Well, because it's bounded, we'll have that K is contained in the ball of radius N centered at zero, *[writes '$K \subset B_N(0)$']* and the ball of radius N is always contained in the cube, right? *[writes '$\subset [-N, N]^n$']*

Mündliches Protokoll [00:52:33 - 00:54:20]

So, if you want, this is a simple exercise — a nice property of the Euclidean distance. *[writes 'Exercise: Euclidean distance']*

In the plane, you know this very well, but you can show this as an exercise in any dimension. So, you have the ball of radius N is like that, right? *[draws a circle on the board]* And then the cube — this cube of diameter $2N$ — is bigger, contains the ball. *[draws a square around the circle]* It's tangent, actually, here, right? So, the ball is contained in the cube.

In any case, the important thing is that we'll have a closed subset of a compact set. So, if we prove that this is compact, K will be a closed subset of a compact set, and then, by one of the corollaries of sequential compactness, this will also be compact. Since a closed subset of a compact set is compact, *[writes '(closed subset of cpt is cpt)']* this will imply that K is compact. *[writes '$\Longrightarrow$ K is cpt']*

So, we only need to show that the closed cubes like that — these guys, *[points at $[-N, N]^n$]* for any N given — are compact sets. This is our goal, and this is what I will show now.

Sufficiency Strategy: Reducing to the Compactness of Cubes

We prove the reverse implication ($\Longleftarrow$): assume $K \subseteq \mathbb{R}^n$ is closed and bounded.

Lemma 2.27 (*Compactness of Closed Cubes in $\mathbb{R}^n$*): For every $N \in \mathbb{N}$, the closed cube:

$$Q_N := [-N, N]^n = \underbrace{[-N, N] \times [-N, N] \times \cdots \times [-N, N]}_{n \text{ times}} \subset \mathbb{R}^n \tag{2.66}$$

is compact in $(\mathbb{R}^n, d_{\text{Euclidean}})$.

Proof that Lemma 2.27 Implies the Reverse Implication:

(1) Since K is bounded, by Definition 2.25 there exists $N \in \mathbb{N}$ such that $K \subseteq B_N(0)$.

(2) Any point $x = (x_1, \ldots, x_n) \in B_N(0)$ satisfies:

$$\|x\|^2 = \sum_{j=1}^{n} x_j^2 < N^2 \implies |x_j| \leq \sqrt{\sum_{j=1}^{n} x_j^2} < N \quad \text{for all } j \in \{1, \ldots, n\}.$$

Thus, $x_j \in [-N, N]$ for all $j \in \{1, \ldots, n\}$, which gives the inclusion:

$$B_N(0) \subseteq [-N, N]^n. \tag{2.67}$$

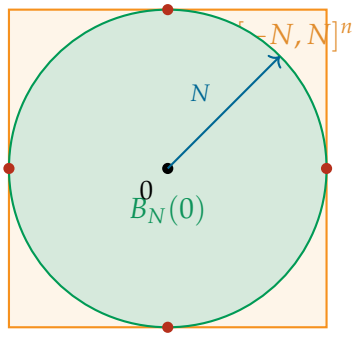
Consequently, $K \subseteq B_N(0) \subseteq Q_N$.

(3) Therefore, K can be written as:

$$K = K \cap Q_N.$$

Since K is closed in $\mathbb{R}^n$ by hypothesis, and Q_N is compact by Lemma 2.27, Corollary 2.21 implies directly that $K = K \cap Q_N$ is compact!

Übung 2.28 (*Inclusion of Euclidean Ball in Cube*): Show that for all $N > 0$ and any dimension $n \geq 1$, the Euclidean ball $B_N(0) \subset \mathbb{R}^n$ is contained in the closed cube $[-N, N]^n$.



Componentwise Extraction via the Bolzano–Weierstrass Theorem

Mündliches Protokoll [00:54:20 - 00:56:36]

And to prove this, the proof reduces to Bolzano–Weierstrass in $\mathbb{R}$. *[writes 'Proof reduces to Bolzano-Weierstrass in $\mathbb{R}$']*

The key is: how do we reduce from this theorem in $\mathbb{R}^n$ to something in $\mathbb{R}$? The idea is to pass to the components of my sequence.

Given a sequence x_k in the cube, *[writes '$(x_k) \subset [-N, N]^n$']* recall that for such a

sequence we have the sequence of the components: we can write $x_k = (x_k^{(1)}, \dots, x_k^{(n)})$, *[writes '$x_k = (x_k^{(1)}, \dots, x_k^{(n)})$']* the n components of my vector.

And each of these is a sequence in $[-N, N]$. So, when I look at one of the components, it is a sequence in a closed and bounded interval of $\mathbb{R}$. So, I can apply Bolzano–Weierstrass to each of them, right?

To make one of the components converge is always easy — well, you need to apply Bolzano–Weierstrass, but you can make one of the components converge. And now the trick is that I will extract a subsequence, and then a subsequence of the subsequence, and then a subsequence of the subsequence of the subsequence, and so on, finitely many times, until I get all the n components to converge at the same time, okay?

Let's start with the first. First, I look at the first component.

Proof of Lemma 2.27: Component Decomposition

Proof of Lemma 2.27 (Compactness of Closed Cubes):

To prove that $Q_N = [-N, N]^n$ is compact, by Theorem 2.17 it suffices to show that Q_N is **sequentially compact**.

Let $(x_k)_{k\geq 0} \subseteq Q_N$ be an arbitrary sequence of points in the cube. Each point $x_k \in \mathbb{R}^n$ decomposes into its n coordinate components:

$$x_k = \left(x_k^{(1)}, x_k^{(2)}, \dots, x_k^{(n)}\right) \in \mathbb{R}^n. \tag{2.68}$$

Since $x_k \in Q_N = [-N, N]^n$, each individual coordinate sequence $(x_k^{(j)})_{k\geq 0}$ is a sequence of real numbers bounded in the compact interval $[-N, N] \subset \mathbb{R}$:

$$x_k^{(j)} \in [-N, N] \quad \text{for all } k \geq 0 \text{ and all } j \in \{1, \dots, n\}.$$

Mündliches Protokoll [00:56:36 - 00:59:08]

1: *[writes '1. Look at sequence']* look at the sequence $(x_k^{(1)})_{k\geq 0}$, *[writes '$(x_k^{(1)})_{k\geq 0}$']* the first component, $k \geq 0$. This is a sequence in $[-N, N]$, *[writes '$\subset [-N, N]$']* a sequence of real numbers. By Bolzano–Weierstrass, *[writes 'by BW']* it has a convergent subsequence.

Let me put it like this, because there will be many subsequences, so I will need a more or less efficient notation for a subsequence of a subsequence. I will take some $k_1(m)$ *[writes '$(x_{k_1(m)}^{(1)})_{m\geq 0}$']* — this will be some increasing sequence in $\mathbb{N}$, *[writes 'increasing sequence in $\mathbb{N}$']* a sequence of indices, and I call it k_1 because there will be more, right? — such that the subsequence $x_{k_1(m)}^{(1)}$ converges to some limit that I call L_1. *[writes '$\to L_1 \in [-N, N]$']*

This is an accumulation point of the first component, right?

So, I'm applying Bolzano–Weierstrass to that sequence of numbers, and then I am inferring the existence of a subsequence. This is how I call the indices of this subsequence, right? There is a sequence of increasing indices, and the first component converges to some number L_1, right?

Step 1: Convergent Subsequence for the First Coordinate Component

Step 1 (First Component): Consider the real-valued sequence of first coordinates:

$$(x_k^{(1)})_{k\geq 0} \subseteq [-N, N] \subset \mathbb{R}.$$

Since $[-N, N]$ is a closed bounded interval in $\mathbb{R}$, by the 1-dimensional **Bolzano–Weierstrass theorem**, there exists a strictly increasing sequence of indices:

$$k_1 \colon \mathbb{N} \to \mathbb{N}, \quad m \mapsto k_1(m) \quad \text{with } k_1(0) < k_1(1) < k_1(2) < \dots$$

such that the corresponding subsequence of first components converges to some limit $L_1 \in [-N, N]$:

$$\lim_{m\to\infty} x^{(1)}_{k_1(m)} = L_1 \in [-N, N]. \tag{2.69}$$

Mündliches Protokoll [00:59:08 - 01:01:08]

Second step. *[writes '2.']*

Now I look at the sequence of second components along this subsequence: *[writes 'Look at']* $(x^{(2)}_{k_1(m)})_{m\geq 0}$. *[writes '$(x^{(2)}_{k_1(m)})_{m\geq 0}$']* This is a different sequence, because here I had the first component. This is associated to the same subsequence of indices, right? I just look at what happens in the second component. And maybe this one is not converging — because I can guarantee that the first component is converging, but the second maybe not.

But then, since this is contained in $[-N, N]$, *[writes '$\subset [-N, N]$']* again, this is a compact set in $\mathbb{R}$, so by Bolzano–Weierstrass there exists a convergent subsequence.

This implies, again using Bolzano–Weierstrass a second time, *[writes '$\implies$ by BW']* that there exists a second sequence of indices that I will call $k_2(m)$, *[writes '$\exists(k_2(m))_{m\geq 0}$']* that is a subsequence of $k_1(m)$, *[writes 'subsequence of $k_1(m)$']* such that $x^{(2)}_{k_2(m)}$ — the second component now — converges to some number L_2; *[writes '$x^{(2)}_{k_2(m)}$']* I call L_2 the limit. *[writes '$\to L_2$']* And this number belongs to $[-N, N]$. *[writes '$\in [-N, N]$']*

Mündliches Protokoll [01:01:08 - 01:02:36]

Now, since this is a subsequence of the previous one, the first component, when I look along this subsequence, also converges to L_1. So, also $x^{(1)}_{k_2(m)}$ converges to L_1 as $m \to \infty$ — I emphasize. *[writes 'also $x^{(1)}_{k_2(m)} \to L_1$']*

Remember this lemma that I said was a bit silly, but that confused me and I gave a wrong statement last Wednesday? This is the correct one, so I'm using the correct one: when you have any sequence that converges to a limit, and you take a subsequence, it will still converge to the same limit. I'm just using that.

And then you continue! You will do this process as many times as components you have. Every time, you extract a subsequence of the previous subsequence, and every time, you make a new component converge. And since you have finitely many components, eventually all of the components converge.

Step 2: Subsequence of the Subsequence for Simultaneous Convergence

Step 2 (Second Component): Now consider the sequence of second coordinates restricted to the index subsequence $k_1(m)$:

$$\left(x^{(2)}_{k_1(m)}\right)_{m\geq 0} \subseteq [-N, N].$$

While this sequence is not guaranteed to converge on its own, it is a bounded sequence in $[-N, N] \subset \mathbb{R}$. Applying the Bolzano–Weierstrass theorem a second time, there exists a strictly increasing subsequence of indices:

$$k_2(m) = k_1(\ell(m)) \quad \text{such that } (k_2(m))_{m\geq 0} \text{ is a subsequence of } (k_1(m))_{m\geq 0},$$

for which the second component converges to some limit $L_2 \in [-N, N]$:

$$\lim_{m\to\infty} x^{(2)}_{k_2(m)} = L_2 \in [-N, N]. \tag{2.70}$$

Simultaneous Convergence of the First Two Components: Crucially, because $(k_2(m))_{m\geq 0}$ is a subsequence of $(k_1(m))_{m\geq 0}$, and every subsequence of a convergent sequence converges to the same limit, the first component also converges along $(k_2(m))$ to L_1:

$$\lim_{m\to\infty} x^{(1)}_{k_2(m)} = L_1 \in [-N, N].$$

Thus, along the single subsequence of indices $(k_2(m))_{m\geq 0}$, **both** the first and second components converge simultaneously to $(L_1, L_2) \in [-N, N]^2$.

Board Transition

The lecturer erases the bottom-left panel of the blackboard to make room for writing the general inductive step (n components) and assembling the limit vector.

Mündliches Protokoll [01:03:34 - 01:05:21]

Continuing this process... *[writes 'Continuing this process']*

We need to repeat this process n times, because we have n components, right? Until we get a last subsequence, $(k_n(m))_{m\geq 0}$, *[writes '$(k_n(m))_{m\geq 0}$']* $m \geq 0$, right?

And then, for this subsequence, we will have $x^{(n)}_{k_n(m)}$... *[writes '$x^{(n)}_{k_n(m)}$']* as m goes to infinity, this will converge to some number L_n *[writes '$\xrightarrow{m\to\infty} L_n$']* that is in $[-N, N]$, right? *[writes '$\in [-N, N]$']*

And also, because it's a subsequence of the subsequence of the subsequence of all the previous ones, $x^{(j)}_{k_n(m)}$ — any other component j — also converges to L_j *[writes '$x^{(j)}_{k_n(m)} \to L_j$']* from previous steps, right? *[writes 'from previous steps']*

Inductive Step for All n Coordinate Components

By iterating this extraction process n times, we obtain a nested chain of index subsequences:

$$(k_n(m))_{m\geq 0} \subseteq (k_{n-1}(m))_{m\geq 0} \subseteq \cdots \subseteq (k_1(m))_{m\geq 0} \subseteq \mathbb{N},$$

such that for the final coordinate n:

$$\lim_{m\to\infty} x^{(n)}_{k_n(m)} = L_n \in [-N, N]. \tag{2.71}$$

Furthermore, since $(k_n(m))_{m\geq 0}$ is a subsequence of $(k_j(m))_{m\geq 0}$ for every coordinate index $j \in \{1, \ldots, n-1\}$, every preceding coordinate also converges along this final subsequence to its established limit:

$$\lim_{m\to\infty} x^{(j)}_{k_n(m)} = L_j \in [-N, N] \quad \text{for all } j \in \{1, \ldots, n\}. \tag{2.72}$$

Mündliches Protokoll [01:05:21 - 01:07:14]

And now, these $L_1, \ldots, L_n$ are n different numbers. I can concatenate these numbers and define the point x: *[writes '$x = (L_1, \ldots, L_n) \in [-N, N]^n$']* let me define the point $x = (L_1, \ldots, L_n)$. This is a point that is in this cube, right?

And now I know that this sequence... we have that $x_{k_n(m)} \xrightarrow{m\to\infty} x$. *[writes 'we have that $x_{k_n(m)} \xrightarrow{m\to\infty} x$']*

Why? Because we also gave on one of the first days a lemma that says that a sequence converges in $\mathbb{R}^n$ if and only if each component converges. So, the sequence of vectors converges if and only if this happens.

And this is our convergent subsequence we were looking for. So, we are happy, and then: end of the proof. *[writes 'this is the convergent subsequence we were looking for $\square$']* Okay?

Assembling the Limit Point and Completing the Proof

Constructing the Limit Vector: We define the concatenated vector of coordinate limits:

$$x := (L_1, L_2, \ldots, L_n) \in [-N, N]^n. \tag{2.73}$$

A fundamental property of Euclidean space states that a sequence of vectors converges in $\mathbb{R}^n$ with respect to the Euclidean metric if and only if each coordinate component converges in $\mathbb{R}$:

$$\lim_{m\to\infty} x_{k_n(m)} = \lim_{m\to\infty} \left(x^{(1)}_{k_n(m)}, \ldots, x^{(n)}_{k_n(m)}\right) = (L_1, \ldots, L_n) = x \in [-N, N]^n.$$

Thus, $(x_{k_n(m)})_{m\geq 0}$ is a subsequence of $(x_k)_{k\geq 0}$ that converges to a limit point $x \in [-N, N]^n$. This proves that the closed cube $[-N, N]^n$ is sequentially compact, and hence compact by Theorem 2.17.

Conclusion of the Heine–Borel Theorem (Theorem 2.24): Since every closed and bounded subset $K \subseteq \mathbb{R}^n$ satisfies $K \subseteq B_N(0) \subseteq [-N, N]^n$ for sufficiently large $N \in \mathbb{N}$, we can write:

$$K = K \cap [-N, N]^n.$$

Because K is closed in $\mathbb{R}^n$ and $[-N, N]^n$ is compact, Corollary 2.21 implies that K is compact. This completes the proof of the Heine–Borel theorem in $\mathbb{R}^n$.

Q.E.D.

Counterexample: Non-Compactness of the Half-Open Interval $(0,1]$

Mündliches Protokoll [01:07:31 - 01:09:14]

Okay, so we still have some minutes. I will give a couple of examples of sets that are compact and not compact, because many people asked me the same question, so I see that there is a common difficulty here.

Many people asked: why, for instance, if you take the metric space $X = \mathbb{R}$ — this is the easiest you know —, and I take the set K equal, let's say, $(0,1]$, some open interval, or even that one... *[writes 'X = ℝ' and 'K = (0,1]']* why is this not compact, right? *[writes 'why is this NOT cpt']* Why is this not compact?

And then you can try to contradict the compactness of sequences, or you can try to contradict topological compactness. Here it's easy, for instance, to contradict... so, why is this not compact?

Well, one way to see it is that you can write... let me call this A, because this will not be compact. *[changes K to A]* So, this set A you can write as a union for $n \geq 1$ *[writes 'A = ⋃n≥1 [1/n, 1]']* of $[1/n, 1]$ closed, right?

You have your interval here, *[draws '( ]' on the board with 0 and 1]* and then you can write this interval as a union of this... well, it's not open, it's semi-open, right? Here it's closed, here it's open. And the problem to be compact is, of course, this open end here, right?

Mündliches Protokoll [01:09:14 - 01:11:26]

And you see it here: because it's open, you can write it as a union of intervals that are smaller, right?

Or you can even take this: you can take A contained in... *[writes 'A ⊂ ⋃n≥1 (1/n, 2)']* These are open sets, right? These are open intervals. So, you can make A contained in the union of these intervals, yet for any finite number of them, this will never cover my set. *[writes 'any finite number of them will miss a chunk']*

Any finite number of them will miss a chunk, right? If you take a finite number, there will be a last one, that is the $1/n$ that is the last you took, and this chunk over here is covered by no one, right? So, you always need infinitely many to cover.

And then someone asked also: okay, but what if I take the cover $(-1,2)$, right? *[writes '(−1, 2)']* This is an open set that is covering, right?

But then I answered: okay, but you don't get to choose the cover. Someone else chooses the cover for you, and then you need to be able to produce the finite subcover, right?

Of course, any set you can cover by the whole space. The whole space is always open. So, if you take the whole X, it is always open, so any set you can cover by the whole space. This is not interesting, that there exists some open cover. Every set can be covered by some other finitely many sets.

Non-Compactness of the Half-Open Interval $(0,1]$

Beispiel 2.29 (*Non-Compactness of $(0,1]$ in $\mathbb{R}$*): Consider the metric space $(\mathbb{R}, d_{\text{Euclidean}})$ and the half-open interval:

$$A = (0,1] \subset \mathbb{R}.$$

Although A is bounded, it is not closed in $\mathbb{R}$ because the accumulation point 0 does not belong to A. By the Heine–Borel theorem (Theorem 2.24), A cannot be compact.

Direct Topological Disproof via Open Covers: To demonstrate non-compactness directly from Definition 2.16, we construct an open cover of A that admits no finite subcover:

(1) We express A as an infinite union of closed subintervals:

$$A = (0,1] = \bigcup_{n=1}^{\infty} \left[\frac{1}{n}, 1\right].$$

(2) We expand each closed interval into an open interval in $\mathbb{R}$ to form the family:

$$\mathcal{U} := \{U_n \mid n \in \mathbb{N}_{\geq 1}\} \quad \text{where} \quad U_n := \left(\frac{1}{n}, 2\right). \tag{2.74}$$

For every $x \in (0,1]$, since $x > 0$, by Archimedes' principle there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < x$. Hence $x \in U_n$, which shows that $\mathcal{U}$ is an open cover of A:

$$A \subseteq \bigcup_{n=1}^{\infty} \left(\frac{1}{n}, 2\right).$$

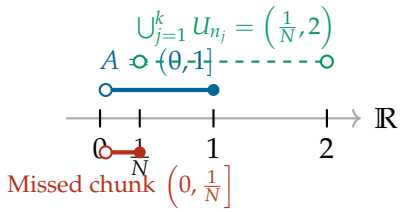
(3) Failure of Finite Subcovers: Let $\{U_{n_1}, \ldots, U_{n_k}\} \subseteq \mathcal{U}$ be an arbitrary finite subcollection. Setting $N := \max\{n_1, \ldots, n_k\}$, the union simplifies to:

$$\bigcup_{j=1}^{k} U_{n_j} = \bigcup_{j=1}^{k} \left(\frac{1}{n_j}, 2\right) = \left(\frac{1}{N}, 2\right).$$

This finite union misses the entire remaining „*chunk*“ $\left(0, \frac{1}{N}\right] \subseteq A$:

$$\left(0, \frac{1}{N}\right] \cap \left(\frac{1}{N}, 2\right) = \emptyset \implies A \not\subseteq \bigcup_{j=1}^{k} U_{n_j}.$$

Thus, no finite subcollection of $\mathcal{U}$ can cover A, showing that $A = (0,1]$ is not compact.



The Student Misconception on Trivial Covers: The lecturer highlights a fundamental logical fallacy that frequently confuses students:

- A student might observe that the single open interval $V = (-1,2)$ covers $A = (0,1]$, and argue that $\{V\}$ is a finite subcover.

- **Why this argument fails:** Compactness demands that **every** open cover must possess a finite subcover ($\forall \mathcal{U}\ \exists \mathcal{U}_{\text{fin}}$). Showing that **one particular** cover can be made finite is mathematically vacuous (indeed, the entire space $X = \mathbb{R}$ covers any set in a single step). To disprove compactness, one only needs to exhibit a **single open cover** $\mathcal{U}$ that admits no finite subcover.

Mündliches Protokoll [01:11:26 - 01:12:47]

Speaking about open and closed, is there any question about compactness? We will see this more and more during the course, especially in the case of $\mathbb{R}^n$. So, you will get to know the notion of compactness much better and interiorize it, but now we needed some starting framework to start to use it.

And then the next thing in the list of topological notions that we want to study — that you also learned a little bit in $\mathbb{R}$, but is much more interesting in a general metric space — is connectedness. This starts to be new, and it's very nice. You will see that this is much more interesting. And new. Connectedness.

Connectedness

Preliminary: „*Clopen*“ Sets

Mündliches Protokoll [01:12:47 - 01:14:49]

Next topic: (X, d) a metric space, and we will define connectedness, right? But before defining connectedness, I need a preliminary.

A preliminary observation *[writes 'preliminary']* is that in any metric space, *[writes 'In any metric space']* the whole set and the empty set *[writes ' $X, \emptyset$ are']* are always open and closed at the same time. *[writes 'always open & closed']*

Here, some authors call these sets „*clopen*“ because they are closed and open: clopen. *[writes '[clopen]']*

In a given space, we will see that it's a rare property that in your space you have sets that are at the same time open and closed. And I'm saying this now because, first, some people asked during these days: is the whole space always open, or is the empty set closed or open? Okay.

Mündliches Protokoll [01:14:49 - 01:16:27]

So, let me say why this is true. First, what was the definition of an open set?

The definition of an open set is that for every point in my set, I can find a ball of positive radius, right? If my set is the whole space, the ball of radius 1 for any point is in my space, tautologically. So, the whole set is always open.

Now, that the empty set is open is more arguable, if you want, right? Some books take it as an axiom that the empty set is open, but actually, if you use the logical conse-

quence of the definition I gave, the empty set is open.

Because the definition I gave was: $U \subset X$ is open *[writes '$U \subset X$ open if']* if for every x in U, *[writes '$\forall x \in U$']* something happens, right? And whenever you have an empty set, for every x in U, anything you want happens, right? So, it is open, right?

Strictly speaking, it is covered by our definition. We don't need to impose any extra axiom. But if you want, just believe as an axiom that the empty set is open. *[boxes the open definition on the board]*

Preliminary: Clopen Sets

In any metric space (X, d), the empty set $\emptyset$ and the whole space X are always both open and closed (often called *„clopen"* sets).

- **The whole space X is open:** For any point $x \in X$, any open ball of radius $r = 1$ is trivially contained within the space:
$$\forall x \in X \; \exists r > 0: \quad B_r(x) \subseteq X.$$
Since X is open, its complement $X^c = X \setminus X = \emptyset$ is closed.
- **The empty set $\emptyset$ is open:** The definition of openness holds vacuously for $\emptyset$:
$$\forall x \in \emptyset \; \exists r > 0: \quad B_r(x) \subseteq \emptyset.$$
Since $\emptyset$ contains no elements, the universal condition over its elements is vacuously true. Since $\emptyset$ is open, its complement $\emptyset^c = X \setminus \emptyset = X$ is closed.

Vacuous Truth in Topology: As the lecturer emphasizes, students often wonder whether $\emptyset$ being open is an arbitrary ad-hoc axiom. In formal first-order logic, the statement $\forall x \in \emptyset: P(x)$ is formally expressed as $\forall x \, (x \in \emptyset \implies P(x))$. Because the premise $x \in \emptyset$ is identically false for all x, the implication is true. Therefore, the openness of the empty set is a direct deductive consequence of the definition of an open set, requiring no extra axioms.

The Partition Problem on the Sphere S^2

Mündliches Protokoll [01:16:27 - 01:17:54]

And now, let me give a question for you. *[writes 'Question:']*

Let's take X to be the two-dimensional sphere, so call it S^2: *[writes '$X = S^2 =$']* it's just the set of points (x, y, z) in $\mathbb{R}^3$ such that $x^2 + y^2 + z^2 = 1$. *[writes '$\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 + z^2 = 1\}$']*

This is the unit sphere. And we take it with the restriction of the Euclidean distance: *[writes '$d_{Eucl}|_X =: X$']* d Euclidean restricted to my set X, right? We have already seen this example on one of the first days.

This is the drawing: a sphere. *[draws a circle with equator line]* I'm just putting this to make an interesting example that by now shouldn't be obvious, and in the next lecture the answer to this question I'm going to make will be obvious.

Mündliches Protokoll [01:17:54 - 01:19:58]

So, we have the sphere, and now I ask you: can you cover this sphere... in this set there are open and closed subsets, right? *[writes 'open subsets']*

Here you have open subsets. And my question is: is it possible to write *[writes 'Is it possible to write']* the sphere — the whole space X, that is my sphere *[writes '$X =$']* — as a disjoint union, $U \cup V$, *[writes '$U \cup V$']* meaning that $U \cap V$ is the empty set, *[writes 'disjoint'] [writes '$(U \cap V = \emptyset)$']* and both are open and non-empty? *[writes 'open, open, non-empty']*

Right?

Can you decompose your sphere as a union of two disjoint open sets? It doesn't look like it, but it's not clear a priori. So, you can intuitively say: no, this will be impossible. But to write a proof of this fact — that for any pair of open sets I give, they will either have some intersection or maybe they don't cover, right? — this looks complicated.

So, we will need more theory to find answers to this type of question. And you will see that this type of question we will use many times in Analysis 2 to prove that functions with zero derivative are constant in a connected set, and this kind of thing.

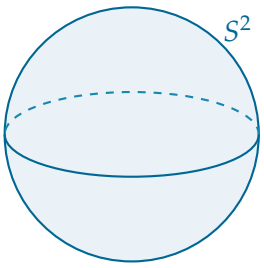
The Sphere Partition Question

Frage 2.30 (*Partitioning the 2-Sphere*): Let $X = S^2 \subset \mathbb{R}^3$ be the two-dimensional unit sphere equipped with the subspace Euclidean metric:

$$S^2 := \left\{(x,y,z) \in \mathbb{R}^3 \mid x^2+y^2+z^2 = 1\right\}, \qquad d_{S^2} := d_{\text{Euclidean}}|_{S^2 \times S^2}.$$

Is it possible to partition S^2 into a disjoint union of two non-empty, open subsets $U, V \subset S^2$? That is:

$$S^2 = U \cup V \quad \text{with} \quad U \cap V = \emptyset, \quad U \neq \emptyset,\ V \neq \emptyset, \quad U, V \text{ open in } S^2. \tag{2.75}$$



Topological Intuition Behind the Question: Intuitively, the sphere S^2 is an unbroken geometric surface consisting of a single continuous piece. Decomposing it into two disjoint open pieces without leaving any boundary seam between them seems impossible because open sets cannot contain their boundary points. This fundamental geometric property of being *„in one piece"* is what the mathematical concept of *„connectedness"* rigorously captures.

The Partition Problem for Intervals in $\mathbb{R}$

Mündliches Protokoll [01:19:58 - 01:21:49]

Okay, so maybe now I can define...

I have put this question on the sphere to make it a bit more difficult to think about, right? And I wanted to give this example to say: something that now is difficult to think of, in a lecture, when you have more theory, you will see the answer clearly; you will have a one-line answer to this.

But you can pose the same question when X is some interval in $\mathbb{R}$: *[writes 'Same question when $X = (0,1) \subset \mathbb{R}$']* when X is, let's say, the interval $(0,1)$ in $\mathbb{R}$.

Can you decompose the interval $(0,1)$ in $\mathbb{R}$ as the disjoint union of two non-empty open sets, right?

If one of the sets is closed and the other open, yes, because you can say: okay, $(0,1)$ I can put as the union of $(0,1/2]$ — close this one — union $(1/2,1)$, right? *[writes '$(0,1) = (0,1/2] \cup (1/2,1)$']* The $1/2$ you can put in this one. But this will not be open.

So, the question is: can we decompose a set into a disjoint union of open sets? And this question motivates the definition of connectedness.

The Partition Problem on Real Intervals

Frage 2.31 (*Partitioning $(0,1)$ into Open Sets*): Can the open interval $(0,1) \subset \mathbb{R}$ be decomposed into two disjoint, non-empty open subsets $U, V \subset (0,1)$?

Analysis of a Candidate Partition: If we drop the requirement that both sets must be open, a disjoint partition is elementary:

$$(0,1) = \left(0, \frac{1}{2}\right] \cup \left(\frac{1}{2}, 1\right).$$

However, while $\left(\frac{1}{2},1\right)$ is open in $(0,1)$, the subset $\left(0,\frac{1}{2}\right]$ is not open in $(0,1)$ because any open ball in $\mathbb{R}$ centered at $x = \frac{1}{2}$ contains points strictly greater than $\frac{1}{2}$. Thus, this does not yield a partition into two open sets.

Board Transition

The lecturer takes a wet sponge and erases the bottom-right blackboard panel from 00:19:40 to 00:20:06 to prepare for writing the formal definitions of disconnected and connected sets.

Formal Definition of Connectedness

Mündliches Protokoll [01:22:15 - 01:24:03]

...is called connected... *[writes 'Def: Given (X,d) metric space, $A \subset X$ (non-empty)']* Sorry, first I will define connected by saying what is disconnected, so what is the contrary. So, this definition will be by opposition to the concept of disconnected.

So, I will define first disconnected: *[writes 'disconnected if']* a subset is disconnected if there exist U and V open, *[writes '$\exists U, V$ open']* disjoint, *[writes 'disjoint']* and non-empty, *[writes 'and non-empty']* such that A is contained in $U \cup V$, *[writes 'such that $A \subset U \cup V$']* and both $A \cap U$ is non-empty *[writes 'and $A \cap U \neq \emptyset$']* and $A \cap V$ is non-empty. *[writes 'and $A \cap V \neq \emptyset$']*

This is the definition of a disconnected set, right?

Definition of Disconnected Subsets

Let (X, d) be a metric space.

Definition 2.32 (*Disconnected Set*): A non-empty subset $A \subseteq X$ is called *„disconnected“* if there exist open subsets $U, V \subseteq X$ such that:

- **(i)** $U \cap V = \emptyset$ (i.e., U and V are disjoint),
- **(ii)** $A \subseteq U \cup V$ (i.e., U and V cover A),
- **(iii)** $A \cap U \neq \emptyset$ and $A \cap V \neq \emptyset$ (i.e., both open sets intersect A).

The pair (U, V) is called a *„separation“* (or disconnection) of A.

Examples of Disconnected Sets

Mündliches Protokoll [01:24:03 - 01:24:33]

And now let's go to the space you control very, very well, that is $\mathbb{R}$, and give me some example of a disconnected set.

What is the easiest example you can think of of a disconnected set?

Student Answer

Open interval from 0 to 1 union open interval from 0 to—from 2 to 3?

Mündliches Protokoll [01:24:37 - 01:25:29]

Yes! So, this is an example. *[writes 'In $\mathbb{R}$, set $A = (0,1) \cup (2,3)$']*

So, this is an open disconnected set: $(0,1) \cup (2,3)$. By construction, this set is the disjoint union of two open sets, right? Non-empty. Then, it is disconnected.

I can also say: if I take the set $\{0\}$ and $\{1\}$, *[writes '$A = \{0,1\}$']* two points, this is always a disconnected set, because I have the set $\{0\}$ and the set $\{1\}$, and I can take two open sets like this that disconnect the set, right? *[draws intervals disconnecting $\{0\}$ and $\{1\}$]*

Examples of Disconnected Sets in $\mathbb{R}$

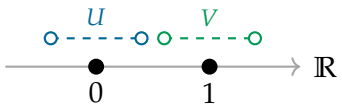
Beispiel 2.33 (*Disconnected Subsets of* $\mathbb{R}$):

(1) Disjoint Open Intervals: Let $A := (0,1) \cup (2,3) \subset \mathbb{R}$. Choosing $U := (0,1)$ and $V := (2,3)$, both U and V are open in $\mathbb{R}$, $U \cap V = \emptyset$, and:

$$A \subseteq U \cup V, \qquad A \cap U = (0,1) \neq \emptyset, \qquad A \cap V = (2,3) \neq \emptyset.$$

Thus, A is disconnected.

(2) A Discrete Two-Point Set: Let $A := \{0,1\} \subset \mathbb{R}$. Choosing open neighborhoods $U := \left(-\frac{1}{2}, \frac{1}{2}\right)$ and $V := \left(\frac{1}{2}, \frac{3}{2}\right)$, we have $U \cap V = \emptyset$, $0 \in U$, $1 \in V$, and $A \subset U \cup V$. Hence, $\{0,1\}$ is disconnected.



Definition of Connected Sets

Mündliches Protokoll [01:25:29 - 01:26:52]

And then I will define a connected set as any set that is not disconnected.

We say A is connected if it is NOT disconnected. *[writes 'We say A is connected if it is NOT disconnected.']*

So, this means that whenever I have two open sets that cover it and are disjoint, one of them will be empty, right?

So, a set is connected if, whenever I can cover it by a union of open sets that are disjoint, necessarily one of them will have empty intersection with my set. And with this definition, we will prove next time several things. *[students applaud and lecturer packs up his notes]*

Definition of Connected Subsets

Definition 2.34 (*Connected Set*): A non-empty subset $A \subseteq X$ is called *„connected“* if it is not disconnected.

Equivalently, A is connected if whenever $U, V \subseteq X$ are open subsets satisfying $U \cap V = \emptyset$ and $A \subseteq U \cup V$, we must have:

$$A \cap U = \emptyset \quad \text{or} \quad A \cap V = \emptyset.$$

Conceptual Summary of Connectedness: Connectedness is formulated as the logical negation of disconnectedness:

- A set is **disconnected** if it can be split into two non-overlapping open pieces.
- A set is **connected** if no such two-piece open separation exists. That is, whenever two disjoint open sets cover A, the set A must be contained entirely within one of them.

Summary of Key Concepts

- **Equivalence of Compactness Notions:** In any metric space (X, d), a subset K is sequentially compact if and only if it is topologically compact (open covers admit finite subcovers).
- **General Metric Consequences:** Compact sets are automatically closed and complete, their closed subsets remain compact, and their continuous images are compact.
- **Extreme Value Theorem:** Every real-valued continuous function $f : K \to \mathbb{R}$ on a compact metric space is bounded and attains its global maximum and minimum on K.
- **Heine–Borel Theorem in $\mathbb{R}^n$:** In Euclidean space, a subset is compact if and only if it is closed and bounded. Boundedness allows reduction to compact cubes $[-N, N]^n$, which are shown to be compact via coordinate-wise Bolzano–Weierstrass extractions.
- **Connectedness:** Subsets that cannot be partitioned into two disjoint, non-empty open sets are connected. The entire space X and the empty set $\emptyset$ are always clopen in any metric space.

CHAPTER 2

Thursday 26 February 2026

Connectedness, Path Connectedness, and the Heine–Cantor Theorem

Course Content Overview

In this lecture, we investigate topological connectedness and uniform continuity in metric spaces:

- **Review of Connectedness:** Definitions of connected and disconnected subsets via disjoint ambient open covers.
- **Connected Subsets of the Real Line:** Characterization of connected subsets of $(\mathbb{R}, d_{\text{Eucl}})$ as intervals, established via contraposition and supremum arguments.
- **Preservation under Continuous Maps:** Proof that continuous maps preserve connectedness, yielding the generalized Intermediate Value Theorem.
- **Path Connectedness:** Definition of continuous curves, paths, loops, and path connectedness; proof that path connectedness implies connectedness.
- **Topologist's Sine Curve:** Construction of a classical counterexample demonstrating that connectedness does not imply path connectedness in $\mathbb{R}^2$.
- **Open Subsets of $\mathbb{R}^n$:** Proof that for open domains in Euclidean space, connectedness and path connectedness are equivalent.
- **Heine–Cantor Theorem:** Proof that every continuous map on a compact metric space is uniformly continuous, using finite open subcovers and radius halving.

Review of Connectedness and Disconnected Sets

Mündliches Protokoll [00:00:00 - 00:02:08]

...reasons to try to persuade you to start on time. None of them worked. One of them was that you are Swiss and you should start on time — or some of you are Swiss. The second reason was the conservation of the syllabus. And the third reason, that is maybe the most compelling: you know, I write the exam, and consciously I would never be influenced by something like not starting on time and how this annoys me. But, you know, the unconscious effects of the mind are very powerful. So, this is appealing. Let's try to start on time.

Last time we saw in a metric space the notion of disconnected sets and connected sets. Intuitively, the word explains itself: a disconnected set has more than one piece

[points at the upper drawing of U_1, U_2 covering E on the board], heuristically, and a connected set has only one piece *[points at the lower drawing of E]*, okay?

And this was the definition: a set is disconnected if we can find two disjoint open sets that cover my set *[writes $E \subset U_1 \cup U_2$ on the board]*. So, if there exist two disjoint open sets such that, first, E needs to be covered by both — $E \subset U_1 \cup U_2$, this is important —, and second, both need to have non-trivial intersection with my set E, okay?

So, this is disconnected, and connected is otherwise. Any set that is not disconnected, we call it connected.

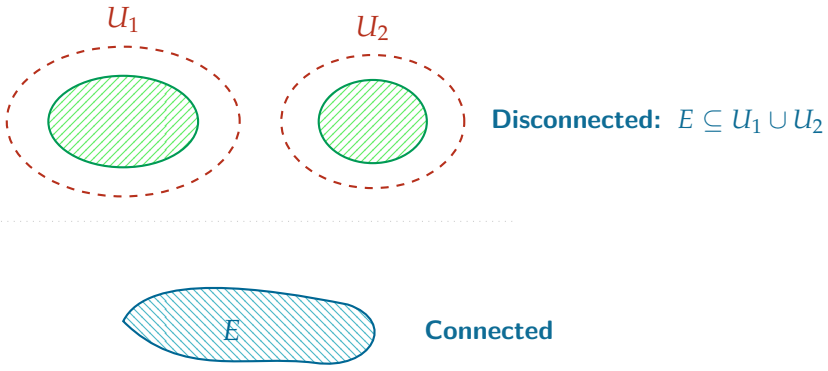
And now we will see, for open sets in Euclidean space, another characterization that is even more, sort of, illuminating. But we will need to first study the connected subsets of $\mathbb{R}$.

Let (X, d) be a metric space.

Definition 2.35 (*Disconnected and Connected Sets*): A subset $E \subseteq X$ is called *„disconnected“* if there exist open sets $U_1, U_2 \subseteq X$ such that:

- **(i)** $U_1 \cap U_2 = \emptyset$ (the sets U_1 and U_2 are disjoint),
- **(ii)** $E \subseteq U_1 \cup U_2$ (the union covers E),
- **(iii)** $E \cap U_1 \neq \emptyset$ and $E \cap U_2 \neq \emptyset$ (both open sets have non-trivial intersection with E).

Otherwise, E is called *„connected“* (i.e., if no such pair of open sets exists).



Topological Intuition of Separation: The formal definition requires that the two open sets U_1, U_2 are mutually disjoint and each captures a non-empty piece of E. The open sets act as ambient *„scissors“* that partition E into two topologically isolated pieces. If no such ambient disconnection can be formed, the set is indivisible and topologically connected.

Connected Subsets of the Real Line

Characterization of Connected Subsets of $\mathbb{R}$

Mündliches Protokoll [00:02:08 - 00:03:33]

The next proposition will be called „*connected subsets of* $\mathbb{R}$" *[writes Prop (connected subsets of* $\mathbb{R}$*)]*.

It says the following: In $\mathbb{R}$, usually with the Euclidean distance, right? *[writes* $(\mathbb{R}, d_{Eucl})$*]* A set E is connected — say $E \subset \mathbb{R}$ *[writes* $E \subset \mathbb{R}$*]* is connected *[writes is connected]* — if and only if *[writes iff]* E is an interval *[writes* E *is an interval]*.

You know from Analysis 1 what an interval is. It can be open or closed from any of the sides; it could be from minus infinity to a value, to plus infinity. Any interval works here. And this is an if and only if. So, these are precisely the connected subsets of $\mathbb{R}$.

Proposition 2.36 (*Connected Subsets of* $\mathbb{R}$): In the Euclidean metric space $(\mathbb{R}, d_{\text{Eucl}})$, a subset $E \subseteq \mathbb{R}$ is connected if and only if E is an interval:

$$E \subseteq \mathbb{R} \text{ is connected} \iff E \text{ is an interval.} \tag{2.76}$$

Mündliches Protokoll [00:03:33 - 00:05:32]

Now, in the proof, I will use a characterization of what is an interval that I leave to you as an exercise to prove. This is easy with what you know from $\mathbb{R}$. It is that a set is an interval if and only if — a subset of $\mathbb{R}$, eh? I'm talking about subsets of $\mathbb{R}$ — is an interval if and only if *[draws a box around the formula in colored chalk]* for every pair of points x_1 and x_2 in E with, let's say, $x_1 \leq x_2$, the closed interval $[x_1, x_2]$ is contained in E.

So, this is a practical characterization of what is an interval, right?

Why? Well, because when you have a set here *[draws a real line with a subset* E *and points* x_1, x_2 *on the chalkboard]* and you tell me that whenever I have two points in the set, all of this interval between the two points needs to be contained in my set, right? Then I can consider the supremum of points such that the interval is contained in my set — the supremum for the right or the infimum for the left —, and then this will give me the endpoints of the interval: the infimum and the supremum. And then there are two possibilities: maybe the endpoint is included, and then the interval is closed from that side, or it is not included. This I can choose.

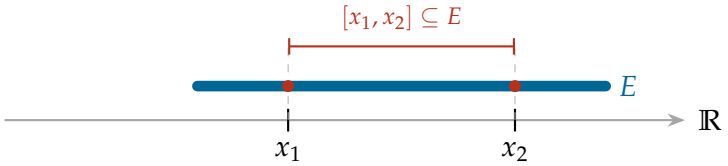
With this characterization, you have all possibilities of intervals, right?

Characterization of Intervals via Convexity

Convexity Characterization of Intervals

A subset $E \subseteq \mathbb{R}$ is an *„interval“* if and only if it contains all intermediate points between any two of its elements:

$$E \text{ is an interval} \iff \forall x_1, x_2 \in E \text{ with } x_1 \leq x_2 \implies [x_1, x_2] \subseteq E. \tag{2.77}$$



The *„No-Gaps“* Property of Intervals: A subset of the real line has no *„holes“* if and only if every segment connecting two members of the set is entirely contained within the set. Geometrically, defining the endpoints as the infimum and supremum of E reproduces all standard interval types in $\mathbb{R}$ (open, closed, half-open, bounded, or unbounded).

Proof of Proposition 2.36

Forward Implication: Connectedness Implies Interval

Mündliches Protokoll [00:05:32 - 00:07:53]

So, let's prove this *[writes proof]*.

There are two implications, no? I want to prove... *[draws $\implies$ and $\impliedby$ next to the equivalence]* There are these two implications, and both are quite similar. This is slightly easier: the implication to the right. If a set is connected, it must be an interval.

I will prove it by contraposition, right?

By contraposition *[writes $\implies$]*, assume E is not an interval, right? *[writes Assume E is not interval]*

Then, using this characterization, there exist two points x_1, x_2 with $x_1 < x_2$ — two points in E, sorry — and a third point, let me call it y... such that $x_1 < y < x_2$. The point y is in the complement of E, sorry, it is not in E; it is in $\mathbb{R} \setminus E$. *[writes $\implies \exists x_1, x_2 \in E$, $\exists y \in \mathbb{R} \setminus E$ such that $x_1 < y < x_2$]*

Let me repeat: If this does not hold, if E is not an interval, it means that there are two points... They cannot be equal, because I need a point in between; they are different points, so I can assume that $x_1 < x_2$. And then there is a third point in between that does not belong to E — it is in the complement of E, right?

Such that $x_1 < y < x_2$.

Mündliches Protokoll [00:07:53 - 00:09:30]

And in this situation, maybe I can draw this: This set E is not an interval, right? This is not an interval. And I'm saying that there is a point here x_1, there is a point here x_2, and there is a point in between, y, right? *[draws a real line with two pieces of E separated by a gap containing y]*

And now I want to prove that if E is not an interval, then E is not connected. So, I need to produce two open sets that disconnect E. And how do I do this? Well, the easy ones are these two, right? *[draws yellow intervals around x_1 and x_2]* To the left and to the right.

So, I take the two open sets: $(-\infty, y)$ open union (y, ∞). These are two disjoint open sets that contain E, and both have non-empty intersection with E, right? Because x_1 belongs here and x_2 belongs here. *[writes $\implies (-\infty, y) \cup (y, \infty) \supset E$]*

x_1 is in this set *[writes x_1 under $(-\infty, y)$ and x_2 under (y, ∞)]*, and x_2 is in this set.

Both have non-empty intersection with E, and then E is disconnected, right? *[writes $\implies$ E disconnected $\square$]*

So, this finishes this implication.

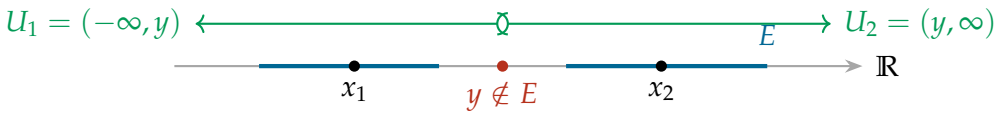
Forward Implication ($\implies$): Contraposition

We prove the forward implication (E connected $\implies$ E is an interval) by contraposition:

$$E \text{ is not an interval} \implies E \text{ is disconnected.}$$

1. Denial of the Interval Property: If $E \subseteq \mathbb{R}$ is not an interval, by the characterization (2.77), the condition fails. Thus, there exist points $x_1, x_2 \in E$ and an intermediate point $y \in \mathbb{R}$ such that:

$$x_1 < y < x_2 \quad \text{with} \quad x_1, x_2 \in E \quad \text{and} \quad y \notin E \quad (y \in \mathbb{R} \setminus E). \tag{2.78}$$



2. Construction of Disconnecting Open Sets: We define the open rays:

$$U_1 := (-\infty, y) \quad \text{and} \quad U_2 := (y, \infty). \tag{2.79}$$

Then:

- U_1 and U_2 are open sets in $(\mathbb{R}, d_{\text{Eucl}})$ and are mutually disjoint: $U_1 \cap U_2 = \emptyset$.
- Since $y \notin E$, every point of E must be strictly less than y or strictly greater than y. Hence:
$$E \subseteq U_1 \cup U_2 = \mathbb{R} \setminus \{y\}.$$
- Since $x_1 \in E \cap U_1$ and $x_2 \in E \cap U_2$, both intersections are non-empty:
$$E \cap U_1 \neq \emptyset \quad \text{and} \quad E \cap U_2 \neq \emptyset.$$

Thus, U_1, U_2 disconnect E, which proves that E is disconnected. By contraposition, if E is connected, then E is an interval.

Backward Implication: Interval Implies Connectedness

Mündliches Protokoll [00:09:30 - 00:11:35]

And now I need to do something similar for the other implication *[writes* $\Longleftarrow$ *|]*.

Right, so we will take this implication also by contraposition.

By contraposition, we will assume that E is not connected, and we need to show that it is not an interval, okay?

Now assume E is disconnected, right? *[writes Assume E disconnected]*

Then there exist, by definition of disconnected, U_1 and U_2 open sets *[writes* $\Longrightarrow$ $U_1, U_2 \subset \mathbb{R}$ *open s.t.* $E \subset U_1 \cup U_2$*]* such that E is contained in the union, right? And they both have non-empty intersection with E.

So, there exist x_i — there exists a point in the intersection for $i = 1$ and 2 *[writes* $\exists x_i \in E \cap U_i,\ i = 1,2$*]*. That they have non-empty intersection means that there is a point in the intersection for each $i = 1$ and 2.

This is the first conclusion.

Now, let me assume... Here I would put: assume without loss of generality that $x_1 < x_2$ *[writes assume w.l.o.g.* $x_1 < x_2$*]*. Because otherwise I change U_1 and U_2, and I can always relabel the sets, right? So, we can assume this.

Setup for Backward Implication

We prove the backward implication (E is an interval $\Longrightarrow$ E is connected) by contraposition:

$$E \text{ is disconnected} \Longrightarrow E \text{ is not an interval.}$$

Assumption: Assume $E \subseteq \mathbb{R}$ is disconnected. By Definition 2.35, there exist open sets $U_1, U_2 \subseteq \mathbb{R}$ such that:

$$U_1 \cap U_2 = \emptyset, \quad E \subseteq U_1 \cup U_2, \quad \text{and} \quad \exists x_1 \in E \cap U_1,\ x_2 \in E \cap U_2. \tag{2.80}$$

Without loss of generality, we may assume $x_1 < x_2$ (otherwise, we simply interchange the labels of U_1 and U_2).

Mündliches Protokoll [00:11:35 - 00:13:35]

And now, what I want to do is to produce a point like here: I want to produce a point that is not in E and that is in between two points in E, which will show that E is not an interval. This is the goal. I need to show that E is not an interval, so I need to produce two points in E and one in between that is not in E, okay?

How do I do that? Because you see, the difficulty is that now the drawing could be like this: you could have a piece here, another smaller piece here, a piece of rationals intersected... So this could be a very dense piece of rationals here, and then another piece here, right? *[draws a fragmented set on the board]*

Sometimes the picture is very complicated. Subsets of $\mathbb{R}$ can be very, very complicated. And it's not that obvious to see: okay, I cut here, right? We need to produce a good cutting point out of our assumption. And the way we will do it is... You see,

U_1 could be like this: U_1 could be the union of these two *[draws U_1 components with parentheses]*, and U_2 could be here in the middle *[draws U_2 interval between U_1 pieces]*. I don't know how U_1 and U_2 are configured.

So, you cannot say, as before, that U_1 is on the left and U_2 is on the right, and then you take the supremum in U_1; because if you take the supremum in U_1, it will be here, and it will not cut your set in two, right?

That is why we will need to do something a bit smarter.

Mündliches Protokoll [00:13:35 - 00:15:10]

What we will do is to consider the sup... We will start at the point... Let's say that x_1 is here, and x_2 is here, right? And we will look at the points that are connected to x_1 inside U_1. So, we will look at the points t — let's say bigger than or equal to x_1, real points — such that the interval $[x_1, t]$ is contained in U_1 *[writes $\{t \geq x_1 \mid [x_1, t] \subset U_1\}$]*.

A point like this could be here. This is U_1, so a point could be here, t, or here, right? These could be the t's.

Because this interval — I don't know if you can see, the interval joining x_1 and t — is all contained in U_1.

And then from this set we will compute the supremum, and this supremum looks nice, because this supremum will be this point over here. We will compute the supremum: t_* is equal to the supremum of this set *[writes $t_* = \sup\{t \geq x_1 \mid [x_1, t] \subset U_1\}$]*.

And this should give us this cutting point over here *[points at the cutting point]*. This would be t_*, hopefully, right?

Definition of the Cutting Point t_*

To prove that E is not an interval, we must find a point $t_* \in \mathbb{R} \setminus E$ strictly between x_1 and x_2:

$$x_1 < t_* < x_2 \quad \text{with} \quad t_* \notin E.$$

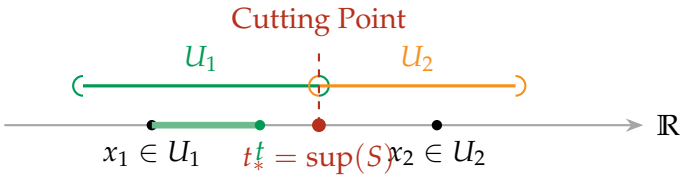
Because U_1 and U_2 can be highly fragmented or intertwined unions of open intervals, we cannot simply take $\sup(U_1)$. Instead, we look at points $t \geq x_1$ such that the entire closed segment $[x_1, t]$ remains contained within U_1.

We define the set:

$$S := \{t \geq x_1 \mid [x_1, t] \subseteq U_1\}. \tag{2.81}$$

Since $x_1 \in U_1$, we have $[x_1, x_1] = \{x_1\} \subseteq U_1$, so $x_1 \in S \neq \emptyset$. We define the candidate cutting point as:

$$t_* := \sup(S) = \sup\{t \geq x_1 \mid [x_1, t] \subseteq U_1\}. \tag{2.82}$$



Mündliches Protokoll [00:15:10 - 00:16:54]

Why is this supremum finite? Because, you know, notice that all the t in this set satisfy that t is less than x_2, right?

Why? Because x_2 is in the second set, U_2. x_2 belongs to U_2. So t needs to be less than this.

This means that this supremum is certainly less than or equal to x_2 *[writes $\leq x_2$]*.

And also, because this is a supremum, even more, right? All these t belong to the complement of U_2, right? Any of these t's that is in the set cannot be in U_2, because then the interval would be outside of U_1. Remember that U_1 and U_2 are disjoint. So these points t that I'm considering here are never in U_2.

And when I take the supremum, because this set is closed — the complement of U_2 is closed —, when I take a supremum of points in a closed set, it belongs to the closed set. So, this t_* belongs to the closed set $\mathbb{R} \setminus U_2$.

So, I proved that t_* is not in U_2.

And this is observation number one: that t_* belongs to $\mathbb{R} \setminus U_2$.

Mündliches Protokoll [00:16:54 - 00:18:05]

Observation number two is that t_* also belongs to the complement of U_1 *[writes **(2)** $t_* \in \mathbb{R} \setminus U_1$]*.

Why? *[writes why?]*

Because U_1 is open. If t_* would belong to U_1, which is open *[writes if $t_* \in U_1$ (open)]*, then all the interval $(t_* - \varepsilon, t_* + \varepsilon)$... Remember the definition of an open set: if a point belongs to an open set, a tiny ball also belongs to the open set. So, this would be contained in U_1 for $\varepsilon > 0$ tiny *[writes $(t_* - \varepsilon, t_* + \varepsilon) \subset U_1$ for $\varepsilon > 0$ tiny]*.

Mündliches Protokoll [00:18:05 - 00:19:25]

There is an ε positive tiny such that this is contained. But then this would contradict the definition of t_*, right? Because I'm saying that t_* is the supremum of the values such that this interval is contained.

Then here it is telling me that besides the interval $[x_1, t_*]$, I can also put the interval $(t_* - \varepsilon, t_* + \varepsilon)$, right? *[writes $[0, t_*] \cup (t_* - \varepsilon, t_* + \varepsilon) \subset U_1$]* Sorry, this is plus ε. And this is still contained in U_1.

So, this would imply that $t_* \geq t_* + \varepsilon$, contradiction, right? *[writes $\Longrightarrow$ $t_* \geq t_* + \varepsilon$ contr.]*

This is a contradiction.

So, this is why t_* cannot belong to U_1.

Board Slip: Interval Basepoint

The lecturer writes $[0, t_*] \cup (t_* - \varepsilon, t_* + \varepsilon)$ on the chalkboard by mistake instead of $[x_1, t_*] \cup (t_* - \varepsilon, t_* + \varepsilon)$. In spoken dialogue, he correctly refers to the interval $[x_1, t_*]$ originating at x_1.

Mündliches Protokoll [00:19:25 - 00:20:33]

The conclusion is that t_* belongs to the complement of U_1, t_* belongs to the complement of U_2, and t_* is between x_1 and x_2. So, t_* is a splitting point, and my set is not an interval.

Consequence: *[writes Consequence:]*

t_* belongs to the complement of $U_1 \cup U_2$ *[writes* $t_* \in \mathbb{R} \setminus (U_1 \cup U_2)$*]*, but remember that these two things cover E, so this is smaller than the complement of E, right? *[writes* $\subset \mathbb{R} \setminus E$*]*

E is contained in this union, so the complements are ordered in the opposite way. When you complement, it changes the order of the inclusion.

And then this proves that, since $x_1 < t_* < x_2$ by construction *[writes* $x_1 < t_* < x_2$*]*, E is not an interval *[writes* $\implies$ *E is not an interval* $\square$*]*.

Okay? Both implications are a bit similar, but the second one is more tricky because we don't know anything about the set E — it could be very complicated — and about the sets U_1 and U_2, and then we need to be careful in defining the cutting point.

Verification: The Cutting Point t_* Leaves the Set E

We establish the following properties for the supremum t_*:

1. Boundedness and Observation 1 ($t_* \notin U_2$):

- For any $t \in S$, we have $[x_1, t] \subseteq U_1$. Since $U_1 \cap U_2 = \emptyset$, no point of $[x_1, t]$ can lie in U_2.
- Since $x_2 \in U_2$, we must have $t < x_2$ for all $t \in S$. Hence, S is bounded from above by x_2:
$$t_* = \sup(S) \leq x_2. \tag{2.83}$$
- Furthermore, every $t \in S$ belongs to $\mathbb{R} \setminus U_2$. Because U_2 is open, its complement $\mathbb{R} \setminus U_2$ is closed. Since the supremum of a subset of a closed set belongs to the closed set:
$$(\mathbf{1}) \quad t_* \in \mathbb{R} \setminus U_2 \implies t_* \notin U_2. \tag{2.84}$$

2. Observation 2 ($t_* \notin U_1$):

- We prove by contradiction that $t_* \in \mathbb{R} \setminus U_1$.
- Assume $t_* \in U_1$. Since U_1 is open, there exists $\varepsilon > 0$ such that the ball $(t_* - \varepsilon, t_* + \varepsilon) \subseteq U_1$.
- By definition of the supremum, there exists $t' \in S$ with $t_* - \varepsilon < t' \leq t_*$. Thus:
$$[x_1, t_* + \varepsilon/2] = [x_1, t'] \cup [t', t_* + \varepsilon/2] \subseteq U_1 \cup (t_* - \varepsilon, t_* + \varepsilon) \subseteq U_1.$$

This implies $t_* + \varepsilon/2 \in S$, which contradicts $t_* = \sup(S)$:

$$t_* \geq t_* + \varepsilon/2 \quad \text{(Contradiction!)}$$

- Therefore:

$$(\mathbf{2}) \quad t_* \in \mathbb{R} \setminus U_1 \implies t_* \notin U_1. \tag{2.85}$$

3. Consequence and Final Contradiction: From (2.84) and (2.85), we conclude:

$$t_* \in (\mathbb{R} \setminus U_1) \cap (\mathbb{R} \setminus U_2) = \mathbb{R} \setminus (U_1 \cup U_2). \tag{2.86}$$

Since $E \subseteq U_1 \cup U_2$, taking complements reverses the inclusion:

$$\mathbb{R} \setminus (U_1 \cup U_2) \subseteq \mathbb{R} \setminus E \implies t_* \notin E. \tag{2.87}$$

Moreover, since $x_1 \in U_1$ and $t_* \notin U_1$, we have $x_1 \neq t_*$, so $x_1 < t_*$. Combined with (2.83) (and since $x_2 \in U_2$ while $t_* \notin U_2$, $t_* < x_2$), we have:

$$x_1 < t_* < x_2 \quad \text{with} \quad x_1, x_2 \in E \quad \text{and} \quad t_* \notin E.$$

Thus, the segment $[x_1, x_2]$ is not contained in E, so by (2.77), E is not an interval. By contraposition, if E is an interval, then E is connected.

Q.E.D.

Board Transition

The lecturer finishes the proof that connected subsets of $\mathbb{R}$ are intervals, wipes the board, lowers the top blackboard, and prepares to introduce continuous mappings between metric spaces.

Preservation of Connectedness under Continuous Maps

Continuous Images of Connected Sets

Mündliches Protokoll [00:21:18 - 00:22:35]

And maybe the last piece, or two, of more abstract mathematics that we will do is... Something that is very useful is the following proposition that says that in a metric space:

Assume that you have a metric space; assume that f from X to Y — so X and Y are both metric spaces — is a continuous map.

Then E connected — so $E \subset X$ connected — implies $f(E) \subset Y$ connected.

So, this is very useful because it's very general: whenever you have a map that is continuous between two metric spaces, you map connected sets to connected sets. And the proof with what you know now is very, very easy.

Proposition 2.37 (*Continuous Image of a Connected Set*)**:** Let (X, d_X) and (Y, d_Y) be metric spaces, and let $f\colon X \to Y$ be a continuous map. If $E \subseteq X$ is connected, then its image $f(E) \subseteq Y$ is also connected:

$$E \subseteq X \text{ connected} \implies f(E) \subseteq Y \text{ connected}. \tag{2.88}$$

Proof of Proposition 2.37

Mündliches Protokoll [00:22:35 - 00:24:35]

The proof is very easy because... What do you know about continuous maps? Now you know that continuous maps are those where the preimages of open sets are open sets. When you take back open sets, they give open sets. And the disconnectedness definition is also related to open sets.

So, what we will do is to prove this by contraposition. We will assume that $f(E)$ is disconnected, we will find two open sets disconnecting this, we will map them back to this side, and then we will find that E is disconnected, right?

So, let's do that. The proof is by contraposition:

Assume that $f(E)$ is disconnected *[writes Assume $f(E)$ is disconnected]*.

Then this by definition means that there exist V_1 and V_2 *[writes $\implies \exists V_1, V_2$]*... V_1 and V_2 open *[writes open]* such that $f(E)$ is contained in their union *[writes s.t. $f(E) \subset V_1 \cup V_2$]* and each of them has non-empty intersection *[writes $V_i \cap f(E) \neq \emptyset$]*: V_i intersected with $f(E)$ is non-empty, for i equal 1 and 2 *[writes $i = 1, 2$]*.

Right?

Well, this is actually the definition of disconnected *[adds $\iff$ before $\exists V_1, V_2$]*: $f(E)$ is disconnected if and only if this happens, by definition.

Proof Setup by Contraposition

We prove the statement by contraposition:

$$f(E) \text{ is disconnected} \implies E \text{ is disconnected.}$$

Assume that $f(E) \subseteq Y$ is disconnected. By definition of disconnectedness (Definition 2.35), there exist open sets $V_1, V_2 \subseteq Y$ such that:

$$V_1 \cap V_2 = \emptyset \qquad \text{(disjoint)}, \tag{2.89}$$

$$f(E) \subseteq V_1 \cup V_2 \qquad \text{(covering the image)}, \tag{2.90}$$

$$V_i \cap f(E) \neq \emptyset \quad \text{for } i \in \{1, 2\} \qquad \text{(non-empty intersections)}. \tag{2.91}$$

Mündliches Protokoll [00:24:35 - 00:26:01]

Now, I will define the sets U_i: $U_1 = f^{-1}(V_1)$ *[writes $U_1 = f^{-1}(V_1)$]*, and $U_2 = f^{-1}(V_2)$ *[writes $U_2 = f^{-1}(V_2)$]*, right?

Now, you notice that E is covered by the preimages, right? Because these two cover the image, so when you map them to the left, you still cover it, right? So, this is $f^{-1}(V_1)$ *[writes $E \subset f^{-1}(V_1)$]* union $f^{-1}(V_2)$ *[writes $\cup f^{-1}(V_2)$]*; this is equal to $U_1 \cup U_2$ by definition *[writes $= U_1 \cup U_2$]*, all right?

And now, these are open *[writes open under $U_1 \cup U_2$]*. These are open.

And these are disjoint *[writes disjoint]*.

And here you need to think one second. You need to think about maps. It's a very easy property, but you need to think one second.

Preimage Pullback Construction

We define the pullbacks in X:

$$U_1 := f^{-1}(V_1) \quad \text{and} \quad U_2 := f^{-1}(V_2). \tag{2.92}$$

Since $f(E) \subseteq V_1 \cup V_2$, taking preimages yields:

$$E \subseteq f^{-1}(f(E)) \subseteq f^{-1}(V_1 \cup V_2) = f^{-1}(V_1) \cup f^{-1}(V_2) = U_1 \cup U_2. \tag{2.93}$$

Furthermore, since $f\colon X \to Y$ is continuous and V_1, V_2 are open in Y, their preimages U_1, U_2 are open in X.

Mündliches Protokoll [00:26:01 - 00:28:02]

When you have two disjoint sets... So, you have an arbitrary map from a set X to a set Y *[draws set X, set Y, and arrow f]*, right? f.

If you take two disjoint sets here *[draws V_1 and V_2 inside Y]*, say V_1 and V_2, and you compute the preimage, this is always disjoint *[draws preimages U_1 and U_2 inside X]*, when you go in this direction *[points left towards X]*, right?

Because a map can send different points to the same point, right? But it never sends one point to multiple images, so it never splits things, right?

What is not true — but I'm not saying that — is that when you take two disjoint sets and you map them here, they could intersect. A map is not bijective, so two different points can be sent to the same point. But in this direction it goes well, right?

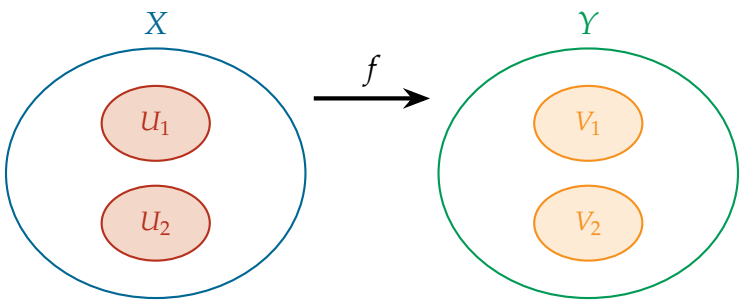
So, you can check this. Since I'm computing preimages of sets, these are disjoint. And open, because the map is continuous.

And also they are non-empty: since this one was non-empty and the other was non-empty, the preimage needs to contain someone, so they are non-empty.

So, non-empty intersection *[writes non-empty intersection with $f^{-1}(f(E)) = E$ on the board]* with the set that is the preimage $f^{-1}(f(E))$, which contains E. All right?

So, E would be disconnected *[writes $\Longrightarrow$ E disconnected]*. Well, it's a contradiction if you assume that E is connected. But we were trying to prove by contraposition, so this is what I wanted: I wanted to prove that $f(E)$ disconnected implies that E is disconnected, all right? *[draws $\square$]*

Disjointness, Non-Emptiness, and Conclusion of Proof



1. **Disjointness of the Preimages:** Since $V_1 \cap V_2 = \emptyset$, for any single-valued map f,

preimages preserve disjointness:

$$U_1 \cap U_2 = f^{-1}(V_1) \cap f^{-1}(V_2) = f^{-1}(V_1 \cap V_2) = f^{-1}(\emptyset) = \emptyset. \tag{2.94}$$

2. Non-Emptiness of the Intersections: For each $i \in \{1,2\}$, by (2.91), there exists $y_i \in V_i \cap f(E)$. Since $y_i \in f(E)$, there exists $x_i \in E$ such that $f(x_i) = y_i \in V_i$. Consequently, $x_i \in f^{-1}(V_i) = U_i$. Since $x_i \in E$, we obtain:

$$x_i \in U_i \cap E \implies U_i \cap E \neq \emptyset \quad \text{for } i \in \{1,2\}. \tag{2.95}$$

3. Conclusion: Properties (2.93), (2.94), and (2.95) establish that the pair (U_1, U_2) disconnects E. Hence, E is disconnected. By contraposition, if E is connected, then $f(E)$ must be connected.

Q.E.D.

The Generalized Intermediate Value Theorem

Mündliches Protokoll [00:28:02 - 00:29:22]

So now, a corollary...

[writes Cor on the board]

Putting together this proposition in arbitrary metric spaces plus the characterization of connected subsets of $\mathbb{R}$, you can prove very easily, in one line, or you can give a new proof of the intermediate value theorem *[writes Cor (Intermediate value thm)]*.

Right? And you can even prove a general version of that for metric spaces.

So, this says that if you have (X,d) a connected metric space *[writes (X,d) connected m.s.]* — or, if you want, a connected subset of a metric space, it's the same — and you have a continuous map f from X to $\mathbb{R}$ *[writes $f: X \to \mathbb{R}$ continuous]*...

Mündliches Protokoll [00:29:22 - 00:31:05]

Right? So these are the assumptions: assume that this is a connected metric space and f is a continuous map.

And assume as well that $f(x) = a$ and $f(y) = b$ *[writes $f(x) = a$, $f(y) = b$, $a \leq b$]*, right? These are real numbers. And let me assume that $a \leq b$.

Then there exists a point z in my space *[writes Then $\exists z \in X$ s.t. $f(z) = c$ $\forall c \in [a,b]$]* such that $f(z) = c$ for every c in the interval $[a,b]$, right?

If you particularize to the case where this is an interval of $\mathbb{R}$, this is exactly the intermediate value theorem. An interval on $\mathbb{R}$ is connected, so if you have a map attaining two different values, you need to attain all intermediate values in between.

This is a theorem that is very related to connectedness, okay?

Korollar 2.38 (*Generalized Intermediate Value Theorem*): Let (X,d) be a connected metric space, and let $f: X \to \mathbb{R}$ be a continuous function. Suppose $x, y \in X$ satisfy $f(x) = a$ and $f(y) = b$ with $a \leq b$. Then for every intermediate value $c \in [a,b]$, there

exists a point $z \in X$ such that:

$$f(z) = c. \tag{2.96}$$

Beweis

Since X is connected and $f\colon X \to \mathbb{R}$ is continuous, by Proposition 2.37, the image $f(X) \subseteq \mathbb{R}$ is a connected subset of $\mathbb{R}$. By Proposition 2.36, every connected subset of $\mathbb{R}$ is an interval. Since $a, b \in f(X)$ and $f(X)$ is an interval, the entire closed interval $[a, b]$ must be contained in $f(X)$ by convexity ((2.77)):

$$[a, b] \subseteq f(X).$$

Therefore, for every $c \in [a, b]$, there exists $z \in X$ such that $f(z) = c$. Q.E.D.

Connection to Classical 1D Calculus: In the special case where $X = [a_0, b_0] \subset \mathbb{R}$ is an interval, X is connected, and Corollary 2.38 immediately recovers the classical Intermediate Value Theorem of Bolzano. The topological formulation reveals that the theorem is not intrinsically about real numbers or derivatives, but purely about the fact that continuous functions preserve connected domains.

Path Connectedness

Curves and Paths in Metric Spaces

Mündliches Protokoll [00:31:05 - 00:32:57]

And now we will see something even more interesting, that is the notion of path connected. This is more intuitive, but first I need to tell you what is a curve in a metric space. And this is something we will use many times in $\mathbb{R}^n$ as well, so it's a good time to introduce it.

[writes Def'n on the board]

So, definition *[writes Def'n]*: If (X, d) is a metric space *[writes (X, d) m.s.]*, a curve *[writes a curve]* — we can call it curve or path *[writes or path]* in X *[writes in X]* — is by definition a map γ from the interval $[0, 1]$ to X *[writes is a map $\gamma\colon [0,1] \to X$]* that is continuous *[writes that is continuous]*, right?

Now, the point $\gamma(0)$ is called the starting point *[writes $\gamma(0)$: starting pt]*, right? And $\gamma(1)$ is called the endpoint of my curve *[writes $\gamma(1)$: end pt]*.

Mündliches Protokoll [00:32:57 - 00:34:27]

And then also, sometimes we say that the curve is closed, or is a loop. So, γ is called closed *[writes γ is called closed (loop)]*, or also loop, when the endpoint and the starting point are the same: so when $\gamma(0) = \gamma(1)$ *[writes when $\gamma(0) = \gamma(1)$]*.

Let me draw a picture, no? This is our abstract space like this *[draws an abstract boundary curve for X]*.

This is the interval $[0,1]$ in $\mathbb{R}$ *[draws the unit interval $[0,1]$]*, right? We draw it like this. Abstract space.

And then a curve will be the map γ, as you can imagine, right? So γ will put this interval in a continuous fashion, will put it here somewhere, no? *[draws continuous curve inside X from $\gamma(0)$ to $\gamma(1)$]* This is $\gamma(0)$, this is $\gamma(1)$.

And then the curve is closed if it closes *[draws a closed loop]*, right?

Definition 2.39 (*Curve / Path and Loop*): Let (X,d) be a metric space.

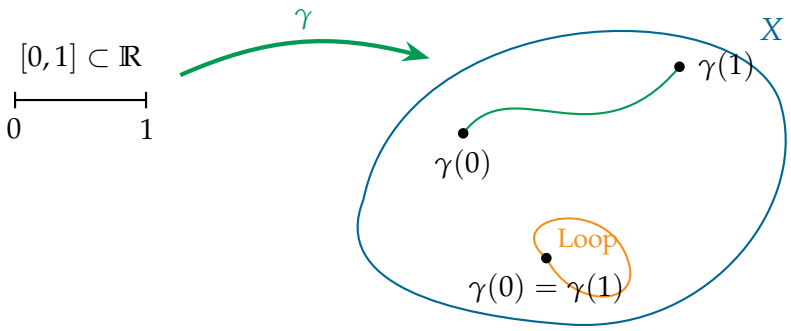
(i) A *„curve“* (or *„path“*) in X is a continuous mapping:

$$\gamma\colon [0,1] \to X. \tag{2.97}$$

The point $\gamma(0) \in X$ is called the *„starting point“* (or initial point), and $\gamma(1) \in X$ is called the *„endpoint“* (or terminal point) of the curve.

(ii) A curve γ is called *„closed“* (or a *„loop“*) if its initial point and endpoint coincide:

$$\gamma(0) = \gamma(1). \tag{2.98}$$



Mündliches Protokoll [00:34:27 - 00:35:42]

Now, to make the drawing, I am drawing this curve very nice and smooth. We still didn't learn what is a smooth curve; we will very soon, no?

But the curve could look very much oscillating and rough. It's just a continuous map. You know that a continuous map could look like that *[draws a jagged, zigzag graph on the board]*, right? This is a continuous map.

So, if you want to imagine a general continuous map, you know the prices of the stock market, right? Something like this. This could be a continuous map that is not smooth.

But we will draw it like this because the intuition is much better, and it will do the same.

Okay. What else? So, this is the definition. And now we will give another definition that I will put on that blackboard.

Maybe I can put this down, and this up.

[adjusts the movable chalkboards]

Smoothness vs. Arbitrary Continuous Curves



Heuristic Drawing: Smooth Path **Arbitrary Continuous Path:** Jagged / Non-Smooth

Topological vs. Differential Curves: A curve $\gamma\colon [0,1] \to X$ in a general metric space only requires topological continuity, not differentiability (which is not even defined in abstract metric spaces). Thus, continuous curves can exhibit fractal-like, nowhere-differentiable behavior (analogous to Brownian motion or financial stock market trajectories). For pedagogical visualization, we draw smooth arcs, as the topological connectedness properties remain identical.

Definition of Path Connectedness

Mündliches Protokoll [00:35:42 - 00:37:27]

Another definition, a very nice one, is the notion of path connected *[writes Def'n (path connectedness)]*:

As always, (X,d) is a metric space *[writes (X,d) m.s.]*.

And $E \subset X$ is a set *[writes $E \subset X$ set]*.

It is called path connected *[writes is called path connected]* if, for every pair of points x, y in my set *[writes if $\forall x, y \in E$]*, there exists a path joining x and y *[writes $\exists$ path joining x and y]*.

More precisely, given the two points x and y, there exists some γ from $[0,1]$ to E continuous *[writes $\implies \exists \gamma\colon [0,1] \to E$ continuous]* such that $\gamma(0) = x$ and $\gamma(1) = y$ *[writes s.t. $\gamma(0) = x$, $\gamma(1) = y$]*.

Mündliches Protokoll [00:37:27 - 00:38:25]

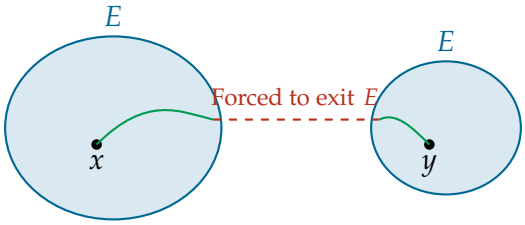
Again, the same drawing: if my set E is like this, and I take the point x here and y here *[draws a disconnected set E consisting of two separate components, with x in one component and y in the other]*, there is no way I'm going to find a continuous curve — at least intuitively — that goes from this point to this point without exiting my set *[draws a curve between x and y that exits E]*, right?

Because the curve would need to go here, and then jump to this point. And continuous curves — this is the point — cannot jump.

But this is an intuition. Now we will need to see what is the connection or the relation between connectedness and path connectedness, right? There is some relation.

Definition 2.40 (*Path Connected Set*): Let (X,d) be a metric space. A subset $E \subseteq X$ is called *„path connected"* if for every pair of points $x, y \in E$, there exists a path within E connecting them. That is:

$$\forall x, y \in E, \quad \exists \gamma\colon [0,1] \to E \text{ continuous such that } \gamma(0) = x \text{ and } \gamma(1) = y. \tag{2.99}$$



Key Distinction: Path in the Subset: Notice that the definition of path connectedness explicitly requires the codomain of the path to be E itself ($\gamma\colon [0,1] \to E$), rather than the ambient space X. In a disconnected set composed of separate pieces, any continuous path attempting to link points in different components must exit the set, violating $\gamma([0,1]) \subseteq E$.

Path Connectedness Implies Connectedness

Mündliches Protokoll [00:38:25 - 00:39:05]

There is one relation that works always, that is the following proposition *[writes Prop]*.

Proposition: Again, in a general metric space *[writes (X,d) m.s.]*, E is path connected *[writes E is path connected]* implies that E is connected *[writes $\implies$ E connected]*.

So, this is the easy implication. And this is always true, right?

[writes proof]

Proposition 2.41 (*Path Connectedness Implies Connectedness*): Let (X,d) be a metric space. If a subset $E \subseteq X$ is path connected, then E is connected:

$$E \subseteq X \text{ path connected} \implies E \text{ connected}. \tag{2.100}$$

Proof of Proposition 2.41

Mündliches Protokoll [00:39:05 - 00:40:24]

Proof.

I know that there are many new techniques and ideas, but now they start to repeat themselves many times, no?

How do I prove this implication? Well, it's good to work with disconnected, so I will prove by contraposition. I will say:

Assume that E is disconnected *[writes proof: Assume that E is disconnected]*, right?

Then this implies that there exist U_1, U_2 open, disjoint *[writes $\Longrightarrow \exists U_1, U_2$ open disjoint]* such that E is contained in the union *[writes $E \subset U_1 \cup U_2$]*, and both of them have non-empty intersection. This means that there exists a point x_i in U_i intersected with E *[writes $\exists x_i \in U_i \cap E$]*.

This I said three times today, so this part should be okay.

Contraposition Setup

We prove the statement by contraposition:

$$E \text{ is disconnected} \implies E \text{ is not path connected.}$$

Assume E is disconnected. By Definition 2.35, there exist open sets $U_1, U_2 \subseteq X$ such that:

$$U_1 \cap U_2 = \emptyset \qquad \text{(disjoint)}, \tag{2.101}$$

$$E \subseteq U_1 \cup U_2 \qquad \text{(covering } E\text{)}, \tag{2.102}$$

$$U_i \cap E \neq \emptyset \quad \text{for } i \in \{1,2\} \qquad \text{(non-empty intersections)}. \tag{2.103}$$

In particular, by (2.103), we can choose points $x_1 \in U_1 \cap E$ and $x_2 \in U_2 \cap E$.

Mündliches Protokoll [00:40:24 - 00:41:35]

And now we want to contradict that this E is path connected.

Assume by contradiction *[writes Assume by contr.]* that there was a map, that there exists a γ *[writes $\exists \gamma$]*, a curve joining x_1 and x_2 *[writes joining x_1 and x_2]*, right?

What does it mean? This means that γ from $[0,1]$ to E *[writes $\gamma\colon [0,1] \to E$]* would be continuous *[writes continuous]*, right? And then $\gamma(0)$ would be x_1 and $\gamma(1)$ would be x_2 *[writes $\gamma(0) = x_1$, $\gamma(1) = x_2$]*.

Hypothesis of a Connecting Path

Suppose for contradiction that E is path connected. Then there exists a continuous path joining x_1 and x_2 within E:

$$\gamma\colon [0,1] \to E \quad \text{continuous with} \quad \gamma(0) = x_1 \quad \text{and} \quad \gamma(1) = x_2. \tag{2.104}$$

Mündliches Protokoll [00:41:35 - 00:43:15]

But U_1 and U_2 are open sets, so we can do the preimage of those open sets by γ, and this would give me... Now I can consider: how do I call them?

I can call them $V_i = \gamma^{-1}(U_i)$ *[writes $V_i = \gamma^{-1}(U_i)$]*, right?

These are subsets of $[0,1]$ *[writes $\subset [0,1]$]*, right? And $[0,1]$ as a metric space.

These would be two subsets of $[0,1]$ that are relatively open — this means open as subsets of this *[writes open]* —, disjoint *[writes disjoint]*, and covering $[0,1]$ *[writes and*

covering $[0,1]$*]*.

And each of them has a point, because one has the zero, and one has the one.

So, this would prove *[writes* $\Longrightarrow$ *]* that $[0,1]$ is disconnected *[writes* $[0,1]$ *disconnected]*.

But we know, because we proved before, that intervals are connected. So, this is a contradiction *[writes contradiction]*.

Pullback Disconnection of the Unit Interval

We define the pullbacks of U_1 and U_2 under γ in the unit interval $[0,1]$:

$$V_1 := \gamma^{-1}(U_1) \quad \text{and} \quad V_2 := \gamma^{-1}(U_2). \tag{2.105}$$

We examine the topological properties of $V_1, V_2 \subseteq [0,1]$:

- **Openness:** Since U_1, U_2 are open in X and $\gamma\colon [0,1] \to X$ is continuous, V_1, V_2 are open in the subspace topology of $[0,1]$.
- **Disjointness:** Since $U_1 \cap U_2 = \emptyset$:
$$V_1 \cap V_2 = \gamma^{-1}(U_1) \cap \gamma^{-1}(U_2) = \gamma^{-1}(U_1 \cap U_2) = \gamma^{-1}(\emptyset) = \emptyset.$$
- **Covering** $[0,1]$**:** Since $\gamma([0,1]) \subseteq E \subseteq U_1 \cup U_2$:
$$[0,1] = \gamma^{-1}(\gamma([0,1])) \subseteq \gamma^{-1}(U_1 \cup U_2) = \gamma^{-1}(U_1) \cup \gamma^{-1}(U_2) = V_1 \cup V_2.$$
- **Non-Emptiness:** Since $\gamma(0) = x_1 \in U_1$, we have $0 \in V_1 \neq \emptyset$. Similarly, since $\gamma(1) = x_2 \in U_2$, we have $1 \in V_2 \neq \emptyset$.

Thus, the pair (V_1, V_2) disconnects the unit interval $[0,1]$:

$$[0,1] \text{ is disconnected.}$$

However, by Proposition 2.36, any interval in $\mathbb{R}$ is connected. This is a **contradiction**. Hence, no such continuous path γ can exist in E. Therefore, E cannot be path connected. By contraposition, if E is path connected, then E is connected.

Mündliches Protokoll [00:43:15 - 00:43:48]

You see, the contradiction comes from the assumption that there is a continuous path joining the points x_1 and x_2 that are here in different, disjoint open sets.

So, this is almost a variation of the proof I gave before, right? That the image of a connected set is connected. *[draws* $\square$ *on the board]*

Q.E.D.

Distinction Between Connectedness and Path Connectedness

A Connected Set That Is Not Path Connected: The Topologist's Sine Curve

Mündliches Protokoll [00:43:48 - 00:45:11]

Okay, so let's continue.

Now we know that in a general metric space, a path connected subset must be connected, right?

But intuitively it looks like the two notions are defining the same type of property. So, what is the difference between the two? When are they equivalent? When are they not?

First I will give you the book example of a set that is connected, but is not path connected, okay? This is a book example. You don't encounter such sets in life very often, but they exist, right?

So, we will take $\mathbb{R}^2$ with the Euclidean distance *[writes $(\mathbb{R}^2, d_{Eucl})$]*.

Why do I need to go to $\mathbb{R}^2$? Because in $\mathbb{R}$, we just proved that connected sets are the same as intervals, and that is the same as path connected. In $\mathbb{R}$, everything is very trivial. But when you go to $\mathbb{R}^2$, you have two directions to play with and things are more interesting.

Mündliches Protokoll [00:45:11 - 00:46:41]

The book example is this set:

This would be y and x *[draws y-axis and x-axis]*. And we will take the set E to be this interval *[writes $E =$]*, this interval over here, right? *[draws a red vertical segment on the y-axis]* The interval $\{0\} \times [0,1]$ *[writes $\{0\} \times [0,1]$]*, this vertical interval, and then the union of the points $(t, \sin(1/t))$ where t is a positive number *[writes $\cup\{(t, \sin(1/t)) \mid t > 0\}$]*.

This is like a curve. You see, t is positive, so I'm parametrizing this curve. When t is very large, $\sin(1/t)$ is very slow. But when t goes to zero, $\sin(1/t)$ will start to oscillate like crazy.

And then it will go like this: here will be super slow, and as you approach zero, it will start to oscillate more and more and more and more and more and more and more *[draws oscillating sine wave approaching the y-axis]*.

The Topologist's Sine Curve: Preliminary Definition

In the Euclidean plane $(\mathbb{R}^2, d_{\text{Eucl}})$, consider the preliminary set:

$$E_{\text{prelim}} := (\{0\} \times [0,1]) \cup \left\{ \left(t, \sin\left(\frac{1}{t}\right)\right) \,\middle|\, t > 0 \right\}.$$

Mündliches Protokoll [00:46:41 - 00:47:25]

Well, do I want to do this? Yes, but then maybe I want to take $[-1,1]$. Then... *[corrects $[0,1]$ to $[-1,1]$ on the board]*

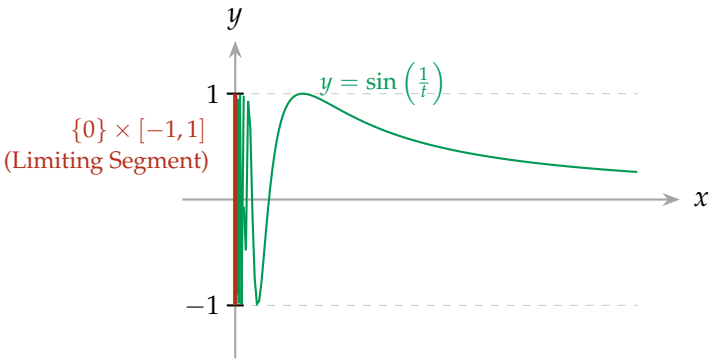
So this would be... yeah. This should hopefully be 1, and this -1 *[labels 1 and -1 on the y-axis and extends the red vertical segment down to -1]*, and the sine oscillates between these two values, right? And you include all this line.

Beispiel 2.42 (*Topologist's Sine Curve*): In the Euclidean space $(\mathbb{R}^2, d_{\text{Eucl}})$, the *„topologist's sine curve"* is the subset $E \subset \mathbb{R}^2$ defined by:

$$E := (\{0\} \times [-1,1]) \cup \left\{ \left(t, \sin\left(\frac{1}{t}\right)\right) \,\middle|\, t > 0 \right\}. \tag{2.106}$$

The set E consists of two parts:

- The limiting vertical segment $L := \{0\} \times [-1,1]$ on the y-axis,
- The oscillating graph $S := \{(t, \sin(1/t)) \mid t > 0\}$.



Mündliches Protokoll [00:47:25 - 00:49:28]

I know that the drawing is very bad, but you can find this everywhere, okay?

So, this is the idea: you take a very oscillating curve, and you put the asymptotic vertical line, no?

Now, this is connected because... I'll tell you the heuristic, and then you need to prove it, hm? Exercise: prove that this is connected.

The idea is that this is a union of two pieces: the curve that is very oscillating — and this curve is connected, right? Because it's path connected, it's a curve — and then the last piece, that is the segment.

Now, you cannot separate this vertical segment from the rest of the curve using open sets, because if you try to put any open set that contains this vertical segment, it will meet some chunk of the curve, because it's very dense: this curve is accumulating to this vertical line.

So, you can never disconnect this piece, the oscillating piece, and the vertical segment using open sets.

Yet, if you try to prove that this set is path connected, you will need a continuous curve that connects this point over here in the vertical segment and this point here.

And you will see that this is impossible: you cannot find a continuous curve from the closed interval $[0,1]$ to here that joins these two points.

Mündliches Protokoll [00:49:28 - 00:49:58]

And the reason is that if you consider such a curve, because a curve is continuous, right? You would have as endpoints of the curve every point on the vertical line.

See, by continuity, you have sequences approaching along the curve: you would have sequences here approaching this point, and sequences here approaching this point — all of the points could be approached.

So then the left endpoint of the curve, or the last endpoint of the curve, should be any point in this segment, which is a contradiction.

To do this carefully is a good exercise: to prove that this set is not path connected. Not an easy exercise, but you can do it now.

Okay, so this is the book example of a set... This set is connected and not path connected *[writes This set is connected and not path connected and boxes the example on the board]*.

Properties of the Topologist's Sine Curve

Proposition 2.43 (*Connectedness vs. Path Connectedness of E*): The topologist's sine curve $E \subset \mathbb{R}^2$ satisfies:

$$E \text{ is connected}, \quad \text{but} \quad E \text{ is not path connected}. \tag{2.107}$$

Heuristic Justification:

- **Why E is connected:** The oscillating piece $S = \{(t, \sin(1/t)) \mid t > 0\}$ is the continuous image of the connected interval $(0, \infty)$ under the continuous map $t \mapsto (t, \sin(1/t))$. Hence, S is connected. The vertical segment $L = \{0\} \times [-1, 1]$ consists entirely of accumulation points of S (since for any $y_0 \in [-1, 1]$, the sequence $t_k = \frac{1}{\arcsin(y_0) + 2\pi k} \to 0^+$ gives $(t_k, \sin(1/t_k)) = (t_k, y_0) \to (0, y_0)$). Thus, $E \subseteq \overline{S}$. Since any set sandwiched between a connected set and its topological closure is connected ($S \subseteq E \subseteq \overline{S}$), E is connected. Any open set containing L necessarily intersects infinitely many oscillations of S, making topological separation by disjoint open sets impossible.
- **Why E is not path connected:** Suppose there were a continuous path $\gamma \colon [0, 1] \to E$ connecting a point on L, say $(0, 0)$, to a point on S, say $(1, \sin(1))$. Let $t_0 = \sup\{t \in [0, 1] \mid \gamma(t) \in L\}$. By continuity of γ, $\gamma(t_0) \in L$. For $t > t_0$, $\gamma(t) \in S$. Writing $\gamma(t) = (\gamma_x(t), \gamma_y(t))$, we have $\gamma_x(t) \to 0^+$ as $t \to t_0^+$. However, the y-component must satisfy $\gamma_y(t) = \sin(1/\gamma_x(t))$, which oscillates wildly between -1 and 1 as $\gamma_x(t) \to 0^+$. Hence, $\lim_{t \to t_0^+} \gamma_y(t)$ cannot exist, contradicting the continuity of γ at t_0.

Equivalence of Connectedness and Path Connectedness for Open Sets in $\mathbb{R}^n$

Theorem Formulation and Equivalence Setup

Mündliches Protokoll [00:49:58 - 00:51:53]

This being said, you know that in analysis there are always these pathological examples that you need to know, but you also need not to be very worried about them, because life in general is much better than what the counterexamples show, right?

So, this being said, there is a theorem that is, for practical purposes, what we will apply all the time: that when you have in $\mathbb{R}^n$ Euclidean, in the Euclidean space *[writes Thm* $(\mathbb{R}^n, d_{Eucl})$*]*, an open subset $U \subset \mathbb{R}^n$ *[writes* $U \subset \mathbb{R}^n$ *open]* — for any open set — U connected is equivalent to U path connected *[writes* U *connected* $\iff$ U *path connected]*.

For open sets of Euclidean space, of $\mathbb{R}^n$, right? And this is what we are going to prove now.

Now, this implication was already proven, because it's general: U path connected implies connected we already proved in a much more general setup *[writes* $\Leftarrow$ ✓*]*. This implication always works.

And what remains to be shown is that implication *[writes* $\Rightarrow$*]*: that in $\mathbb{R}^n$, U connected implies U path connected *[draws a box around* $\Rightarrow$*]*.

Satz 2.44 (*Connectedness Equals Path Connectedness for Open Sets in* $\mathbb{R}^n$):** Let $U \subseteq \mathbb{R}^n$ be an open subset of the Euclidean space $(\mathbb{R}^n, d_{\mathrm{Eucl}})$. Then U is connected if and only if U is path connected:

$$U \text{ is connected} \iff U \text{ is path connected.} \tag{2.108}$$

Proof of Theorem 2.44

Reduction to a Fixed Basepoint

Mündliches Protokoll [00:51:53 - 00:54:10]

Okay, so goal... *[writes Goal:]*

How will we prove that? We will have our connected set U... *[draws an open connected domain* U *on the chalkboard]*

This U, I draw it connected, but just to make the drawing, no? And then I will pick a point x_0 in U *[marks* x_0 *in* U*]*.

Pick x_0 in U and prove we can join it with any other point x in U by a path *[writes Pick* $x_0 \in U$ *and prove we can join it with any other point* $x \in U$ *by a path, and boxes it in yellow chalk]*.

My goal is: I will fix this point, and now I will take any point x here *[marks point* x *in* U*]*, and I will show that I can always join x_0 and x by some path *[draws a path from* x_0 *to* x*]*.

And since this is arbitrary, now you can tell me: okay, is this enough? Why, if you have this, you can join x and y, right? Imagine that now you want to join x and y *[marks point* y *in* U*]*.

So, if I have a path going from x_0 to x, and another path going from x_0 to y *[draws dashed path from x_0 to y]*, there is always a way to construct a path from x to y.

How do you do it? You take the path from x_0 to x, you reverse it, and then when you finish that, you compound it with the other path.

Reduction to Connecting to a Fixed Basepoint

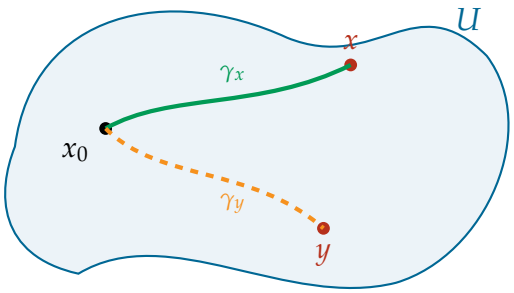
The implication U path connected $\implies$ U connected holds in every metric space by Proposition 2.41. It remains to prove the forward direction:

$$U \subseteq \mathbb{R}^n \text{ open and connected} \implies U \text{ is path connected.}$$

Goal Formulation via Fixed Basepoint

Fix an arbitrary basepoint $x_0 \in U$. We claim that it suffices to prove:

$$\forall x \in U, \quad \exists \text{ path in } U \text{ joining } x_0 \text{ to } x. \tag{2.109}$$



Why Connecting to a Basepoint Suffices: If every point in U can be joined to the fixed point x_0, then for any two points $x, y \in U$:

(1) There exists a path γ_x from x_0 to x.

(2) There exists a path γ_y from x_0 to y.

By reversing γ_x (running backwards from x to x_0) and concatenating it with γ_y (running from x_0 to y), we obtain a continuous path running from x to y entirely within U.

Operations on Paths: Reversal and Composition

Mündliches Protokoll [00:54:10 - 00:55:55]

Let me write this analytically. This is the composition of paths.

Composing paths *[writes Composing paths]* is something that I will use in this proof many times.

It says: if γ_1 from $[0,1]$ to X *[writes $\gamma_1 \colon [0,1] \to X$]* satisfies that $\gamma_1(0) = x$ *[writes $\gamma_1(0) = x$]* and $\gamma_1(1) = y$ *[writes $\gamma_1(1) = y$]*... Let me call this first path γ_1.

And the second path, γ_2, is a path from $[0,1]$ to X *[writes $\gamma_2 \colon [0,1] \to X$]* joining y and z *[writes $\gamma_2(0) = y$, $\gamma_2(1) = z$]*, right?

Let me do the drawing again. This is my abstract space X *[draws boundary of X]*, and I have a path going from x to y *[draws path from x to y]* — this is γ_1 — and another path going from y to z *[draws path from y to z]*.

The arrow is the direction of motion *[draws directional arrow on γ_1]*, and this is γ_2 *[draws directional arrow on γ_2]*, right?

Mündliches Protokoll [00:55:55 - 00:57:45]

Then I can always do an operation. I can first reverse any of the paths. I can define γ_1 reversed *[writes γ_1^* (reversed)]* — let's say γ_1^*, to call it something. This is the reversed path, that is defined as $\gamma_1^*(t) = \gamma_1(1-t)$ *[writes $\gamma_1^*(t) = \gamma(1-t)$]*.

You see, this is composing γ_1 with a map that is also continuous, right? The composition of continuous maps will be continuous.

By the way, exercise: show this *[writes [Show comp. of continuous maps is cont.]]*. Show that the composition of continuous maps is continuous.

Now you will see that you know many tools to show things very quickly, because if I want to show that the composition of continuous maps is continuous, I can check: okay, what is my definition of continuous? Preimage of an open set is an open set.

So, if I take two preimages: preimage of an open set is an open set, preimage of an open set is an open set, and I'm done, right?

So now it's very easy to prove many things when you have a lot of equivalent definitions of continuity, for instance.

If you believe me that the composition of continuous maps is continuous, this is a new continuous map from $[0,1]$ to X, and you see that now it is going from y to x, because you are going backwards in time. Yeah.

Student Question

The length?

Mündliches Protokoll [00:58:09 - 00:58:35]

The length? Yeah, we will see the length, but not yet. The length is a property mainly of differentiable paths, or piecewise differentiable paths.

For a continuous path, the length can be very large; it can be infinite. But we will define the length later on.

Mündliches Protokoll [00:58:35 - 01:00:18]

So, this is just reversing the path. And what you can always do is compose the two

paths, right?

You can do the composition, let's say γ_3 *[writes $\gamma_3(t) =$]*. And how do you do the composition?

Okay, you say: for t between 0 and $1/2$ *[writes $t \in [0, 1/2]$]*, I will take $\gamma_1(2t)$ *[writes $\gamma_1(2t)$]*, right?

And for t between $1/2$ and 1 *[writes $t \in [1/2, 1]$]*, I will define γ_3 as $\gamma_2(2t-1)$ *[writes $\gamma_2(2t-1)$]*.

And if you think of what it's doing, this is doing this first path γ_1 at twice the speed. Now you will tell me: you don't know what the speed is in this context. But I'm just saying intuitively what it's doing — this is informal language.

So, you are doing this path at twice the speed, and this path at twice the speed, and then in one second you are joining x to z. So this is the composition of paths, right?

Given any path, you can do the inverse and the composition of paths.

And that's why, whenever you can join x_0 to x and x_0 to y, you can also join x to y. You just do the composition of the inverse of this and this, okay? Is this clear? It is intuitive also.

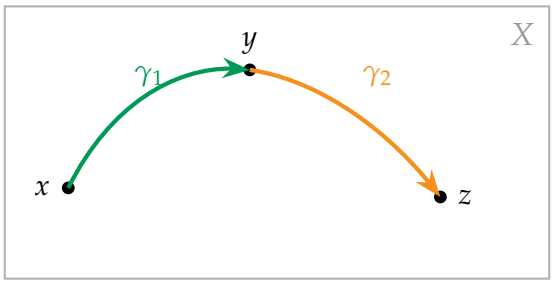
Definition 2.45 (*Reversed Path*): Let $\gamma\colon [0,1] \to X$ be a continuous path from x to y. The *„reversed path"* $\gamma^*\colon [0,1] \to X$ is defined by:

$$\gamma^*(t) := \gamma(1-t). \tag{2.110}$$

The path γ^* is continuous and runs from $y = \gamma^*(0)$ to $x = \gamma^*(1)$.

Definition 2.46 (*Composition / Concatenation of Paths*): Let $\gamma_1\colon [0,1] \to X$ be a path with $\gamma_1(0) = x$, $\gamma_1(1) = y$, and let $\gamma_2\colon [0,1] \to X$ be a path with $\gamma_2(0) = y$, $\gamma_2(1) = z$. The *„concatenated path"* $\gamma_3 = \gamma_1 * \gamma_2\colon [0,1] \to X$ is defined by:

$$\gamma_3(t) := \begin{cases} \gamma_1(2t) & \text{if } t \in [0, 1/2], \\ \gamma_2(2t-1) & \text{if } t \in [1/2, 1]. \end{cases} \tag{2.111}$$



Kinematic Intuition of Concatenation: The piecewise formula traverses the first path γ_1 on the time interval $[0, 1/2]$ at double speed $(2t)$, arriving at the junction point $\gamma_1(1) = y$ at $t = 1/2$. On $[1/2, 1]$, it traverses γ_2 at double speed $(2t-1)$, starting at $\gamma_2(0) = y$ at $t = 1/2$ and terminating at $\gamma_2(1) = z$ at $t = 1$.

Proof of Path Reachability via Connectedness of the Base Component

Mündliches Protokoll [01:00:18 - 01:01:53]

Okay, so now we need to prove our goal over here.

And now we will see a strategy of proof that will appear at least three times in this course. At least three times! It's very typical. And I would say that the first time you see it, it looks super smart, this strategy of proof.

Later, when you see it three times, it starts to appear more natural. But it's a nice strategy of proof.

It's how you use in practice the assumption of connectedness.

Remember that we have a set that is connected, and we want to use this assumption of connectedness to prove that it is path connected. How will you use the assumption of connectedness in practice to derive a conclusion?

So, we will do that: we will define... Proof of the yellow claim, proof of that *[writes proof [box]]*.

Student Question

Uh, so, if you take the γ_1 and γ_2 and you just put one-half in, it seems like you don't get the same result, so I'm confused about that.

Mündliches Protokoll [01:02:05 - 01:02:44]

So, if you put... You mean here?

For this path to be continuous, what do you need? At the point $1/2$, you need to get the same. This is what you're asking.

But you see, if you put $t = 1/2$, you get $\gamma_1(1)$, and $\gamma_1(1)$ was y.

And if you put $\gamma_2(1/2)$, you get $\gamma_2(0)$, that is also y.

Student Response

Okay, thank you.

Mündliches Protokoll [01:02:45 - 01:03:02]

The two paths are correctly compounded. But it's a fair question, and we checked that it was correct, okay?

Mündliches Protokoll [01:03:02 - 01:04:40]

Okay, so to prove this, the trick is to define the set...

Define the set $G \subset U$ *[writes Define set $G \subset U$]*.

G is, by definition, the set of points $x \in U$ such that there exists a path γ from $[0,1]$ to U with $\gamma(0) = x_0$ and $\gamma(1) = x$ *[writes $G := \{x \in U \mid \exists\, path\ \gamma\colon [0,1] \to U\ with\ \gamma(0) = x_0\ and\ \gamma(1) = x\}$]*.

The point x_0 is fixed throughout the proof; the point x_0 will be fixed all the time, it's the same. I picked any point in U.

And now I consider a set G that is the set of points that can be joined by a continuous path with the given point x_0, right?

And my goal is to prove that G is everything, is the whole U *[writes goal $G = U$]*.

If I prove that G is the whole U, I'm happy, because then I prove that any point can be joined to x_0, which is what I claimed, right?

And now this is the cool trick: how we use connectedness.

Path-Component Set G and Proof Goal

Fix an arbitrary basepoint $x_0 \in U$. We define the set of points in U that can be connected to x_0 via a continuous path in U:

$$G := \{x \in U \mid \exists \text{ path } \gamma\colon [0,1] \to U \text{ such that } \gamma(0) = x_0 \text{ and } \gamma(1) = x\}. \tag{2.112}$$

Goal: Prove that $G = U$.

Mündliches Protokoll [01:04:40 - 01:06:17]

So, what we will prove is that:

First, G is open *[writes 1) G open]*. We will prove that G is open.

And we will prove that the complement of G, $U \setminus G$, is also open *[writes 2) $U \setminus G$ open]*.

And you see that G always contains the point x_0 *[writes $\ni x_0$ next to 1)]*. The point x_0 always can be joined by a path that is constant. A constant path is always a nice path. So the point x_0 belongs to G.

And the only possibility, since U is connected *[draws a bracket and writes U connected]*... Using that U is connected, these two things will imply that $U \setminus G$ is empty *[writes $\Longrightarrow\ U \setminus G = \emptyset$]*, right?

A connected set cannot be split as the union of two disjoint non-empty open sets. One of the two needs to be empty; they cannot both be non-empty. This is the very definition of connected, okay?

This is how the strategy goes. So, I need to prove **1)** and **2)**: that G is open, and that $U \setminus G$ is open.

Proof Strategy: Clopen Components in a Connected Set

To show that $G = U$, we demonstrate that G is both open and closed in the relative topology of U (a *„clopen"* subset of U):

1) G is open in $\mathbb{R}^n$, with $x_0 \in G$ (via the constant path $\gamma(t) \equiv x_0$).
2) $U \setminus G$ is open in $\mathbb{R}^n$.

Reduction via Connectedness: Since $x_0 \in G$, we have $G \neq \emptyset$. The pair $(G, U \setminus G)$ partitions U into disjoint open sets:

$$U = G \cup (U \setminus G), \quad G \cap (U \setminus G) = \emptyset.$$

If $U \setminus G$ were non-empty, $(G, U \setminus G)$ would form a disconnection of U, contradicting the hypothesis that U is connected (Definition 2.35). Therefore, we must have:

$$U \setminus G = \emptyset \implies G = U.$$

Mündliches Protokoll [01:06:17 - 01:08:17]

And the key observation for both **1)** and **2)** *[writes Key observation for 1), 2)]* is that if I take any point... This is my set U *[draws boundary of U]*, this is x_0, this is x, right?

For all x in U *[writes $\forall x \in U$]*, we know that there exists a ball of radius r centered at x that is contained in U *[writes $\exists B_r(x) \subset U$ for $r > 0$ (because U open)]*, right?

This would be the ball of radius r centered at x *[draws dashed circle $B_r(x)$ around x]*.

And now, a point y in this ball *[marks $y \in B_r(x)$]*: y in the ball of radius r centered at x belongs to G *[writes $[y \in B_r(x)\ (\in G \iff x \in G)]$]* if and only if x belongs to G.

This is the key observation.

Give me any point x. Give me a ball around it, and there is a y. y belongs to G — that is, y can be connected to x_0 — if and only if x can be connected to x_0.

Why so? Because the points in a ball can always be joined by a segment, which is a path.

Why? *[writes Why]*

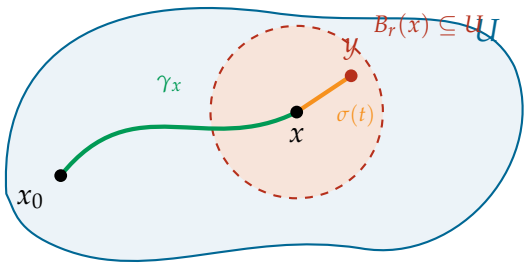
Key Observation: Ball Invariance of Path Reachability

Key Observation for 1) and 2): Since $U \subseteq \mathbb{R}^n$ is open, for every $x \in U$, there exists $r > 0$ such that the Euclidean open ball $B_r(x)$ is entirely contained in U:

$$\forall x \in U, \quad \exists r > 0 \quad \text{such that} \quad B_r(x) \subseteq U. \tag{2.113}$$

Then, for every point $y \in B_r(x)$, we have the equivalence:

$$y \in B_r(x) \implies \big(y \in G \iff x \in G\big). \tag{2.114}$$



Mündliches Protokoll [01:08:17 - 01:10:27]

Because the map that goes from $t \in [0,1]$ is mapped to... *[writes because the map $t \in [0,1] \mapsto (1-t)x + ty$]*

Remember, we are in an open set of $\mathbb{R}^n$, we are in a ball of $\mathbb{R}^n$. So I will use now that this is $\mathbb{R}^n$, so I have the algebraic properties of $\mathbb{R}^n$. I can add points, and can multiply points; $\mathbb{R}^n$ is a vector space.

My map will be $tx + (1-t)y$.

If you think of what this map is doing in $\mathbb{R}^n$, this is a continuous map, right? Each coordinate is continuous: it's a polynomial, so it is a linear map.

And then also, you see that for $t = 0$, this gives me... Maybe I want to do the opposite: I want to do t here and $1-t$ there. It's the same, but I want one of the two.

[rewrites the convex combination as $(1-t)x + ty$]

So if I put $t = 0$, it gives me x, and if I put $t = 1$, it gives me y, right?

So, the center of the ball with any other point of the ball can also be joined with a path.

And then, compounding paths, I can prove this, right?

If I could join this point with y, then compounding with the radial path, I would join this to x. And the opposite: if I could join this point with x, compounding with the opposite path, I would join to y, right?

So, by composition of paths, this is true.

Convexity of Balls in $\mathbb{R}^n$ and Path Concatenation

Justification of the Key Observation via Convexity: In the vector space $\mathbb{R}^n$, the Euclidean ball $B_r(x)$ is convex. For any $y \in B_r(x)$, the straight line segment connecting x to y is parametrized by:

$$\sigma\colon [0,1] \to B_r(x), \quad \sigma(t) := (1-t)x + ty. \tag{2.115}$$

The map σ is affine (linear in each coordinate), hence continuous, with:

$$\sigma(0) = x \quad \text{and} \quad \sigma(1) = y.$$

Since $B_r(x)$ is convex, $\sigma([0,1]) \subseteq B_r(x) \subseteq U$. Therefore, σ is a path in U joining x and y.

- **Forward Direction ($x \in G \implies y \in G$):** If $x \in G$, there exists a path $\gamma_x\colon [0,1] \to U$ from x_0 to x. Concatenating γ_x with σ (Definition 2.46):
$$\gamma_y := \gamma_x * \sigma\colon [0,1] \to U$$
yields a continuous path in U from x_0 to y. Thus, $y \in G$.
- **Backward Direction ($y \in G \implies x \in G$):** If $y \in G$, there exists a path $\gamma_y\colon [0,1] \to U$ from x_0 to y. Concatenating γ_y with the reversed path $\sigma^*(t) = \sigma(1-t)$:
$$\gamma_x := \gamma_y * \sigma^*\colon [0,1] \to U$$
yields a continuous path in U from x_0 to x. Thus, $x \in G$.

This establishes the equivalence $y \in G \iff x \in G$ for all $y \in B_r(x)$.

Mündliches Protokoll [01:10:27 - 01:11:57]

And now, this proves that G is open, right? Because what is the definition of an open set? If a point is in G, all the ball is in G, right? So G is open.

The complement is also open, because if a point is not in G — this is an if and only if, no? — so if a point is not in G, also the ball is not in G.

So, this is open, and this is open.

And now, I repeat how we end the proof: this contains a point, so this needs to be empty, because U is connected.

So, this is how one does this proof, and we will see the same mechanics of proof over and over again.

Whenever we have the assumption that some subset of Euclidean space, some subset of $\mathbb{R}^n$, is connected, many times to use this assumption in the proof, we will consider the set of points in this subset that satisfies some property, and show that this set is open and that the complement is open. And then every point needs to satisfy the property. You will see that we will repeat this several times. This is the first one.

Conclusion of the Proof of Theorem 2.44

Using (2.114), we verify conditions **1)** and **2)**:

1) G is open in $\mathbb{R}^n$: Let $x \in G$. Choose $r > 0$ such that $B_r(x) \subseteq U$. By (2.114), for every $y \in B_r(x)$, since $x \in G$, we have $y \in G$. Therefore, $B_r(x) \subseteq G$. This proves that G is open. Furthermore, $x_0 \in G$, so $G \neq \emptyset$.

2) $U \setminus G$ is open in $\mathbb{R}^n$: Let $x \in U \setminus G$. Choose $r > 0$ such that $B_r(x) \subseteq U$. By (2.114), if any $y \in B_r(x)$ were in G, it would imply $x \in G$, a contradiction. Thus, no point of $B_r(x)$ belongs to G, meaning $B_r(x) \subseteq U \setminus G$. This proves that $U \setminus G$ is open.

Conclusion: Since U is connected and $U = G \cup (U \setminus G)$ with $G \cap (U \setminus G) = \emptyset$ and $G \neq \emptyset$, the open set $U \setminus G$ must be empty:

$$U \setminus G = \emptyset \implies G = U.$$

Consequently, every point $x \in U$ can be joined to x_0 by a path in U. By path inversion and concatenation, any two points $x, y \in U$ can be joined by a continuous path $\gamma_x^* * \gamma_y$ in U. Therefore, U is path connected.

Q.E.D.

Uniform Continuity on Compact Metric Spaces

The Heine–Cantor Theorem

Mündliches Protokoll [01:11:57 - 01:13:20]

Okay, so we have some minutes still left, and I wanted to give you a proof of a result that I forgot to prove last time, and it is an important one.

It is about compact sets. To finish, I will go back to compact sets, and I will prove the

following proposition...

[lowers the right chalkboard and writes Prop.]

Proposition:

Suppose that (X,d) is any metric space *[writes (X,d) ms]*.

(Y,d) is also another metric space *[writes (Y,d) ms, d_X, d_Y if you want]*.

And let f from X to Y be a continuous map *[writes and let $f: X \to Y$ continuous map]*.

Mündliches Protokoll [01:13:20 - 01:15:07]

Now, assume that K is a subset of X that is compact *[writes Assume $K \subset X$ is cpt]*, right?

Then the restriction of f to K, $f|_K$, is uniformly continuous *[writes $f|_K$ is uniformly continuous]*.

We have seen the difference between a continuous map and a uniformly continuous map. We know that uniformly continuous is stronger than continuous.

But on compact sets, the two notions are the same, right? Which is very useful for some proofs.

And moreover, not only this very result, which we will apply many times, but also the idea behind the proof of this result: the fact that when you have something non-uniform or possibly non-uniform, and you restrict to a compact set, things become uniform.

This idea is very powerful, and we will use it also several times to make our life easier throughout the proofs of the course.

Proposition 2.47 (*Heine–Cantor Theorem: Uniform Continuity on Compact Sets*): Let (X,d_X) and (Y,d_Y) be metric spaces, and let $f: X \to Y$ be a continuous mapping. If $K \subseteq X$ is compact, then the restricted map:

$$f|_K: K \to Y$$

is *„uniformly continuous“*.

Proof of Proposition 2.47

Formulation of the Uniform Modulus Goal

Mündliches Protokoll [01:15:07 - 01:16:18]

Let me prove this *[writes proof]*.

In this way, we also review the definition of uniformly continuous, no?

What does it mean, uniformly continuous?

It means that for every ε *[writes $\forall \varepsilon > 0$]*...

Let me write what is the goal.

Goal *[writes (Goal:)]*: I need to prove that, given $\varepsilon > 0$ *[writes given $\varepsilon > 0$]*, there exists $\delta > 0$ *[writes $\exists \delta > 0$]* such that for every x in my compact set *[writes s.t. $\forall x \in K$]*, the ball of radius δ centered at x is mapped by f into the ball of radius ε *[writes $f(B_\delta(x)) \subset B_\varepsilon(f(x))$]*.

And the point of uniform continuity is that this needs to work for every x.

You first give me the ε, and I answer with the δ before knowing which x you pick, okay?

So, this is the goal.

Goal: Uniform Continuity Formulation

Goal: Given $\varepsilon > 0$, show there exists $\delta > 0$ (independent of x) such that:

$$\forall x \in K, \quad f(B_\delta(x) \cap K) \subseteq B_\varepsilon(f(x)). \tag{2.116}$$

Equivalently:

$$\forall x, y \in K, \quad d_X(x, y) < \delta \implies d_Y(f(x), f(y)) < \varepsilon.$$

Open Covering via Local Radii Halving

Mündliches Protokoll [01:16:18 - 01:18:10]

How do we prove this using compactness?

Well, by usual continuity *[writes Usual continuity:]*, we know that for every x in my space — in K in particular — there exists a δ that depends on x *[writes $\forall x \in K \, \exists \delta(x)$ s.t.]* such that the ball of radius $\delta(x)$ centered at x, when I compute where it is mapped, f of this ball, will be contained in the ball of radius $\varepsilon/2$ *[writes $f(B_{\delta(x)}(x)) \subset B_{\varepsilon/2}(f(x))$]*.

ε is arbitrary, so I will put here $\varepsilon/2$ to have a bit of extra space with the ε, right?

So, this is standard continuity. Not uniform.

And the only little problem is that this δ depends on x.

Local Pointwise Continuity with Target Radius $\varepsilon/2$

Let $\varepsilon > 0$ be given. By standard pointwise continuity of f at each point $x \in K$, there exists a local radius $\delta(x) > 0$ such that:

$$f(B_{\delta(x)}(x)) \subseteq B_{\varepsilon/2}(f(x)). \tag{2.117}$$

Because continuity is only assumed pointwise, the radius $\delta(x)$ depends a priori on x.

Mündliches Protokoll [01:18:10 - 01:20:08]

And now, what is the trick?

I will use that K is compact, and I will take the open cover of K, $\mathcal{U}$ *[writes open cover $\mathcal{U}$]*, which is the union of all these balls: the ball of radius $\delta(x)/2$ *[writes $\mathcal{U} = \{B_{\delta(x)/2}(x) \mid x \in K\}$]*.

Here I also divide by 2 for convenience; you will see why I need the division by 2. This is still a ball, with $x \in K$.

Since all these centers of the balls cover K, and all of these are open balls, this is an open cover of K. The only problem is that this could contain many, many balls, right?

But now, since K is compact, this cover has a finite subcover *[writes has finite subcover because K cpt]*.

With this finite subcover, I can immediately tell you what is the right δ, right?

So now, let me put it here.

You see where compactness is helping us, no? First we had one different value of δ for every point x — infinitely many, probably —, and now we pass to only finitely many, which allows us to pick the smallest δ and make one uniform δ work for everyone, okay?

So we have that K is contained in the union for $i = 1$ to capital N *[writes* $K \subset \bigcup_{i=1}^{N} B_{\delta(x_i)/2}(x_i)$ *]*, a finite number of balls of radius $\delta(x_i)/2$ centered at the points x_i, right?

And now, what is my δ? I pick δ equal to the minimum of all these $\delta(x_i)/2$ *[writes* $\delta = \min\{\frac{\delta(x_1)}{2}, \ldots, \frac{\delta(x_N)}{2}\}$ *]*.

Now I define this δ, and I check that this δ works well for me.

Construction of the Finite Subcover and the Uniform Delta

Consider the collection of open balls with halved radii:

$$\mathcal{U} := \left\{ B_{\delta(x)/2}(x) \;\middle|\; x \in K \right\}. \tag{2.118}$$

Since each $x \in K$ is the center of its own ball, $\mathcal{U}$ is an open cover of K:

$$K \subseteq \bigcup_{x \in K} B_{\delta(x)/2}(x).$$

Since K is compact, there exists a finite subcover. That is, there exist finitely many points $x_1, \ldots, x_N \in K$ such that:

$$K \subseteq \bigcup_{i=1}^{N} B_{\delta(x_i)/2}(x_i). \tag{2.119}$$

Definition of the Uniform δ: We define $\delta > 0$ as the minimum of the halved radii:

$$\delta := \min\left\{ \frac{\delta(x_1)}{2}, \frac{\delta(x_2)}{2}, \ldots, \frac{\delta(x_N)}{2} \right\} > 0. \tag{2.120}$$

Since this is the minimum of a finite collection of strictly positive numbers, δ is strictly positive ($\delta > 0$) and independent of x.

Verification via Double Triangle Inequality

Mündliches Protokoll [01:20:08 - 01:22:10]

How do I check this?

Check it works *[writes Check it works:]*:

If x belongs to K and y belongs to the ball of radius this δ centered at x *[writes if $x \in K$ and $y \in B_\delta(x)$ then]*, right?

Then, because all these balls cover, we have x belongs to the ball of radius $\delta(x_i)/2$ centered at x_i for some i *[writes $x \in B_{\delta(x_i)/2}(x_i)$ for some i]*, right?

Then y... Since the distance between x and y *[writes $d(x,y) \leq \delta$]* is less than δ, which is less than or equal to — because it's the minimum — $\delta(x_i)/2$ *[writes $\leq \frac{\delta(x_i)}{2}$]*, then by the triangle inequality, the distance between x_i and y *[writes $d(x_i,y) \leq d(x_i,x) + d(x,y) < \frac{\delta(x_i)}{2} + \frac{\delta(x_i)}{2} = \delta(x_i)$]* is less than $\delta(x_i)$, because it's less than $\delta(x_i)/2 + \delta(x_i)/2$.

Verification Step 1: Inclusions in the Domain Space X

Let $x \in K$ and $y \in K$ satisfy $d_X(x,y) < \delta$ (i.e., $y \in B_\delta(x)$).

1. Finding the Covering Center x_i: By (2.119), there exists an index $i \in \{1,\dots,N\}$ such that:

$$x \in B_{\delta(x_i)/2}(x_i) \implies d_X(x,x_i) < \frac{\delta(x_i)}{2}. \tag{2.121}$$

2. Trapping y in the Full Ball $B_{\delta(x_i)}(x_i)$: By definition of δ in (2.120), we have $d_X(x,y) < \delta \leq \frac{\delta(x_i)}{2}$. Applying the triangle inequality in (X,d_X):

$$d_X(y,x_i) \leq d_X(y,x) + d_X(x,x_i) < \frac{\delta(x_i)}{2} + \frac{\delta(x_i)}{2} = \delta(x_i). \tag{2.122}$$

Consequently, both points x and y lie in the larger ball of radius $\delta(x_i)$ around x_i:

$$x \in B_{\delta(x_i)}(x_i) \quad \text{and} \quad y \in B_{\delta(x_i)}(x_i).$$

In particular, $B_\delta(x) \cap K \subseteq B_{\delta(x_i)}(x_i)$.

Mündliches Protokoll [01:22:10 - 01:23:42]

And now I use the definition of $\delta(x_i)$. This implies that the ball of radius δ centered at x — which, by this reasoning, is contained in the ball of radius $\delta(x_i)$ centered at x_i *[writes $B_\delta(x) \subset B_{\delta(x_i)}(x_i)$]*...

When I compute the image, f of this will be contained in f of that *[writes $f(B_\delta(x)) \subset f(B_{\delta(x_i)}(x_i))$]*, and by definition of $\delta(x_i)$, this was contained in the ball of radius $\varepsilon/2$ centered at $f(x_i)$ *[writes $\subset B_{\varepsilon/2}(f(x_i))$]*.

Right?

Now $\varepsilon/2$ is important, right? Because with this $\varepsilon/2$, you see that the diameter of this ball of radius $\varepsilon/2$ is ε.

So, this ball is contained in the ball of radius ε centered at $f(x)$ *[writes $\subset B_\varepsilon(f(x))$]*.

That is what you wanted to show.

Verification Step 2: Inclusions in the Codomain Space Y

By construction of $\delta(x_i)$ in (2.117), the entire ball $B_{\delta(x_i)}(x_i)$ maps into the ball of radius

$\varepsilon/2$ around $f(x_i)$:

$$f(B_\delta(x) \cap K) \subseteq f(B_{\delta(x_i)}(x_i)) \subseteq B_{\varepsilon/2}(f(x_i)). \tag{2.123}$$

In particular, since both $x, y \in B_{\delta(x_i)}(x_i)$:

$$d_Y(f(x), f(x_i)) < \frac{\varepsilon}{2} \quad \text{and} \quad d_Y(f(y), f(x_i)) < \frac{\varepsilon}{2}.$$

Applying the triangle inequality in (Y, d_Y):

$$d_Y(f(x), f(y)) \leq d_Y(f(x), f(x_i)) + d_Y(f(x_i), f(y)) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \tag{2.124}$$

Therefore, $f(B_\delta(x) \cap K) \subseteq B_\varepsilon(f(x))$. This establishes that $f|_K$ is uniformly continuous.

Mündliches Protokoll [01:23:42 - 01:25:10]

Let me do the picture. Everything when I put these inclusions, you can check, every time is just the triangle inequality.

[draws a circle representing the ball of radius $\varepsilon/2$ centered at $f(x_i)$ with $f(x)$ and $f(y)$ inside]

If you want to see what's going on here, it is just this, no? We have $f(x_i)$, and we have this ball of radius $\varepsilon/2$.

And now we could have two different points: $f(x)$ is here, and $f(y)$ maybe is there.

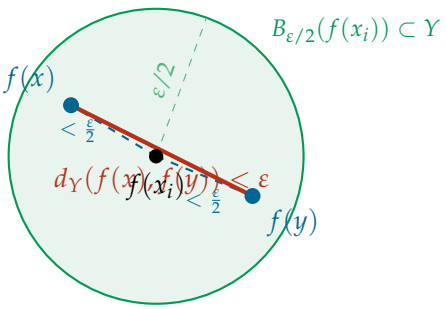
And what is the distance between $f(x)$ and $f(y)$? It could be up to ε, because that is the diameter of the ball. By the triangle inequality, the distance between these two is less than the distance between these two plus the distance between these two, that is $\varepsilon/2 + \varepsilon/2 = \varepsilon$.

From the drawing to the argument, it is always a simple application of the triangle inequality. Some will see it more analytically, doing what these balls are and computing the distances; others will see it better doing a small drawing and seeing where the points are.

So, you need to find your way to visualize these things.

And we continue soon. Okay? *[students applaud, lecturer packs up equipment]*

Geometric Visualization of the $\varepsilon/2$-Ball Diameter Argument



The *„Halving Trick“* in Metric Space Topology: This derivation showcases a fundamental strategy in metric topology:

- **Target Halving ($\varepsilon/2$ in Codomain):** Because the image $f(x)$ and $f(y)$ are only known to land in the same ball around $f(x_i)$, their mutual separation is bounded by the **diameter** of the ball:
$$\operatorname{diam}(B_{\varepsilon/2}(f(x_i))) = 2 \cdot \frac{\varepsilon}{2} = \varepsilon.$$
Setting the radius to $\varepsilon/2$ guarantees $d_Y(f(x), f(y)) < \varepsilon$.
- **Covering Halving ($\delta(x_i)/2$ in Domain):** Halving the covering radii guarantees that whenever $d_X(x,y) < \delta \leq \frac{\delta(x_i)}{2}$, the point y cannot *„escape“* the larger ball $B_{\delta(x_i)}(x_i)$ where continuity is controlled.

Q.E.D.

Summary of Key Concepts

- **Connected Subsets of $\mathbb{R}$:** A subset $E \subseteq \mathbb{R}$ is connected if and only if it is an interval, characterized by convexity ($\forall x_1 \leq x_2 \in E \implies [x_1, x_2] \subseteq E$).
- **Continuous Invariance & Generalized IVT:** Continuous maps preserve connectedness (E connected $\implies$ $f(E)$ connected). For real-valued continuous maps on connected metric spaces, this immediately implies the Intermediate Value Theorem.
- **Curves, Paths, and Path Connectedness:** A path is a continuous map $\gamma\colon [0,1] \to X$. Path connectedness always implies topological connectedness in general metric spaces.
- **The Topologist's Sine Curve:** The set $E = (\{0\} \times [-1,1]) \cup \{(t, \sin(1/t)) \mid t > 0\} \subset \mathbb{R}^2$ is connected (as a dense extension of a connected curve) but not path connected, showing the converse fails in general.
- **Equivalence on Open Subsets of $\mathbb{R}^n$:** For open sets $U \subseteq \mathbb{R}^n$, connectedness is equivalent to path connectedness. The proof relies on showing that path-connected components in U are open (using local convexity of Euclidean balls) and closed in U.
- **Heine–Cantor Theorem:** Every continuous map on a compact metric space is uniformly continuous. The proof demonstrates the power of compactness: passing from infinitely many pointwise moduli $\delta(x)$ to a finite subcover and taking their minimum $\delta > 0$ via the radius-halving trick.